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MIL-STD-40051
31 July 1996

SUPERSEDING
(See Section 6.4)

DEPARTMENT OF DEFENSE STANDARD PRACTICE

TECHNICAL MANUAL PREPARATION



AMSC A7194 AREA TMSS

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FOREWORD

1. This standard is approved for use by the Department of the Army and is available for use by all Departments and Agencies of the Department of Defense (DoD).

2. This MIL-STD-40051 series establishes the technical content requirements and mandatory style and format requirements for the preparation of paper and digital page-oriented technical manuals (TMs) and revisions required to support the various types of equipment and weapon systems within the Department of the Army. The requirements contained in this standard cover operation and maintenance at all levels through depot.

3. This 8-part book form consists of the following parts.

MIL-STD-40051	-	Technical Manual Preparation
MIL-STD-40051-1	-	General Preparation and Assembly Information
MIL-STD-40051-2	-	Introductory Information With Theory of Operation
MIL-STD-40051-3	-	Operator Instructions
MIL-STD-40051-4	-	Troubleshooting Procedures
MIL-STD-40051-5	-	Maintenance Instructions
MIL-STD-40051-6	-	Repair Parts and Special Tools List (RPSTL)
MIL-STD-40051-7	-	Supporting Information

4. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Executive Director, USAMC Logistics Support Activity, ATTN: AMXLS-AP, Redstone Arsenal, AL 35898-7466, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of the document or by letter.

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1. SCOPE.

1.1 Scope. This standard establishes the technical content, style and format requirements for all technical manuals (TMs) covering operation and maintenance, at all levels through depot. The requirements are applicable for both paper and digital page-oriented TMs.

2. APPLICABLE DOCUMENTS.

2.1 General. The documents listed in this section are specified in sections 3, 4, and 5 of MIL-STD-40051-1 through MIL-STD-40051-7. This section does not include documents cited in other sections of this multipart standard or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in section 3, 4, and 5 of MIL-STD-40051-1 through MIL-STD-40051-7, whether or not they are listed..

2.2 Government documents.

2.2.1 Specifications, standards and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

DEPARTMENT OF DEFENSE

MIL-PRF-28000	—	Digital Representation for Communications of Product Data: IGES Application Subsets and IGES Application Protocols.
MIL-PRF-28001	—	Markup Requirements and Generic Style Specification for Electronic Printed Output and Exchange of Text.
MIL-PRF-28002	—	Raster Graphics Representation in Binary Format, Requirements for.
MIL-PRF-28003	—	Digital Representation for Communications of Illustration Data: CGM Application Profile.
MIL-T-31000	—	Technical Data Packages, General Specification for.

STANDARDS

DEPARTMENT OF DEFENSE

MIL-STD-12	—	Abbreviations for Use on Drawings and in Specifications, Standards, and Technical Documents.
MIL-STD-17-1	—	Mechanical Symbols (Other Than Aeronautical, Aerospacecraft, and Spacecraft).

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MIL-STD-17-2 — Mechanical Symbols for Aeronautical, Aerospacecraft, and Spacecraft Use.

MIL-STD-100 — Engineering Drawing Practices

MIL-STD-1309 — Definition of Terms for Testing, Measurement and Diagnostics.

MIL-STD-1686 — Electrostatic Discharge Control Program for Protection of Electrical and Electronic Parts, Assemblies, and Equipment (Excluding Electrically Initiated Explosive Devices).

MIL-STD-1840 — Automated Interchange of Technical Information.

HANDBOOKS

DEPARTMENT OF DEFENSE

MIL-HDBK-113 — Guide for the Selection of Lubricants, Functional Fluids, Preservatives and Specialty Products for use in Ground Equipment Systems.

MIL-HDBK-263 — Electrostatic Discharge Control Handbook for Protection of Electrical and Electronic Parts, Assemblies and Equipment, Excluding Electrically Initiated Explosive Devices (Metric).

MIL-HDBK-275 — Guide for Selection of Lubricants, Fluids, and Compounds for Use in Flight Vehicles and Components.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from Standardization Documents Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5904.)

H4/H8 — Cataloging Handbook: Commercial and Government Entity Code (United States and Canada) - Name to Code.

H6 — Federal Supply Cataloging Handbook.

(Copies of Handbooks H4/H8 and H6 are available from the Commander, Defense Logistics Services Center, Battle Creek, MI 49017-3084.)

2.2.2 Other Government documents and publications. The following other Government documents and publications form a part of this document to the extent specified herein. Unless specified otherwise, the issues are those cited in the solicitation.

DA PAM 738-751 — Functional Users Manual for The Army Maintenance Management System-Aviation (TAMMS-A).

(Application for copies should be addressed to U.S. Army Publications Distribution Center, 2800 Eastern Blvd., Baltimore, MD 21220-2896.)

DOD 5200.1-R — DoD Information Security Program.

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DOD 5200.20	—	Distribution Statements (Other Than Security) on Technical Documents.
DOD 5220.22-M	—	National Industrial Security Program for Operating Manual.
DOD 5230.24	—	Distribution Statements on Technical Documents.

(Copies of DOD 5200.1-R are available from the National Technical Information Service, U.S. Department of Commerce, 5285 Port Royal Road, Springfield, VA 22161. Copies of DOD 5220.22-M are available from the U.S. Government Printing Office, ATTN: Superintendent of Documents, Washington, DC 20402-0001. Copies of DOD 5230.24 and DOD 5230.20 are available from Standardization Documents Order Desk, Bldg 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5904.)

CTA 50-970	—	Expendable/Durable Items (Except Medical, Class V, Repair Parts, and Heraldic Items).
DESCOM-R 702-1	—	Depot Quality Systems.
FM 1-500	—	Army Aviation Maintenance.
TB 750-93-1	—	Functional Grouping Codes: Combat, Tactical, and Support Vehicles and Special Purpose Equipment.
TM 9-1300-206	—	Ammunition and Explosives Standards.
TM 55-1500-335-23	—	Nondestructive Inspection Methods.
TM 55-1500-342-23	—	Army Aviation Maintenance Engineering Manual, Weight and Balance.
TM 1-1500-204-23	—	Aviation Unit Maintenance (AVUM) and Aviation Intermediate Maintenance (AVIM) for General Aircraft Maintenance.

(Copies of these publications are available from the U.S. Army Publications Distribution Center, 2800 Eastern Blvd, Baltimore, MD 21220-2896.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified therein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI Y14.15-1966 (R1973)	—	Electrical and Electronic Diagrams.
ANSI Y32.10	—	Diagrams, Fluid Power, Graphic Symbol for.
ISO 8879	—	Information Processing - Text and Office Systems - Standard Generalized Markup Language (SGML).

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(Application for copies should be addressed to the American National Standards Institute Inc., 1430 Broadway, New York, NY 10018-3308.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E380-91 — The International System of Units for Metric System.

(Application for copies should be addressed to the American Society for Testing Materials, 1916 Race Street, Philadelphia, PA 19103, or from the Standardization Documents Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE 91-1984 — Graphic Symbols for Logic Functions.

IEEE 200-75 — Reference Designators for Electrical and Electronics Parts and Equipments.

IEEE 260-78 — IEEE Standard Letter Symbols for Units of Measurement.

IEEE 280-85 — Letter Symbols for Use in Electrical Science and Electrical Engineering.

IEEE 315A-86 — Graphic Symbols for Electrical and Electronic Diagrams.

IEEE 945-84 — IEEE Recommended Practice for Preferred Metric Units for Use in Electrical and Electronics, Science and Technology.

(Application for copies should be addressed to the Institute of Electrical and Electronics Engineers, Inc., 345 East 47th Street, New York, NY 10017 or from the Standardization Documents Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. DEFINITIONS.

3.1 Additional Authorization List (AAL) items. Items are optional (discretionary), are not essential to operate the end item, and are not listed on engineering drawings. It is not turned in with the end item.

3.2 Adjust. To maintain or regulate within prescribed limits, by bringing into proper position, or by setting the operating characteristics to specified parameters.

3.3 Align. To adjust specified variable elements of an item to bring about optimum or desired performance.

3.4 American National Standards Institute (ANSI). A private sector organization which plans, develops, establishes or coordinates standards, specifications, handbooks or related documents.

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3.5 Army Master Data File (AMDF). The files required to record, maintain, and distribute supply management data between and from Army commands to requiring activities.

3.6 Army Oil Analysis Program (AOAP). Effort to detect impending equipment component failure and determine lubricant condition through periodic analytical evaluation of oil samples.

3.7 Assembled item. An item source coded AO, AF, AH, AL, or AD that is not stocked as an assembly but is assembled from its constituent repair parts.

3.8 Assembly. Two or more parts or subassemblies joined together to perform a specific function and capable of disassembly (e.g., brake assembly, fan assembly, audio frequency amplifier).

NOTE

The distinction between an assembly and subassembly is determined by the individual application. An assembly in one instance may be a subassembly in another where it forms a portion of an assembly.

3.9 Auxiliary equipment. Equipment, accessories, or devices which, when used with basic equipment, extend or increase its capability (e.g., Modified Table of Organization and Equipment (MTOE) items, etc.).

3.10 Aviation Intermediate Maintenance (AVIM). The next higher maintenance level after Unit-Organizational. Aircraft maintenance at this level is the responsibility of, and is performed by, designated maintenance activities for direct support of the using organizations. Its phases normally consist of: calibration, repair, or replacement of damaged or unserviceable parts, components or assemblies; emergency manufacture of nonavailable parts; and technical assistance to using organizations.

3.11 Aviation Unit Maintenance (AVUM). Aircraft maintenance which is the responsibility of, and is performed by, the using organization on its assigned equipment. Its phases normally consist of inspecting, servicing, lubricating, adjusting, and replacing parts, minor assemblies and subassemblies.

3.12 Basic Issue Items (BII). Equipment essential for operation, do emergency repairs, but not listed on the drawings. These items are shipped separately packaged, but must be with the equipment when it is operated or transferred between property accounts.

3.13 Basis of Issue (BOI). The quantity of an item (special tool) authorized for the end item density spread or for the unit level specified.

3.14 Block diagram. A modified schematic diagram in which each group of maintenance-significant components that together perform one or more functions is represented by a single symbol or block. The block or symbol representing the group of components shows simplified relevant input and output signals pertinent to the subject diagram.

3.15 Built-in Test Equipment (BITE). Any identifiable device that is a part of the supported end item and is used for testing that supported end item.

3.16 Bulk material. Material issued in bulk for manufacture or fabrication of support items (e.g., sheet metal, pipe tubing, bar stock, or gasket material); excludes expendable items.

3.17 Calibrate. To determine and cause corrections or adjustments to be made to instruments or test, measuring, and diagnostic equipment used in precision measurement. Consists of comparisons of two

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instruments, one of which is a certified standard of known accuracy, to detect and adjust any discrepancy in the accuracy of the instrument being compared.

3.18 Callout. Anything placed on an illustration to aid in identifying the objects being illustrated, such as index numbers, nomenclature, leader lines, and arrows.

3.19 CALS raster. Compressed scanned raster images (CCITT, Group 4) in accordance with MIL-PRF-28002.

3.20 Commercial and Government Entity Code (CAGEC). A five character code assigned to commercial activities that manufacture or supply items used by the Federal Government and to Government activities that control design or are responsible for the development of certain specifications, standards, or drawings which control the design of Government items. CAGE Code assignments are listed in the H4/H8 CAGE Publications.

3.21 Commodity Command Standard System (CCSS). A system that standardizes the wholesale logistics operations performed by the major subordinate commands of the U.S. Army Materiel Command in the management of secondary items and repair parts.

3.22 Complete part number. Consists of the CAGEC and part number; used for requisition processing. The CAGEC is entered on a requisition form first, followed by the part number.

3.23 Complete repair. Maintenance capacity, capability, and authority to perform all the corrective maintenance tasks of the repair function in a use or user environment in order to restore serviceability to a failed item. Excludes the prescriptive maintenance functions, overhaul and rebuild.

3.24 Component. A constituent part not normally considered to be capable of independent operation; a piece part.

3.25 Components of End Item (COEI). Items identified on the engineering drawing tree which are physically separated and distinct from the end item.

3.26 Comprehensibility. The completeness with which a user in the target audience understands the information in the TM.

3.27 Continuous Acquisition and Life-cycle Support (CALS). A DoD initiative to transition from paper-intensive, non-integrated weapon systems design, manufacturing, and support processes to a highly automated and integrated mode of operation. This transition will be facilitated by acquiring, managing, and using technical data in standardized digital form.

3.28 Computer Graphics Metafile (CGM). One of three standard digital forms for graphics preparation. Defined by MIL-PRF-28003.

3.29 Continuous tone photographs or drawings. Continuous tone photographs or drawings have a continuous gradation of tonal values ranging from light (white) to dark (black), including gray. These tonal values are not created by lines or dots.

3.30 Contracting activity. The Department of Defense (DoD) component, activity, or organization of a using military service, or that organization delegated by a using service, that is responsible for the selection and determination of requirements for TMs.

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3.31 Corrosion Prevention and Control (CPC). Systematic maintenance steps/procedures taken to prevent or retard the gradual destruction and/or pitting of a metal surface or other materials, such as rubber and plastic, due to chemical attack.

3.32 "Current as of" date. Indicates the date that all data in the Repair Parts and Special Tools List (RPSTL) were verified as being current prior to forwarding for printing.

3.33 Degradation. The reduction in systems/subsystems/components performance capability.

3.34 Department of Defense (DoD). The Office of the Secretary of Defense (OSD) (including all boards and councils), the Military Departments (Army, Navy, and Air Force), the Organization of the Joint Chiefs of Staff (OJCS), the Unified and Specified Commands, and the Defense Agencies.

3.35 Department of Defense Ammunition Code (DODAC). An eight character code developed to indicate interchangeability of ammunition and explosive items in Federal Supply Classification (FSC) Group 13. This eight-character code is divided into two parts. The two parts are separated by a hyphen. The first four digits represent the FSC; the letter and last three numerals represent the DoD Identification Code that is assigned to items that are interchangeable in function and use. The eight-character DoD ammunition code is used for such ammunition operations as worldwide stock status reporting and requisitioning when specific items are not required.

3.36 Department of Defense Index of Specifications and Standards (DODISS). The DoD publication that lists unclassified Federal and military specifications and standards, related standardization documents, and voluntary standards approved for use by DoD.

3.37 Depot-level maintenance. Maintenance that is beyond the capability of the unit, direct support, and general support activities. Depot-level maintenance normally consists of overhaul, recondition, manufacture, repair, or modification and requires technical assistance beyond lower maintenance level capability.

3.38 Depot Maintenance Work Requirement (DMWR). A maintenance serviceability document for depot maintenance operations. The document prescribes the essential factors to ensure that an acceptable and cost-effective product is obtained.

3.39 Digital graphics forms. Any of three standard graphics forms acceptable for graphics preparation under this standard. These forms include Computer Graphics Metafile (CGM), CALS raster, and Initial Graphics Exchange Specification (IGES).

3.40 Direct Support (DS) maintenance. Maintenance accomplished on a component, accessory, assembly, subassembly, plug-in unit, or other portion either on the system or after it is removed. The replace function for this level of maintenance is indicated by the letter "F" appearing in the third position of the Source, Maintenance, Recoverability (SMR) code. An "F" appearing in the fourth position of the SMR code indicates complete repair is possible at the direct support maintenance level.

3.41 Disassemble. The step-by-step taking apart (or breakdown) of a spare or functional group-coded item to the level of its least componency identified as maintenance-significant (i.e., assigned an SMR code for the category of maintenance under consideration).

3.42 Document instance. The part of a Standard Generalized Markup Language (SGML) document which contains the markup textual data.

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3.43 Document Type Definition (DTD). The definition of the markup rules for a given class of documents. A DTD or reference to one should be contained in any SGML conforming document.

3.44 Electronic Countermeasures (ECM). Electronic surveillance equipment for detecting and adverting threatening enemy weapons systems.

3.45 Electrostatic Discharge (ESD). Static electricity. A transfer of electrostatic charge between objects of different potentials caused by direct contact or induced by an electrostatic field. Devices such as integrated circuits and discrete devices (e.g., resistors, transistors, and other semiconductor devices) are susceptible to damage from electrostatic discharge.

3.46 End Item Code (EIC). A final combination of end products, component parts, or materials that is ready for its intended use (e.g., tank, mobile machine shop, aircraft, receiver, rifle, recorder).

3.47 Equipment. One or more units capable of performing specified functions.

3.48 Equipment Improvement Recommendation (EIR). Solicitation of suggestions from end item users/operators for means to improve the operation and effectiveness of equipment. SF 368 is the instrument by which suggested improvements are forwarded to the proponent agency.

3.49 Essential. Those systems/subsystems/components that are required for a designated mission or system operation.

3.50 Evacuation. A combat service support function which involves the movement of recovered material from a main supply route; maintenance collection material may be returned to the user, to the supply system for reissue, or to property disposal activities.

3.51 Expendable items. Items, other than repair parts, that are consumed in use (e.g., paint, lubricants, wiping rags, tape, cleaning compounds, sandpaper).

3.52 Final Reproducible Copy (FRC). The final document ready for reproduction and publication as an authenticated TM, including all necessary changes made as a result of validation/verification and acquisition activity conditions of acceptance or approval. The delivery media includes, but is not limited to, reproducible camera-ready copy, direct image copies, negatives, disks, tapes, etc., as specified. For Army, FRC equates to Final Draft Equipment Publication (FDEP).

3.53 Flight safety hazard. An existing or potential condition that can result in a flight mishap.

3.54 Footer. One or more lines of standard text that appear at the bottom of each page (also called feet and running feet).

3.55 Formatting Output Specification Instance (FOSI). The FOSI interprets the style and formatting requirements of the Output Specification (OS). The FOSI can include font, leading, hyphenation characteristics, etc.

3.56 Functional diagram. A type of illustration in which symbols are connected by lines to show relationships among the symbols. The symbols may be rectangles or other shapes, standard electronic

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symbols representing components or functions, or pictorials representing equipment or components. Where appropriate, voltage readings shall be shown. The lines may represent procedures or processes, such as signal or logic flow, and physical items, such as wires. Functional diagram includes schematics, wiring and piping diagrams, logic diagrams, flow charts, and block diagrams.

3.57 Functional Group Code (FGC). A basic (usually two-position) group code assigned to identify major components, assemblies, and subassemblies to a functional system. Subordinate subfunctional groups/subassemblies are coded to relate back to the basic (top position) FGC in a sequential, Next Higher Assembly (NHA) relationship (i.e., top-down breakdown structure).

3.58 General Support (GS) maintenance. Maintenance accomplished on a component, accessory, assembly, subassembly, plug-in unit, or other portion either on the system or after it is removed. The replace function for this level of maintenance is indicated by the letter "H" appearing in the third position of the SMR code. An "H" appearing in the fourth position of the SMR code indicates complete repair is possible at the general support maintenance level.

3.59 Grade Level (GL). Level of reading comprehensibility to which a document is written. The required reading grade level of a document is specified by the contracting activity. For example a level of about seventh grade may be required for materials of a technical nature to be included in maintenance manuals.

3.60 Graphic(s). Any type of presentation or representation which gives a clear visual impression.

3.61 Hardness Critical Item (HCI). A support item that provides the equipment with special protection from electromagnetic pulse (EMP) damage during a nuclear attack.

3.62 Hardness Critical Process (HCP). A process affecting a mission critical item which could degrade system survivability in a nuclear, biological, or chemical hostile environment if hardness were not considered. Nuclear HCPs are processes, finishes, specifications, manufacturing techniques, and/or procedures which are hardness critical, and which, if changed, could degrade nuclear hardness.

3.63 Hardtime scheduled maintenance. Hardtime maintenance is scheduled maintenance conducted at predetermined fixed intervals because of age, calendar, or usage such as operating time, flying hours, miles driven, or rounds fired.

3.64 Header. One or more lines of standard text that appear at the top of each page (also called heads and running heads).

3.65 Horizontal (Landscape) TM format. Positioning of technical manual so that page horizontal (width) dimensions are greater than vertical (height) dimensions.

3.66 Icon. Pictorial representation; visual image to give immediate recognition of a hazard or to provide essential information.

3.67 Illustration. A general term meaning graphic presentations of all types. Illustrations include pictorials, functional diagrams, and line graphs. This term is used instead of such terms as figure, graphic, drawing, diagram, and artwork.

3.68 Initial Graphics Exchange Specification (IGES). One of three standard digital forms for graphics preparation. Defined by MIL-PRF-28000.

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3.69 Inspect. To determine the serviceability of an item by comparing its physical, mechanical, and/or electrical characteristics with established standards through examination (e.g., by sight, sound, or feel).

3.70 Institute of Electrical and Electronics Engineers (IEEE). Membership organization that includes engineers, scientists and students in electronics and allied fields. Founded in 1963, it has over 300,000 members and is involved with setting standards for computers and communications.

3.71 Interchangeability. Defined in this specification as above, the scope of classic interchangeability. The intent/purpose of this specification is to allow fully innovative fixes/repairs to the aircraft. This includes minor modifications that can be made to achieve interchangeability. Capable of being put or used in place of each other.

3.72 Landscape mode. To print an image sideways on the page so that the longest edge of the form corresponds to the horizontal axis.

3.73 Leak rate. The speed or rate of flow of accidental escape of fluid or gas from a system which is caused by damage processes. The leak rate is influenced by such factors as the hole size, internal/external pressures, and fluid level.

3.74 Legend. A tabular listing and explanation of the numbers or symbols on a figure or an illustration.

3.75 Limited repair. Scope of corrective repair authorized to be performed by a level of maintenance lower than the level of authorized complete repair.

3.76 Line Replacement Unit (LRU). An item normally removed and replaced as a single unit to correct a deficiency or malfunction on a weapon system or end item of equipment.

3.77 List of Applicable Publications (LOAP). A listing of publications which are applicable to a piece of equipment or a group of equipment.

3.78 Logic tree. Diagram comprised of a branching series of questions, resulting in a "yes" or "no" answer, leading to determination and resolution of problem.

3.79 Logistics Support Analysis (LSA). The selective application of scientific and engineering efforts undertaken during the acquisition process, as part of the systems engineering process, to assist in acquiring the required support, and providing the required support during the operational phase at minimum cost.

3.80 Maintenance Allocation Chart (MAC). A list of equipment maintenance functions showing maintenance level. The MAC is arranged in functional group code sequence or in top-down, breakdown sequence in the logical order of disassembly following the RPSTL order of assembly/subassembly listings.

3.81 Maintenance level. The separation of maintenance activities or functions in the U.S. Army according to the required skills and available facilities.

3.82 Maintenance task. A series of related maintenance procedures with a definite beginning and end.

3.83 Maximum Time to Repair (MTR). The maximum corrective maintenance downtime within which a specified percent (normally 90 or 95 percent) of all corrective maintenance actions can be accomplished.

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3.84 Meantime Between Corrective Maintenance (MTBCM). For a particular interval, the total functional life of a population of an item divided by the total number of failures within the population during the measurement interval. The definition holds for time, rounds, miles, events, or other measure of life units. (Used only when referring to depot level maintenance.)

3.85 Meantime Between Failures (MTBF). For a particular interval, the total functional life of a population of an item divided by the total number of failures within the population during the measurement interval. The definition holds for time, rounds, miles, events, or other measure of life units.

3.86 Modified Table of Organization and Equipment (MTOE). A modified version of a TOE that prescribed the unit organization, personnel, and equipment needed to perform an assigned mission in a specific geographical or operational environment.

3.87 Modification Word Order (MWO). Detailed instructions (including text and graphics) for making changes/improvements to a particular system in order to bring the system up to date and/or to improve its overall efficiency.

3.88 Module. A subassembly that, in the area of electronic systems, may be removed and replaced without use of soldering equipment or special tools; a module may be encapsulated.

3.89 National Item Identification Number (NIIN). The last nine digits of the National/NATO stock number. The first two digits of the NIIN identify the country assigning the number and the remaining seven digits are a serially assigned number.

3.90 National Stock Number (NSN). A 13-digit number assigned to a repair part to be used for requisitioning purposes.

3.91 Next Higher Assembly. Assembly or subassembly of which subject component(s) or subassembly are a subpart.

3.92 Nomenclature. The approved name or alphanumeric identifier assigned to an item, equipment, or component in agreement with an organized designation system.

3.93 Nondestructive Testing Inspection (NDTI). Testing of a nature which does not impair the useability of the item.

3.94 Nuclear, Biological, and Chemical (NBC). Reference to decontamination procedures performed on equipment and/or personnel exposed to nuclear, biological, and chemical weapons.

3.95 On-condition maintenance. Maintenance performed or an item replacement action performed based upon condition of the item as determined by an evaluation of each item on a scheduled basis.

3.96 Operator maintenance. Consists of inspecting, servicing, lubricating, adjusting, replacing, and repairing those items authorized by Logistic Support Analysis (LSA) and/or Maintenance Allocation Chart (MAC).

3.97 Orphan. Last line of a paragraph pushed to a new page, stranded alone (orphaned) at the top of the page without the rest of its paragraph.

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3.98 Overall Grade Level (OGL). Computed average reading comprehensibility of specified number of document text samples.

3.99 Overhaul. That maintenance effort (service/action) prescribed to restore an item to a completely serviceable/operational condition as required by maintenance standards in appropriate technical publications. Overhaul does not normally return an item to like new condition.

3.100 Overhaul Inspection Procedure (OIP). Routine maintenance inspection conducted just prior to period specified for removal of aircraft is for overhaul or retirement.

3.101 Part Number (P/N). A primary number used to identify an item used by the manufacturer (individual, company, firm, corporation, or Government activity) that controls the design, characteristics, and production of the item by means of its engineering drawings, specifications, and inspection requirements.

3.102 Phased maintenance inspection (aircraft). A thorough and searching examination of the aircraft and associated equipment. Removal of access plates, panels, screens, and some partial disassenbly of the aircraft is required to complete the inspection. Inspections are due after an appointed number of flying hours since new or from the completion of the last inspection.

3.103 Pictorial. A type of illustration showing the physical appearance of equipment or component parts. This term is used instead of such general terms as illustration, drawing, and diagram.

3.104 Portrait mode. To print an image the regular way so that the longest edge of the form corresponds to the vertical axis.

3.105 Prescreening. A process in which a clear material with a dot pattern or crossing opaque lines is used through which an image is photographed in making a halftone.

3.106 Preshop analysis. To determine, prior to beginning maintenance activities, the extent of maintenance required to return the end item, assembly, subassembly, or component to a serviceable condition as specified by the depot level maintenance instructions.

3.107 Preventive maintenance (scheduled maintenance). The performance of scheduled inspections and maintenance functions necessary to keep the equipment in serviceable condition and ready for its primary mission.

3.108 Preventive Maintenance Checklist (PMC). A listing of all before, during, and after operation preventive maintenance checks, including tactical and safety checks, that the operator or crew performs to ensure that the equipment is mission capable and in good operating condition.

3.109 Preventive maintenance daily (aircraft). Inspection of aircraft and associated equipment after the last flight of the mission day or before the first flight of the next day. Some operational checks and removal of screens, panels, and inspection plates may be required to accomplish the inspection.

3.110 Preventive maintenance services inspection (aircraft). Special recurring inspection of aircraft and associated equipment after an appointed number of flying hours or days whichever occurs first (e.g. 10 flying hours or 14 days). Some operational checks and removal of screens, panels and inspection plates may be required to accomplish the inspection.

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3.111 Preventive Maintenance Checks and Services (PMCS). Periodic inspection and maintenance at scheduled intervals to ensure that the equipment and its components remain mission capable and in good operating condition. In aircraft, checks are required of mandatory safety-of-flight items. Lubrication is part of PMCS.

3.112 Procuring activity. The agency responsible for the purchasing of supplies or services.

3.113 Proponent. An Army organization or staff which has been assigned primary responsibility for material or subject matter in its area of interest.

3.114 Quality Assurance (QA). A planned and systematic pattern of all actions necessary to provide adequate confidence that the item or product conforms to established technical requirements.

3.115 Reading Grade Level (RGL). A measurement of reading difficulty of text related to grade levels (such as ninth grade level, fourteenth grade level, etc.).

3.116 Rebuild. Consists of those services/actions necessary for the restoration of unserviceable equipment to a like new condition in accordance with original manufacturing tolerances.

3.117 Reference designator. Letters or numbers, or both, used to identify and locate discreet units, portions thereof, and basic parts of a specific equipment, assembly, or subassembly.

3.118 Reliability, Availability, Maintainability (RAM). Requirements imposed on materiel systems to ensure that they are operationally ready for use when needed, will successfully perform assigned functions, and can be economically operated and maintained within the scope of logistic concepts and policies.

3.119 Reliability Centered Maintenance (RCM). A systematic approach for identifying preventive maintenance tasks for an equipment end item in accordance with a specified set of procedures and for establishing intervals between maintenance tasks.

3.120 Remove/install. To remove and install the same item when required to perform service or other maintenance functions. Install may be the act of emplacing, seating, or fixing into position a spare, repair part, or module (component or assembly) in a manner to allow the proper functioning of an equipment or system.

3.121 Repair. The application of maintenance services (inspect, test, service, adjust, align, calibrate, and/or replace), including fault location/troubleshooting, removal/installation, and disassembly/assembly procedures, and maintenance actions to identify troubles and restore serviceability to an item by correcting specific damage, fault, malfunction, or failure in a part, subassembly, module (component or assembly), end item, or system. Repair is authorized by the LSA/MAC and the assigned maintenance level is shown as the fourth position code of the SMR code.

3.122 Repair part. Any component required for maintenance/repair of an end item/equipment.

3.123 Repair Parts and Special Tools List (RPSTL). The technical document which contains an introduction, list of repair parts, list of special tools, NSN index, part number index, and reference designator index for a specified end item of equipment.

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3.124 Replace. To remove an unserviceable spare or repair part and install a serviceable counterpart in its place. Replace is authorized by the LSA/MAC and the assigned maintenance level is shown as the third position code of the SMR code.

3.125 Revision. A revision is a second or subsequent edition of a manual that supersedes the preceding edition with all its changes.

3.126 Schematic diagram. A graphic representation showing the interrelationship of each component or group of components in the system/equipment. The essential characteristic of these diagrams is that every maintenance-significant functional component is separately represented. Also, where appropriate, voltage readings shall be shown.

3.127 Service. Operations required periodically to keep an item in proper operating condition (i.e., to clean (includes decontaminate, when required), to preserve, to drain, to paint, or to replenish fuel, lubricants, chemical fluids, or gases).

3.128 Set. A unit and necessary assemblies, subassemblies, and parts connected together or used in association to perform an operational function (e.g., radio receiving set, measuring set, radar, or homing set which includes parts, assemblies, and units such as cables, microphones, and measuring instruments).

3.129 Source, Maintenance, and Recoverability (SMR) code. The five-position code containing supply/requisitioning information, maintenance level authorization criteria, and disposition instruction. The first two positions of the SMR code determine how to get an item. The third position represents who can install, replace, or use the item. The fourth position dictates who can do complete repair on the item. The fifth position represents who determines disposition action on unserviceable items.

3.130 Spare part. Any reparable and recoverable component required for the maintenance or repair of an end item or equipment (will have a recoverability code other than "Z" and will be assigned an FGC or subfunctional group code in the MAC).

3.131 Special tools. Those tools that have single or peculiar application to a specific end item/system.

3.132 Specialized Repair Activity (SRA). A level of maintenance usually characterized by the capability to perform maintenance functions requiring specialized skills, disciplined quality control, highly sophisticated and expensive special tools, and TMDE. Its phases normally consist of adjustments, calibration, alignment, testing, troubleshooting, assembly, disassembly, fault isolation, and repair of unserviceable parts, modules, and printed circuit boards (PCB).

3.133 Standard Generalized Markup Language (SGML). A language for document representation that formalizes markup and frees it of system and processing dependencies.

3.134 Standard Generalized Markup Language (SGML) declaration. Defines which characters are used in a document instance, in which syntax the DTD is written, which SGML features are used, etc.

3.135 Subassembly. Two or more parts that form a portion of an assembly or a component replaceable as a whole, but having a part or parts that are individually replaceable (e.g., gun mount stand, window recoil mechanism, floating piston, intermediate frequency strip, mounting board with mounted parts).

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3.136 Supply Catalog (SC). The DA publication which is the configuration control document that provides the user identification of an (SKO) and its components. It also provides user supply management data and is an accountability aid.

3.137 System. A group of items united or regulated by interaction or interdependence to accomplish a set of specific functions.

3.138 Tags. Descriptive markup, as in a start-tag and end-tag.

3.139 Tailoring. The process of evaluating individual potential requirements to determine their pertinence and cost effectiveness. The tailoring of data requirements is limited to the exclusion of information requirement provisions and selecting or specifying applicable requirements.

3.140 Task. A sequence of user actions with a beginning and an end. User tasks relate to installation, checkout, operation, and maintenance of systems or equipment.

3.141 Technical Manuals (TMs). Manuals that contain instructions for the installation, operation, maintenance, and support of weapon systems, weapon system components, and support equipment. TM information may be presented, according to prior agreement between the contractor and the Government, in any form or characteristic, including hard printed copy, audio and visual displays, electronic imbedded media, disks, other electronic devices, or other media. They normally include operational and maintenance instructions, parts lists, and related technical information or procedures exclusive of administrative procedures.

3.142 Test. To verify serviceability by measuring the mechanical, pneumatic, hydraulic, electrical or electronic characteristics of an item and comparing those characteristics with prescribed standards.

3.143 Test, Measurement, and Diagnostic Equipment (TMDE). Any system or device used to evaluate the operational condition of an end item or subsystem thereof, or to identify and/or isolate any actual or potential malfunction. TMDE includes diagnostic and prognostic equipment, semiautomatic and automatic test equipment (with issued software), and calibration test or measurement equipment.

3.144 Time Between Overhaul (TBO) items. Those items having a definite retirement schedule within a defined overhaul interval, e.g., those items which must be replaced within a system assembly, subassembly, or component between scheduled overhauls.

3.145 Title Block Page. The first page after the warning summary in the front matter portion of a TM. It identifies the TM by publication number, date, title and NSN/part number/model of equipment covered in the manual.

3.146 Top-down breakdown. The pyramidal breakdown of an end item, with the top item being the complete end item. The process of breakdown is established from the engineering drawing structure in an NHA progression until the lowest reparable in each family tree group is identified. All nonreparables (spare parts) can be identified in like manner to establish their NHA relationships.

3.147 Unit maintenance. The responsibility of a using organization to perform maintenance on its assigned equipment. It normally consists of inspecting, servicing, lubricating, adjusting, and replacing parts, minor assemblies, and subassemblies. The replace function for this level of maintenance is indicated by the letter "O" in the third position of the SMR code. An "O" appearing in the fourth position of the SMR code indicates complete repair is possible at the unit maintenance level.

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3.148 Usable On Code (UOC). A three-position alphanumeric code representing the applicable configuration in which an item is used. When an item is used on all configurations or when only one configuration is covered by the RPSTL, UOCs shall not be shown.

3.149 User. A person supported by the technical manual.

3.150 Wiring diagram. Diagram illustrating signal flow or wiring connections. Where appropriate, voltage readings shall be shown.

3.151 Work Packages (WPs). Presentation of information functionally divided into individual task packages in the logical order of work sequence. These WPs shall be stand alone general information, operating, maintenance, troubleshooting, parts, and supporting information units containing all information required for directing task performance.

4. GENERAL REQUIREMENTS.

4.1 General. Technical content, style and format requirements for the preparation of technical manuals are provided in MIL-STD-40051, Parts 1 through 7. They contain general technical content information, mandatory style and format requirements, and TM assembly instructions for the preparation and delivery of all TMs and revisions covering operation and maintenance, at all levels through depot.

4.2 Types of technical manuals. Appendix A, Technical Manual Content Selection Matrixes, of MIL-STD-40051 lists specific technical content requirements for each type of maintenance manual, including multilevel TMs, covered by this standard. Each type of TM shall provide in detail the maintenance coverage prescribed for the applicable maintenance level(s) by the Maintenance Allocation Chart (MAC) and SMR-coded items.

4.3 Selective application and tailoring. MIL-STD-40051 contains some requirements that may not be applicable to the preparation of all technical manuals. Selective application and tailoring of requirements contained in MIL-STD-40051 are the responsibility of the contracting activity and shall be accomplished through the use of Appendix A, Technical Manual Content Selection Matrixes, of MIL-STD-40051. The applicability of some requirements is also designated by one of the following statements: unless specified otherwise by the contracting activity; as/when specified by the contracting activity; or when specified by the procuring or proponent activity.

4.4 Style and format. Style and format contained in this standard are considered mandatory and are intended for compliance. The examples provided at the rear of this Part of the standard are an accurate interpretation of the mandatory style and format requirements contained herein and shall be followed to ensure that the conforming Document Type Definitions (DTDs) can be used to develop digital data in accordance with MIL-PRF-28001.

4.4.1 Non-mandatory style and format requirements. Preferred general style and format requirements for the preparation of Army TMs are provided by the procuring activity.

4.5 Preparation of digital data for electronic delivery. Technical manual data prepared in work package format and delivered digitally in accordance with this standard shall be SGML tagged and assembled using modular Document Type Definitions (DTDs) and Formatting Output Specification Instances (FOSIs). The DTDs and FOSIs have been developed in accordance with MIL-PRF-28001 and ISO 8879. Refer to 4.6 for information on obtaining or accessing the modular DTDs and FOSIs.

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4.5.1 Use of the DTDs/FOSIs. The modular DTDs referenced in this standard interpret the technical content and structure for the functional requirements contained in this standard and are mandatory for use. The modular FOSIs referenced herein interprets the style and format. As specified by the contracting activity, FOSIs or style sheets may be used to produce final reproducible paper copy for all TMs prepared in accordance with this standard.

4.6 Obtaining the modular DTDs/FOSIs. The DTDs, FOSIs and associated tag and attribute descriptions, which are SGML constructs, may be obtained from the Army SGML Registry and Library (ASRL). The ASRL assets may be obtained using the methods described in 4.6.1 through 4.6.3.

4.6.1 World Wide Web (WWW). The homepage for the ASRL may be accessed through the Uniform Resource Locator (URL). The URL will be specified by the contracting activity.

4.6.2 Bulletin Board System (BBS). The ASRL BBS may be accessed at (1-888) 880-ASRL. From the Main Menu, select the Files Menu to access the various File Areas in which the DTDs and FOSIs are categorized and available for download. For those who wish to access and browse the ASRL BBS through a graphical user interface, a self-extracting, self-installing Windows-based application, "Wildcat Navigator", is available for free. The file, *wcnav.exe*, is downloadable from "File Area 10 - Navigator" from the Files Menu.

4.6.3 Mail. Requested files will be mailed on 3.5" DOS formatted diskettes or on 1/4" UNIX tar formatted tape.

a. Written request:

Commander, USAPPC
Attn.: ASQZ - PDP
2461 Eisenhower Avenue
Alexandria, VA 22331

b. Telephone request:

Commercial: 1 (703) 428-0508/0504
DSN: 328-0508/0504

5. DETAILED REQUIREMENTS.

This section is not applicable to this Part.

6. NOTES.

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The MIL-STD-40051 series prescribes requirements applicable to various types of technical manuals, and the revisions for these manuals as prepared by or for the Department of the Army.

6.2 Issue of DODISS. When this document is used in acquisition, the applicable issue of the DODISS must be cited in the solicitation. (See 2.2.1.)

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6.3 Tailoring guidance. The contracting activity should tailor any required options offered herein in accordance with Appendix A, Technical Manual Content Selection Matrixes.

6.4 Supersession data. MIL-STD-40051 consolidates, standardizes, and supersedes the requirements contained in the following documents,

MIL-M-63008	—	Manuals, Technical: Acquisition/Preparation Requirements
MIL-M-63014(AV)	—	Manuals, Technical: For Phased Maintenance Checklist
MIL-M-63030(AV)	—	Manuals, Technical: Preventive Maintenance Services and Preventive Maintenance Daily
MIL-STD-361-1	—	Writing Style and Format Requirements.
MIL-STD-361-2	—	Comprehensibility Requirements.
MIL-STD-361-3	—	Requirements for Graphics.
MIL-STD-361-4	—	Quality Assurance Requirements.
MIL-STD-361-5	—	Requirements for Packaging.
MIL-STD-361-6	—	Introductory Information with Theory of Operations Requirements.
MIL-STD-361-7	—	Operator and Preventive Maintenance Instructions.
MIL-STD-361-8	—	Maintenance Instructions.
MIL-STD-361-9	—	Troubleshooting Procedures.
MIL-STD-361-10	—	Lubrication Instructions.
MIL-STD-361-11	—	Repair Parts and Special Tools List (RPSTL).

6.5 Subject term (key word) listing. The following terms are to be used to identify the MIL-STD-40051 series documents during retrieval searches,

Additional authorization list (AAL)
Basic issue items (BII)
Basis of issue (BOI)
CALS raster
Computer graphics metafile (CGM)
Components of end item (COEI)
Depot maintenance work requirement (DMWR)
Expendable and durable items list
Illustrations
Initial Graphics Exchange Specification (IGES)
Introductory information

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Maintenance allocation chart (MAC)
Maintenance instructions
Operator instructions
Quality assurance (QA)
Repair parts and special tools lists (RPSTL)
Security classification
Standard Generalized Markup Language (SGML)
Supporting information
Theory of operation
Troubleshooting procedures
Work package (WP)
Work package identification number

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APPENDIX A
TECHNICAL MANUAL CONTENT SELECTION MATRIXES

A.1 SCOPE

A.1.1 Scope. This appendix contains tables that list all applicable technical content requirements for the development of the following maintenance level technical manuals (TMs). This Appendix is a mandatory part of this standard. The information contained herein is intended for compliance. Copies of the applicable tables will be completed and added as an attachment to the Document Summary List of the contract.

a. Operator level (-10) TM assembly information.

b. Unit maintenance level (-20) and unit maintenance level with Repair Parts and Special Tools List (RPSTL) (-20&P) TM assembly information.

c. Direct support maintenance level (-30) and direct support maintenance level with RPSTL (-30&P) TM assembly information.

d. General support maintenance level (-40) and general support maintenance level with RPSTL (- 40&P) TM assembly information.

e. Combined operator and unit maintenance levels (-12) and operator and unit maintenance levels with RPSTL (-12&P) TM assembly information.

f. Combined operator, unit, and direct support maintenance levels (-13) and operator, unit, and direct support maintenance levels with RPSTL (-13&P) TM assembly information.

g. Combined operator, unit, direct support, and general support maintenance levels (-14) and operator, unit, direct support, and general support maintenance levels with RPSTL (-14&P) TM assembly information.

h. Combined unit and direct support maintenance levels (-23) and unit and direct support maintenance levels with RPSTL (-23&P) TM assembly information.

i. Combined unit, direct support, and general support maintenance levels (-24) and unit, direct support, and general support maintenance levels with RPSTL (-24&P) TM assembly information.

j. Combined direct support and general support maintenance levels (-34) and direct support and general support maintenance levels with RPSTL (-34&P) TM assembly information.

k. Depot Maintenance Work Requirements (DMWR) and DMWR with RPSTL assembly information.

l. Aviation Unit Maintenance (AVUM) level (-20) and AVUM level with RPSTL (-20&P) **(aircraft only)** TM assembly information.

m. Aviation Intermediate Maintenance (AVIM) level (-30) and AVIM level with RPSTL (-30&P) **(aircraft only)** TM assembly information.

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n. Combined AVUM and Aviation Intermediate Maintenance (AVIM) levels (-23) and AVUM and AVIM levels with RPSTL (-23&P) **(aircraft only)** TM assembly information.

o. All RPSTLs (-P) TM assembly information.

p. Aircraft troubleshooting manual.

q. Aircraft preventive maintenance services (PMS).

A.2 APPLICABLE DOCUMENTS.

This section is not applicable to this appendix.

A.3 DEFINITIONS.

This section is not applicable to this appendix.

A.4 GENERAL REQUIREMENTS.

This section is not applicable to this appendix.

A.5 DETAILED REQUIREMENTS.

A.5.1 General. Tables A.1 through A.8 simplify tailoring the technical content requirements of technical manuals prepared using this standard as a guide. The tables indicate which parts of MIL-STD-40051 are applicable and list the content requirements for each type of TM. The content requirements for each applicable TM shall be arranged in the order presented in the tables. Inclusion of the applicable tables of this appendix is mandatory and is intended for compliance.

A.5.2 Intended use. First determine the types of TMs required for each acquisition and then duplicate the table(s) that contain the content requirements for those types of TMs. Indicate the types of TMs needed by filling in the blank after "TM Requirements Matrix for" at the top of each matrix. For each type of TM selected, indicate in the open blocks the "TM Content" desired by entering an "R" for "REQUIRED" content, an "NR" for content that is "NOT REQUIRED", or an "O" for "OPTIONAL" content that may be required in the TM later by the Government, but can not be determined at the time of the contract. All blocks for the selected TM types in Tables A.1 through A.8 must be completed with an "R" or "NR" or "O" for each TM acquisition.

A.5.3 Acquisition requirements. The properly executed Technical Manual Content Selection Matrix table becomes contractually binding when it is made part of the contract, statement of work or any other contractual instrument.

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Table A.1 TM Requirements Matrix for _____

TM Content	-10	-12 -12&P	-13 -13&P	-14 -14&P	MIL-STD-40051 Reference	Element Name	Req'd / Not Req'd
FRONT MATTER	R	R	R	R	1- 5.8.1	<frnt>	
Front cover	R	R	R	R	1- 5.8.1.1	<frntcover>	
Warning summary					1- 5.8.1.2	<warnsum>	
List of effective pages / work packages					1- 5.8.1.3		
Revision transmittal page	R	R	R	R	1-5.8.1.4	<chgsheet>	
Title block page	R	R	R	R	1- 5.8.1.5	<titleblk>	
Table of contents	R	R	R	R	1- 5.8.1.6	<contents>	
How to use this manual					1- 5.8.1.7	<howtouse>	
CHAPTER 1. INTRODUCTORY INFORMATION WITH THEORY OF OPERATION	R	R	R	R	2- 5.1	<gim>	
GENERAL INFORMATION WORK PACKAGE	R	R	R	R	2- 5.1.1	<ginfowp>	
Scope	R	R	R	R	2- 5.1.1.1	<scope>	
Maintenance forms, records, and reports	R	R	R	R	2- 5.1.1.2	<mfrr>	
Reporting equipment improvement recommendations (EIR)	R	R	R	R	2- 5.1.1.3	<eir>	
Hand receipt manuals					2- 5.1.1.4	<handreceipt>	
Corrosion prevention and control (CPC)	R	R	R	R	2- 5.1.1.5	<cpcdata>	
Destruction of Army materiel to prevent enemy use	R	R	R	R	2- 5.1.1.6	<destructmat>	
Preparation for storage or shipment	R	R	R	R	2- 5.1.1.7	<pssref>	
Warranty information					2- 5.1.1.8	<wrntyref>	
Nomenclature cross-reference list					2- 5.1.1.9	<nomenreflist>	
List of abbreviations					2- 5.1.1.10	<loa>	
Quality assurance (QA) (aviation only)					2- 5.1.1.11	<qainfo>	

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Table A.1 TM Requirements Matrix for _____

TM Content	-10	-12 -12&P	-13 -13&P	-14 -14&P	MIL-STD-40051 Reference	Element Name	Req'd / Not Req'd
Safety, care, and handling					2- 5.1.1.12	<sftyinfo>	
Nuclear hardness					2- 5.1.1.13	<hcp>	
Security measures for electronic data					2- 5.1.1.14	<secref>	
Calibration					2- 5.1.1.15	<calref>	
Copyright credit line					2- 5.1.1.19	<copyrt>	
EQUIPMENT DESCRIPTION AND DATA WORK PACKAGE	R	R	R	R	2- 5.1.3	<descwp>	
Equipment characteristics, capabilities, and features	R	R	R	R	2- 5.1.3.1	<eqpinfo>	
Location and description of major components	R	R	R	R	2- 5.1.3.2	<locdesc>	
Differences between models					2- 5.1.3.3	<eqpdiff>	
Equipment data	R	R	R	R	2- 5.1.3.4	<eqpdata>	
Equipment configuration					2- 5.1.3.5	<eqpconfig>	
THEORY OF OPERATION WORK PACKAGE	R	R	R	R	2- 5.1.4	<thrywp>	
SUPPORTING DATA WORK PACKAGE FOR REPAIR PARTS, SPECIAL TOOLS, TMDE, AND SUPPORT EQUIPMENT					2- 5.1.5	<supdatawp>	
CHAPTER X. OPERATOR INSTRUCTIONS	R	R	R	R	3- 5.1	<opim>	
DESCRIPTION AND USE OF OPERATOR CONTROLS AND INDICATORS WORK PACKAGE	R	R	R	R	3- 5.1.1	<ctrlindwp>	
OPERATION UNDER USUAL CONDITIONS WORK PACKAGE	R	R	R	R	3- 5.1.2	<opusualwp>	
Siting requirements					3- 5.1.2.2	<site>	
Shelter requirements					3- 5.1.2.3	<shelter>	
Assembly and preparation for use					3- 5.1.2.4	<assem>	
Initial adjustments, before use and self-test					3- 5.1.2.5	<initial>	
Operating procedures	R	R	R	R	3- 5.1.2.6	<oper>	

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Table A.1 TM Requirements Matrix for _____

TM Content	-10	-12 -12&P	-13 -13&P	-14 -14&P	MIL-STD-40051 Reference	Element Name	Req'd / Not Req'd
Decals and instruction plates					3- 5.1.2.6.3	<instructplt>	
Operating auxiliary equipment					3- 5.1.2.7	<operaux>	
Preparation for movement					3- 5.1.2.8	<prepmove>	
OPERATION UNDER UNUSUAL CONDITIONS WORK PACKAGE	R	R	R	R	3- 5.1.3	<opunuwp>	
Unusual environment / weather					3- 5.1.3.2	<unusualenv>	
Fording and swimming					3- 5.1.3.3	<fording>	
Interim nuclear, biological, and chemical (NBC) decontamination procedures					3- 5.1.3.4	<decon>	
Jamming and electronic countermeasures (ECM) procedures					3- 5.1.3.5	<ecm>	
Emergency procedures					3- 5.1.3.6	<emergency>	
CHAPTER X. TROUBLESHOOTING PROCEDURES		R	R	R	4- 5.4	<tim>	
INTRODUCTION WORK PACKAGE		R	R	R	4- 5.5.1	<tsintrowp>	
MALFUNCTION / SYMPTOM INDEX WORK PACKAGE					4- 5.5.2	<symndxwp>	
TROUBLESHOOTING PROCEDURES WORK PACKAGE		R	R	R	4- 5.5.4	<tswp>	
CHAPTER X. MAINTENANCE INSTRUCTIONS **NOTE** All maintenance work packages shall include a scope of task, initial setup, and all maintenance tasks applicable to the equipment.	R	R	R	R	5- 5.3	<mim>	
SERVICE UPON RECEIPT WORK PACKAGE	NR	R	R	R	5- 5.4.5.1	<surwp>	
Siting					5- 5.4.5.1.1	<siting>	
Shelter requirements					5- 5.4.5.1.2	<shltr>	
Service upon receipt of materiel					5- 5.4.5.1.3	<surmat>	

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Table A.1 TM Requirements Matrix for _____

TM Content	-10	-12 -12&P	-13 -13&P	-14 -14&P	MIL-STD-40051 Reference	Element Name	Req'd / Not Req'd
Installation instructions					5- 5.4.5.1.4	<install>	
Preliminary servicing of equipment					5- 5.4.5.1.5	<preserv>	
Preliminary checks and adjustment of equipment					5- 5.4.5.1.6	<prechkadj>	
Preliminary calibration of equipment					5- 5.4.5.1.7	<precal>	
Circuit alignment					5- 5.4.5.1.8	<calign>	
Ammunition markings					5- 5.4.5.1.9	<markings>	
Classification of defects					5- 5.4.5.1.10	<defect>	
Ammunition handling					5- 5.4.5.1.11	<handling>	
Procedures to activate ammunition					5- 5.4.5.1.12	<arm>	
EQUIPMENT / USER FITTING INSTRUCTIONS WORK PACKAGE (personal use equipment)	NR				5- 5.4.5.2	<perseqpwp>	
PMCS, INCLUDING LUBRICATION INSTRUCTIONS, WORK PACKAGE	R	R	R	R	5- 5.4.5.3	<pmcswp>	
Introduction	R	R	R	R	5- 5.4.5.3.1	<pmcsintro>	
PMCS procedures (table)	R	R	R	R	5- 5.4.5.3.2	<pmcstable>	
MAINTENANCE WORK PACKAGES **NOTE** As applicable, the following maintenance tasks shall be presented in the general order listed below:	R	R	R	R	5- 5.4.5.8	<maintwp>	
Assembly and preparation for use (aviation only)					5- 5.4.5.8.1.3	<assem>	
Servicing					5- 5.4.5.8.1.4	<service>	
Ground handling					5- 5.4.5.8.1.5	<groundtsk>	
Operational check					5- 5.4.5.8.1.6	<opchk>	
Inspection of installed items					5- 5.4.5.8.1	<inspinstitm>	
Removal					5- 5.4.5.8.1	<remove>	
Disassembly					5- 5.4.5.8.1	<disassem>	

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Table A.1 TM Requirements Matrix for _____

TM Content	-10	-12 -12&P	-13 -13&P	-14 -14&P	MIL-STD-40051 Reference	Element Name	Req'd / Not Req'd
Cleaning					5- 5.4.5.8.1	<clean>	
Inspection - acceptance and rejection criteria					5- 5.4.5.8.1.7	<acptrejinsp>	
Nondestructive testing inspection (NDTI)					5- 5.4.5.8.1.8	<ndti>	
Repair or replacement					5- 5.4.5.8.1.9	<repair-rplc>	
Alignment					5- 5.4.5.8.1	<align>	
Painting					5- 5.4.5.8.1	<paint>	
Lubrication					5- 5.4.5.8.1	<lube>	
Assembly					5- 5.4.5.8.1	<assem>	
Test and inspection					5- 5.4.5.8.1.10	<test-inspect>	
Installation					5- 5.4.5.8.1	<install>	
Adjustment					5- 5.4.5.8.1	<adjust>	
Calibration					5- 5.4.5.8.1	<calibration>	
Radio interference suppression					5- 5.4.5.8.1	<ris>	
Placing in service					5- 5.4.5.8.1.11	<pis>	
Testing					5- 5.4.5.8.1.12	<test-pass>	
Preparation for storage or shipment					5- 5.4.5.8.1.16	<pss>	
Classification of defects					5- 5.4.5.8.1.17	<defect>	
Handling ammunition					5- 5.4.5.8.1.18	<handling>	
Ammunition markings					5- 5.4.5.8.1	<marking>	
Procedures for ammunition activation					5- 5.4.5.8.1.19	<arm>	
AMMUNITION MAINTENANCE WORK PACKAGE					5- 5.4.5.6	<ammowp>	
AUXILIARY EQUIPMENT MAINTENANCE WORK PACKAGE					5- 5.4.5.7	<auxeqpwp>	

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Table A.1 TM Requirements Matrix for _____

TM Content	-10	-12 -12&P	-13 -13&P	-14 -14&P	MIL-STD-40051 Reference	Element Name	Req'd / Not Req'd
CHAPTER X. SUPPORTING INFORMATION **NOTE** Applicable supporting information work packages shall be arranged in the order in which they are presented here and numbered accordingly.	R	R	R	R	7- 5.1	<sim>	
REFERENCES	R	R	R	R	7- 5.2	<refwp>	
MAC	NR	R	R	R	7- 5.3	<macwp>	
RPSTL *(-10 through -14)* *(-12&P through -14&P)*	NR	NR R	NR R	NR R	7- 5.4	<rpstlwp>	
COMPONENTS OF END ITEM (COEI) AND BASIC ISSUE ITEMS (BII) LISTS	R	R	R	R	7- 5.5	<coeibiiwp>	
ADDITIONAL AUTHORIZATION LIST (AAL)					7- 5.6	<aalwp>	
EXPENDABLE AND DURABLE ITEMS LIST					7- 5.7	<explistwp>	
STOWAGE AND DECAL / DATA PLATE GUIDE					7- 5.8	<stowagewp>	
ON-VEHICLE EQUIPMENT LOADING PLAN					7- 5.9	<eqploadwp>	
TOOL IDENTIFICATION LIST	NR				7- 5.10	<toolidwp>	
ILLUSTRATED LIST OF MANUFACTURED ITEMS	NR				7- 5.11	<manuwp>	
TORQUE LIMITS	NR				7- 5.12	<torquewp>	
MANDATORY REPLACEMENT PARTS	NR				7- 5.13	<mrplwp>	
AMMUNITION MARKING INFORMATION	NR				7- 5.14	<ammowp>	
FOREIGN AMMUNITION (NATO)	NR				7- 5.15	<natowp>	
WIRING DIAGRAMS	NR				7- 5.22	<wiringwp>	
ADDITIONAL SUPPORTING WORK PACKAGES					7- 5.23	<genwp>	

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Table A.1 TM Requirements Matrix for _____

TM Content	-10	-12 -12&P	-13 -13&P	-14 -14&P	MIL-STD-40051 Reference	Element Name	Req'd / Not Req'd
REAR MATTER	R	R	R	R	1- 5.8.2	<rear>	
Glossary					1- 5.8.2.1	<glossary>	
Alphabetical index					1- 5.8.2.2	<aindx>	
Foldout pages					1- 5.8.2.3	<foldsect>	
DA Form 2028	R	R	R	R	1- 5.8.2.4	<da2028>	
Authentication page	R	R	R	R	1- 5.8.2.5	<authent>	
Back cover	R	R	R	R	1- 5.8.2.6	<back>	

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Table A.2 TM Requirements Matrix for _____

TM Content	-20 -20&P	-30 -30&P	-40 -40&P	MIL-STD-40051 Reference	Element Name	Req'd / Not Req'd
FRONT MATTER	R	R	R	1- 5.8.1	<frnt>	
Front cover	R	R	R	1- 5.8.1.1	<frntcover>	
Warning summary				1- 5.8.1.2	<warnsum>	
List of effective pages / work packages				1- 5.8.1.3		
Revision transmittal page	R	R	R	1-5.8.1.4	<chgsheet>	
Title block page	R	R	R	1- 5.8.1.5	<titleblk>	
Table of contents	R	R	R	1- 5.8.1.6	<contents>	
How to use this manual				1- 5.8.1.7	<howtouse>	
CHAPTER 1. INTRODUCTORY INFORMATION WITH THEORY OF OPERATION	R	R	R	2- 5.1	<gim>	
GENERAL INFORMATION WORK PACKAGE	R	R	R	2- 5.1.1	<ginfowp>	
Scope	R	R	R	2- 5.1.1.1	<scope>	
Maintenance forms, records, and reports	R	R	R	2- 5.1.1.2	<mfrr>	
Reporting equipment improvement recommendations (EIR)	R	R	R	2- 5.1.1.3	<eir>	
Hand receipt manuals				2- 5.1.1.4	<handreceipt>	
Corrosion prevention and control (CPC)	R	R	R	2- 5.1.1.5	<cpcdata>	
Destruction of Army materiel to prevent enemy use	R	R	R	2- 5.1.1.6	<destructmat>	
Preparation for storage or shipment	R	R	R	2- 5.1.1.7	<pssref>	
Warranty information				2- 5.1.1.8	<wrntyref>	
Nomenclature cross-reference list				2- 5.1.1.9	<nomenreflist>	
List of abbreviations				2- 5.1.1.10	<loa>	
Quality assurance (QA) (aviation only)				2- 5.1.1.11	<qainfo>	

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Table A.2 TM Requirements Matrix for _____

TM Content	-20 -20&P	-30 -30&P	-40 -40&P	MIL-STD-40051 Reference	Element Name	Req'd / Not Req'd
Safety, care, and handling				2- 5.1.1.12	<sftyinfo>	
Nuclear hardness				2- 5.1.1.13	<hcp>	
Security measures for electronic data				2- 5.1.1.14	<secref>	
Calibration				2- 5.1.1.15	<calref>	
Copyright credit line				2- 5.1.1.19	<copyrt>	
EQUIPMENT DESCRIPTION AND DATA WORK PACKAGE	R	R	R	2- 5.1.3	<descwp>	
Equipment characteristics, capabilities, and features				2- 5.1.3.1	<eqpinfo>	
Location and description of major components				2- 5.1.3.2	<locdesc>	
Differences between models				2- 5.1.3.3	<eqpdiff>	
Equipment data				2- 5.1.3.4	<eqpdata>	
Equipment configuration				2- 5.1.3.5	<eqpconfig>	
THEORY OF OPERATION WORK PACKAGE				2- 5.1.4	<thrywp>	
SUPPORTING DATA WORK PACKAGE FOR REPAIR PARTS, SPECIAL TOOLS, TMDE, AND SUPPORT EQUIPMENT	R	R	R	2- 5.1.5	<supdatawp>	
CHAPTER X. TROUBLESHOOTING PROCEDURES	R	R	R	4- 5.4	<tim>	
INTRODUCTION WORK PACKAGE	R	R	R	4- 5.5.1	<tsintrowp>	
MALFUNCTION / SYMPTOM INDEX WORK PACKAGE				4- 5.5.2	<symndxwp>	
TROUBLESHOOTING PROCEDURES WORK PACKAGE	R	R	R	4- 5.5.3	<tswp>	

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Table A.2 TM Requirements Matrix for _____

TM Content	-20 -20&P	-30 -30&P	-40 -40&P	MIL-STD-40051 Reference	Element Name	Req'd / Not Req'd
CHAPTER X. MAINTENANCE INSTRUCTIONS **NOTE** All maintenance work packages shall include a scope of task, initial setup, and all maintenance tasks applicable to the equipment.	R	R	R	5- 5.3	<mim>	
SERVICE UPON RECEIPT WORK PACKAGE	R	NR	NR	5- 5.4.5.1	<surwp>	
Siting				5- 5.4.5.1.1	<siting>	
Shelter requirements				5- 5.4.5.1.2	<shltr>	
Service upon receipt of materiel				5- 5.4.5.1.3	<surmat>	
Installation instructions				5- 5.4.5.1.4	<install>	
Preliminary servicing of equipment				5- 5.4.5.1.5	<preserv>	
Preliminary checks and adjustment of equipment				5- 5.4.5.1.6	<prechkadj>	
Preliminary calibration of equipment				5- 5.4.5.1.7	<precal>	
Circuit alignment				5- 5.4.5.1.8	<calign>	
Ammunition markings				5- 5.4.5.1.9	<markings>	
Classification of defects				5- 5.4.5.1.10	<defect>	
Ammunition handling				5- 5.4.5.1.11	<handling>	
Procedures to activate ammunition				5- 5.4.5.1.12	<arm>	
EQUIPMENT / USER FITTING INSTRUCTIONS WORK PACKAGE (personal use equipment)				5- 5.4.5.2	<perseqpwp>	
PMCS, INCLUDING LUBRICATION INSTRUCTIONS, WORK PACKAGE	R	R	R	5- 5.4.5.3	<pmcswp>	
Introduction	R	R	R	5- 5.4.5.3.1	<pmcsintro>	
PMCS procedures (table)	R	R	R	5- 5.4.5.3.2	<pmcstable>	

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Table A.2 TM Requirements Matrix for _____

TM Content	-20 -20&P	-30 -30&P	-40 -40&P	MIL-STD-40051 Reference	Element Name	Req'd / Not Req'd
MAINTENANCE WORK PACKAGES **NOTE** As applicable, the following maintenance tasks shall be presented in the general order listed below:	R	R	R	5- 5.4.5.8	<maintwp>	
Assembly and preparation for use (aviation only)				5- 5.4.5.8.1.3	<assem>	
Servicing				5- 5.4.5.8.1.4	<service>	
Ground handling				5- 5.4.5.8.1.5	<groundtsk>	
Operational check				5- 5.4.5.8.1.6	<opchk>	
Inspection of installed items				5- 5.4.5.8.1	<inspinstitm>	
Removal				5- 5.4.5.8.1	<remove>	
Disassembly				5- 5.4.5.8.1	<disassem>	
Cleaning				5- 5.4.5.8.1	<clean>	
Inspection - acceptance and rejection criteria				5- 5.4.5.8.1.7	<acptrejinsp>	
Nondestructive testing inspection (NDTI)				5- 5.4.5.8.1.8	<ndti>	
Repair or replacement				5- 5.4.5.8.1.9	<repair-rplc>	
Alignment				5- 5.4.5.8.1	<align>	
Painting				5- 5.4.5.8.1	<paint>	
Lubrication				5- 5.4.5.8.1	<lube>	
Assembly				5- 5.4.5.8.1	<assem>	
Test and inspection				5- 5.4.5.8.1.10	<test-inspect>	
Installation				5- 5.4.5.8.1	<install>	
Adjustment				5- 5.4.5.8.1	<adjust>	
Calibration				5- 5.4.5.8.1	<calibration>	
Radio interference suppression				5- 5.4.5.8.1	<ris>	
Placing in service				5- 5.4.5.8.1.11	<pis>	
Testing				5- 5.4.5.8.1.12	<test-pass>	
Preparation for storage or shipment				5- 5.4.5.8.1.16	<pss>	

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Table A.2 TM Requirements Matrix for _____

TM Content	-20 -20&P	-30 -30&P	-40 -40&P	MIL-STD-40051 Reference	Element Name	Req'd / Not Req'd
Classification of defects				5- 5.4.5.8.1.17	<defect>	
Handling ammunition				5- 5.4.5.8.1.18	<handling>	
Ammunition markings				5- 5.4.5.8.1	<marking>	
Procedures for ammunition activation				5- 5.4.5.8.1.19	<arm>	
AMMUNITION MAINTENANCE WORK PACKAGE				5- 5.4.5.6	<ammowp>	
AUXILIARY EQUIPMENT MAINTENANCE WORK PACKAGE				5- 5.4.5.7	<auxeqpwp>	
CHAPTER X. SUPPORTING INFORMATION **NOTE** Applicable supporting information work packages shall be arranged in the order in which they are presented here and numbered accordingly.	R	R	R	7- 5.1	<sim>	
REFERENCES	R	R	R	7- 5.2	<refwp>	
MAC	R	NR	NR	7- 5.3	<macwp>	
RPSTL *(-20 through -40)* *(-20&P through -40&P)*	NR R	NR R	NR R	7- 5.4	<rpstlwp>	
EXPENDABLE AND DURABLE ITEMS LIST				7- 5.7	<explistwp>	
TOOL IDENTIFICATION LIST				7- 5.10	<toolidwp>	
ILLUSTRATED LIST OF MANUFACTURED ITEMS				7- 5.11	<manuwp>	
TORQUE LIMITS				7- 5.12	<torquewp>	
MANDATORY REPLACEMENT PARTS				7- 5.13	<mrplwp>	
AMMUNITION MARKING INFORMATION				7- 5.14	<ammowp>	
FOREIGN AMMUNITION (NATO)				7- 5.15	<natowp>	
WIRING DIAGRAMS				7- 5.22	<wiringwp>	

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Table A.2 TM Requirements Matrix for _____

TM Content	-20 -20&P	-30 -30&P	-40 -40&P	MIL-STD-40051 Reference	Element Name	Req'd / Not Req'd
ADDITIONAL SUPPORTING WORK PACKAGES				7- 5.23	<genwp>	
REAR MATTER	R	R	R	1- 5.8.2	<rear>	
Glossary				1- 5.8.2.1	<glossary>	
Alphabetical index				1- 5.8.2.2	<aindx>	
Foldout pages				1- 5.8.2.3	<foldsect>	
DA Form 2028	R	R	R	1- 5.8.2.4	<da2028>	
Authentication page	R	R	R	1- 5.8.2.5	<authent>	
Back cover	R	R	R	1- 5.8.2.6	<back>	

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Table A.3 TM Requirements Matrix for _____

TM Content	-23 -23&P	-24 -24&P	-34 -34&P	MIL-STD-40051 Reference	Element Name	Req'd / Not Req'd
FRONT MATTER	R	R	R		<frnt>	
Front cover	R	R	R	1 - 5.8.1.1	<frntcover>	
Warning summary				1 - 5.8.1.2	<warnsum>	
List of effective pages / work packages				1 - 5.8.1.3		
Revision transmittal page	R	R	R	1 - 5.8.1.4	<chgsheet>	
Title block page	R	R	R	1 - 5.8.1.5	<titleblk>	
Table of contents	R	R	R	1 - 5.8.1.6	<contents>	
How to use this manual				1 - 5.8.1.7	<howtouse>	
CHAPTER 1. INTRODUCTORY INFORMATION WITH THEORY OF OPERATION	R	R	R	2 - 5.1	<gim>	
GENERAL INFORMATION WORK PACKAGE	R	R	R	2 - 5.1.1	<ginfowp>	
Scope	R	R	R	2 - 5.1.1.1	<scope>	
Maintenance forms, records, and reports	R	R	R	2 - 5.1.1.2	<mfrr>	
Reporting equipment improvement recommendations (EIR)	R	R	R	2 - 5.1.1.3	<eir>	
Hand receipt manuals				2 - 5.1.1.4	<handreceipt>	
Corrosion prevention and control (CPC)	R	R	R	2 - 5.1.1.5	<cpcdata>	
Destruction of Army materiel to prevent enemy use	R	R	R	2 - 5.1.1.6	<destructmat>	
Preparation for storage or shipment	R	R	R	2 - 5.1.1.7	<pssref>	
Warranty information				2 - 5.1.1.8	<wrntyref>	
Nomenclature cross-reference list				2 - 5.1.1.9	<nomenreflist>	
List of abbreviations				2 - 5.1.1.10	<loa>	
Quality assurance (QA) (aviation only)				2 - 5.1.1.11	<qainfo>	

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Table A.3 TM Requirements Matrix for _____

TM Content	-23 -23&P	-24 -24&P	-34 -34&P	MIL-STD-40051 Reference	Element Name	Req'd / Not Req'd
Safety, care, and handling				2 - 5.1.1.12	<sftyinfo>	
Nuclear hardness				2 - 5.1.1.13	<hcp>	
Security measures for electronic data				2 - 5.1.1.14	<secref>	
Calibration				2 - 5.1.1.15	<calref>	
Copyright credit line				2 - 5.1.1.19	<copyrt>	
EQUIPMENT DESCRIPTION AND DATA WORK PACKAGE	R	R	R	2 - 5.1.3	<descwp>	
Equipment characteristics, capabilities, and features	R	R	R	2 - 5.1.3.1	<eqpinfo>	
Location and description of major components	R	R	R	2 - 5.1.3.2	<locdesc>	
Differences between models				2 - 5.1.3.3	<eqpdiff>	
Equipment data	R	R	R	2 - 5.1.3.4	<eqpdata>	
Equipment configuration	R	R	R	2 - 5.1.3.5	<eqpconfig>	
THEORY OF OPERATION WORK PACKAGE	R	R	R	2 - 5.1.4	<thrywp>	
SUPPORTING DATA WORK PACKAGE FOR REPAIR PARTS, SPECIAL TOOLS, TMDE, AND SUPPORT EQUIPMENT	R	R	R	2 - 5.1.5	<supdatawp>	
CHAPTER X. TROUBLESHOOTING PROCEDURES	R	R	R	4 - 5.4	<tim>	
INTRODUCTION WORK PACKAGE	R	R	R	4 - 5.5.1	<tsintrowp>	
MALFUNCTION / SYMPTOM INDEX WORK PACKAGE				4 - 5.5.2	<symndxwp>	
TROUBLESHOOTING PROCEDURES WORK PACKAGE	R	R	R	4 - 5.5.3	<tswp>	

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Table A.3 TM Requirements Matrix for _____

TM Content	-23 -23&P	-24 -24&P	-34 -34&P	MIL-STD-40051 Reference	Element Name	Req'd / Not Req'd
CHAPTER X. MAINTENANCE INSTRUCTIONS **NOTE** All maintenance work packages shall include a scope of task, initial setup, and all maintenance tasks applicable to the equipment.	R	R	R	5 - 5.3	<mim>	
SERVICE UPON RECEIPT WORK PACKAGE	R	R	NR	5 - 5.4.5.1	<surwp>	
Siting			NR	5 - 5.4.5.1.1	<siting>	
Shelter requirements			NR	5 - 5.4.5.1.2	<shltr>	
Service upon receipt of materiel			NR	5 - 5.4.5.1.3	<surmat>	
Installation instructions			NR	5 - 5.4.5.1.4	<install>	
Preliminary servicing of equipment			NR	5 - 5.4.5.1.5	<preserv>	
Preliminary checks and adjustment of equipment			NR	5 - 5.4.5.1.6	<prechkadj>	
Preliminary calibration of equipment			NR	5 - 5.4.5.1.7	<precal>	
Circuit alignment			NR	5 - 5.4.5.1.8	<calign>	
Ammunition markings			NR	5 - 5.4.5.1.9	<markings>	
Classification of defects			NR	5 - 5.4.5.1.10	<defect>	
Ammunition handling			NR	5 - 5.4.5.1.11	<handling>	
Procedures to activate ammunition			NR	5 - 5.4.5.1.12	<arm>	
EQUIPMENT / USER FITTING INSTRUCTIONS WORK PACKAGE (personal use equipment)				5 - 5.4.5.2	<perseqpwp>	
PMCS, INCLUDING LUBRICATION INSTRUCTIONS, WORK PACKAGE	R	R	R	5 - 5.4.5.3	<pmcswp>	
Introduction	R	R	R	5 - 5.4.5.3.1	<pmcsintro>	
PMCS procedures (table)	R	R	R	5 - 5.4.5.3.2	<pmcstable>	

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Table A.3 TM Requirements Matrix for _____

TM Content	-23 -23&P	-24 -24&P	-34 -34&P	MIL-STD-40051 Reference	Element Name	Req'd / Not Req'd
MAINTENANCE WORK PACKAGES **NOTE** As applicable, the following maintenance tasks shall be presented in the general order listed below:	R	R	R	5 - 5.4.5.8	<maintwp>	
Assembly and preparation for use (aviation only)				5 - 5.4.5.8.1.3	<assem>	
Servicing				5 - 5.4.5.8.1.4	<service>	
Ground handling				5 - 5.4.5.8.1.5	<groundtsk>	
Operational check				5 - 5.4.5.8.1.6	<opchk>	
Inspection of installed items				5 - 5.4.5.8.1	<inspinstitm>	
Removal				5 - 5.4.5.8.1	<remove>	
Disassembly				5 - 5.4.5.8.1	<disassem>	
Cleaning				5 - 5.4.5.8.1	<clean>	
Inspection - acceptance and rejection criteria				5 - 5.4.5.8.1.7	<acptrejinsp>	
Nondestructive testing inspection (NDTI)				5 - 5.4.5.8.1.8	<ndti>	
Repair or replacement				5 - 5.4.5.8.1.9	<repair-rplc>	
Alignment				5 - 5.4.5.8.1	<align>	
Painting				5 - 5.4.5.8.1	<paint>	
Lubrication				5 - 5.4.5.8.1	<lube>	
Assembly				5 - 5.4.5.8.1	<assem>	
Test and inspection				5 - 5.4.5.8.10	<test-inspect>	
Installation				5 - 5.4.5.8.1	<install>	
Adjustment				5 - 5.4.5.8.1	<adjust>	
Calibration				5 - 5.4.5.8.1	<calibration>	
Radio interference suppression				5 - 5.4.5.8.1	<ris>	
Placing in service				5 - 5.4.5.8.11	<pis>	
Testing				5 - 5.4.5.8.12	<test-pass>	
Preparation for storage or shipment				5 - 5.4.5.8.16	<pss>	

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Table A.3 TM Requirements Matrix for _____

TM Content	-23 -23&P	-24 -24&P	-34 -34&P	MIL-STD-40051 Reference	Element Name	Req'd / Not Req'd
Classification of defects				5 - 5.4.5.8.17	<defect>	
Handling ammunition				5 - 5.4.5.8.18	<handling>	
Ammunition markings				5 - 5.4.5.8.1	<marking>	
Procedures for ammunition activation				5 - 5.4.5.8.19	<arm>	
AMMUNITION MAINTENANCE WORK PACKAGE				5 - 5.4.5.6	<ammowp>	
AUXILIARY EQUIPMENT MAINTENANCE WORK PACKAGE				5 - 5.4.5.7	<auxeqpwp>	
CHAPTER X. SUPPORTING INFORMATION **NOTE** Applicable supporting information work packages shall be arranged in the order in which they are presented here and numbered accordingly.	R	R	R	7 - 5.1	<sim>	
REFERENCES	R	R	R	7 - 5.2	<refwp>	
MAC	R	R	NR	7 - 5.3	<macwp>	
RPSTL *(-23, -24, -34)* *(-23&P, -24&P, -34&P)*	NR R	NR R	NR R	7 - 5.4	<rpstlwp>	
EXPENDABLE AND DURABLE ITEMS LIST				7 - 5.7	<explistwp>	
TOOL IDENTIFICATION LIST				7 - 5.10	<toolidwp>	
ILLUSTRATED LIST OF MANUFACTURED ITEMS				7 - 5.11	<manuwp>	
TORQUE LIMITS				7 - 5.12	<torquewp>	
MANDATORY REPLACEMENT PARTS				7 - 5.13	<mrplwp>	
AMMUNITION MARKING INFORMATION				7 - 5.14	<ammowp>	
WIRING DIAGRAMS				7 - 5.22	<wiringwp>	
ADDITIONAL SUPPORTING WORK PACKAGES				7 - 5.23	<genwp>	

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Table A.4 TM Requirements Matrix for _____

TM Content	AVUM -20 -20&P	AVIM -30 -30&P	AVUM/ AVIM -23 -23&P	MIL-STD-40051 Reference	Element Name	Req'd / Not Req'd
FRONT MATTER	R	R	R	1 - 5.8.1	<frnt>	
Front cover	R	R	R	1 - 5.8.1.1	<frntcover>	
Warning summary				1 - 5.8.1.2	<warnsum>	
List of effective pages / work packages				1 - 5.8.1.3		
Revision transmittal page	R	R	R	1 - 5.8.1.4	<chgsheet>	
Title block page	R	R	R	1 - 5.8.1.5	<titleblk>	
Table of contents	R	R	R	1 - 5.8.1.6	<contents>	
How to use this manual				1 - 5.8.1.7	<howtouse>	
CHAPTER 1. INTRODUCTORY INFORMATION WITH THEORY OF OPERATION	R	R	R	2 - 5.1	<gim>	
GENERAL INFORMATION WORK PACKAGE	R	R	R	2 - 5.1.1	<ginfowp>	
Scope	R	R	R	2 - 5.1.1.1	<scope>	
Maintenance forms, records, and reports	R	R	R	2 - 5.1.1.2	<mfrr>	
Reporting equipment improvement recommendations (EIR)	R	R	R	2 - 5.1.1.3	<eir>	
Hand receipt manuals				2 - 5.1.1.4	<handreceipt>	
Corrosion prevention and control (CPC)	R	R	R	2 - 5.1.1.5	<cpcdata>	
Destruction of Army materiel to prevent enemy use	R	R	R	2 - 5.1.1.6	<destructmat>	
Preparation for storage or shipment	R	R	R	2 - 5.1.1.7	<pssref>	
Warranty information	R	R	R	2 - 5.1.1.8	<wrntyref>	
Nomenclature cross-reference list				2 - 5.1.1.9	<nomenreflist>	
List of abbreviations				2 - 5.1.1.10	<loa>	
Quality assurance (QA) (aviation only)				2 - 5.1.1.11	<qainfo>	

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Table A.3 TM Requirements Matrix for _____

TM Content	-23 -23&P	-24 -24&P	-34 -34&P	MIL-STD-40051 Reference	Element Name	Req'd / Not Req'd
REAR MATTER	R	R	R	1 - 5.8.2	<rear>	
Glossary				1 - 5.8.2.1	<glossary>	
Alphabetical index				1 - 5.8.2.2	<aindx>	
Foldout pages				1 - 5.8.2.3	<foldsect>	
DA Form 2028	R	R	R	1 - 5.8.2.4	<da2028>	
Authentication page	R	R	R	1 - 5.8.2.5	<authent>	
Back cover	R	R	R	1 - 5.8.2.6	<back>	

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Table A.4 TM Requirements Matrix for _____

TM Content	AVUM -20 -20&P	AVIM -30 -30&P	AVUM/ AVIM -23 -23&P	MIL-STD-40051 Reference	Element Name	Req'd / Not Req'd
Safety, care, and handling				2 - 5.1.1.12	<sftyinfo>	
Nuclear hardness				2 - 5.1.1.13	<hcp>	
Security measures for electronic data				2 - 5.1.1.14	<secref>	
Calibration				2 - 5.1.1.15	<calref>	
Copyright credit line				2 - 5.1.1.19	<copyrt>	
EQUIPMENT DESCRIPTION AND DATA WORK PACKAGE	R	R	R	2 - 5.1.3	<descwp>	
Equipment characteristics, capabilities, and features				2 - 5.1.3.1	<eqpinfo>	
Location and description of major components				2 - 5.1.3.2	<locdesc>	
Differences between models				2 - 5.1.3.3	<eqpdiff>	
Equipment data				2 - 5.1.3.4	<eqpdata>	
Equipment configuration				2 - 5.1.3.5	<eqpconfig>	
THEORY OF OPERATION WORK PACKAGE				2 - 5.1.4	<thrywp>	
SUPPORTING DATA WORK PACKAGE FOR REPAIR PARTS, SPECIAL TOOLS, TMDE, AND SUPPORT EQUIPMENT	R	R	R	2 - 5.1.5	<supdatawp>	
CHAPTER X. TROUBLESHOOTING PROCEDURES	R	R	R	4 - 5.4	<tim>	
INTRODUCTION WORK PACKAGE	R	R	R	4 - 5.5.1	<tsintrowp>	
MALFUNCTION / SYMPTOM INDEX WORK PACKAGE				4 - 5.5.2	<symndxwp>	
TROUBLESHOOTING PROCEDURES WORK PACKAGE	R	R	R	4 - 5.5.3	<tswp>	

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Table A.4 TM Requirements Matrix for _____

TM Content	AVUM -20 -20&P	AVIM -30 -30&P	AVUM/ AVIM -23 -23&P	MIL-STD-40051 Reference	Element Name	Req'd / Not Req'd
CHAPTER X. MAINTENANCE INSTRUCTIONS **NOTE** All maintenance work packages shall include a scope of task, initial setup, and all maintenance tasks applicable to the equipment.	R	R	R	5 - 5.3	<mim>	
SERVICE UPON RECEIPT WORK PACKAGE	R	NR	R	5 - 5.4.5.1	<surwp>	
Siting				5 - 5.4.5.1.1	<siting>	
Shelter requirements				5 - 5.4.5.1.2	<shltr>	
Service upon receipt of materiel				5 - 5.4.5.1.3	<surmat>	
Installation instructions				5 - 5.4.5.1.4	<install>	
Preliminary servicing of equipment				5 - 5.4.5.1.5	<preserv>	
Preliminary checks and adjustment of equipment				5 - 5.4.5.1.6	<prechkadj>	
Preliminary calibration of equipment				5 - 5.4.5.1.7	<precal>	
Circuit alignment				5 - 5.4.5.1.8	<calign>	
Ammunition markings				5 - 5.4.5.1.9	<markings>	
Classification of defects				5 - 5.4.5.1.10	<defect>	
Ammunition handling				5 - 5.4.5.1.11	<handling>	
Procedures to activate ammunition				5 - 5.4.5.1.12	<arm>	
EQUIPMENT / USER FITTING INSTRUCTIONS WORK PACKAGE (personal use equipment)				5 - 5.4.5.2	<perseqpwp>	
PMCS, INCLUDING LUBRICATION INSTRUCTIONS, WORK PACKAGE	R	R	R	5 - 5.4.5.3	<pmcswp>	
Introduction	R	R	R	5 - 5.4.5.3.1	<pmcsintro>	
PMCS procedures (table)	R	R	R	5 - 5.4.5.3.2	<pmcstable>	

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Table A.4 TM Requirements Matrix for _____

TM Content	AVUM -20 -20&P	AVIM -30 -30&P	AVUM/ AVIM -23 -23&P	MIL-STD-40051 Reference	Element Name	Req'd / Not Req'd
MAINTENANCE WORK PACKAGES **NOTE** As applicable, the following maintenance tasks shall be presented in the general order listed below:	R	R	R	5 - 5.4.5.8	<maintwp>	
Assembly and preparation for use (aviation only)				5 - 5.4.5.8.1.3	<assem>	
Servicing				5 - 5.4.5.8.1.4	<service>	
Ground handling				5 - 5.4.5.8.1.5	<groundtsk>	
Operational check				5 - 5.4.5.8.1.6	<opchk>	
Inspection of installed items				5 - 5.4.5.8.1	<inspinstitm>	
Removal				5 - 5.4.5.8.1	<remove>	
Disassembly				5 - 5.4.5.8.1	<disassem>	
Cleaning				5 - 5.4.5.8.1	<clean>	
Inspection - acceptance and rejection criteria				5 - 5.4.5.8.1.7	<acptrejinsp>	
Nondestructive testing inspection (NDTI)				5 - 5.4.5.8.1.8	<ndti>	
Repair or replacement				5 - 5.4.5.8.1.9	<repair-rplc>	
Alignment				5 - 5.4.5.8.1	<align>	
Painting				5 - 5.4.5.8.1	<paint>	
Lubrication				5 - 5.4.5.8.1	<lube>	
Assembly				5 - 5.4.5.8.1	<assem>	
Test and inspection				5 - 5.4.5.8.1.10	<test-inspect>	
Installation				5 - 5.4.5.8.1	<install>	
Adjustment				5 - 5.4.5.8.1	<adjust>	
Calibration				5 - 5.4.5.8.1	<calibration>	
Radio interference suppression				5 - 5.4.5.8.1	<ris>	
Placing in service				5 - 5.4.5.8.1.11	<pis>	
Testing				5 - 5.4.5.8.1.12	<test-pass>	
Preparation for storage or shipment				5 - 5.4.5.8.1.16	<pss>	

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Table A.4 TM Requirements Matrix for _____

TM Content	AVUM -20 -20&P	AVIM -30 -30&P	AVUM/ AVIM -23 -23&P	MIL-STD-40051 Reference	Element Name	Req'd / Not Req'd
Classification of defects				5 - 5.4.5.8.1.17	<defect>	
Handling ammunition				5 - 5.4.5.8.1.18	<handling>	
Ammunition markings				5 - 5.4.5.8.1	<marking>	
Procedures for ammunition activation				5 - 5.4.5.8.1.19	<arm>	
AMMUNITION MAINTENANCE WORK PACKAGE				5 - 5.4.5.6	<ammowp>	
AUXILIARY EQUIPMENT MAINTENANCE WORK PACKAGE				5 - 5.4.5.7	<auxeqpwp>	
CHAPTER X. SUPPORTING INFORMATION **NOTE** Applicable supporting information work packages shall be arranged in the order in which they are presented here and numbered accordingly.	R	R	R	7 - 5.1	<sim>	
REFERENCES	R	R	R	7 - 5.2	<refwp>	
MAC	R	NR	R	7 - 5.3	<macwp>	
RPSTL *(AVUM, AVIM, AVUM/AVIM)* *(AVUM&P, AVIM&P, AVUM/AVIM&P)*	NR R	NR R	NR R	7 - 5.4	<rpstlwp>	
EXPENDABLE AND DURABLE ITEMS LIST				7 - 5.7	<explistwp>	
TOOL IDENTIFICATION LIST	R	R	R	7 - 5.10	<toolidwp>	
ILLUSTRATED LIST OF MANUFACTURED ITEMS	R	R	R	7 - 5.11	<manuwp>	
TORQUE LIMITS	R	R	R	7 - 5.12	<torquewp>	
MANDATORY REPLACEMENT PARTS	R	R	R	7 - 5.13	<mrplwp>	
AMMUNITION MARKING INFORMATION				7 - 5.14	<ammowp>	
FOREIGN AMMUNITION (NATO)				7 - 5.15	<natowp>	
AIRCRAFT INVENTORY MASTER GUIDE	R	R	R	7 - 5.16	<inventorywp>	

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Table A.4 TM Requirements Matrix for _____

TM Content	AVUM -20 -20&P	AVIM -30 -30&P	AVUM/ AVIM -23 -23&P	MIL-STD-40051 Reference	Element Name	Req'd / Not Req'd
STORAGE OF AIRCRAFT	R	R	R	7 - 5.17	<storagewp>	
WEIGHT AND BALANCE		R		7 - 5.18	<wtbalwp>	
WIRING DIAGRAMS				7 - 5.22	<wiringwp>	
ADDITIONAL SUPPORTING WORK PACKAGES				7 - 5.23	<genwp>	
REAR MATTER	R	R	R	1 - 5.8.2	<rear>	
Glossary				1 - 5.8.2.1	<glossary>	
Alphabetical index				1 - 5.8.2.2	<aindx>	
Foldout pages				1 - 5.8.2.3	<foldsect>	
DA Form 2028	R	R	R	1 - 5.8.2.4	<da2028>	
Authentication page	R	R	R	1 - 5.8.2.5	<authent>	
Back cover	R	R	R	1 - 5.8.2.6	<back>	

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Table A.5 TM Requirements Matrix for _____

TM Content	RPSTL	MIL-STD-40051 Reference	Element Name	Req'd / Not Req'd
FRONT MATTER	R	6 - 5.2	<frnt>	
Front cover	R	1 - 5.8.1.1	<frntcover>	
List of effective pages / work packages		1 - 5.8.1.3		
Revision transmittal page	R	1 - 5.8.1.4	<chgsheet>	
Title block page	R	1 - 5.8.1.5	<titleblk>	
Table of contents	R	1 - 5.8.1.6	<contents>	
CHAPTER X. REPAIR PARTS AND SPECIAL TOOLS LIST FOR (enter equipment name)	R	6 - 5.1	<pim>	
INTRODUCTION WORK PACKAGE	R	6 - 5.4	<introwp>	
REPAIR PARTS LIST WORK PACKAGES	R	6 - 5.5	<plwp>	
SPECIAL TOOLS LIST WORK PACKAGE	R	6 - 5.6	<stlwp>	
NSN INDEX WORK PACKAGE	R	6 - 5.7.1	<nsnindxwp>	
P/N INDEX WORK PACKAGE	R	6 - 5.7.2	<pnindxwp>	
REFERENCE DESIGNATOR INDEX WORK PACKAGE		6 - 5.7.3	<refdesindxwp>	
REAR MATTER	R	6 - 5.9	<rear>	
DA Form 2028	R	1 - 5.8.2.4	<da2028>	
Authentication page	R	1 - 5.8.2.5	<authent>	
Back cover	R	1 - 5.8.2.6	<back>	

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Table A.6 TM Requirements Matrix for _____

TM Content	DMWR and DMWR with RPSTL	MIL-STD-40051 Reference	Element Name	Req'd / Not Req'd
FRONT MATTER	R	1 - 5.8.1	<frnt>	
Front cover	R	1 - 5.8.1.1	<frntcover>	
Warning summary		1 - 5.8.1.2	<warnsum>	
List of effective pages / work packages		1 - 5.8.1.3		
Revision transmittal page	R	1 - 5.8.1.4	<chgsheet>	
Title block page	R	1 - 5.8.1.5	<titleblk>	
Table of contents	R	1 - 5.8.1.6	<contents>	
How to use this manual	NR	1 - 5.8.1.7	<howtouse>	
CHAPTER 1. INTRODUCTORY INFORMATION WITH THEORY OF OPERATION	R	2 - 5.1	<gim>	
GENERAL INFORMATION WORK PACKAGE	R	2 - 5.1.1	<ginfowp>	
Scope	R	2 - 5.1.1.1	<scope>	
Maintenance forms, records, and reports	R	2 - 5.1.1.2	<mfrr>	
Reporting equipment improvement recommendations (EIR)	R	2 - 5.1.1.3	<eir>	
Hand receipt manuals		2 - 5.1.1.4	<handreceipt>	
Corrosion prevention and control (CPC)		2 - 5.1.1.5	<cpcdata>	
Destruction of Army materiel to prevent enemy use		2 - 5.1.1.6	<destructmat>	
Preparation for storage or shipment		2 - 5.1.1.7	<pssref>	
Warranty information		2 - 5.1.1.8	<wrntyref>	
Nomenclature cross-reference list		2 - 5.1.1.9	<nomenreflist>	
List of abbreviations		2 - 5.1.1.10	<loa>	
Quality assurance (QA) (aviation only)		2 - 5.1.1.11	<qainfo>	

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Table A.6 TM Requirements Matrix for _____

TM Content	DMWR and DMWR with RPSTL	MIL-STD-40051 Reference	Element Name	Req'd / Not Req'd
Safety, care, and handling		2 - 5.1.1.12	<sftyinfo>	
Nuclear hardness		2 - 5.1.1.13	<hcp>	
Security measures for electronic data		2 - 5.1.1.14	<secref>	
Calibration		2 - 5.1.1.15	<calref>	
Engineering change proposals (ECP)	R	2 - 5.1.1.16	<ecp>	
Deviations and exceptions	R	2 - 5.1.1.17	<deviation>	
Mobilization requirements	R	2 - 5.1.1.18	<mobreq>	
Copyright credit line		2 - 5.1.1.19	<copyrt>	
EQUIPMENT DESCRIPTION AND DATA WORK PACKAGE	R	2 - 5.1.3	<descwp>	
Equipment characteristics, capabilities, and features	R	2 - 5.1.3.1	<eqpinfo>	
Location and description of major components		2 - 5.1.3.2	<locdesc>	
Differences between models		2 - 5.1.3.3	<eqpdiff>	
Equipment data		2 - 5.1.3.4	<eqpdata>	
Equipment configuration		2 - 5.1.3.5	<eqpconfig>	
SUPPORTING DATA WORK PACKAGE FOR REPAIR PARTS, SPECIAL TOOLS, TMDE, AND SUPPORT EQUIPMENT	R	2 - 5.1.5	<supdatawp>	
CHAPTER X. DEPOT MAINTENANCE INSTRUCTIONS **NOTE** All maintenance work packages shall include a scope of task, initial setup, and all maintenance tasks applicable to the equipment.	R	5 - 5.3	<mim>	
MAINTENANCE WORK PACKAGES **NOTE** As applicable, the following maintenance tasks shall be presented in the general order listed below:	R	5 - 5.4.5.8	<maintwp> <maintsk>	
Preshop analysis	R	5 - 5.4.5.8.1.2	<pshopanal>	

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Table A.6 TM Requirements Matrix for _____

TM Content	DMWR and DMWR with RPSTL	MIL-STD-40051 Reference	Element Name	Req'd / Not Req'd
Servicing		5 - 5.4.5.8.1.4	<service>	
Operational check		5 - 5.4.5.8.1.6	<opchk>	
Inspection of installed items		5 - 5.4.5.8.1	<inspinstitm>	
Removal		5 - 5.4.5.8.1	<remove>	
Disassembly		5 - 5.4.5.8.1	<disassem>	
Cleaning		5 - 5.4.5.8.1	<clean>	
Inspection - acceptance and rejection criteria		5 - 5.4.5.8.1.7	<acptrejinsp>	
Nondestructive testing inspection (NDTI)		5 - 5.4.5.8.1.8	<ndti>	
Repair or replacement		5 - 5.4.5.8.1.9	<repair-rplc>	
Alignment		5 - 5.4.5.8.1	<align>	
Painting		5 - 5.4.5.8.1	<paint>	
Lubrication		5 - 5.4.5.8.1	<lube>	
Assembly		5 - 5.4.5.8.1	<assem>	
Test and inspection		5 - 5.4.5.8.1.10	<test-inspect>	
Installation		5 - 5.4.5.8.1	<install>	
Adjustment		5 - 5.4.5.8.1	<adjust>	
Calibration		5 - 5.4.5.8.1	<calibration>	
Radio interference suppression		5 - 5.4.5.8.1	<ris>	
Placing in service		5 - 5.4.5.8.1.11	<pis>	
Testing		5 - 5.4.5.8.1.12	<test-pass>	
Quality assurance requirements	R	5 - 5.4.5.8.1.13	<qawp>	
Preservation, packaging, and marking	R	5 - 5.4.5.8.1.14	<ppm>	
Preparation for storage or shipment		5 - 5.4.5.8.1.16	<pss>	

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Table A.6 TM Requirements Matrix for _____

TM Content	DMWR and DMWR with RPSTL	MIL-STD-40051 Reference	Element Name	Req'd / Not Req'd
CHAPTER X. SUPPORTING INFORMATION **NOTE** Applicable supporting information work packages shall be arranged in the order in which they are presented here and numbered accordingly.	R	7 - 5.1	<sim>	
REFERENCES	R	7 - 5.2	<refwp>	
RPSTL *(Required for DMWR with RPSTL only)*		7 - 5.4	<rpstlwp>	
EXPENDABLE AND DURABLE ITEMS LIST		7 - 5.7	<explistwp>	
DEPOT MOBILIZATION REQUIREMENTS	R	7 - 5.19	<mobilwp>	
COMPONENT CHECKLIST	R	7 - 5.20	<compchklistwp>	
QUALITY ASSURANCE REQUIREMENTS	R	7 - 5.21	<qawp>	
ADDITIONAL SUPPORTING WORK PACKAGES		7 - 5.23	<genwp>	
REAR MATTER	R	1 - 5.8.2	<rear>	
Glossary		1 - 5.8.2.1	<glossary>	
Alphabetical index		1 - 5.8.2.2	<aindx>	
Foldout pages		1 - 5.8.2.3	<foldsect>	
DA Form 2028	R	1 - 5.8.2.4	<da2028>	
Authentication page	R	1 - 5.8.2.5	<authent>	
Back cover	R	1 - 5.8.2.6	<back>	

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Table A.7 TM Requirements Matrix for _____

TM Content	Aircraft Troubleshooting	MIL-STD-40051 Reference	Element Name	Req'd / Not Req'd
FRONT MATTER	R	1 - 5.8.1	<frnt>	
Front cover	R	1 - 5.8.1.1	<frntcover>	
Warning summary		1 - 5.8.1.2	<warnsum>	
List of effective pages / work packages		1 - 5.8.1.3		
Revision transmittal page	R	1 - 5.8.1.4	<chgsheet>	
Title block page	R	1 - 5.8.1.5	<titleblk>	
Table of contents	R	1 - 5.8.1.6	<contents>	
How to use this manual	R	1 - 5.8.1.7	<howtouse>	
CHAPTER X. TROUBLESHOOTING PROCEDURES	R		<tim>	
INTRODUCTION WORK PACKAGE			<tsintrowp>	
EQUIPMENT DESCRIPTION AND DATA WORK PACKAGE			<descwp>	
Equipment characteristics, capabilities, and features			<eqpinfo>	
Location and description of major components			<locdesc>	
Differences between models			<eqpdiff>	
Equipment data			<eqpdata>	
Equipment configuration			<eqpconfig>	
Safety, care, and handling			<sftyinfo>	
Controls and indicators			<ctrlindwp>	
THEORY OF OPERATION WORK PACKAGE			<thrywp>	
MALFUNCTION / SYMPTOM INDEX WORK PACKAGE			<symndxwp>	
TROUBLESHOOTING PROCEDURES WORK PACKAGE	R		<tswp>	
REAR MATTER	R	1 - 5.8.2	<rear>	
Glossary		1 - 5.8.2.1	<glossary>	
Alphabetical index		1 - 5.8.2.2	<aindx>	

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Table A.7 TM Requirements Matrix for _____

TM Content	Aircraft Troubleshooting	MIL-STD-40051 Reference	Element Name	Req'd / Not Req'd
Foldout pages		1 - 5.8.2.3	<foldsect>	
DA Form 2028	R	1 - 5.8.2.4	<da2028>	
Authentication page	R	1 - 5.8.2.5	<authent>	
Back cover	R	1 - 5.8.2.6	<back>	

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Table A.8 TM Requirements Matrix for _____

TM Content	Aircraft PMS	MIL-STD-40051 Reference	Element Name	Req'd / Not Req'd
FRONT MATTER	R	1 - 5.8.1		
Cover page	R	1 - 5.8.1.1		
Warning summary	R	1 - 5.8.1.2	<warnsum>	
CHAPTER 1. GENERAL INFORMATION CHAPTER	R	2 - 5.1	<gim>	
GENERAL INFORMATION WORK PACKAGE	R	2 - 5.1.2	<pms-ginfowp>	
Scope	R	2 - 5.1.2.1	<scope>	
Inspection requirements	R	2 - 5.1.2.2	<inspec-req>	
Maintenance activities	R	2 - 5.1.2.3	<scope>	
General information	R	2 - 5.1.2.4	<geninfo>	
CHAPTER X. MAINTENANCE INFORMATION CHAPTER	R	5 - 5.3	<mim>	
PMS INSPECTION WORK PACKAGE	R	5 - 5.4.5.10	<pms-inspecwp>	
Checklist data	R	5 - 5.4.5.10	<pms-form>	

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NOT MEASUREMENT SENSITIVE

MIL-STD-40051-1

DEPARTMENT OF DEFENSE STANDARD PRACTICE

TECHNICAL MANUALS

GENERAL PREPARATION AND ASSEMBLY INFORMATION

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conforming Document Type Definitions (DTDs) can be used to develop digital data in accordance with MIL-PRF-28001.

4.5.1 Non-mandatory style and format requirements. Preferred general style and format requirements for the preparation of Army TMs shall be provided by the procuring activity.

4.6 Standard tables. Various standard tables required are noted throughout the text of this standard in bold and in parentheses (i.e., **(standard table)**). The formats and table heading names of these standard tables shall have no deviations. A list of standard tables is provided in 5.3.11.3.

4.7 Preparation of digital data for electronic delivery. Technical manual data prepared in work package format and delivered digitally in accordance with this standard shall be Standard Generalized Markup Language (SGML)tagged and assembled using the modular Assembly DTD and Formatting Output Specification Instance (FOSI). The DTD and FOSI have been developed in accordance with MIL-PRF-28001 and ISO 8879. Refer to MIL-STD-40051 for information on obtaining or accessing the modular DTD and FOSI. SGML tags used in the modular DTD are noted throughout the text of this standard in bracketed, bold characters (i.e., **<maintwp>**) as a convenience for the TM author and to ensure that the tags are used correctly when developing a document instance.

4.7.1 Use of the DTDs / FOSIs. The modular DTDs referenced in this standard interpret the technical content and structure for the functional requirements contained in this standard and are mandatory for use. The modular FOSIs referenced herein interpret the style and format. As specified by the contracting activity, FOSIs or style sheets may be used to produce final reproducible paper copy for all TMs prepared in accordance with this standard.

5. DETAILED REQUIREMENTS.

5.1 Technical content preparation. Technical manual data developed in accordance with this standard shall be task oriented and fully consistent with the maintenance concepts derived from the baseline documents described below.

a. Logistic Support Analysis Record (LSAR). The technical data and instructions developed by the requirements of Logistics Support Analysis and Department of Defense (DoD) Requirements for a Logistic Support Analysis Record (LSAR), (including the maintenance allocation chart (MAC)) shall be used as the baseline to prepare TMs.

b. MAC. For equipment that does not have LSAR data available, either a Preliminary Maintenance Allocation Chart (PMAC) or the MAC shall be used as the baseline to prepare TMs.

c. TB 750-93-1. If a waiver of standard requirements is granted by the contracting activity, TB 750-93-1 may be used as the baseline for preparing TMs.

d. Additional source data. Available engineering drawings shall be used with the other required data. Sound engineering principles and techniques, available engineering analyses, service experience, performance data on the item and on similar items, and all other Reliability, Availability, and Maintainability (RAM) data available shall be used in the preparation of specific instructions.

5.2 TM organization. TMs prepared in accordance with this standard shall describe in accurate and appropriate detail the unit, direct support, intermediate (aviation), general support, and depot maintenance authorized by the MAC. The operator and maintenance procedures shall be organized in MAC Functional

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1. SCOPE.

1.1 Scope. This standard establishes the requirements for the preparation of front matter, rear matter and mandatory style and format requirements for all technical manuals (TMs) covering operation and maintenance, at all levels through depot maintenance. The requirements are applicable for both paper and digital page-oriented TMs. In addition, this standard provides instructions for assembling individual information work packages into complete TMs. Electronic delivery of TMs is accomplished through the use of the Assembly Document Type Definition (DTD). The DTD is available in a digital format. Refer to MIL-STD-40051 for information on obtaining this DTD. The style and format requirements provided in this standard shall be used for text and graphics development for all TMs.

2. APPLICABLE DOCUMENTS.

The applicable documents in section 2 of MIL-STD-40051 apply to this Part.

3. DEFINITIONS.

The definitions in section 3 of MIL-STD-40051 apply to this Part.

4. GENERAL REQUIREMENTS.

4.1 General. The general information, style and format requirements and TM assembly instructions provided in this standard supplement the technical content requirements in the other parts of this 8-part standard. It contains general technical content information, mandatory style and format requirements, and TM assembly instructions for the preparation and delivery of all TMs and revisions covering operation and maintenance, at all levels through depot.

4.2 Types of technical manuals. Appendix A, Technical Manual Content Selection Matrixes, of MIL-STD-40051 lists specific technical content requirements for each type of maintenance manual, including multilevel TMs, covered by this standard. Each type of TM shall provide in detail the maintenance coverage prescribed for the applicable maintenance level(s) by the Maintenance Allocation Chart (MAC) and SMR-coded items.

4.3 Work package development. TM data developed in accordance with this standard shall be divided into individual, stand alone units of information work packages. A work package shall be task oriented and fully consistent with the maintenance concept as detailed by the MAC. Work packages shall contain descriptive, operational, maintenance, troubleshooting, support, or parts information for the weapon system or equipment.

4.4 Selective application and tailoring. MIL-STD-40051 contains some requirements that may not be applicable to the preparation of all technical manuals. Selective application and tailoring of requirements contained in MIL-STD-40051 are the responsibility of the contracting activity and shall be accomplished through the use of Appendix A, Technical Manual Content Selection Matrixes, of MIL-STD-40051. The applicability of some requirements is also designated by one of the following statements: unless specified otherwise by the contracting activity; as/when specified by the contracting activity; or when specified by the procuring activity.

4.5 Style and format. Style and format contained in this standard are considered mandatory and are intended for compliance. The examples provided at the rear of this Part are an accurate interpretation of the mandatory style and format requirements contained herein and shall be followed to ensure that the

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Group Code (FGC) order. The TM shall be organized in accordance with the applicable technical manual content selection matrix, provided in Appendix A of MIL-STD-40051.

5.2.1 TM divisions. TMs shall be divided into volumes, chapters, work packages (WPs), paragraphs, subparagraphs, and steps.

5.2.1.1 Volume size and content.

a. Division into volumes shall occur when the number of printed pages (excluding pocket TMs) exceeds 1,500 pages or 750 sheets. Each volume shall not exceed 1,500 pages or 750 sheets. A pocket TM (4 by 5-1/2 by 4 inches) or a pocket TM volume shall not exceed 200 pages or 100 sheets.

b. There shall be no page limitations for TMs prepared solely for digital display.

c. Each volume of a series shall display the TM number on its cover. Each volume of a series shall contain a title block page and table of contents.

d. Separate volumes shall not be used to distinguish between models of equipment (e.g., -10 for basic model, -10-1 for model A, -10-2 for model B, etc.).

5.2.1.2 Chapters. Chapters shall be used to divide TM data into specific functional information. Chapter types include General Information, Operating Instructions, Maintenance Information, Troubleshooting Information, Parts Information and Supporting Information. Each chapter shall be made up of one or more work packages.

5.2.1.3 Work packages. Work packages shall be used to logically divide TM data into functional task-oriented information. Work packages shall begin on a right-hand page.

5.2.1.3.1 Work package content. Work packages may contain a scope of tasks, initial setups, descriptive information, operating tasks, and maintenance tasks. These data types can be further divided into paragraphs, procedural steps, tables, lists, warnings, cautions and notes, and supporting illustrations. Refer to MIL-STD-40051-2 through MIL-STD-40051-7 for the specific content requirements for each of the functional work package types (i.e., description information, operator's instructions, maintenance, troubleshooting, repair parts, and supporting information).

5.3 Style and format. Style and format requirements provided in 5.3.1 through 5.7.5 are considered mandatory and shall be used in the preparation of all TMs. Additional preferred, non-mandatory style and format requirements shall be provided by the procuring activity.

5.3.1 Type size and style. Type style, size, and spacing shall be in accordance with best commercial practices for technical publications. Type shall be proportionally spaced (non mono spaced). Fonts shall be selected for a balance between readability and economy of space. Setting text in all capital letters shall be limited to appropriate uses, such as major headings, acronyms, equipment markings, and other instances as indicated in the samples provided at the rear of this Part.

5.3.2 Page size and orientation. The TM shall be prepared in a size selected from table 1 and specified by the contracting activity. Placement of margins, headers, and footers shall be in accordance with best commercial practices. Orientation of pages, either vertical (portrait) or horizontal (landscape), shall be consistent throughout a given manual for ease of use. The growing prevalence of TMs used in electronic display mode (instead of paper) makes this consistency extremely important. Exceptions may be made only

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if essential for proper grouping of information for the user's benefit. Otherwise, information shall be formatted or reformatted so that all pages have the same orientation.

TABLE I. Manual styles and trim sizes.

Style	Trim Size	Orientation
Pocket	4 x 5.5 5.5 x 4	Vertical Horizontal
Logbook	6.5 x 9.5 9.5 x 6.5	Vertical Horizontal
Standard	8.5 x 11 11 x 8.5	Vertical Horizontal
Double Standard	17 x 11	Horizontal

5.3.3 Foldout pages.

a. Foldout pages, if needed, shall be the same height as regular pages in the manual, and shall be folded 2, 4, or 6 times, depending on the width necessary. Each foldout shall have a blank apron wide enough for the user to look at the data while reading text elsewhere in the TM. Foldouts shall not be used in repair parts and special tools lists (RPSTL) or operator-only TMs.

b. Work packages shall not contain foldouts. Foldout pages shall follow the last work package, the glossary, or the alphabetical index, whichever forms the last portion of the TM or volume.

5.3.4 Final reproducible copy (FRC). FRC shall be a direct output of the contractor's digital TM files. The master copy of any TM is a set of digital files, not the hard-copy results. There are no peculiar layout requirements for FRC distinct from those for nonfinal drafts or proofs. The only special criterion for FRC is reproducibility: Its resolution and contrast must be sufficient for creation of offset plates or raster page images without loss of detail that would be noticeable to users.

5.3.5 Warnings, cautions, and notes.

5.3.5.1 Use and placement.

a. A warning shall precede the text of any procedure involving a clear danger to the person doing that procedure. A caution shall precede the text of any procedure involving a clear risk of damage to the equipment. A note, used to highlight essential procedures, conditions, or statements may either precede or follow the text. If multiple warnings, cautions, or notes apply to the same text, the warnings shall appear first, cautions second, notes last.

b. The header **WARNING**, **CAUTION**, or **NOTE** shall be bold and centered above the appropriate text. Headers shall not be numbered. When a warning, caution, or note consists of two or more paragraphs, the header **WARNING**, **CAUTION**, or **NOTE** shall not be repeated above each paragraph. W/C/N on unrelated topics may not be contained under one heading.

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c. All lines of warnings, cautions, and notes shall be indented five spaces or characters from both left and right margins.

d. Layout shall not result in warnings, cautions, and notes divided so first lines or groups of icons appear on one page and remaining lines or groups of icons on another.

e. Layout shall not result in warnings, cautions, and notes on a different page than the paragraph they apply to.

5.3.5.2 Icons. Use of standardized icons to improve readers' recognition of hazards is encouraged. Warning icons used shall be defined in the warning summary, and no icon shall be used without prior approval of the procuring activity.

5.3.5.3 Hazardous materials warnings and icons. Hazardous materials warnings may be presented in the standard warnings format (5.3.5.1), as an icon, or a combination of icons. Hazards that result from a combination of materials shall clearly indicate that mixing or combining the materials creates the hazard.

5.3.5.3.1 Format for hazardous materials warnings and icons. Hazardous materials warnings with icons shall consist of the icons and the nomenclature of the hazardous material. They shall immediately precede the text to which they apply. The header **WARNING** shall not be required.

5.3.5.3.2 Health hazards. Warnings shall be used when exposure to hazardous chemicals, adverse health factors, or use of the equipment cannot be eliminated.

5.3.6 Chapters.

5.3.6.1 Chapter title page **<titlpg>**. Each chapter shall begin with a chapter title page. See figure 1 for the contents of a chapter title page. A chapter title page shall always be a right-hand page.

5.3.6.2 Chapter numbering. Chapters shall be numbered in sequential order throughout the TM using Arabic numerals. Chapters shall not be renumbered in separate volumes.

5.3.7 Work packages.

5.3.7.1 Work package titles.

a. Work packages shall have titles. The title shall describe the subject or task.

b. Titles shall stand-alone (i.e., are not run-in with text) and shall begin at the left margin. (Refer to figure 2).

c. Titles shall be in capital letters and boldface type. A horizontal line shall be placed above and below the work package title to separate it from the body of the technical information.

d. When work packages are continued on subsequent pages, the work package title shall be continued at the top of those pages (for example, **FUEL PUMP MAINTENANCE - Continued**).

e. When TMs are acquired and specified by the Army for joint use with another or other Services (Joint Service TMs), work packages in joint publications which do not apply to all Services concerned shall

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be marked to indicate the Services to which they apply (for example, **LANDING GEAR MAINTENANCE (ARMY ONLY)**).

f. Work packages shall not contain foldouts. Foldout pages shall follow the last work package, the glossary, or the alphabetical index, whichever forms the last portion of the manual or volume. (Refer to 5.8.2.3.)

5.3.7.2 Work package sequential numbering. To maintain a sequential order in the TM and to facilitate referencing, each work package (WP) shall be assigned a six digit number beginning with the number 0001 00. There shall be one blank space between the forth and fifth numerals. The work package sequence numbers shall run consecutively throughout the TM. For example, the first work package in Chapter 2 will be assigned the number immediately following the last work package number in Chapter 1 (e.g., if 0010 00 is the last WP in Chapter 1, 0011 00 will be the first work package in Chapter 2). WP sequence numbers shall be assigned in numerical sequence, initially by the first four digits, then by the last two digits. (Refer to figure 2).

5.3.7.2.1 Assignment of the last two digits of the work package sequence number. The last two digits of the WP sequence number shall be reserved to permit unlimited expansion of the TM to incorporate new configuration data without affecting the WP sequence numbers already assigned. The last two digits of a WP sequence number may be assigned for the following reasons.

a. To permit adding one or more WPs between any two existing WPs during any revision cycle. The placement within the TM shall depend on the technical content task arrangement and its relationship to the existing WP and the WP(s) to be added.

b. To insert new WPs between two WPs already assigned WP sequence numbers so as not to cause renumbering just prior to producing the basic manual.

During any revision cycle, the first WP sequence number to be assigned after an existing basic WP sequence number shall be identified as "01"; for example, "0029 00" shall be followed by "0029 01". Subsequent WP sequence numbers to be assigned after "01" shall be "02" through "99"; for example, "0029 02" through "0029 99".

5.3.7.2.2 Assignment of work package sequence numbers in volumized TMs. When a TM is divided into two or more volumes, the WP sequence number shall continue in sequence. The first volume shall contain as many WPs as necessary beginning with 0001 00. The work packages contained in the second and subsequent volumes shall be numbered consecutively beginning with the number immediately following the last work package sequence number in the preceding volume.

5.3.7.3 Work package page numbering. Each work package shall be page numbered consecutively using the six digit work package sequence number followed by -1, -2, -3, etc. (e.g., 0001 00-1, 0001 00-2, etc.). Page numbers shall be centered at the bottom of the page. Even - numbers shall be assigned to the left-hand pages and odd - numbers to right-hand pages. (Refer to figure 2).

5.3.7.4 Work package identification number. For data base retrieval purposes, a unique number shall be assigned to each work package. This WP identification number will not appear on the printed or screen display page and should not be confused with the WP sequence number in 5.3.7.2. It shall be assigned when preparing the document instance in accordance with the modular DTDs and shall not be changed throughout the life of the WP. The WP identification number shall consist of an alpha designation for the type of

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information contained in the work package, a five digit block number assigned by the procuring activity, and the TM number less the maintenance level dash numbers.

a. The following alpha designators shall be assigned to the specific types of information contained within the work packages.

G - Introductory information with theory of operation

I - Inspection

O - Operator instructions

T - Troubleshooting procedures

M - Maintenance instructions

R - Repair Parts and Special Tools List (RPSTL)

S - Supporting Information

b. Examples of work package data base identification numbering are shown below.

M00432-9-1425-646

M	Identifies a WP containing maintenance instructions.
00432	Identifies the 432nd work package containing specific maintenance instructions for the M270 Armored Vehicle Mounted Rocket Launcher.
9-1425-646	Identifies the M270 Armored Vehicle Mounted Rocket Launcher TM.

T02000-1-1520-238

T	Identifies a WP containing troubleshooting procedures.
02000	Identifies the 2000th work package containing specific troubleshooting procedures for the AH-64A Helicopter.
1-1520-238	Identifies the AH-64A Helicopter.

5.3.8 Maintenance tasks and descriptive information. Procedural maintenance tasks or descriptive information contained in a WP shall have a paragraph title. When it is necessary to divide a maintenance task into subtasks, for clarity, subparagraph titles may be used. Refer to paragraph 5.3.9.2. The words "END OF TASK" shall be placed below the last data item (i.e., text, illustration, etc.) in any WP containing procedural maintenance tasks. Refer to figure 2.

5.3.9 Paragraphs.

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5.3.9.1 Paragraph numbering. Paragraphs and subparagraphs within a WP shall be unnumbered.

5.3.9.2 Paragraphs and subparagraph titles. Paragraphs and subparagraphs within a work package may have titles. If titled, the title shall begin at the left margin. A first level paragraph title shall be in all capital letters. Lower level paragraph titles shall have the first letter of the first word and of each principle word capitalized. When titled paragraphs are continued on subsequent pages, the first level paragraph title shall be placed at the top of those pages (e.g., REMOVAL - Continued). All titles shall be in boldface type. Paragraph text shall begin flush left and stand alone. Refer to figure 2.

5.3.10 Procedural steps. Procedural steps shall be used to present detailed step-by-step instructions for performing an operational or maintenance task. Procedural steps may be further divided into substeps and subordinate substeps.

5.3.10.1 Procedural step numbering.

a. Procedural steps shall be numbered consecutively with Arabic numerals.

b. If different levels of substeps and subordinate substeps are required to enhance clarity, these substeps shall also be numbered. Each level of substep numbering used shall be different. Once a numbering sequence is established, it shall remain consistent throughout the TM. (Refer to figure 2).

c. Procedural steps and substeps shall not be allowed beyond the fifth level of subordinate substeps.

5.3.10.2 Procedural step titles. Procedural steps may have paragraph or subparagraph titles. Procedural substeps and subordinate substeps shall have no titles. Refer to paragraph 5.3.9.2.

5.3.10.3 Procedural step text. Procedural step text, including the title if required, shall begin two line spaces below the preceding text and be flush left. Substeps shall be indented to enhance clarity.

5.3.11 Tables and lists.

5.3.11.1 Table numbering. Tables shall be numbered consecutively within each work package in the order of their reference. The table numbers shall begin with the Arabic numeral 1.

5.3.11.2 Table titles.

a. Each table shall have a title describing the content or purpose of the table. The table number and title shall be repeated when a table continues on subsequent pages.

b. The table title shall consist of the word "TABLE" followed by the table number, and the table title. Table titles shall be located above the table.

5.3.11.3 Table format. Tables in MIL-STD-40051-1 through MIL-STD-40051-7 are in either a standard or non-standard format. Tables designated as standard shall have no deviations. Standard tables are indicated by "**(standard table)**" within the applicable content requirements. All other tables are considered non-standard tables. A list of standard tables is provided below.

a. Controls and Indicators (MIL-STD-40051-3)

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b. Troubleshooting Symptom/System Index (MIL-STD-40051-4)

c. Flow Diagram References (MIL-STD-40051-4)

d. Symptom/Malfunction-Base Troubleshooting (MIL-STD-40051-4)

e. Operational Test-Base Troubleshooting (MIL-STD-40051-4)

f. Message Word Index (MIL-STD-40051-4)

g. Fault Report Index (MIL-STD-40051-4)

h. Operational Check Setup of a Test Module (MIL-STD-40051-4)

i. Preventive Maintenance Checks and Services (PMCS) (MIL-STD-40051-5)

j. Repair Parts List (MIL-STD-40051-6)

k. Special Tools List (MIL-STD-40051-6)

l. NSN Index (MIL-STD-40051-6)

m. Part Number Index (MIL-STD-40051-6)

n. Reference Designator Index (MIL-STD-40051-6)

o. Maintenance Allocation Chart (MAC) (MIL-STD-40051-7)

p. Aviation Maintenance Allocation Chart (AMAC) (MIL-STD-40051-7)

q. Tools and Test Equipment Requirements for MAC/AMAC (MIL-STD-40051-7)

r. Expendable and Durable Items List (MIL-STD-40051-7)

s. Mandatory Replacement Parts List (MIL-STD-40051-7)

t. Index of Manufactured Items (MIL-STD-40051-7)

u. Component of End Items (COEI) List (MIL-STD-40051-7)

v. Basic Issue Items (BII) List (MIL-STD-40051-7)

w. Additional Authorization List (AAL) (MIL-STD-40051-7)

x. Tools Identification List (MIL-STD-40051-7)

5.3.11.3.1 Graphics in tables. Graphics may be placed in tables. Graphics may be placed within a column or may span one or more columns.

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5.3.11.4 Table footnotes.

a. For numbering footnotes to tables, consecutive superior numbers beginning with 1 shall be used (unless numbers would cause confusion). (For example, footnote[1] , footnote[2] .) The numbering system shall be per table. Superior lower case letters, asterisks, or other designations may be used where numbers would cause confusion.

b. Footnotes to tables shall be placed below the closing line of the applicable table unless the table is continued. If a table is continued onto other pages, all footnotes shall be placed at the bottom of the page on which they are referenced.

5.3.12 Placement of text.

a. For 8-1/2 by 11 manuals, unless otherwise specified by the procuring activity, lines of text shall be in a single (page wide) column at the chapter and work package level. Text is single spaced (double spaced between procedural steps).

b. Procedural step text shall not be placed on an illustration.

c. Text shall always be positioned within the image area. Text shall not be wrapped around an illustration.

5.3.13 Placement of illustrations. Illustrations shall be placed as close to their reference in text as possible. Foldout illustrations shall not be included in work packages, but shall follow the last work package, the glossary, or the alphabetical index, whichever forms the last portion of the manual or volume.

5.3.13.1 Rotating illustrations. When necessary, illustrations may be placed sideways on a page (rotated 90 degrees counterclockwise).

5.3.14 Margin data. Margin data (usually headers and footers) shall be placed outside the area of the page used for either text, full-page tabular data, or full-page illustrations, but within the printing area dimensions of the page. (Refer to 5.3.14.1 and 5.3.14.2.) Complete headers and footers shall be prepared for all pages except covers and title block pages. (Refer to figure 2).

5.3.14.1 Headers. Headers shall consist of the TM number centered at the top of each page and other information as specified by the procuring activity. TM numbers for pocket TMs are required on front and back covers only. The work package sequence number shall be placed below the TM number on foldout pages. (Refer to figure 3.)

5.3.14.1.1 TM number and location. For all TMs, excluding pocket TMs, the TM number shall be centered at the top of each page and the top outer segment of each foldout page. If the manual is jointly used by two or more Services, only the contracting activity's TM number shall be placed on each page. (Refer to figure 2.)

5.3.14.2 Footers. Footers shall include the security classification markings (5.3.20), if any, the page numbers, (refer to figure 2) and other information as specified by the contracting activity (i.e., revision designator).

5.3.14.2.1 Page numbering. For all TMs page numbers shall be centered at the bottom of the page. Even numbers shall be assigned to left-hand pages and odd numbers to right-hand pages. For horizontal TMs, the upper pages shall have even numbers, and the lower pages shall have odd numbers. Page numbers shall be in

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boldface type. (Refer to figure 2.) Page numbering for RPSTLs shall also be in accordance with this paragraph and paragraphs 5.3.14.2.1.1 through 5.3.14.2.1.3.

5.3.14.2.1.1 Front matter. Page numbering for front matter shall be as follows.

a. Front cover. Front covers shall be unnumbered.

b. Warning summary. The pages of the warning summary shall be numbered consecutively using lowercase letters (i.e., a, b, c, etc.).

c. List of effective pages/work packages. When a list of effective pages / work packages is prepared, it shall be numbered with upper case letters (i.e., A, B, etc.).

d. Revision transmittal page. The revision transmittal page shall be unnumbered.

e. Title block page, table of contents, and the How to Use This Manual section These pages shall be numbered consecutively using lower case Roman numerals beginning with i (i.e., i, ii, iii, etc.).

5.3.14.2.1.2 Rear matter. DA Form 2028s, authentication pages, metric conversion charts (on the inside of the back cover), and back covers shall be unnumbered.

5.3.14.2.1.3 Blank pages. A blank page shall be assigned a number, but it shall appear on the preceding or following page. For example, if page 0001 00-10 of a work package is blank, page 0001 00-9 shall have the number 0001 00-9/10 blank; or if page 0001 00-9 of a work package is blank, page 0001 00-10 shall have the number 0001 00-9 blank/10.

5.3.15 Abbreviations and acronyms.

a. The first use of abbreviations and acronyms shall have the word(s) spelled out completely with the abbreviation or acronym in parentheses immediately after the word(s). Abbreviations and acronyms which are accepted as words (radar, sonar, laser, etc.) need not be spelled out.

b. Abbreviations and acronyms used shall be in accordance with MIL-STD-12, except that abbreviations may be plural (s) or possessive ('s). New abbreviations and acronyms shall not duplicate those presently listed in MIL-STD-12 where possible.

c. All nonstandard abbreviations and acronyms (excluding acronyms for Electrostatic Discharge (ESD) and Hardness-Critical Processes (HCP)) shall be defined in the "list of abbreviations/acronyms" paragraph of the general information work package. (Refer to MIL-STD-40051-2.)

d. Abbreviations and acronyms used in tables, but not found in the text or in any other portion of the TM, shall be spelled out in a footnote to the applicable table. Abbreviations and acronyms used in illustrations or figures, but not found in the text or in any other portion of the TM, shall be spelled out in a note to the applicable illustration or figure.

e. When abbreviations or acronyms are used as markings on the equipment (placarding), the same abbreviations or acronyms shall be used in the TM.

5.3.16 Symbols.

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5.3.16.1 General information for symbols. All nonstandard symbols (excluding icons) shall be defined in the "list of abbreviations/acronyms" paragraph of the general information work package. (Refer to MIL-STD-40051-2.) New symbols shall not duplicate those presently listed in MIL-STD-12 where possible.

5.3.16.2 Metric symbols. Metric symbols shall be in accordance with ASTM E380A-91 and IEEE 945-84.

5.3.17 Nuclear hardness (hardness-critical processes) marking. All Hardness-Critical Processes shall be marked with the acronym **HCP** as shown in subparagraph "b" below. The acronym shall be prepared in boldface type and in the same style and size as the adjacent text. The acronym shall not be shown with the titles in the table of contents. Use of the acronym is as follows.

a. When the entire task and all subordinate paragraphs and steps relate to establishing nuclear hardness, the acronym **HCP** shall precede the task title. (For example, **HCP DISASSEMBLY**.)

b. When the entire task and all subordinate paragraphs and steps do not contribute to establishing nuclear hardness, only those which do contribute shall be annotated with the acronym **HCP**. For example,

SERVICING

1. _____.

2. **HCP** _____.

c. Operating or maintenance actions which could degrade hardness, but which are not directly involved in establishing nuclear hardness, shall not be annotated with the acronym, but shall be preceded by a caution.

5.3.18 Electrostatic Discharge (ESD) sensitive marking

a. All paragraphs addressing handling or maintenance which could damage ESD sensitive parts shall be identified by the acronym **ESD** as shown below. Preparation and use of the acronym shall be similar to the requirements of 5.3.17. For example,

REMOVAL

1. _____.

2. **ESD** _____.

b. Handling or maintenance actions which could damage ESD sensitive parts, but which are not directly related to handling or maintenance of ESD sensitive parts, shall not be annotated with the acronym ESD, but shall be preceded by a caution.

5.3.19 Quality Assurance (QA) symbol. Depot and aviation maintenance procedures which have a major quality assurance effect shall be identified by the acronym **QA** in boldface letters. Only procedures at the step level shall be labelled with **QA**. (For example, 1. **QA** _____.)

5.3.20 Security classification, emergency page and protective markings

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5.3.20.1 Security classification markings. When specified by the procuring activity, a classified TM shall be prepared. The security classification markings for classified TMs, titles of parts, chapters, work packages, appendixes, paragraphs, illustrations, tables, and their contents, shall be identified in accordance with DoD 5200.1-R and DoD 5220.22-M. Additional instructions applicable to security classification markings are described in paragraphs 5.3.20.2 through 5.3.20.4.

5.3.20.2 Overall security classification. The overall security classification assigned to a TM shall agree with the highest security classification assigned to any portion within, and shall be marked accordingly at the top and bottom of the front and back cover sheets. The security classification markings for pages, including those for unclassified pages, shall be bold and at the top and bottom center of each page. (Refer to figure 4.)

5.3.20.3 Security classification placement for foldouts. For foldouts, the security classification shall be in boldface type, placed 3/4 inch from the right-hand edge of the page, and repeated continuously to the left with four inches of space between each placement.

5.3.20.4 Blank page backing a classified page. Blank pages normally require no copy. However, if the reverse side of a blank page contains classified material, security markings for the blank page shall be bold and at the top and bottom center of the blank page. The blank page shall reflect the highest classification of the reverse side, and include the statement "This page is unclassified".

5.3.20.5 Emergency page markings. When specified by the contracting activity, emergency pages shall be prepared. Pages containing emergency information shall have a broken black border placed on the three unbound edges. The border, the black marking, and the space between markings shall each be 3/16 inch wide. The angle of markings shall be 45 degrees. (Refer to figure 5.)

5.3.20.6 Protective markings. When specified by the contracting activity, a FOR OFFICIAL USE ONLY protectively marked manual shall be prepared. In FOR OFFICIAL USE ONLY manuals, the protective marking shall be bold and at the bottom of each page.

5.3.20.7 COMSEC protective markings. Unless requirement is specifically excluded by the contracting activity, Army Communications Security (COMSEC) Equipment Manuals shall contain the protective marking "FOR OFFICIAL USE ONLY".

5.3.21 Referencing.

5.3.21.1 Other documents. Reference shall be made only to other documents available and authorized to the user. Reference shall be to the publication number and, when necessary, to the work package sequence number. References to pending publication actions shall not be made.

5.3.21.2 Government specifications and standards. Reference shall be made to the basic number for Government specifications and standards.

5.3.21.3 Within the TM. Reference made within the TM shall be to the necessary location data only. For example, if the entire chapter is applicable, only the chapter number shall be referenced; if the data needed is contained in another work package or within the same work package, reference shall be made to the work package sequence number, to maintenance task titles, to titled procedures, to steps, to titled figures, and to titled tables.

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5.3.21.4 Repeating information. Repeating information shall be allowed to ensure the work package information is complete. Information, two pages or less, may be repeated; information more than two pages shall be referenced.

5.3.21.5 Equipment, components, and parts. Reference to parts of the equipment and to equipment components may be by nomenclature, model, type, reference designator, and figure and item number, as applicable. Reference shall be made only to models or types of equipment covered by the manual. To facilitate coverage of modified or additional models or types at a later date, references shall be held to a minimum.

5.3.21.6 National Stock Numbers (NSNs) and Part Numbers (P/Ns). Reference to NSNs shall be made only in tables, other tabular material, and lists. Reference to NSNs shall not be made on illustrations or in illustration legends. Reference to P/Ns shall not be made in the narrative portions of the TM, procedural steps, illustrations, or legends, except when essential for identification. Reference to P/Ns may be made in tables, other tabular material, and lists.

5.3.21.7 Equipment panel markings (placarding). Reference shall be made to panel markings and switch positions exactly as marked on the equipment. However, symbols on panel markings shall be spelled out when they cannot be produced by the software, composing equipment, or printers used in producing the manual, such as the symbol for ohm, infinity, etc.

5.3.21.8 Metric and U.S. standard measurements. Unless specified otherwise by the contracting activity, all measurements shall be expressed in both U.S. standard units (e.g., ounces, pounds, gallons, inches, feet, knots, miles, etc.) and metric units. U.S. standard measurements shall be followed by the metric conversion in parentheses unless the equipment, instrument, or tool, etc., is calibrated in metric units. In that case, metric units shall be first, followed by the U.S. standard units. (For example, "169.5 Nm (125 lb-ft)".)

5.3.21.9 Temperature. Reference shall be made to temperature readings as calibrated on the equipment. If other than Fahrenheit, the equivalent in Fahrenheit shall follow in parentheses. General temperature references, such as room temperature, shall be given in degrees Fahrenheit (for example, 78°F).

5.3.21.10 TM divisions. Reference shall be made to any major division of the manual. (For example, Volume 5, Chapter 6, Table of Contents, Glossary, Index, etc.) or by an abbreviation (WP, etc.).

5.3.21.11 Volumes. References to information in another volume within the TM shall include the volume number.

5.3.21.12 Work packages. References to work packages within the same TM shall be to the work package sequence number (i.e., WP 0125 00, etc.). .

5.3.21.13 Maintenance tasks, procedures, and paragraphs. Reference to maintenance tasks, procedures, and paragraphs shall be by work package sequence number and reference to title, as necessary (i.e., WP 0025 00, Disassembly; WP 0012 00, Equipment Data).

5.3.21.14 Tables. Reference shall be made to tables by table number (for example, table 2) within a work package. Reference shall be made to tables in a different work package by work package sequence number and table number (for example, WP 0012 00, table 2). Reference shall be made only to tables within the same manual or another volume of the same manual.

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5.3.21.15 Footnotes. Reference shall be made to footnotes when essential for reference, explanation, comments, or other information.

5.3.21.16 Figures and multisheet figures. Reference shall be made to figures within a work package by figure number (for example, figure 2) and the sheet number for multisheet illustrations, when applicable (for example, figure 17, sheet 1). Reference shall be made to figures in a different work package by work package sequence number and figure number (for example, WP 0012 00, figure 2). References shall be made only to figures within the same manual or another volume of the same manual.

5.3.21.17 Index numbers.

a. Reference shall be made to figure numbers first, followed by the index number; for example, (figure 6, 34). However when multiple references refer to the same figure, only the first reference shall indicate the figure number. When the index numbers continue on successive pages, the figure number shall be repeated with the first index number reference on each succeeding page. For example,

"1. Unscrew safety disc retainer (figure 2, 1) from valve body (2).

2. Remove safety disc (3) and safety disc washer (4) from valve body (2)."

b. When the index numbers continue in sequence for procedures requiring two or more pages, the figure number shall be repeated in the first reference on each succeeding page.

c. If two or more figures are involved in the same sequence (procedure or task), the figure with the greater number of items shall be cited as described above. When the reference changes, the new figure reference shall be cited and also have the index number follow the figure number, i.e., "(figure 5, 21)." In such cases, the paragraph lead-in shall contain a statement similar to the following:

"NOTE

Item numbers below refer to figure 4 unless otherwise indicated."

d. For illustrations which do not have figure numbers, reference shall be made to the index numbers only.

5.3.21.18 Items on diagrams. Reference shall be made to parts on diagrams by enough of their description or reference designator to identify the item (for example, resistor A6R11).

5.3.22 Equations. The use of equations shall be held to the minimum use required by the needs of the TM user.

NOTE

MATHPACK 911001 as included in MIL-PRF-28001 shall be used for preparing equations. The use of some equations may be limited by the Mathpack and the output system.

5.3.23 Graphics.

5.3.23.1 Graphic format. All graphics developed in accordance with this standard shall be delivered in one of the three graphic formats: MIL-PRF-28003, Computer Graphic Metafile (CGM); MIL-PRF-28002,

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Continuous Acquisition and Life-cycle Support (CALS) Raster; or MIL-PRF-28000, Initial Graphics Exchange Specification (IGES).

a. The CGM file format is the preferred graphics file format.

b. All graphics files for a particular TM should be applied in the same graphics format if practical. Otherwise, files may be delivered in any combination of the allowable formats.

c. Appropriate header and identification information shall be included in each graphics file. Refer to the applicable specification for the specific requirements.

5.3.23.2 Types of graphics. The following types of graphics shall be used in the preparation of TMs.

a. Line drawings.

b. Photographs.

c. Engineering drawings.

d. Diagrams.

e. Charts and graphs.

5.3.23.2.1 Line drawings. Line drawings including exploded views, locator views, and detailed views shall be used to support the operational and maintenance procedures, and the RPSTLs.

a. When index numbers are used to locate and identify equipment components or parts, the index numbers shall be assigned sequentially (clockwise, disassembly, or in the order mentioned in text).

b. To assist the maintenance technician or operator in locating major components, controls and indicators, etc., locator views may be included. Refer to figure 6.

c. When the illustration does not adequately or clearly depict the subject matter or part(s), specific detailed views may be included to support the main illustration. Refer to figure 7.

5.3.23.2.2 Multiview and multisheet illustrations. Multiview and multisheet illustrations may be used to clarify, identify significant features, or further detail equipment assemblies, subassemblies, and detailed parts.

5.3.23.2.3 Photographs. When approved by the procuring activity, photographs may be used in lieu of line drawings. When approved for use by the procuring activity, prescreened photographs are acceptable as final reproducible copy provided they have been screened only once, and the screen on the final sized illustrations shall be as specified by the procuring activity. When prescreened photographs are used, they shall be clearly marked to indicate prescreening. As specified by the procuring activity, unscreened continuous tone photographs and/or original illustrations shall be supplied with final reproducible copy.

5.3.23.2.4 Engineering drawings. Unless specified otherwise by the contracting activity, engineering drawings shall not be used as illustrations. When used,

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a. They shall be in accordance with MIL-STD-100, and MIL-T-31000 (required for new designs after 01 July 1990) and shall be modified, as necessary, to meet the content, style, arrangement, legibility, format, and production requirements described in this document and the contract.

b. All unnecessary data that would reduce the comprehension or clarity of the illustration shall be removed.

c. They must be reduced or redrawn to meet page size restrictions.

5.3.23.2.5 Diagrams.

5.3.23.2.5.1 Diagram specifications. Diagrams shall be prepared in accordance with the specifications listed below.

Subject	Equipment Covered	Specification
Abbreviations	All	MIL-STD-12
Drafting Practices	Mechanical, Electrical and Electronic	ANSI Y14.15-1966 (R1973)
Engineering Drawing Practices	All	MIL-STD-100, MIL-T-31000
Graphic Symbols	Electrical and Electronic	IEEE 315A-86, IEEE 280-85
	Mechanical	MIL-STD-17-1 and -2
	Digital (Logic)	IEEE 91-84
	Fluid Power	ANSI Y32.10
Reference Designators	Electrical and Electronic	IEEE 200-75
Unit Symbols	All	IEEE 260-78
Logic	All	IEEE 91-84

5.3.23.2.5.2 Types of diagrams. The following types of diagrams may be included in the TM.

a. Block diagrams.

b. Schematic diagrams.

c. Pictorial diagrams.

d. Cutaway diagrams.

e. Logic diagrams.

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f. Wiring diagrams/wire lists.

g. Cable diagrams.

h. Piping diagrams.

i. Test setup diagrams.

5.3.23.2.6 Charts and graphs. Charts and graphs shall be prepared as illustrations. Instructions shall be provided for use and interpretation of complex graphs.

5.3.23.3 Graphic techniques. Graphic techniques provided in 5.3.23.3.1 through 5.3.23.6, shall be used for the preparation of U.S. Army TMs.

5.3.23.3.1 Illustration figure numbers. Figure numbers for illustrations contained in work packages shall be avoided whenever possible. However, figure numbers for Depot Maintenance Work Requirements (DMWRs), Repair Parts and Special Tool Lists (RPSTLs), and foldouts are required. Figure numbers shall be placed on the illustration and be an integral part of the illustration.

5.3.23.3.2 DMWR figure numbers. Figures shall be numbered consecutively within each work package starting with the Arabic numeral 1.

5.3.23.3.3 RPSTL figure numbering. Figures for RPSTL TMs shall be numbered sequentially within a RPSTL chapter (not within each work package), using Arabic numerals beginning with 1. Figures in RPSTL supporting information work packages shall also be numbered sequentially within a RPSTL chapter (not within each work package), using Arabic numerals beginning with 1.

5.3.23.3.4 Foldout figure numbering. The figure numbers for foldouts shall be in accordance with 5.3.23.3.1 and shall be placed preceding the figure title under the illustration. (Refer to figure 3.)

5.3.23.3.5 Multisheet numbering. Multisheet figures shall be consecutively sheet numbered following the title; for example, "Figure 2. Wing Hydraulic Assembly (Sheet 1)." or"Figure 1. Cable Assembly W12 Wiring Diagram (Sheet 1)." Remaining sheets shall be numbered in consecutive order, Sheet 2, Sheet 3, etc.

5.3.23.3.6 Illustration figure titles. When illustration figure numbers are used, a figure title is required. The title shall describe the subject or content of the figure, and follow the figure number; for example, “Figure 2. Cannon Assembly.” Figure titles shall be centered below the illustration.

5.3.23.3.7 Illustration legends. When text legends are prepared to identify index numbers on illustrations, the legend shall become part of the illustration and not be placed with the text of the TM or work package.

5.3.23.3.8 Illustration identification numbers. Each illustration shall be assigned a unique identification number provided by the proponent activity.

a. Contractor's identification number may be used when approved by the proponent activity.

b. When the identification number is to be printed in the TM, such number shall be approximately 4- to 6- point type and placed in the lower right-hand corner of the illustration (within the graphics area) sufficiently removed to avoid being confused as part of the illustration.

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5.3.23.4 Illustration consistency. A standard referencing system for associated text, signal flow and equipment nomenclature shall be used between illustrations and text.

a. Standard graphic symbols shall be used when possible.

b. If special graphic symbols are required, they shall be made visually distinctive from other graphic symbols used and included in a special symbols chart.

c. Official nomenclature shall be used for hardware, controls, indicators, switches, etc.; consistent, standard nomenclature shall be used for functions, signals, etc.

5.3.23.4.1 Portraying signal flow. Signal flow, especially for electrical and electronic equipment, critically affects the understandability of diagrams. To assist the TM user in following the diagram, where possible, major signal or pressure flow shall be from left to right, and feedback or return flow shall be from right to left.

5.3.23.5 Color in illustrations. Unless specified otherwise by the contracting activity, black and shades of black (one color) shall be used for TMs. Prior approval for color will be obtained by the contracting activity from the U.S. Army Publications and Printing Command (USAPPC). The contracting activity will provide written approval, designating color(s) to be used.

a. When color (other than black) is required, it shall be held to the minimum absolutely necessary to highlight or clarify important information.

b. The number of colors shall be kept to a minimum by use of various techniques such as tints, patterns, cross-hatching, and dots.

c. Any number of shades of a primary color used shall be considered as one color (e.g., a two-color printing could consist of black and three shades of red).

d. When color is approved/specified, the primary colors of red and blue shall be used first.

e. Yellow shall not be used alone.

5.3.23.6 Hazard icons. Standard icons may be used in warnings when applicable. Hazard icons may be used in technical manual warnings either singly or in combinations. The list of standard icons shall be provided by the procuring activity.

5.4 Copyrights, proprietary names, and advertising.

5.4.1 Copyright/copyright credit line. TMs shall not contain copyrighted material except as specified in the Federal Acquisition Regulations (FAR) and Defense Federal Acquisition Regulation (DFAR) Supplement. When copyrighted material is to be included in a TM, the preparer shall obtain prior written permission from the copyright owner or authorized agent for its use. The signed, written permission shall be delivered with the FRC when it is delivered. The written permission shall contain a statement declaring whether or not a copyright credit line is required.

5.4.2 Proprietary names. Trade names, copyrighted names, or other proprietary names applying exclusively to the product of one company shall not be used unless the items cannot be adequately described because of the technical involvement, construction, or composition. In such instances, one, and if possible, several

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commercial products shall be listed, followed by the words "or equal." The same shall apply to manufacturers' part numbers or drawing numbers for minor parts where it is impractical to specify the exact requirements. Insofar as practical, the particular characteristics required for the "or equal" products shall be defined.

5.4.3 Advertising. Publication material prepared in accordance with this standard shall contain no advertising matter (other than that allowed in 5.4.2).

5.5 Clarity. TMs shall be written for the target audience. Reading grade level shall be as specified by the procuring activity.

5.6 Nomenclature.

5.6.1 Nomenclature consistency and applicability. Nomenclature, other terms, and names shall be consistent within a manual and throughout the RPSTL, MAC, and other directly related manuals. Statements that explain applicability for individual items of equipment shall use specific serial numbers, block designations, model designations, or similar identification. Such terms as "on later equipment" and "on early serial numbers" shall not be used.

5.6.2 Official/approved nomenclature. Unless specified otherwise by the procuring activity, only approved names and official nomenclature shall be used. (Official nomenclature shall be the nomenclature used in the RPSTL.) If unofficial nomenclature (common name) is approved, an appropriate nomenclature cross-reference list shall be prepared for the TM. (Refer to MIL-STD-40051-2.) Shortened versions of the approved nomenclature are not considered deviations. Approved nomenclature shall be used wherever the use of a common name might be ambiguous.

5.6.3 Military terms. Military terms used shall be in accordance with Joint Pub 1-02, or any approved dictionary or glossary of Army military terms.

5.6.4 Automatic electronic test and checkout terminology. Terms used for automatic electronic test and checkout shall be in accordance with MIL-STD-1309.

5.7 Revisions. Revisions shall consist of a revision transmittal page and applicable revision pages or work packages. For the content and format of a revision transmittal page, refer to figure 8.

a. Each revision to a TM shall be numbered in sequence beginning with 1.

b. Front matter, work package, and rear matter revision pages shall conform to the style and format of the basic TM and shall incorporate all approved information.

5.7.1 Revised work packages. When updates to a work package are made the entire work package shall be revised and reissued.

5.7.2 Revised front and rear matter pages. When updates to the front and rear matter of a TM are required, only the effected pages shall be revised and reissued. Copy shall be prepared for both sides of the printed page on which an update is made, even when an update is made to one side only.

5.7.3 Revision symbols. Revision symbols shall be inserted to identify technical updates in text, illustrations and tables.

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a. Updates to the text and tables shall be indicated by a vertical line (revision bar) opposite the updated, deleted, or added text (except as noted below). The revision bar shall be placed in the margin opposite the binding edge. Exception: Pages with emergency markings (black diagonal lines around three edges) shall have the revision bars placed along the inner margins.

b. When tables are updated or added, the revision bar shall also be placed to the left of the table number and title.

c. The method(s) used shall be explained in the revision transmittal page.

d. Revision symbols from a previous revision shall be deleted when a page is subsequently updated. Symbols shall show current updates only.

e. If the composing equipment is incapable of producing a vertical line, another symbol may be used as specified by the contracting activity (for example, a number sign "#;" plus sign "+;" black circle; black square; the letters "C," "R," or "X").

f. Revision symbols are not required for the following.

(1) Indexes and tabular data where the update cannot be identified.

(2) Blank space resulting from the deletion of text, an illustration, part of an illustration, or a table.

(3) Correction of minor inaccuracies, such as spelling, punctuation, relocation of material, renumbering, etc., unless such correction changes the meaning of the information.

5.7.4 Revision symbols for illustrations. Unless specified otherwise by the procuring activity, a miniature pointing hand shall be used for illustrations (other than diagrams and schematics) to highlight the area containing the revised information. (Refer to figure 9).

a. Revisions confined to the same general area shall be indicated only once on the illustration.

b. A vertical line next to revised text and callouts on illustrations may be used in lieu of a pointing hand.

c. A vertical line next to revised material may be used on a chart or graph.

d. If a callout is deleted from an illustration, the word "DELETED" shall be placed after the appropriate number in the legend, if applicable. If a callout is deleted from an illustration without a legend, such as those used to supplement a RPSTL, the word "DELETED" shall be placed on the illustration at the end of the leader line. NOTE: Consideration should be given to showing the callout with a strikeout to indicate that it has been deleted.

e. When an illustration is revised, index numbers added between existing numbers shall be the same as the preceding index number with added alpha characters (e.g., 22A, 22B). This system shall also be used in basic manuals when errors are discovered so late in preparation that renumbering of all following index numbers would delay submittal. Index numbers with added alpha characters shall be eliminated for a revision.

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f. As specified by the procuring activity, screens (shading), screened (shaded) boxes, or miniature pointing hands shall be used to highlight updated areas of diagrams and schematics. (Refer to figure 9.)

g. When a figure has a figure number and title, a vertical line shall be placed to the left of the figure number and/or title.

5.7.5 Revisions to RPSTL supporting information work packages and TMs Requirements shall apply with the following exceptions.

a. Deleted figures and items. When figures and items have been deleted, the cross-reference indexes work packages shall be revised as necessary.

b. Item changes. Unless specified otherwise by the procuring activity, an asterisk shall be placed to the left of the item no. column in the list adjacent to the line item indicating that an update has been made to the item and is reflected in the associated text, illustration, P/N index, or reference designator index.

5.8 Technical manual assembly. Requirements for the preparation of front and rear matter necessary to supplement the technical content chapters and associated work packages are provided in 5.8.1 through 5.8.2.6. Appendix A, Technical Manual Content Selection Matrixes, of MIL-STD-40051 provides detailed assembly and content requirements for all TMs covering operation, maintenance and parts information, at all levels through depot.

5.8.1 Front matter **<frnt>**. As applicable, material preceding the first text page shall consist of the following in the order specified below.

a. Front cover.

b. Warning summary.

c. List of effective pages/work packages.

d. Revision transmittal page.

e. Title block page.

f. Table of contents.

g. "How To Use This Manual" information.

5.8.1.1 Front cover **<frntcover>**. A front cover shall be prepared for each TM and DMWR. The format of the TM front cover is shown in figure 10. The format of the DMWR front cover is shown in figure 11. The front cover shall contain the following content information.

a. Security classification (when required).

b. TM number **<tminfono>**.

c. TM title **<prtitle>**.

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d. National stock number (NSN) **<nsn>** for item(s) covered.

e. End Item Code (EIC) **<eic>**, as specified in the Army Master Data File (AMDF).

f. Subtitle when required **<stitle>**.

g. Equipment illustration when required **<graphic>**.

h. Availability statement **(depot only)** **<avail>**.

i. Supersedure notice for revisions only **<super>**.

j. Disclosure notice **<disclos>**.

k. Distribution statement **<dist>**.

l. Export control notice warning **<export>**.

m. Destruction notice **<destr>**.

n. Service nomenclature **<servnomen>**.

o. TM date.

p. Revision designator **<chgno>** and date **<chgdate>** for revisions only.

q. Reproduction notice **(depot only)**.

r. Distribution restriction **(depot only)**.

Additional detailed requirements for the above front cover content information is described in 5.8.1.1.1 through 5.8.1.1.8.

5.8.1.1.1 TM number for joint service TMs. If the manual is jointly used, each Service's number shall be placed on the front cover and only the proponent activity's TM number shall be placed on each page within the TM. The numbers shall be prefixed with the word Air Force, Army, Marine Corps, or Navy (NAVSEA or NAVAIR), as applicable. The contracting activity's (proponent activity's) name **<servbranch>** and manual number **<tmno>** shall be placed first. The TM number(s) for the other Services shall be in alphabetical sequence following the contracting activity's name and manual number. For example,

"ARMY TM 11-1510-204-34
AIR FORCE TO 21M-LGM30G-12
MARINE CORPS TM 12345A-15/1
NAVY (NAVAIR) AI-F18AA-WRM-070
NAVY (NAVSEA) SE211-FA-MMA-010/SPS-10A"

5.8.1.1.2 Availability statement **(depot only)** **<avail>**. For DMWRs only, the front cover shall contain the following availability statement: "This publication is not available through the AG publications centers. This publication is available through *(insert the name and address of the proponent activity*)."

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5.8.1.1.3 Disclosure notice **<disclos>**. Unless specified otherwise by the contracting activity, the following disclosure notice shall be placed on the front cover of all classified and unclassified TMs, except those with distribution statement A: "This information is furnished upon the condition that it will not be released to another nation without the specific authority of the Department of the Army of the United States, that it will be used for military purposes only, that individual or corporate rights originating in the information, whether patented or not, will be respected, that the recipient will report promptly to the United States, any known or suspected compromise, and that the information will be provided substantially the same degree of security afforded it by the Department of Defense of the United States. Also, regardless of any other markings on the document, it will not be downgraded or declassified without written approval of the originating United States agency."

5.8.1.1.4 Distribution statement **<dist>**. All TMs shall have a distribution statement placed on the front cover for each manual or revision. (Refer to figure 10.) The appropriate distribution statement shall be provided by the contracting activity as selected from DOD 5230.24.

5.8.1.1.5 Destruction notice **<destr>**. All TMs marked with distribution statements "B", "C", "D", "E", "F", or "X" shall be marked with the destruction notice provided by the contracting activity from DoD 5230.24. (Refer to figure 10.)

5.8.1.1.6 TM date. TMs shall be dated. The TM date shall be the date at which the last material to be included was received (copy freeze date, provided by the contracting activity). The day, month, and year shall be given in that sequence (for example, 10 JULY 1988). (Refer to figure 10.)

5.8.1.1.7 Revision designator and date **<chgno> <chgdate>**. When updates are prepared, the revision designator and date shall be shown on the front cover (refer to figure 10), the title block page, and the revision sheet. Unless specified otherwise by the contracting activity, the revision date shall be the date at which the material to be included was received (copy freeze date, provided by the contracting activity).

5.8.1.1.8 For Army COMSEC manuals use. Unless specified otherwise by the contracting activity, classified TMs shall contain the notice FOR OFFICIAL USE ONLY. It shall be placed above the security classification at the bottom center of the front cover and all TM pages.

5.8.1.2 Warning summary (including first aid data)**<warnsum>**. A warning summary shall be prepared for all TMs containing warnings and cautions. The content and format of the warning summary is shown in figure 12. The warning summary shall appear on the first right-hand page immediately after the front cover. The warning summary shall include each general type of warning and warning symbol**<warninfo>** and hazardous materials warnings **<hazmat>**.

5.8.1.2.1 Preventive maintenance services manual warning summary**(aviation only)**. For preventive maintenance services only, the following statement shall be placed on the front page.

"WARNING

Certain inspections are Mandatory Safety-of-Flight requirements, and the inspection intervals cannot be exceeded. In the event these inspections cannot be accomplished at the specified interval, the aircraft condition status symbol will be changed to a red X. Mandatory Safety-of-Flight inspection items are printed in bold face type.

NOTE

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Inspection items contained in this manual are considered the minimum requirements for performing phased maintenance and must be performed. The cumulative effects of inspection deferrals are unknown and could result in catastrophic failure or increased maintenance at a later date. Therefore, the use of special lettering to emphasize Mandatory Safety-of-Flight Items is not to be construed as authority for deferral of other inspections."

5.8.1.3 List of effective pages/work packages. When directed by the procuring activity, a list of effective pages/work packages shall be prepared in accordance with figure 13. The list of effective pages/work packages shall be prepared and transmitted with the basic version of the TM and each subsequent revision. It shall be located immediately following the warning summary.

5.8.1.4 Revision transmittal page **<chgsheet>**. A revision transmittal page shall be prepared and accompany each revision to a TM. The revision transmittal page shall not be page numbered and shall be located following the warning summary, or, the list of effective pages/work packages, if included. (Refer to figure 8).

5.8.1.5 Title block page **<titleblk>**. A title block page (refer to figure 14) shall be prepared and follow the list of effective pages/work packages, if any. The title block page shall include the reporting errors and recommended improvements statement **<reporting>**. For RPSTLs, the "current as of" date shall be inserted. When depot level repair parts are included in a lower level RPSTL, the following statement shall be added to the RPSTL title: "(Including Depot Maintenance Repair Parts)."

5.8.1.5.1 Reporting errors and recommending improvements statement **<reporting>**. A reporting errors and recommending improvements statement (refer to figure 14) shall appear below the prime title, NSN, EIC, and subtitle (if any) on the title block page. The mailing address, e-mail address and fax number of the responsible proponent shall be inserted in the statement.

a. Unclassified/standard TM. Except for classified TMs, oversize TMs, pocket size TMs, and TMs with less than eight pages, the following statement shall precede the table of contents title.

"REPORTING ERRORS AND RECOMMENDING IMPROVEMENTS

You can help improve this manual. If you find any mistakes or if you know of a way to improve the procedures, please let us know. Mail your letter, DA Form 2028 (Recommended Changes to Publications and Blank Forms), located in the back of this manual direct to: (*name and address of proponent*). You may also send in your recommended changes via electronic mail or by fax. Our fax number is (*insert DSN and commercial number of proponent*). Our e-mail address is (*insert address of proponent*). A reply will be furnished to you."

b. Pocket size TMs, oversize TMs, and TMs with less than eight pages. For pocket-size TMs, oversize TMs, and TMs with less than eight pages, the following statement shall precede the table of contents title.

"REPORTING ERRORS AND RECOMMENDING IMPROVEMENTS

You can help improve this manual. If you find any mistakes or if you know of a way to improve the procedures, please let us know. Mail your letter or DA Form 2028 (Recommended Changes to Publications and Blank Forms) direct to: (*name and address of proponent*). You may also send in your recommended changes via electronic mail or by fax. Our fax number is (*insert DSN and commercial number of proponent*). Our e-mail address is (*insert address of proponent*). A reply will be furnished to you."

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c. Classified TMs. For classified TMs, the following statement shall precede the table of contents title:

"REPORTING ERRORS AND RECOMMENDING IMPROVEMENTS

You can help improve this manual. If you find any mistakes or if you know of a way to improve this manual, write and tell us about it. Address your correspondence to (*name and address of proponent*). When dealing with classified information, make sure that your correspondence is properly marked and is handled in accordance with current security regulations."

5.8.1.6 Table of contents <**contents**>. A table of contents listing chapters and work packages in the same order and with the exact titles as used in the body text shall be prepared for all TMs. The content and format of the table of contents is shown in figure 15. When space permits, the table of contents shall begin on the title block page (5.8.1.5) below the reporting of errors and recommending improvements statement. If space does not permit, the table of contents shall begin on a right-hand page following the title block page.

a. The security classification, if any, of chapters and work packages shall be indicated.

b. Each volume of a multivolume manual shall contain its own table of contents and shall reference companion volumes for the same TM. Volume 1 shall contain a complete table of contents covering the entire set. Entries shall indicate the volume in which the referenced material appears for example, Operator Instructions, Vol 1.

c. The following requirements are applicable to a RPSTL table of contents.

(1) Titles of RPSTL work packages, including the Functional Group Codes (FGCs) as applicable, shall be listed by the same nomenclature and in the same sequence in which they appear in the tabular listings. The WP sequence number shall be referenced with each WP title. (Refer to figure 16.)

(2) NSN, P/N and (as applicable) reference designator cross-reference indexes shall be listed.

(3) Only the work packages applicable to the maintenance level(s) for which the RPSTL is prepared shall be listed.

(4) The table of contents shall begin on a separate, right-hand page.

5.8.1.7 "How To Use This Manual" information<**howtouse**>.

a. "How to Use This Manual" information shall be located after the table of contents and before the first chapter of the TM. "How to Use This Manual" information shall begin on a separate, right-hand page.

b. Information to familiarize the user with special or unusual features of the TM shall be prepared. Coverage shall lead the user through the TM and explain important features of the organization and content. For example, the format is explained; operating, troubleshooting, Preventive Maintenance Checks and Services (PMCS) are explained; and repair, maintenance instructions, and other pertinent information are explained.

c. Any peculiarities in the basic arrangement of the TM shall be described. "How To Use This Manual" information shall not repeat instructions given within the chapters and/or work packages.

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Check the source to verify that this is the current version before use.

d. For all TMs (excluding operators) the "How To Use This Manual" information shall include reference to the associated RPSTL and an explanation on how to use the RPSTL in conjunction with the manual.

e. For all TMs with a glossary, reference to the glossary shall be made and an explanation of its features and use shall be provided.

5.8.1.7.1 International standardization agreements. When specified by the procuring activity, the "How To Use This Manual" shall contain the following.

"NOTE

Certain provisions of this technical manual (*identify by chapter, work package, paragraph, or similar manner, if appropriate*) are the subject of international standardization agreement (*insert the ABCA or ASCC standard number; the NATO, STANAG, NETR, or NEPR number; or appropriate documentary reference*). When revision or cancellation of this technical manual is proposed which will modify the international agreement concerned, the technical manual management activity will take appropriate action through international standardization channels, including departmental standardization offices, to change the agreement or make other appropriate accommodations."

5.8.2 Rear matter **<rear>**. As applicable, material following the last text page shall consist of the following: glossary **<glossary>**, alphabetical index **<aindx>**, foldout pages **<foldsect>**, reporting errors and recommending improvements DA Forms 2028 **<da2028>**, the authentication page **<authent>**, and back cover **<back>**. For RPSTLs, the glossary and alphabetical index is not required.

5.8.2.1 Glossary **<glossary>**. A glossary shall be prepared for TMs only when the terms are uncommon and are not adequately defined in the text or in the Army, DoD, or standard dictionary. The glossary shall include a list of terms **<term>** followed by definitions **<def>**. The terms shall be listed in alphabetical order. If a glossary is required, it shall begin on a separate, right-hand page and immediately precede the alphabetical index, if any. (Refer to figure 17.)

5.8.2.2 Alphabetical index **<aindx>**.

a. An alphabetical index shall be prepared unless specified otherwise by the procuring activity. The index may be an index of work packages only or it may be a detailed index, as applicable.

b. Detailed indexes shall include entries for every subject which may be useful to the user. "See" and "See also" references may be included to guide the user to other entries.

c. All applicable work package references for each entry shall be indicated, regardless of the type of index being prepared. Page references may be included in a detailed index.

d. The index shall be located at the end of the TM but shall precede the sample DA Form 2028. Indexes shall begin on a separate, right-hand page. (Refer to figure 18.)

5.8.2.3 Foldout pages **<foldsect>**. If foldout pages are approved by the procuring activity, such pages shall follow the last work package, the glossary, or the alphabetical index, whichever forms the last portion of the manual or volume.

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Check the source to verify that this is the current version before use.

5.8.2.4 Reporting errors and recommending improvements DA Form 2028<**da2028**>. One filled-out sample copy of DA Form 2028, provided by the contracting activity, and three blank DA Forms 2028 with the TM number, date, and title shall be included at the back of every unclassified TM (except for oversize TMs, pocket-size TMs, and TMs with less than eight pages). The filled out sample shall include guidelines for completing the form.

5.8.2.5 Authentication page <**authent**>. The authentication page, provided by the contracting activity, shall be the last printed text page of the TM. (Refer to figure 19.)

5.8.2.6 Back cover <**back**>. The outside back cover shall be blank, except for pocket-size TMs and classified TMs. For pocket-size TMs, the outside back cover shall include the TM number. For classified TMs, security classification markings shall be included on the back cover. When applicable, a metric conversion table, covering applicable units included in the TM, shall be placed on the inside back cover. (Refer to figure 20 .)

6. NOTES.

The notes in section 6 of MIL-STD-40051 apply to this Part.

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TM 9-2350-314-24

CHAPTER 6

UNIT MAINTENANCE INSTRUCTIONS FOR 155 MM, M109A6 HOWITZER

FIGURE 1. Example of a chapter title page.

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TM 11-6625-3178-14

RIFLE HOLDER ASSEMBLY REPLACEMENT ← *WORK PACKAGE TITLE* 0310 00

THIS WORK PACKAGE COVERS:
Removal, Installation

WORK PACKAGE SEQUENCE NUMBER

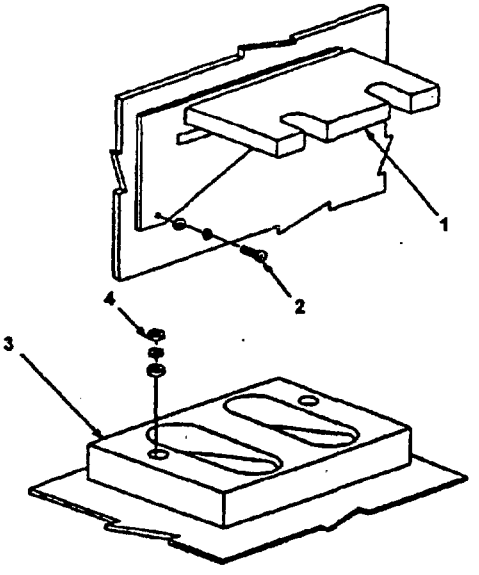
INITIAL SETUP:

Maintenance Level
Unit
Tools and Special Tools
Tool Kit, General Mechanics (Item 2, WP 1462 00)

Materials/Parts
Rifle Barrel Holder, 107236
Rifle Stock Holder, 107242

REMOVAL ← *MAINTENANCE TASK TITLE (PARAGRAPH TITLE)*

Rifle Barrel Holder Removal ← *PROCEDURAL STEP TITLE (SUBPARAGRAPH TITLE)*



PROCEDURAL STEP

1. On rifle barrel holder assembly (1), remove two screws, lock washers and flat washers (2) holding rifle barrel holder assembly to shelter wall and remove rifle barrel holder assembly.

Rifle Stock Holder Removal

1. On rifle stock holder assembly (3), remove two nuts, lock washers and flat washers (4) holding rifle holder assembly to shelter floor and remove rifle holder assembly.

0310 00-1 ← *WORK PACKAGE PAGE NUMBER*

FIGURE 2. Example of work package, paragraph, and procedural step titles and numbering

Source: https://assist.dla.mil -- Downloaded: 202640-26T08:04Z
Check the source to verify that this is the current version before use.

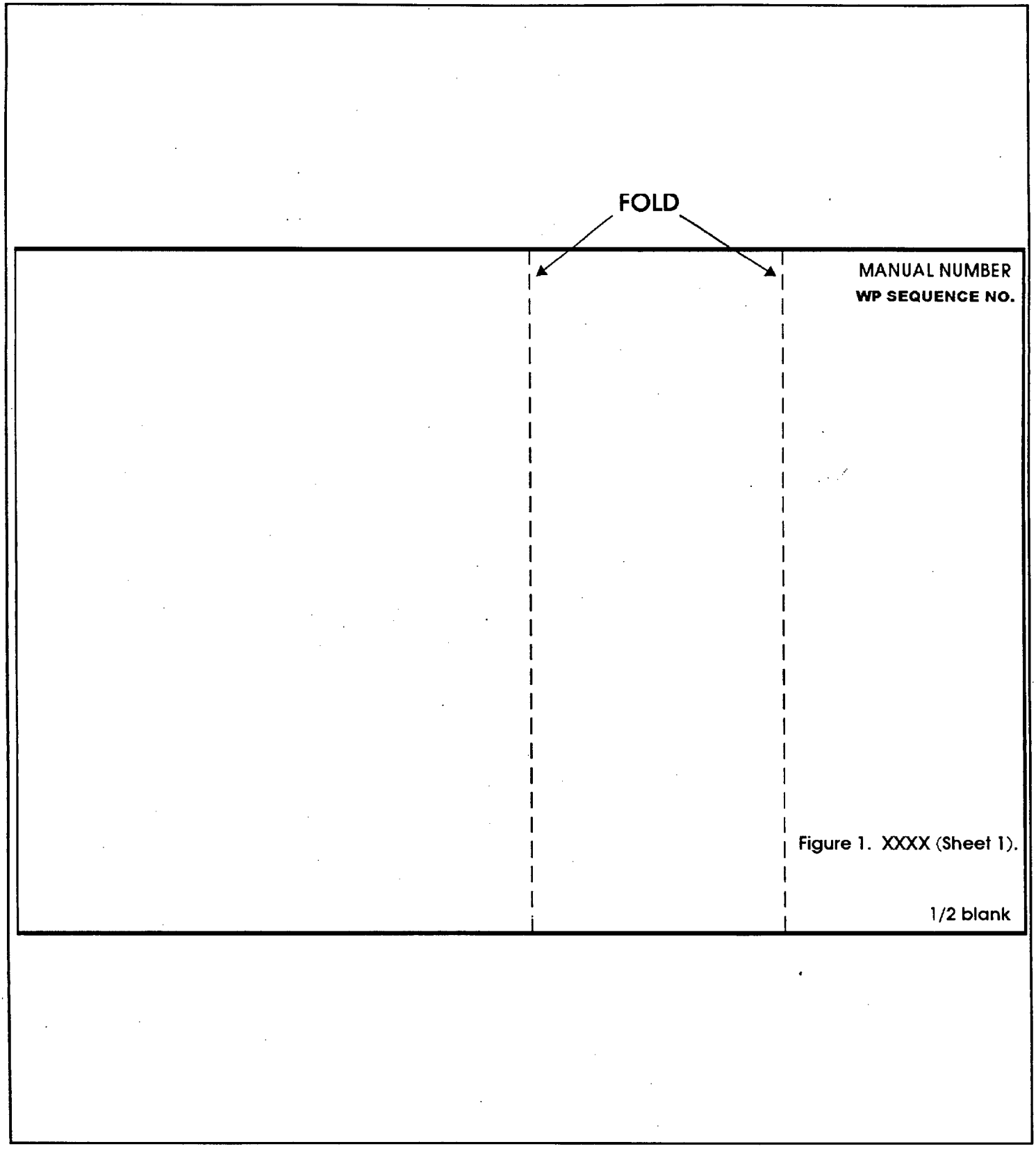


FIGURE 3. Example of a foldout page.

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CONFIDENTIAL

TM 9-1025-211-10

M198 HOWITZER OPERATION UNDER USUAL CONDITIONS **0017 00**

GENERAL

The personnel of the howitzer section consist of the following:

1. A chief of section (CS) whose duties and responsibilities are the following:
   a. Training and efficiency of personnel.
   b. Performance of the section in training; firing, testing, and adjusting fire control equipment; and inspection and maintenance of all section equipment, including the prime mover.
   c. Observance of safety precautions.
   d. Preparation of field fortifications for protection of equipment, ammunition, and personnel.
   e. Camouflage discipline and local security, and radiological, biological, and chemical security discipline.
   f. Maintenance of forms in the equipment record folder.
   g. Policing the section area.
2. A gunner (G) who assists the chief of section in carrying out the duties specified in step 1. above. The gunner's specific duties are described in this manual.
3. An ammunition team chief (ATC) who leads and directs the handling of ammunition and assists the chief of section in the supervision of the howitzer section. The ATC also performs duties as listed in this manual and other duties as directed.
4. An assistant gunner (AG) who assists the gunner and, in an emergency, acts as the gunner. The assistant gunner's specific duties are described in this manual.
5. Five cannoneers, numbered 1 to 5, who perform duties as listed in this manual and other duties as directed.

0017 00-1

CONFIDENTIAL
(This page is UNCLASSIFIED)

FIGURE 4. Example of a page with security classification markings

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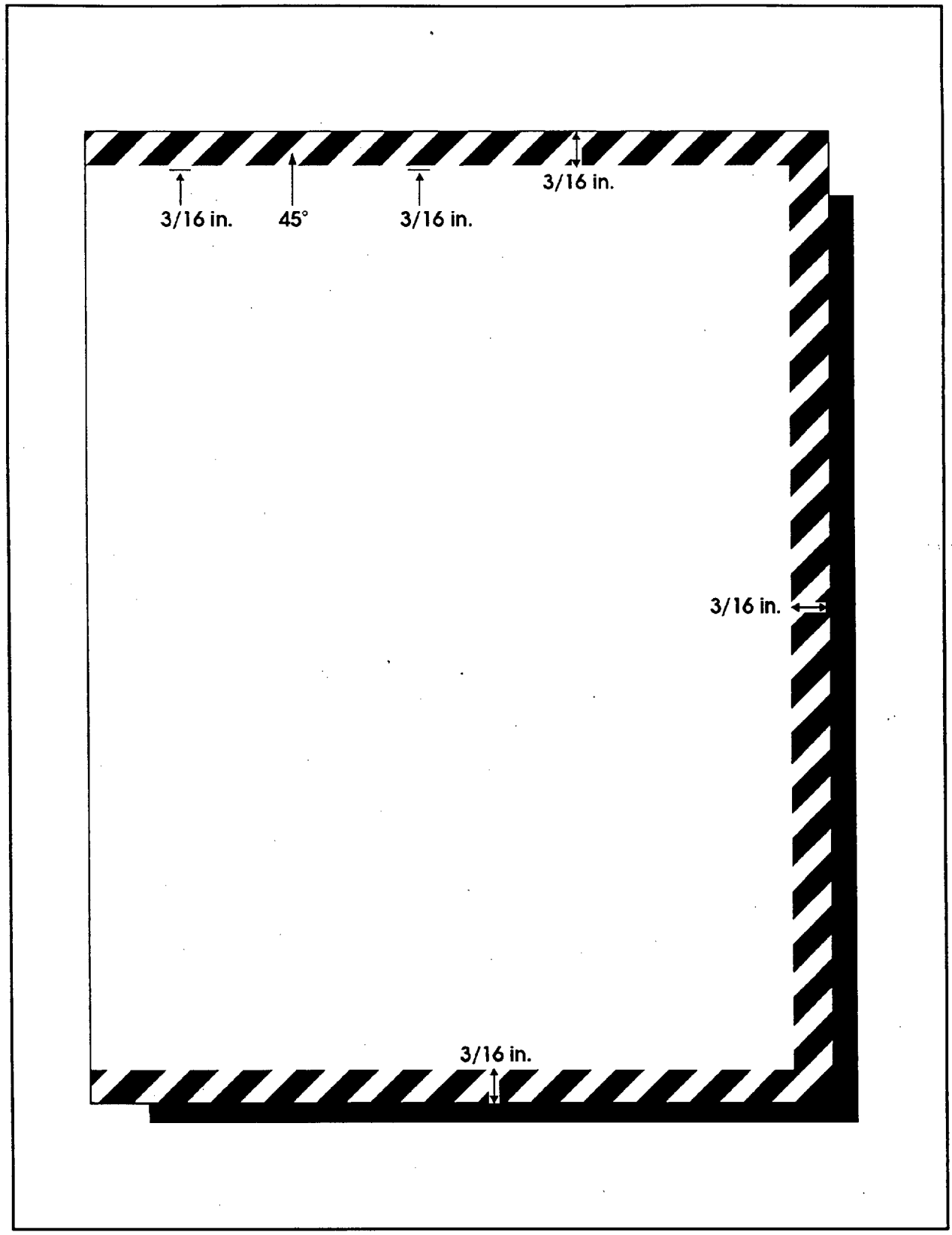


FIGURE 5. Example of emergency page markings.

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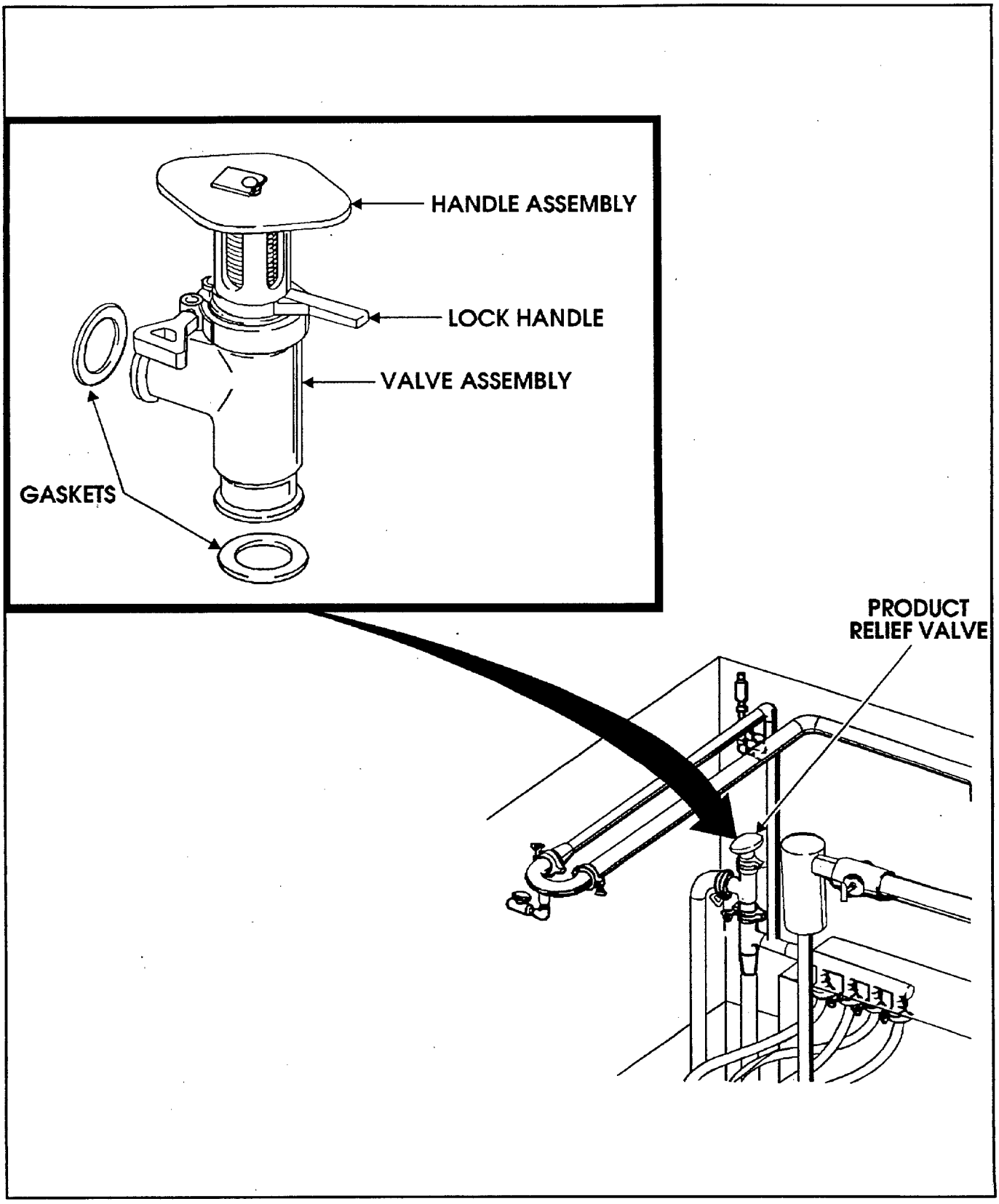


FIGURE 6. Example of a locator illustration

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Check the source to verify that this is the current version before use.

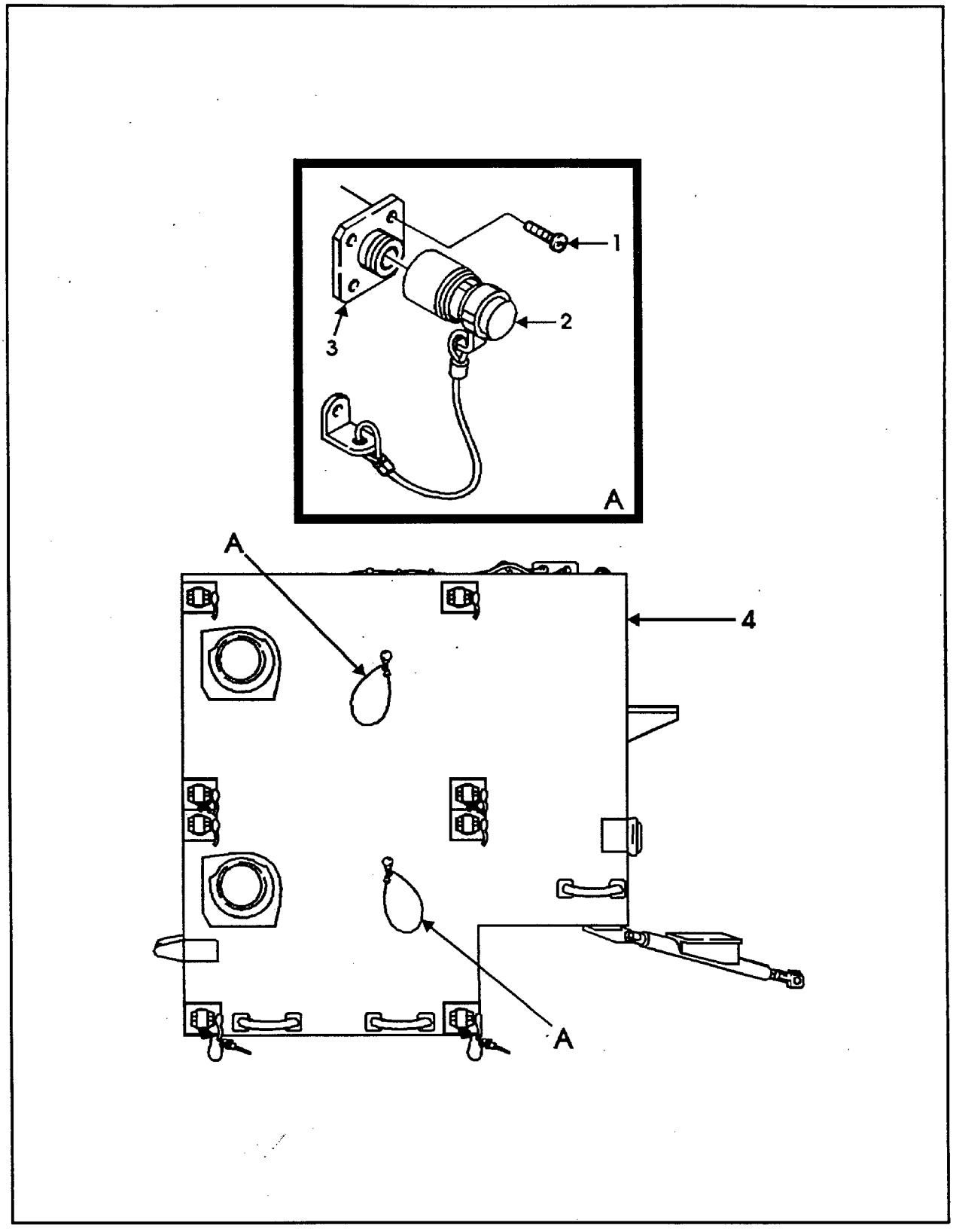


FIGURE 7. Example of a detail view illustration

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MANUAL NUMBER

REVISION NO. 1

HEADQUARTERS, DEPARTMENT OF THE ARMY
WASHINGTON, DC, 31 AUG 1993

TECHNICAL MANUAL

OPERATOR'S, UNIT, AND
DIRECT SUPPORT MAINTENANCE MANUAL
FOR

TEST SET
RADAR AN/TPM-22
NSN 4931-00-707-1229 (EIC D42)

DISTRIBUTION STATEMENT A - Approved for public release; distribution is unlimited.

TM X-XXX-XXXX-XX, 5 June 1987, is updated as follows:

1. File this sheet in front of the manual for reference.
2. This revision is a result of new preventive maintenance checks and services procedures and new expendable/durable supplies and materials.
3. New or updated text is indicated by a vertical bar in the outer margin of the page.
4. Added illustrations are indicated by a vertical bar adjacent to the figure number. Revised illustrations are indicated by a miniature pointing hand adjacent to the updated area and a vertical bar adjacent to the figure number.
5. Remove old pages and insert new pages as indicated below.

Remove Pages	Insert Pages
a through d	a through d
None	e through g / (h blank)

6. Replace the following work packages with their revised version.

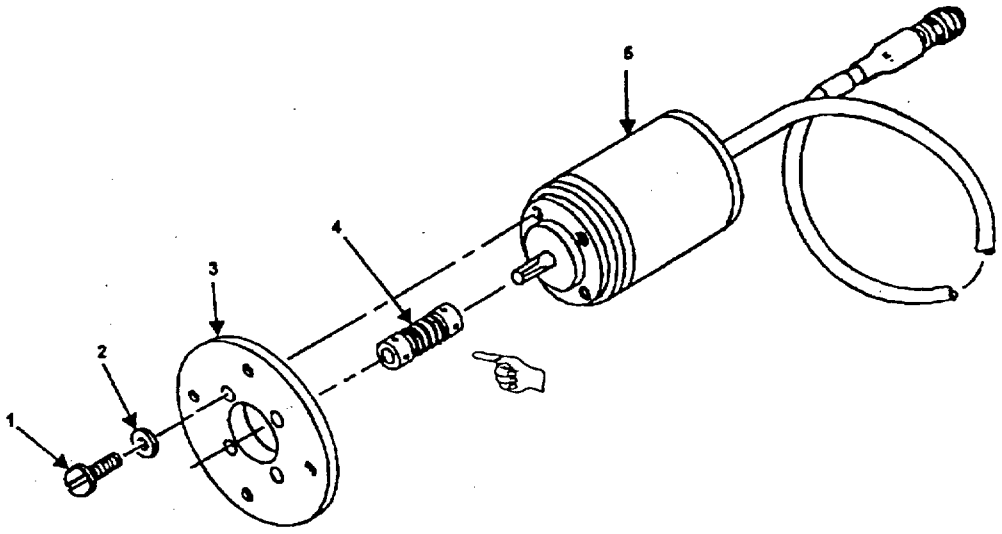
Work Package Number
WP 0042 00
WP 0154 00

7. Add the following new work packages.

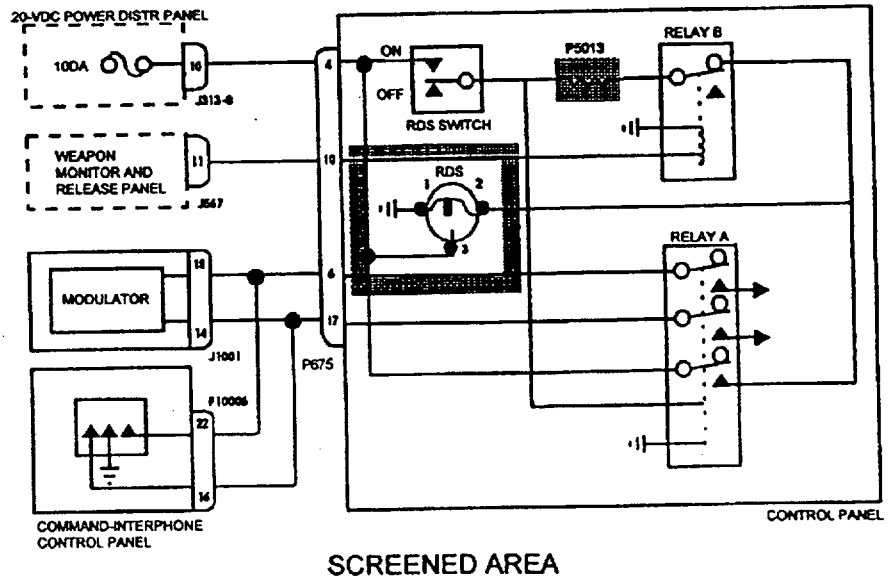
Work Package Number
WP 1625 01
WP 1700 01

FIGURE 8. Example of a revision transmittal page.

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MINIATURE POINTING HAND



SCREENED AREA

FIGURE 9. Example of revision symbols.

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SECURITY CLASSIFICATION

TM NUMBER

TECHNICAL MANUAL

TYPE OF PUBLICATION
MAINTENANCE LEVELS FOR

NOMENCLATURE OF EQUIPMENT
TYPE, MODEL, PART NUMBER
NATIONAL STOCK NUMBER (EIC)
OR
SUBJECT

SUBTITLE

ILLUSTRATION

AVAILABILITY STATEMENT
SUPERSEDURE NOTICE
DISCLOSURE NOTICE
DISTRIBUTION STATEMENT
WARNING
DESTRUCTION NOTICE

SERVICE NOMENCLATURE

REVISION - DATE TM DATE

SECURITY CLASSIFICATION

FIGURE 10. Example of a TM front cover.

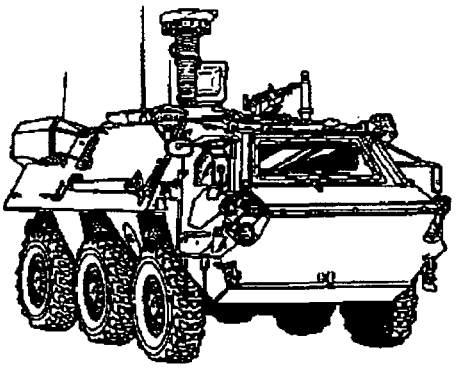
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TM 3-6665-339-10

TECHNICAL MANUAL

OPERATOR'S MANUAL FOR NUCLEAR-BIOLOGICAL-CHEMICAL RECONNAISSANCE SYSTEM (NBCRS) FOX M93A1

NSN 6665-01-372-1303 (EIC Y60)



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HEADQUARTERS, DEPARTMENT OF THE ARMY

1 June 1996

FIGURE 10. Example of a TM front cover - Continued

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DMWR 11-5895-532-2

DEPOT MAINTENANCE WORK REQUIREMENT

FOR

INTERROGATOR SETS

AN/TPX-46(V)1 (NSN 5895-00-423-1693) (EIC: IZA)
AN/TPX-46(V)2 (NSN 5895-00-423-1694) (EIC: IZB)
AN/TPX-46(V)3 (NSN 5895-00-423-1696) (EIC: IZC)
AN/TPX-46(V)4 (NSN 5895-00-423-1700) (EIC: IZD)
AN/TPX-46(V)6 (NSN 5895-00-423-1702) (EIC: IZE)
AN/TPX-46A(V)1 (NSN 5895-01-162-5237) (EIC: N/A)
AN/TPX-46A(V)2 (NSN 5895-01-162-5239) (EIC: N/A)
AN/TPX-46A(V)3 (NSN 5895-01-163-3646) (EIC: N/A)
AN/TPX-46A(V)4 (NSN 5895-01-162-5240) (EIC: N/A)
AN/TPX-46A(V)6 (NSN 5895-01-163-1235) (EIC: N/A)

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US ARMY COMMUNICATIONS-ELECTRONICS COMMAND, FORT MONMOUTH, NJ
15 MAY 1992

FIGURE 11. Example of a DMWR front cover.

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TM 3-6665-339-10

WARNING SUMMARY

WARNING

RADIOACTIVE MATERIAL



RADIATION HAZARD

RULES AND REGULATIONS

Copies of the following rules and regulations are maintained at HQ, U.S. Army Armament and Chemical Acquisition and Logistics Activity (ACALA), Rock Island, IL 61299-7630. Copies may be requested, or information pertinent to these rules and regulations obtained, by contacting the ACALA Radiological Protection Officer (RPO), Defense Switched Network (DSN) 793-2965/2962/2995, Commercial (309) 782-2965/2962/2995. Depleted Uranium (DU) armor licensee is the U.S. Army Tank-Automotive Command (TACOM), Warren, MI 48397-5000. Radiation safety support may be obtained from the TACOM Safety Office at DSN 786-6121/8529 or Commercial (313) 574-6121/8529.

10CFR Part 19 - Notices, Instructions and Reports to Workers: Inspections.
10CFR Part 20 - Standards for Protection Against Radiation.
10CFR Part 21 - Reporting of Defects and Noncompliance.
NRC license, license conditions and license applications.

SAFETY PRECAUTIONS

Identification

Radioactive self-luminous sources are identified by means of radioactive warning labels. These labels should not be defaced or removed, and should be replaced immediately when necessary. Refer to the local RPO or the ACALA RPO for instructions on handling, storage, or disposal.

Storage

All radioactively illuminated instruments which are defective will be evacuated to a depot maintenance activity. These items must be placed in a plastic bag and packaged in the shipping container from which the replacement was taken before evacuation to a higher echelon is made. Spare equipment must be stored in the shipping container, as received, until installed on the weapon. Storage of these items is recommended to be in an outdoor shed-type storage or unoccupied building.

First Aid

For further information on first aid, see FM 21-11.

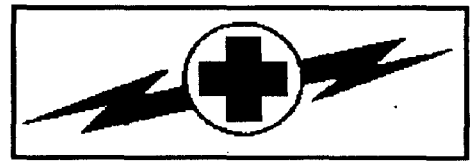
a

FIGURE 12. Example of a warning summary.

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TM 3-6665-339-10

WARNING



HIGH VOLTAGE is used in the operation of this equipment.
DEATH ON CONTACT may result if personnel fail to observe safety precautions

Never work on electronic equipment unless there is at least one other person nearby who is familiar with the operation and hazards of the equipment. That person should also be competent in giving first aid. Ask maintenance personnel about extremely hazardous areas of the NBCRS vehicle prior to doing any maintenance.

Whenever possible, vehicle master power must be shut off before performing any maintenance. Use extreme caution around any electronic components of the NBCRS vehicle. Some components store energy that can injure personnel even with vehicle master power turned off.

Be careful not to contact high-voltage connections when removing, installing, or operating this equipment.

Whenever possible, keep one hand away from the equipment to reduce hazard of current flowing through vital organs of the body.

Do not be misled by the term low voltage. Voltages as low as 50 volts may cause death.

For artificial respiration, refer to FM 21-11.

b

FIGURE 12. Example of a warning summary - Continued

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TM 9-1440-433-24

INSERT LATEST UPDATED PAGES / WORK PACKAGES. DESTROY SUPERSEDED DATA.

LIST OF EFFECTIVE PAGES / WORK PACKAGES

NOTE: The portion of text affected by the updates is indicated by a vertical line in the outer margins of the page. Updates to illustrations are indicated by miniature pointing hands. Updates to wiring diagrams are indicated by shaded areas.

Dates of issue for original and updated pages / work packages are:

Original .. 0 .. 31 Mar 96

Revision .. 1 .. 30 Sep 96

TOTAL NUMBER OF PAGES FOR FRONT AND REAR MATTER IS 47 AND TOTAL NUMBER OF WORK PACKAGES IS 380 CONSISTING OF THE FOLLOWING:

Page / WP No.	*Revision No.	Page / WP No.	*Revision No.	Page / WP No.	*Revision No.
Title	0				
A - B	1				
a - d	0				
i - iii	0				
iv - vi	1				
vii - x	1				
WP 0001 00 - 0081 00	0				
WP 0082 00	1				
WP 0083 00 - 0085 00	0				
WP 0086 00 - 0098 00	1				
WP 0099 00 - 0371 00	0				
WP 0372 00 - 0378 00	1				
WP 0379 00 - 0380 00	0				
Index-1 - Index-15	0				
Index-16 Blank	0				

* Zero in this column indicates an original page or work package.

Revision 1 A

FIGURE 13. Example of a list of effective pages/work packages.

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TM 3-6665-339-10

TECHNICAL MANUAL

OPERATOR'S MANUAL

NUCLEAR-BIOLOGICAL-CHEMICAL
RECONNAISSANCE SYSTEM (NBCRS)
FOX M93E1
NSN 6665-01-372-1303 (EIC Y60)

REPORTING ERRORS AND RECOMMENDING IMPROVEMENTS

You can help improve this manual. If you find any mistakes or if you know of a way to improve the procedures, please let us know. Mail your letter, DA Form 2028 (Recommended Changes to Publications and Blank Forms), located in the back of this manual, direct to: Technical Director, Edgewood Research Development and Engineering Center, ATTN: SCBRD-ENL-V, Aberdeen Proving Ground, MD 21010-5423. You may also send in your recommended changes via electronic mail or by fax. Our fax number is (*insert DSN and commercial number of proponent).* Our e-mail address is (*insert address of proponent).* A reply will be furnished to you.

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HEADQUARTERS, DEPARTMENT OF THE ARMY

1 JUNE 1996

FIGURE 14. Example of a title block page, and reporting of errors and recommending improvements statement.

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TM 3-6665-339-10

TABLE OF CONTENTS

WP Sequence No.

WARNING SUMMARY

HOW TO USE THIS MANUAL

CHAPTER 1 - INTRODUCTORY INFORMATION WITH THEORY OF OPERATION
General Information 0001 00
Equipment Description and Data 0002 00
Introduction 0003 00
Powerpack Theory of Operation 0004 00
Fuel System Theory of Operation 0005 00
Electrical System Theory of Operation 0006 00
Hydraulic System Theory of Operation 0007 00
Amphibious System Theory of Operation 0008 00
Bilge Pumps and Drain Valves Theory of Operation 0009 00
Fire Extinguisher System Theory of Operation 0010 00

CHAPTER 2 - OPERATOR INSTRUCTIONS
Description and Use of Controls and Indicators 0019 00
Operate Intercom 0020 00
Operate NBC Collective Protection System (Operate Microclimatic System And M42 Mask) 0021 00
Operate Domelight 0022 00
Operate Portable Fire Extinguisher 0023 00
Enter Driver's Ststion 0024 00
Adjust Seat, Safety Belt, and Mirrors 0025 00
Power Up Hull Systems 0026 00
Install Periscopes 0027 00
Operate Ballistic Shield 0028 00

FIGURE 15. Example of table of contents.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:04Z
Check the source to verify that this is the current version before use.

TM 9-1007-215-24P

TABLE OF CONTENTS

		WP Sequence No.
INTRODUCTION		0001 00
GROUP 00	20-mm self-propelled air defense artillery gun M163A1 repair parts list	0002 00
GROUP 01	20-mm air defense gun cannon M168 repair parts list	0003 00
GROUP 0101	Recoil adapter assembly repair parts list	0004 00
GROUP 0102	Breech bolt assembly repair parts list	0005 00
GROUP 0103	Center clamp assembly repair parts list	0006 00
GROUP 02	20-mm gun mount assembly M157A1 repair parts list	0007 00
GROUP 0201	Storage battery repair parts list	0008 00
GROUP 0202	Ammunition chute repair parts list	0009 00
GROUP 0203	Element chute repair parts list	0010 00
GROUP 0204	Mount component assembly detail illustrations repair parts list	0011 00
GROUP 020401	Turret drum assembly repair parts list	0012 00
GROUP 020402	Azimuth drive assembly repair parts list	0013 00
GROUP 020403	Azimuth drive friction clutch assembly repair parts list	0014 00
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GROUP 0302	Combining glass and gimbal repair parts list	0017 00
GROUP 0303	Caging device and cable assembly repair parts list	0018 00
GROUP 0304	Housing support assembly repair parts list	0019 00
GROUP 09	Special tools (repair parts)	0045 00
GROUP 0901	Boresight repair parts list	0046 00
GROUP 0902	Storage drum slot gauge with case repair parts list	0047 00
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National stock number index		0051 00
Part number index		0052 00
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FIGURE 16. Example of a RPSTL table of contents.

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GLOSSARY

Term	Definition
Arming Levers	Each lever arms one-half row of mines.
Canisters Remaining Panel	Tells user number of canisters in each launch rack.
Crossmember Assembly	Provides lateral support to female and male beam assemblies.
DCU Cable Tube Assembly	Provides support to which cabling may be tied.
DCU Card File Assembly	Houses all DCU CCAs.
DCU Chassis Assembly	Contains connectors and CCAs used in mine dispenser system.
DCU/Helicopter Interface Cable	Interface between DCU and helicopter controls.
DCU Information Panel	Contains indicators required to set up, launch, and provide readouts of Built In Testing (BIT).
DCU Pallet Assembly	DCU mount that rests on ground vehicle bed floor.
DIM Dial	Allows operator to brighten or dim readout displays on DCU.
Dispenser Control Unit (DCU)	Houses the electronics that control the mine dispensing system and a control panel for operating the system.
Display Cables	Input/output cables between Display Panel to CCAs.
Display Interface CCA	Manages inputs and outputs between CCAs and display board.
Driver CCA	Provides control over circuit functions of mine dispenser.
Driver Logic CCA	Contains logic circuitry required by Driver CCAs.
EMER JETTISON Toggle Switch	Backup jettison switch.
ESSS Porthole Grommet	Prevents chafing of jettison and launcher rack cable.

FIGURE 17. Example of a glossary.

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TM X-XXXX-XXX-XX

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Assembly	0012 00-5
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Disassembly	0012 00-1
Inspection	0012 00-3
Repair	0012 00-4
Adapter, Compressor Repair	0022 00-9
Afterburner	
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Breather Pressurizing Valve	
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FIGURE 18. Example of an alphabetical index.

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DMWR 9-1005-318-4

FOR THE COMMANDER:

LARRY D. BACHELOR
Colonel, GS
Chief of Staff

OFFICIAL:

R.D. HUSSON
Director of Maintenance

DISTRIBUTION:

COMMANDER RED RIVER ARMY DEPOT ATTN: SDSRR-ME TEXARKANA, TX 75507-5000	5 CYS
COMMANDER U.S. ARMY ARMAMENT, MUNITIONS AND CHEMICAL COMMAND	
ATTN: AMSMC-MAE-WA	2 CYS
AMSMC-MAW-AD	2 CYS
AMSMC-QAW	2 CYS
AMSMC-MAS-B	15 CYS
AMSMC-MAF-AP ROCK ISLAND, IL 61299-6000	2 CYS
COMMANDER AMC LOGISTICS SUPPORT ACTIVITY ATTN: AMXMD-MP REDSTONE ARSENAL, AL 35898-7466	2 CYS

FIGURE 19. Example of an authentication page.

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METRIC CONVERSION CHART

APPROXIMATE CONVERSION FACTORS

TO CHANGE	TO	MULTIPLY BY
Inches	Centimeters	2.540
Feet	Meters	0.305
Yards	Meters	0.914
Miles	Kilometers	1.609
Square Inches	Square Centimeters	6.451
Square Feet	Square Meters	0.093
Square Yards	Square Meters	0.836
Square Miles	Square Kilometers	2.590
Acres	Square Hectometers	0.405
Cubic Feet	Cubic Meters	0.028
Cubic Yards	Cubic Meters	0.765
Fluid Ounces	Milliliters	29.573
Pints	Liters	0.473
Quarts	Liters	0.946
Gallons	Liters	3.785
Ounces	Grams	28.349
Pounds	Kilograms	0.454
Short Tons	Metric Tons	0.907
Pound-Feet	Newton-Meters	1.356
Pounds per Square Inch	Kilopascals	6.895
Miles per Gallon	Kilometers per Liter	0.425
Miles per Hour	Kilometers per Hour	1.609

TO CHANGE	TO	MULTIPLY BY
Centimeters	Inches	0.394
Meters	Feet	3.280
Meters	Yards	1.094
Kilometers	Miles	0.621
Square Centimeters	Square Inches	0.155
Square Meters	Square Feet	10.764
Square Meters	Square Yards	1.196
Square Kilometers	Square Miles	0.386
Square Hectometers	Acres	2.471
Cubic Centimeters	Cubic Inches	0.060
Cubic Meters	Cubic Feet	35.315
Cubic Meters	Cubic Yards	1.308
Milliliters	Fluid Ounces	0.034
Liters	Pints	2.113
Liters	Quarts	1.057
Liters	Gallons	0.264
Grams	Ounces	0.035
Kilograms	Pounds	2.205
Metric Tons	Short Tons	1.102
Newton-Meters	Pound-Feet	0.738
Kilopascals	Pounds per Square Inch	0.145
Kilometers per Liter	Miles per Gallon	2.354
Kilometers per Hour	Miles per Hour	0.621

TEMPERATURE CONVERSIONS

5/9 (°F-32) = °C
212° Fahrenheit is equivalent to 100 Celsius
90° Fahrenheit is equivalent to 32.2 Celsius
32° Fahrenheit is equivalent to 0 Celsius
9/5 C° + 32 = F°

FIGURE 20. Example of a metric conversion chart on inside of back cover

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NOT MEASUREMENT SENSITIVE

MIL-STD-40051-2

DEPARTMENT OF DEFENSE STANDARD PRACTICE

TECHNICAL MANUALS

INTRODUCTORY INFORMATION WITH THEORY OF OPERATION

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1. SCOPE.

1.1 Scope. This standard establishes the technical content requirements for the preparation of introductory information with theory of operation for weapon systems and equipment Technical Manuals (TMs) and Depot Maintenance Work Requirements (DMWRs). These requirements are applicable for both paper and digital page-oriented TMs. Electronic delivery of TMs is accomplished through the use of the Introductory Information With Theory of Operation modular Document Type Definition (DTD). The DTD is available in a digital format. Refer to MIL-STD-40051 for information on obtaining this DTD. Introductory information with theory of operation requirements include applicable forms and records, descriptive information about the equipment or weapon system and an explanation of how the equipment or weapon system accomplishes its mission.

2. APPLICABLE DOCUMENTS.

The applicable documents in section 2 of MIL-STD-40051 apply to this Part.

3. DEFINITIONS.

The definitions in section 3 of MIL-STD-40051 apply to this Part.

4. GENERAL REQUIREMENTS.

4.1 General. Introductory information with theory of operation shall be prepared for weapon systems, major equipment, components and applicable support and interface equipment. Information required to complete all applicable forms and records, provide the user with a physical description, and functionally explain how the weapon system or equipment operates shall be included. This information shall be contained in work packages and become a part of a General Information chapter.

4.2 Maintenance level applicability. Requirements contained in this standard are applicable to all maintenance levels unless specifically noted in bold and in parentheses (i.e.,**-34 only**). The labeled requirements shall be applicable to all TMs containing that maintenance level. For example, a**(-20 only)** requirement would only be applicable to the following TMs: -12, -13, -14, -20, -23, and -24.

4.3 Standard tables. Various standard tables required are noted throughout the text of this standard in bold and in parentheses (i.e.,**(standard table)**). The formats and table heading names of these standard tables shall have no deviations.

4.4 Preparation of digital data for electronic delivery. Technical manual data prepared in work package format and delivered digitally in accordance with this standard shall be SGML tagged and assembled using the modular Assembly DTD and Formatting Output Specification Instance (FOSI). The DTD and FOSI have been developed in accordance with MIL-PRF-28001 and ISO 8879. Refer to MIL-STD-40051 for information on obtaining or accessing this modular DTD and FOSI. SGML tags used in the modular DTD are noted throughout the text of this standard in bracketed, bold characters (i.e.,**<ginfowp>**) as a convenience for the TM author and to ensure that the tags are used correctly when developing a document instance.

4.4.1 Use of the DTDs / FOSIs. The modular DTDs referenced in this standard interpret the technical content and structure for the functional requirements contained in this standard and are mandatory for use. The modular FOSIs referenced herein interpret the style and format. As specified by the contracting activity, FOSIs or style sheets may be used to produce final reproducible paper copy for all TMs prepared in accordance with this standard.

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4.5 Content structure and format. The examples provided at the rear of this Part are an accurate representation of the content structure and format requirements contained herein and shall be followed to permit the effective use of the modular DTD for Introductory Information With Theory of Operation.

4.6 Style and format. Style and format requirements for the preparation of Department of Army TMs are contained in MIL-STD-40051-1; they are considered mandatory and are intended for compliance. Preferred general style and format requirements for Army TMs shall be provided by the proponent activity.

4.7 Work package development. Technical manual data developed in accordance with this standard shall be divided into individual, stand alone units of information work packages. A work package shall consist of descriptive, operational, maintenance, troubleshooting, support, or parts information for the weapon system or equipment.

4.8 Selective application and tailoring. MIL-STD-40051 contains some requirements that may not be applicable to the preparation of all technical manuals. Selective application and tailoring of requirements contained in MIL-STD-40051 are the responsibility of the contracting activity and shall be accomplished through the use of Appendix A, Technical Manual Content Selection Matrixes, of MIL-STD-40051. The applicability of some requirements is also designated by one of the following statements: unless specified otherwise by the contracting activity; as/when specified by the contracting activity; or when specified by the procuring activity.

5. DETAILED REQUIREMENTS.

5.1 Preparation of introductory information with theory of operation. Introductory information with theory of operation shall be prepared as a General Information Chapter**<gim>**. This chapter shall be subdivided into individual work packages to provide the user with information for completing forms and records, descriptive data about the weapon system or equipment, and an explanation of how the weapon system or equipment works. Information shall be divided into the following work packages.

a. General information work package**<ginfowp>**.

b. General information work package**<pms-ginfowp>** (Preventive Maintenance Services Manual only).

c. Equipment description and data work package**<descwp>**.

d. Theory of operation work package**<thrywp>**.

e. Support data work package**<supdatawp>** for repair parts; special tools; Test, Measurement, and Diagnostic Equipment (TMDE); and support equipment.

5.1.1 General information work package**<ginfowp>**. This work package shall contain the content requirements provided in 5.1.1.1 through 5.1.1.19, as applicable, for the weapon system equipment.

5.1.1.1 Scope **<scope>**. A brief statement shall be prepared to tell what is covered in the TM (Refer to figure 1.) As applicable, the following information shall also be included.

a. Type of manual.

b. Model number(s) and equipment name(s).

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c. Purpose of equipment.

d. Special inclusions in the manual, such as drill procedures or on-vehicle loading plans.

5.1.1.2 Maintenance forms, records, and reports **<mfrr>**. The following statement shall be included.

"MAINTENANCE FORMS, RECORDS, AND REPORTS

Department of the Army forms and procedures used for equipment maintenance will be those prescribed by (as applicable) DA PAM 738-750, Functional Users Manual for the Army Maintenance Management System (TAMMS); DA PAM 738-751, Functional Users Manual for the Army Maintenance Management Systems - Aviation (TAMMS-A); or AR 700-138, Army Logistics Readiness and Sustainability."

In addition, for **conventional and chemical ammunition TMs**, add the following statement.

"Accidents involving injury to personnel or damage to material will be reported on DA Form 285 (Accident Report) in accordance with AR 385-40. Explosives and ammunition malfunctions will be reported in accordance with AR 75-1."

When applicable, add references to SB 742-1, Ammunition Surveillance Procedures.

5.1.1.3 Reporting equipment improvement recommendations **<eir>**. The following statement shall be included.

"REPORTING EQUIPMENT IMPROVEMENT RECOMMENDATIONS (EIR)

If your (insert equipment short item name) needs improvement, let us know. Send us an EIR. You, the user, are the only one who can tell us what you don't like about your equipment. Let us know why you don't like the design or performance. Put it on an SF 368 (Product Quality Deficiency Report). Mail it to the address specified in DA PAM 738-750, Functional Users Manual for the Army Maintenance Management System (TAMMS), or as specified by the contracting activity. We will send you a reply."

5.1.1.4 Hand receipt (HR) manuals **<handreceipt>**. The following statement may be included in operator's/unit maintenance manuals.

"HAND RECEIPT (HR) MANUALS

This manual has a companion document with a TM number followed by "-HR" (which stands for Hand Receipt). TM X-XXXX-XXX-10-HR consists of preprinted hand receipts that list end item related equipment (i.e., COEI, BII, and AAL) that must be accounted for. As an aid to property accountability, additional HR manuals may be requisitioned through normal publication channels."

5.1.1.5 Corrosion prevention and control **<cpcdata>**. A statement similar to the following shall be prepared.

"CORROSION PREVENTION AND CONTROL (CPC)

Corrosion Prevention and Control (CPC) of Army materiel is a continuing concern. It is important that any corrosion problems with this item be reported so that the problem can be corrected and improvements can be made to prevent the problem in future items.

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While corrosion is typically associated with rusting of metals, it can also include deterioration of other materials, such as rubber and plastic. Unusual cracking, softening, swelling, or breaking of these materials may be a corrosion problem.

If a corrosion problem is identified, it can be reported using SF 368, Product Quality Deficiency Report. Use of key words such as "corrosion," "rust," "deterioration," or "cracking" will ensure that the information is identified as a CPC problem.

The form should be submitted to the address specified in DA PAM 738-750, Functional Users Manual for the Army Maintenance Management System (TAMMS)."

For **aircraft TMs**, this information shall include a reference to TM 55-1500-343-23 (Organizational and Intermediate Avionic Cleaning and Corrosion Prevention/Control).

5.1.1.6 Destruction of Army materiel to prevent enemy use **<destructmat>**. Reference shall be made to the appropriate TM(s) covering the destruction of Army materiel to prevent enemy use as provided by the proponent activity.

5.1.1.7 Preparation for storage or shipment **<pssref>**. Reference shall be made to the preparation for storage or shipment procedures, including packaging and administrative storage, found in the applicable maintenance instructions work package.

5.1.1.8 Warranty information **<wrntyref>**. When the TM covers equipment that is under warranty and a Warranty Technical Bulletin (WTB) is published, the applicable WTB shall be referenced. When a WTB is not published, the following statement shall be included.

"WARRANTY INFORMATION

The (insert name of equipment) is warranted for (insert miles or other timeframe as appropriate). The warranty starts on the date found in block 23 of DA Form 2408-9, Equipment Control Record. Report all defects to your supervisor, who will take appropriate action."

5.1.1.9 Nomenclature cross-reference list **<nomenreflist>**. A cross-reference list shall be prepared when unofficial nomenclature (common name) is approved by the proponent activity. (Refer to figure 1.)

5.1.1.10 List of abbreviations / acronyms **<loa>**. A list shall be prepared, consisting of all abbreviations, acronyms, signs, or symbols used in the TM. (Refer to figure 1.) For **aircraft TMs only**, a statement shall be prepared that abbreviations are in accordance with MIL-STD-12, except when the abbreviation stands for a marking actually found in the aircraft.

5.1.1.11 Quality Assurance (QA) **(depot and aviation only) <qainfo>**. When specified by the contracting activity, reference shall be made to the pertinent QA TM(s) or include the appropriate general QA information. If QA information is not referenced but is included in the TM, it shall be stated that the text of each quality assurance procedure or step in the TM is preceded (and highlighted) by the addition of "**QA**." The abbreviation "QA" shall be defined either in a note or in the text. For **aircraft maintenance TMs**, include a reference to FM 1-500.

5.1.1.12 Safety, care, and handling **<sftyinfo>**. The following general precautions and safety regulations shall be prepared.

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a. For **ammunition TMs**, information shall be prepared to comply with AR 385-62. References to applicable Army Regulations (ARs) for range safety and danger zones during training and combat shall be included. Explanations and official definitions shall be prepared for such safety-related terms as "misfire," "hangfire," and "cook-off," which describe characteristics associated with the specific items(s) covered by the TM under preparation. A reference to TM 9-1300-206 shall be made for general ammunition care, handling, and safety.

b. For TMs covering equipment with radioactive parts or components, information shall be prepared to comply with Nuclear Regulatory Commission provisions, and references to applicable ARs and safety TMs on radioactive materials shall be included. If additional coverage on radioactive materials is needed, but is not included in applicable TMs, instructions shall be prepared as required. In addition, the following information shall be prepared for inclusion throughout the TM.

(1) Nuclear warning notices shall be placed at the beginning of any instruction covering procedures that will expose personnel to a nuclear radiation hazard.

(2) Procedures to be followed prior to maintenance actions, or in the event of breakage of radioactive parts or components, including safety, care, and handling instructions.

(3) Radioactive parts or components shall be shown and identified on a parts location diagram or illustration, and warning notices.

(4) A list of radioactive parts or components and the type and quantity of radioactive material involved shall be included as part of equipment data. (Refer to 5.1.3.)

(5) Instructions for the disposal of radioactive material, such as the requirement to double bag in plastic all broken tritium sources.

c. Electrostatic Discharge (ESD) control standards for the protection of electrical and electronic parts, assemblies, and equipment shall be prepared. The ESD classes shall be identified. (Refer to MIL-STD-1686 and MIL-HDBK-263.) (For classifications of ESD marking procedures, refer to MIL-STD-40051-1.)

5.1.1.13 Nuclear hardness **<hcp>**. If equipment covered in the TM has nuclear survivability requirements (i.e., overpressure and burst, thermal radiation, electromagnetic pulse, or transient radiation effects on electronics), it shall be so stated. (Refer to MIL-STD-40051-1 for marking Hardness Critical Process (HCP) procedures.) The following statement shall be included.

"NUCLEAR HARDNESS

All hardness critical procedures in this manual are marked with the acronym **HCP** as follows:

1. When an entire task, including all paragraphs and procedures, is considered hardness critical, only the task title will be marked by the acronym **HCP**, placed before the title.

2. When only certain processes and steps within the work package are hardness critical, only the applicable processes and steps will be marked by placement of the acronym **HCP** between each applicable step number and the text."

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5.1.1.14 Security measures for electronic data **<secref>**. Instructions for handling, loading, purging, overwriting, or unloading classified electronic data under usual or unusual conditions, shall be developed. Instructions shall meet the requirements of current regulations as they pertain to automation security.

5.1.1.15 Calibration **<calref>**. Equipment requiring calibration shall be identified, and reference shall be made to the publication containing the applicable calibration procedure.

5.1.1.16 Engineering Change Proposals (ECPs) **(depot only) <ecp>**. The following statement shall be included.

"ENGINEERING CHANGE PROPOSALS

Engineering Change Proposals (ECPs) will be submitted using DD Form 1693 (Engineering Change Proposal [Short Form]). (Refer to MIL-STD-973, Configuration Management, for instructions.) Completed forms should be mailed directly to (enter the name and address of the responsible command or activity). A reply will be furnished to you."

5.1.1.17 Deviations and exceptions **(depot only) <deviation>**. The following statement shall be included.

"DEVIATIONS AND EXCEPTIONS

Requests for deviations or exceptions to this Depot Maintenance Work Requirement (DMWR) will be processed in accordance with DESCOM-R 702-1 (Depot Quality Systems)."

5.1.1.18 Mobilization requirements **(depot only) <mobreq>**. The following statement shall be included.

"MOBILIZATION REQUIREMENTS

All requirements of this DMWR will be exempted or revised in the event of mobilization. Only those procedures necessary to return the (insert equipment name) to a serviceable condition will be performed. The exemptions and revisions are explained in supporting information work package (insert appropriate work package sequence number)."

5.1.1.19 Copyright credit line **<copyrt>**. When a copyright credit line is required (refer to MIL-STD-40051-1), the information shall appear as the last paragraph of the general information work package.

5.1.2 General information work package **(Preventive Maintenance Services Manual only) <pms-ginfowp>**. This work package shall be prepared for Preventive Maintenance Services manuals and shall contain the content requirements provided in 5.1.2.1 through 5.1.2.4.

5.1.2.1 Scope. Refer to paragraph 5.1.1.1 for scope requirements.

5.1.2.2 Inspection requirements **<inspec-req>**. The following statement shall be included.

"This manual contains complete requirements for intermediate (when applicable) and periodic inspection for (insert aircraft model) aircraft. It does not contain instructions for repair, adjustment, or other means of rectifying conditions, nor does it contain instruction for troubleshooting to find causes for malfunctioning. Specific tolerances, limits, etc., can be found in the applicable maintenance manuals. Use of the alphabetical index in the applicable manuals will facilitate locating the required information."

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5.1.2.3 Maintenance activities <**scope**>. The following statement shall be included.

"The inspections prescribed by this manual will be performed at a specific period by Aviation Unit Maintenance (AVUM) activities with assistance of Aviation Intermediate Maintenance (AVIM) and Depot Maintenance activities when required."

5.1.2.4 General information <**geninfo**>. The following statement shall be included.

"**INSPECTION REQUIREMENTS.**

The inspection requirements contained herein are stated in such a manner as to establish when certain equipment is to be inspected and what conditions are desired/undesired (figure 2). Compliance with the provisions outlined herein is required in order to ensure that latent defects are discovered and corrected before malfunctioning or serious trouble results. Inspection requirements are arranged, as nearly as possible, according to the manner in which they will be performed. The requirements are divided into groups and listed under area headings (figure 4).

INSPECTION INTERVALS

The inspection intervals designated herein will not be exceeded except in actual operational emergencies as explained herein. It is the commander's responsibility to determine (on an individual aircraft basis) when inspection intervals may be exceeded. For this purpose, operational emergencies are conditions of combat, or conditions of disaster which necessitate flight to evacuate aircraft or personnel. When aircraft are operated beyond the normal inspection due-time because of such emergency situation, a circled red X status symbol and an appropriate statement (to include authority) must be entered in block 16 and 17 of DA Form 2408-13 (Aircraft Status Information Record) until such time as the inspection is complete. Since safety may be jeopardized when inspections are delayed to meet emergency requirements, commanders will assure that the aircraft status symbol reverts to a red "X" and that delayed inspections are accomplished immediately upon termination of the actual emergency. When unusual local conditions of environment, utilization, mission, experience of flight crew and maintenance personnel, periods of inactivity, etc., are encountered, the maintenance officer will, at his discretion, increase the scope and/or frequency of maintenance of inspections as necessary to ensure safe flight.

SPECIFIC NON-INSTALLED EQUIPMENT ON AIRCRAFT

This manual may contain inspection requirements applicable to specific equipment not installed on your aircraft. Those requirements should be disregarded.

DA FORMS

DA Form 2408-13 will be used to record all deficiencies or shortcomings discovered during the daily inspection. DA Form 2404, Equipment Inspection and Maintenance Worksheet, will be used as a work sheet to record all deficiencies or shortcomings discovered during accomplishment of the intermediate or periodic inspection. Upon completion of the inspection, all uncorrected deficiencies or shortcomings listed on DA Form 2404 will be entered on DA Form 2408-13 prepared for that date or DA Form 2408-14, Uncorrected Fault Record (Aircraft), (For intermediate and periodic inspection.)

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SPECIAL INSTRUCTIONS

a. The columns headed I and P are used to indicate the requirements for Intermediate (when applicable) and Periodic inspections respectively. When item is required with a frequency other than that indicated by these column headings, the proper frequency is indicated in the appropriate column. For example, when a given item is required at each 3rd intermediate inspection, "3rd" is entered in the I column, opposite the pertinent item. To use the chart, perform all preventive maintenance service items indicated in the pertinent inspection column. The types of inspections are defined as follows:

I - Intermediate inspection which is performed at (insert flying hours as specified by the aircraft maintenance schedule) flying hours after completion of the periodic inspection.

P - Periodic inspection is performed every (insert flying hours as specified by the aircraft maintenance schedule) flying hours.

b. Work time requirements to accomplish each inspection are stated at top of the checklist. The work time requirements for individual items within the checklist are stated in the W/T column opposite the pertinent item.

REPORTING ERRORS AND RECOMMENDING IMPROVEMENTS

You can help improve this manual. If you find any mistakes or if you know of a way to improve the procedures, please let us know. Mail your letter, DA Form 2028 (Recommended Changes to Publications and Blank Forms), located in the back of the applicable aircraft maintenance manual (when using the 2028 from the maintenance manual, ensure the publication number and title refer to this PMS) directly to (*insert address of proponent*). You may also send in your recommended changes via electronic mail or by fax. Our fax number is (*insert DSN and commercial number of proponent*). Our e-mail address is: (*insert address of proponent*). A reply will be furnished to you.

INSPECTION AREAS

Inspection areas are shown in figure 1."

5.1.3 Equipment description and data work package<**descwp**>. This work package shall contain the descriptive data requirements listed in 5.1.3.1. through 5.1.3.5, as applicable. For**depot only**, an end item DMWR shall identify components of the end item/system that are not covered by the DMWR, but are covered in a lower level TM(s).

5.1.3.1 Equipment characteristics, capabilities, and features <**eqpinfo**>. An overall description of the equipment <**eqpdesc**> shall be prepared, including general capabilities, special features, and other like information (e.g., applications, limitations) which will be helpful in the operation and maintenance of the equipment. The information may be in narrative format or may be tabular. (Refer to figure 2.)

a. The equipment type shall be stated, as shall the following equipment features: portability or mobility, operational and special environment, and remote control.

b. Components and their functions shall not be described unless essential to continuity. For functional data, reference shall be made to theory of operation.

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c. When equipment covered varies in scope and application or has several applications within an end item, a brief explanation of the multiple usage and a simple diagram showing all aspects of a typical application shall be prepared.

d. For **ammunition TMs**, packing and packaging information shall be prepared, including number of rounds per pack.

5.1.3.2 Location and description of major components **<locdesc>**. Equipment location information shall be prepared including external and internal views of the equipment used to show general features and all major components. (Refer to figure 2.) This information shall not duplicate information contained in the equipment data requirements and the equipment characteristics, capabilities, and features.

a. The equipment and weapon systems configuration shall be described as follows.

(1) A description of system areas and compartments shall be prepared, and the system equipment and components contained in the areas shall be identified. To identify and locate the listed system equipment the configuration description shall be supported by separate illustrations of each compartment and area. (Refer to figure 2.) For **aircraft only**, a station diagram showing fuselage station, water line, and butt line, etc. shall be included. (Refer to figure 3.)

(2) The subsystems or equipment comprising the system shall be identified and described. Other equipment which is installed in the subject system compartments and areas need not be listed in the text or called out in the illustrations if they do not directly affect the operation or maintenance of the subject system. Descriptions of operator-attended equipment shall include general statements about the nature and purpose of the controls and indicators. The text shall be supported by illustrations.

(3) Descriptions and illustrations of associated-system equipment shall be limited to the major units of that equipment. The descriptions shall be more concise than those of the subject system equipment; otherwise, the same requirements shall apply. In the descriptions, emphasis shall be placed on associated systems equipment that constitutes operational or functional interfaces with the subject system. Such units shall be included in the system illustrations.

b. When the equipment is designed for use with other equipment, one or more diagrams shall be prepared to illustrate the use of the equipment. Only information pertaining to the user shall be prepared.

c. Location and contents of end item and major component identification plates shall be illustrated. Modification information, and warranty plates, stencils, or location of serial numbers shall be illustrated.

5.1.3.3 Differences between models **<eqpdiff>**. Significant differences affecting interchangeability shall be identified. Specifically, differences associated with equipment models or units of the same model shall be indicated that would affect operator or maintenance actions. These differences shall be related explicitly to equipment model, part number, or serial number ranges in such a manner that the TM user can identify the specific equipment configuration involved. When model differences exist but have no effect on operation or maintenance, this fact shall be stated. (Refer to figure 2.)

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5.1.3.4 Equipment data **<eqpdata>**.

a. Performance data shall be prepared, including numerical and other standard-related data applying to operational and maintenance functions. The equipment data shall summarize the specific capabilities and limitations of the equipment and other critical data needed by the TM user for maintenance of the equipment. Vehicle and cargo space dimensions and metric and other equivalents shall be included. (Refer to figure 2.)

b. For systems, a list of the environmental control requirements, such as limited temperature, humidity, or other limited conditions shall be prepared. Reference shall be made to the work package(s) containing information on damage to be expected from exceeding these limits and procedures for minimizing the damage.

c. A summary shall be prepared that lists the effects of weather conditions on equipment affecting system capability or causing equipment damage. This summary shall include references to any special servicing procedures that must be accomplished because of climatic changes, such as adding antifreeze to coolants.

d. Instructions for the use, transportation, handling, storage, or disposal of such substances as fuels, toxic and hazardous substances, chemicals, ordnance, and munitions shall be prepared. These instructions shall meet the applicable requirements of the Federal Environmental Protection Standards (standards to be provided by the contracting activity).

e. The energy efficiency rating shall be included for products that directly consume energy in normal operations and that commonly have a method of expressing energy efficiency.

5.1.3.5 Equipment configuration **<eqpconfig>**. For equipment which can be configured several different ways, information on the various possible configurations shall be prepared. (Refer to figure 2.)

5.1.4 Theory of operation work package **<thrywp>**. Theory of operation shall be prepared to provide the maintenance technician with adequate background information to support and perform maintenance tasks and troubleshooting on the weapon system, equipment or components. The amount of detail and complexity of the theory of operation presentation shall be in accordance with the Logistic Support Analysis (LSA) maintenance concept, the Maintenance Allocation Chart (MAC) or an approved maintenance plan. Theory of operation shall be provided as described in 5.1.4.1.

5.1.4.1 Theory presentation. Theory of operation shall consist of a functional narrative to explain the weapon system, equipment and component operation (electrical/electronic, hydraulic, pneumatic, and mechanical). Block diagrams, functional flow diagrams, schematics and other illustrations shall be included to support the text. Basic theory, normally found in textbooks, shall not be included. If the TM covers more than one model of equipment, or more than one configuration of weapon system, differences shall be explained or separate work packages may be used. Additional theory requirements are outlined below. (Refer to figures 4, 5 and 6.)

a. When necessary, introductory general information**<geninfo>** may precede the theory of operation narrative.

b. For simple systems or equipment/components, all theory may be included in a single work package.

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c. If the relative complexity of the weapon system/equipment is such that it is reasonable to first present the theory of the end item as a unit and then present the theory of its major systems **<systhry>**, subsystems **<ssysthry>**, and components (**<lruthry>** or **<sruthry>**), it shall be presented in a series of work packages. This shall be done to facilitate use and understanding.

d. Theory narrative shall be to a depth necessary to support the technician in fault isolation to the level directed by the LSA and or maintenance plan. The operation of the weapon system and related systems/components shall be presented in a logical flow. Significant input, output, and control signals, supply voltages and power supply output voltages shall be identified. If the equipment operates in more than one mode, each mode shall be explained and supported by functional block diagrams. Theory of operation shall describe detailed circuitry of all repairable components as directed by the LSA/maintenance plan. Internal circuits, their relationship to each other, input and output signals, waveforms and time-phase relationship to significant waveforms shall be included when required to understand detailed equipment operation. Theory shall not be prepared for non-repairable, throw-away components.

5.1.5 Supporting data work package for repair parts, special tools, TMDE, and support equipment **<supdatawp>**.

NOTE

The following work package applies only to unit (-20), Aviation Unit Maintenance (AVUM), direct support (-30), Aviation Intermediate Maintenance (AVIM), general support (-40), and depot (overhaul) maintenance, and any combination of those maintenance levels (i.e., -23, -34, etc.).

This work package shall include a reference to the common tools and equipment; special tools, TMDE, and support equipment; and the repair parts as shown in 5.1.5.1 through 5.1.5.3.

5.1.5.1 Common tools and equipment. The following statement shall be included.

"COMMON TOOLS AND EQUIPMENT

For authorized common tools and equipment, refer to the Modified Table of Organization and Equipment (MTOE), CTA 50-970, Expendable/Durable Items (Except: Medical, Class V, Repair Parts, and Heraldic Items), or CTA 8-100, Army Medical Department Expendable/Durable Items, as applicable to your unit."

5.1.5.2 Special tools, TMDE, and support equipment. A reference to the Repair Parts and Special Tools List (RPSTL) and MAC shall be included. When no special tools or equipment are required, it shall be so stated. If tools are to be fabricated, reference to the Illustrated List of Manufactured Items work package shall be made.

5.1.5.3 Repair parts. One of the two following statements shall be included.

"Repair parts are listed and illustrated in the repair parts and special tools list (*insert TM number of RPSTL*)."

or

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"Repair parts are listed and illustrated in supporting information work package (*insert appropriate work package sequence number*) of this manual."

6. **NOTES.**

The notes in section 6 of MIL-STD-40051 apply to this Part.

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TM 11-6625-3178-14

TRANSPORTABLE ELECTRONIC SHOP
GENERAL INFORMATION

0021 00

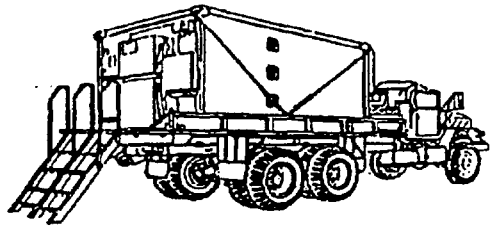
SCOPE

This technical manual contains instructions for operation, checks, and adjustments, and corrective maintenance for the AN/TSM-191(V)* Transportable Electronic Shop (shelter). The shelter provides a field level direct maintenance level, protected test facility which is used to support and improve the operational readiness of Army electronic and avionic systems.

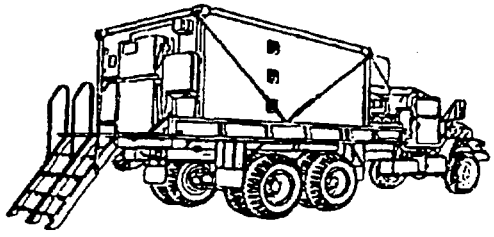
Type of Manual: Operator, Unit, Direct, and General Support Maintenance.

Model Number and Equipment Names: AN/TSM-191(V)2 Transportable Electronic Shop, AN/TSM-192(V)3 Transportable Electronic Shop, AN/TSM-191(V)4 Transportable Electronic Shop.

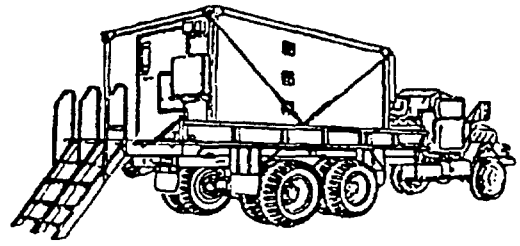
Purpose of Equipment: The shelter provides a field level direct support maintenance level, protected test facility which is used to support and improve the operational readiness of Army electronic and avionic systems.



AN/TSM-191(V)2



AN/TSM-191(V)3



AN/TSM-191(V)4

0021 00-1

FIGURE 1. Example of a general information work package

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TM11-6625-3178-14

TRANSPORTABLE ELECTRONIC SHOP 0021 00
GENERAL INFORMATION -Continued

MAINTENANCE FORMS, RECORDS, AND REPORTS

Department of the Army forms and procedures used for equipment maintenance will be those prescribed by (as applicable) DA PAM 738-750, Functional Users Manual for the Army Maintenance Management System (TAMMS); DA PAM 738-751, Functional Users Manual for the Army Maintenance Management System - Aviation (TAMMS-A); or AR 700-138, Army Logistics Readiness and Sustainability.

REPORTING EQUIPMENT IMPROVEMENT RECOMMENDATIONS (EIR)

If your shelter needs improvement, let us know. Send us an EIR. You, the user, are the only one who can tell us what you don't like about your equipment. Let us know why you don't like the design or performance. Put it on an SF 368 (Product Quality Deficiency Report). Mail it to the address specified in DA PAM 738-750, or as specified by the contracting activity. We will send you a reply.

CORROSION PREVENTION AND CONTROL (CPC)

Corrosion Prevention and Control (CPC) of Army materiel is a continuing concern. It is important that any corrosion problems with this item be reported so that the problem can be corrected and improvements can be made to prevent the problem in future items.

While corrosion is typically associated with rusting of metals, it can also include deterioration of other materials, such as rubber and plastic. Unusual cracking, softening, swelling, or breaking of these materials may be a corrosion problem.

If a corrosion problem is identified, it can be reported using SF 368, Product Quality Deficiency Report. Use of key words such as "corrosion", "rust", "deterioration", or "cracking" will ensure that the information is identified as a CPC problem.

The form should be submitted to the address specified in DA PAM 738-750.

DESTRUCTION OF ARMY MATERIEL TO PREVENT ENEMY USE

For procedures to destroy this equipment to prevent its use by the enemy, refer to TM 750-244-2, Procedures for Destruction of Electronic Materiel to Prevent Enemy Use.

NOMENCLATURE CROSS-REFERENCE LIST

Common Name	Official Nomenclature
Shelter	Transportable Electronic Shop (all models)
AN/TSM-191(*)	Transportable Electronic Shop (all models)

0021 00-2

FIGURE 1. Example of a general information work package - Continued

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TM11-6625-3178-14

TRANSPORTABLE ELECTRONIC SHOP
GENERAL INFORMATION - Continued

0021 00

LIST OF ABBREVIATIONS/ACRONYMS

Abbreviation/Acronym

ac	Alternating Current
BIT	Built-in Test
C	Celsius
ECU	Environmental Control Unit
MCPE	Modular Collective Protection Equipment

SAFETY, CARE, AND HANDLING

The following procedures should be observed when handling all ESD sensitive components and units containing such components. Failure to observe all of these precautions can cause permanent damage to the electrostatic device. This damage can cause the device to fail immediately or at a later date when exposed to an adverse environment.

1. Turn off and/or disconnect all power, signal sources and loads used with the unit.
2. Place the unit on a grounded non-conductive work surface.
3. Ground the repair operator using a non-conductive wrist strap or other device using 1 megaohm series resistor to protect the operator.
4. Ground any tools (including soldering equipment) that will contact the unit. Contact with the operator's hand provides a sufficient ground for tools that are otherwise electrically isolated.
5. All electrostatic sensitive replacement components are shipped in non-conductive foam or tubes and must be stored in the original shipping container until installed.
6. When these devices and assemblies are removed from the unit, they should be placed on the non-conductive work surface or in non-conductive containers.
7. When not being worked on, place deconnected circuit boards in plastic bags that have been coated or impregnated with a non-conductive material.
8. Do not handle these devices unnecessarily or remove them from their packages until actually used or tested.

0021 00-3

FIGURE 1. Example of a general information work package - Continued

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TM 11-6625-3178-14

TRANSPORTABLE ELECTRONIC SHOP EQUIPMENT DESCRIPTION AND DATA **0042 00**

EQUIPMENT CHARACTERISTICS, CAPABILITIES, AND FEATURES

The Transportable Electronic Shop, AN/TSM-191(*) is a truck mounted, modified S-280C/G electrical equipment shelter which contains an Electrical-Electronic Equipment Test Station AN/USM- 632(*), an Environmental Control Unit (ECU), Modular Collective Protection Equipment (MCPE) (applies to AN/TSM-191(V)2 and AN/TSM-191(V)3 only), a power distribution system and interconnecting cables. The shelter is a front-line automatic test system used to perform diagnostic checks and troubleshoot designated electronic susystems. The modified S-280C/G shelter provides housing, equipment and work space for the AN/USM-632(*). The AN/USM-632(*) performs computer controlled testing on suspected faulty Units Under Test (UUTs).

The test station is installed in a truck mounted shelter. The shelter cooling and heating is provided by a built in ECU which allows operation under a wide range of weather conditions. Environmental protection against chemical and biological contamination is provided by the MCPE (applies to AN/TSM-191(V)2 and AN/TSM-191(V)3 only). The shelter is powered by an external power source. The test station is capable of running self-test and self-alignment programs with fault isolation routines. Tests and isolates faults in UUTs automatically via computer controlled test programs.

LOCATION AND DESCRIPTION OF MAJOR COMPONENTS

EXTERIOR COMPONENTS

The following paragraph applies to both models of ECU's.

NOTE

Horizontal ECU is used only on AN/TSM-191(V)3. Vertical ECU is used only on AN/TSM-191(V)2 and AN/TSM-191(V)4.

ENVIRONMENTAL CONTROL UNIT (ECU)

The ECU is a compact, electric motor driven unit which provides filtered, cooled air (and heated air, as required) to maintain the service conditions of the AN/USM-632(*). The horizontal unit has a cooling compacity of 36,000 BTU/hr and a heating capacity of 31,200 BTU/hr. The input power requirements for the horizontal ECU are three phase, 208 delta volts ac, 50/60Hz at 13.5 kw. The vertical unit has a cooling capacity of 36,000 BTU/hr and a heating capacity of 28,600 BTU/hr. The input power requirements for the vertical ECU are three phase, 208 delta volts ac, 400 Hz at 11.0kw. A detailed description of both model air conditioners can be found in the technical manuals referenced in WP 0205 00

MODULAR COLLECTIVE PROTECTION EQUIPMENT (MCPE)

The MCPE removes toxic gases and dust from air supplied to the shelter in a Nuclear, Biological Chemical (NBC) environment. The MCPE consists of an XM20 external Integrated Protective Entrance (external IPE) and an XM93-100 CFM Gas-Particulate Filter Unit (GPFU). The external IPE is a collapsible assembly which provides a pressurized transition area between the shelter and the outside contaminated environment. The external IPE assembly consists of top and bottom pan assemblies, support assemblies and a biological and chemical resistant fabric assembly that attaches to the pan assemblies. The fabric forms the entrance assembly walls and has a zipper sewn in which forms the access door to the protective entrance from the outside contaminated zone. A door assembly consists of inner and outer door assemblies. The protective entrance assembly mounts to the outside of the outer door. During non-NBC operation, the entrance is collapsed against the outer door. In this operating mode, the outer door functions as a normal shelter door to allow access to the shelter and ventilation for shelter personnel. During NBC operation, the entrance assembly is erected and access to the shelter is through the inner door and the zipper door on the entrance assembly. A Protective Entrance Module (PEM) is mounted in the roof shell of the top pan assembly. When the entrance assembly is erect, the PEM provides white or blackout (red) lighting, decontamination purge timing and low pressure warning for the protective entrance.

0042 00-1

FIGURE 2. Example of an equipment description and data work package

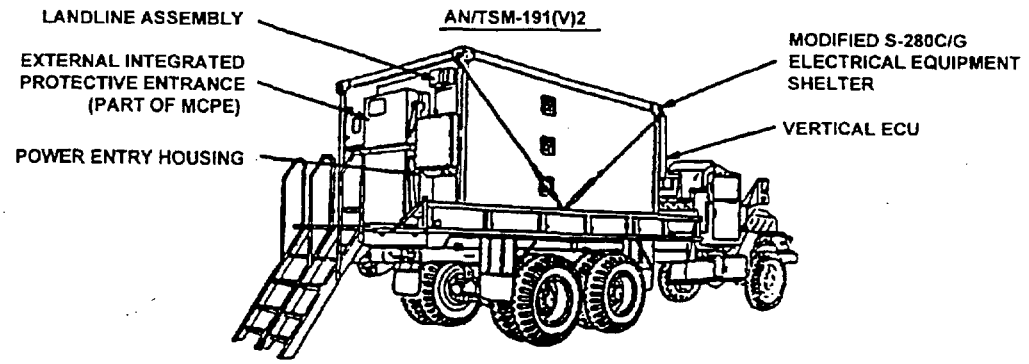
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TM 11-6625-3178-14

TRANSPORTABLE ELECTRONIC SHOP EQUIPMENT
DESCRIPTION AND DATA - Continued

0042 00

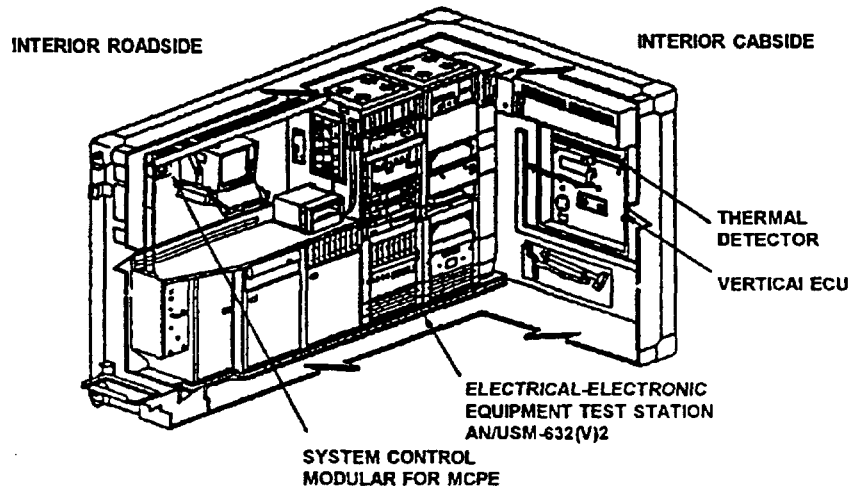
EXTERIOR COMPONENTS- Continued



INTERIOR COMPONENTS

The AN/USM-632(*) is an automatic test system used to perform diagnostic checks and troubleshoot designated electronic subsystems. It is mounted to the base assembly on the roadside of the modified shelter. A detailed description of the AN/USM-632(*) is provided in TM 11-6625-3173-12-1.

A thermal detector and alarm assembly is mounted to the inside wall of the shelter near the air conditioner. This unit monitors the air temperature inside the shelter via a thermocouple mounted inside the air conditioner, behind the return air vent. A meter on the front panel provides a visual indication of shelter temperature in degrees Fahrenheit and a horn provides an audible alarm to alert shelter personnel of a potentially damaging condition for the AN/USM-632(*) equipment due to a rising ambient temperature or a temperature below the minimum operating level. The horn will sound when the temperature is above 90 degrees F or below 50 degrees F.



0042 00-2

FIGURE 2. Example of an equipment description and data work package - Continued

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DIFFERENCES BETWEEN MODELS

Table 1 summarizes the difference between the three models of the shelter covered in this manual.

Table 1. Differences Between Models.

EQUIPMENT DESCRIPTION	AN/TSM-191(V)2 P/N A3070220-4	AN/TSM-191(V)3 P/N A3100860-2	AN/TSM-191(V)4 P/N A3143250
Electrical-Electronic Equipment Test Station Model AN/USM-632(V)2	X	—	X
Electrical-Electronic Equipment Test Station Model AN/USM0-632(V)3	—	X	—
Air Conditioner, Verticle, Compact, Keco Model: F36T4-2SB	X	—	X
Air Conditioner, Horizontal, Compact, Keco Model: F36H-2S	—	X	—
Frequency Converter, Avionic Instruments, Model: 4A3000-1B-HRP	X	—	X
Filter Unit, Gas-Particulate, 100 CFM, Model: XM93	X	X	—
Entrance, Protective, Integrated, Extenal, Pressurized, Model: XM20	X	X	—

EQUIPMENT DATA

Table 2 provides information pertaining to the operational, electrical, mechanical, and environmental characteristics of the shelter.

Table 2. Equipment Data.

ITEM	AN/TSM-191(V)2	AN//TSM-191(V)4	AN/TSM-191(V)3
Mechanical Characteristics:			
Length	186.00 in. exterior	166.00 in. exterior	200.00 in. exterior
	138.00 in. exterior	138.00 in. interior	138.00 in. interior
Width	87.00 in. exterior	87.00 in. exterior	87.00 in. exterior
	81.50 in. interior	81.50 in. interior	81.50 in. interior
Height	83.38 in. exterior	83.38 in. exterior	86.38 in. exterior
	77.12 in. interior	77.12 in. interior	77.12 in. interior
Weight	7,805 lb	7,416 lb	7,630 lb
Cubage	787.09 cu. ft.	735.09 in. cu. ft.	870.09 in. cu. ft.
Electrical Characteristics:			

FIGURE 2. Example of an equipment description and data work package - Continued

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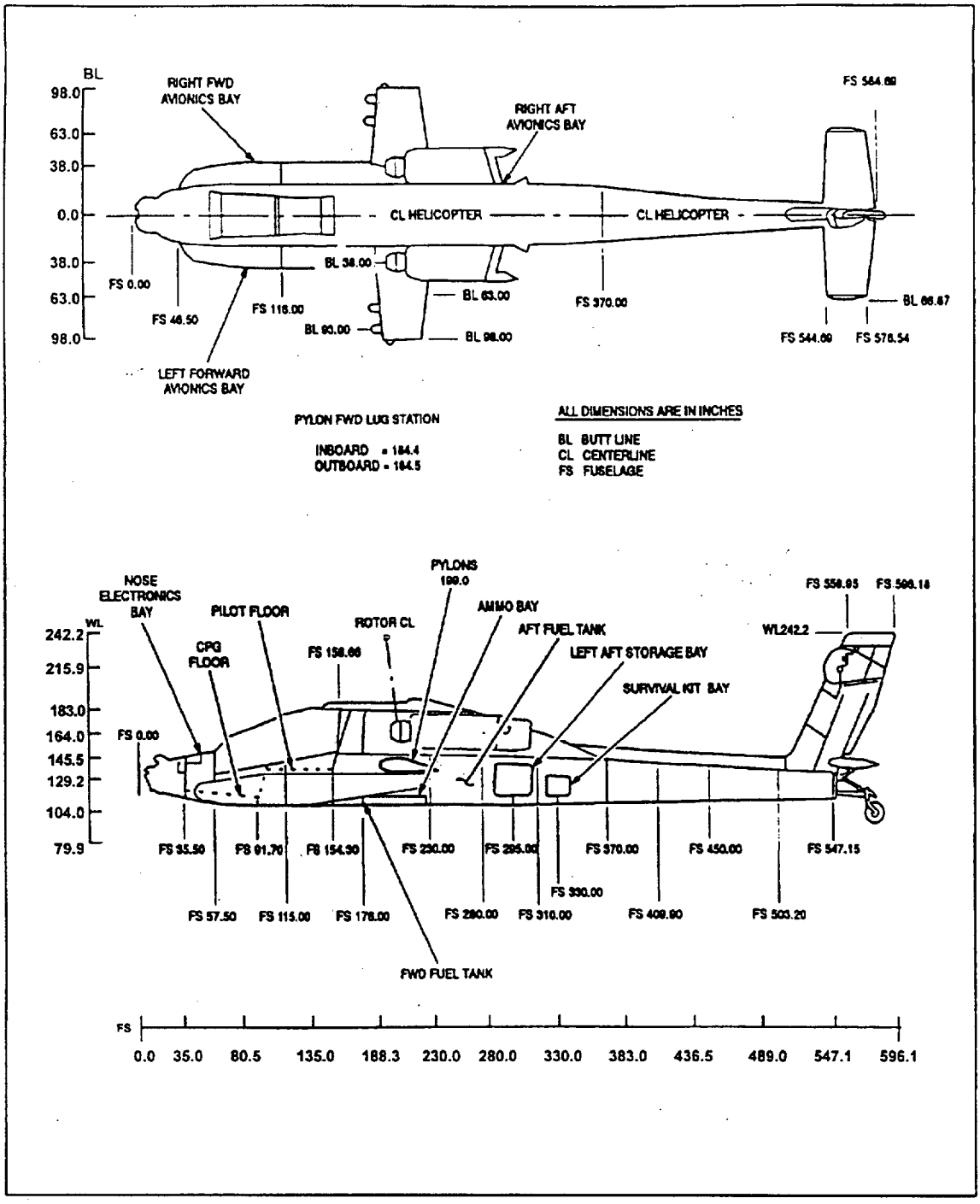


FIGURE 3. Example of a station diagram.

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TM 11-6625-3178-14

TRANSPORTATION ELECTRONICS SHOP CONTROLS THEORY OF OPERATION 0015 00

SYSTEM OPERATION

POWER ENTRY HOUSING AND SECONDARY INPUT POWER FILTER

Three phase input power from an external source, i.e. diesel powered genarator or facility power, is cabled to the power entry housing and connected to the power input connector. Inside the housing, each phase line and the neutral line are wired to four independent surge arresters and the primary input power filter. The surge arresters protect the circuits within the shelter from transients caused by lightning, induction, switching surges and Electro Magnetic Pulse (EMP). When a surge voltage exceeds the sparkover voltage of the arrester, the arrester becomes a short circuit and remains so until the transient has been by-passed and the line automatically returns to normal. The primary input power filter suppresses condition of noise caused by Electro Magnetic Interference (EMI), into and out of the shelter over the input power lines. A secondary input power filter is connected in series with the primary input power filter to ensure protection of shelter circuits from transient voltages below the sparkover voltage of the EMP protectors coming in through the power lines.

POWER DISTRIBUTION ASSEMBLY (PDA)

The PDA consists of a base assembly which houses the circuit protection and the power distribution panel which contains circuit breakers and indicators. The circuit protection devices consist of an over/under voltage-relay, a phase sequence relay and an over current relay. If a condition exists in a circuit that does not meet the parameters of any one of these devices, that fault will cause the MAIN POWER circuit breaker to reset to the OFF position, thereby removing power to the internal circuits of the shelter and causes the AC POWER FAULT light to come on. When set to ON, the MAIN POWER circuit breaker applies power to the inputs of the other breakers and three green power indicators showing power applied for each phase will come on. As each of the remaining circuit breakers are set to ON, the corresponding indicators will come on providing a visual indication of power applied to each circuit.

ENVIRONMENTAL CONTROL UNIT (ECU)

(Horizontal ECU is used only on the AN/TSM-191(V)3 model of the shelter. Vertical ECU is used only on the AN/TSM-191(V)2 and AN/TSM-191(V)4 models of the shelter.) Once started, the air conditioner operates automatically due to the relationship of the components, controls and instruments. With the model selector switch in the OFF position, all electrical components are isolated from electrical power except for the crankcase heater. This device must be energized for 30 minutes prior to operation in the coolmode. The following operating modes of the ECU are controlled by the mode selector switch.

Ventilation

Ventilation is provided in the VENT position by energizing the fan motor which forces air out of the evaporator discharge louver. The amount of outdoor air used for ventilation is determined by the position of the fresh air damper.

Cooling

With the mode selector in the COOL position, the fan motors are energized and 30 seconds later the compressor is energized. Once started, all the fan motors and compressor run continuously. The temperature selector controls the amount of cooling.

0015 00-1

FIGURE 4. Example of system theory of operation.

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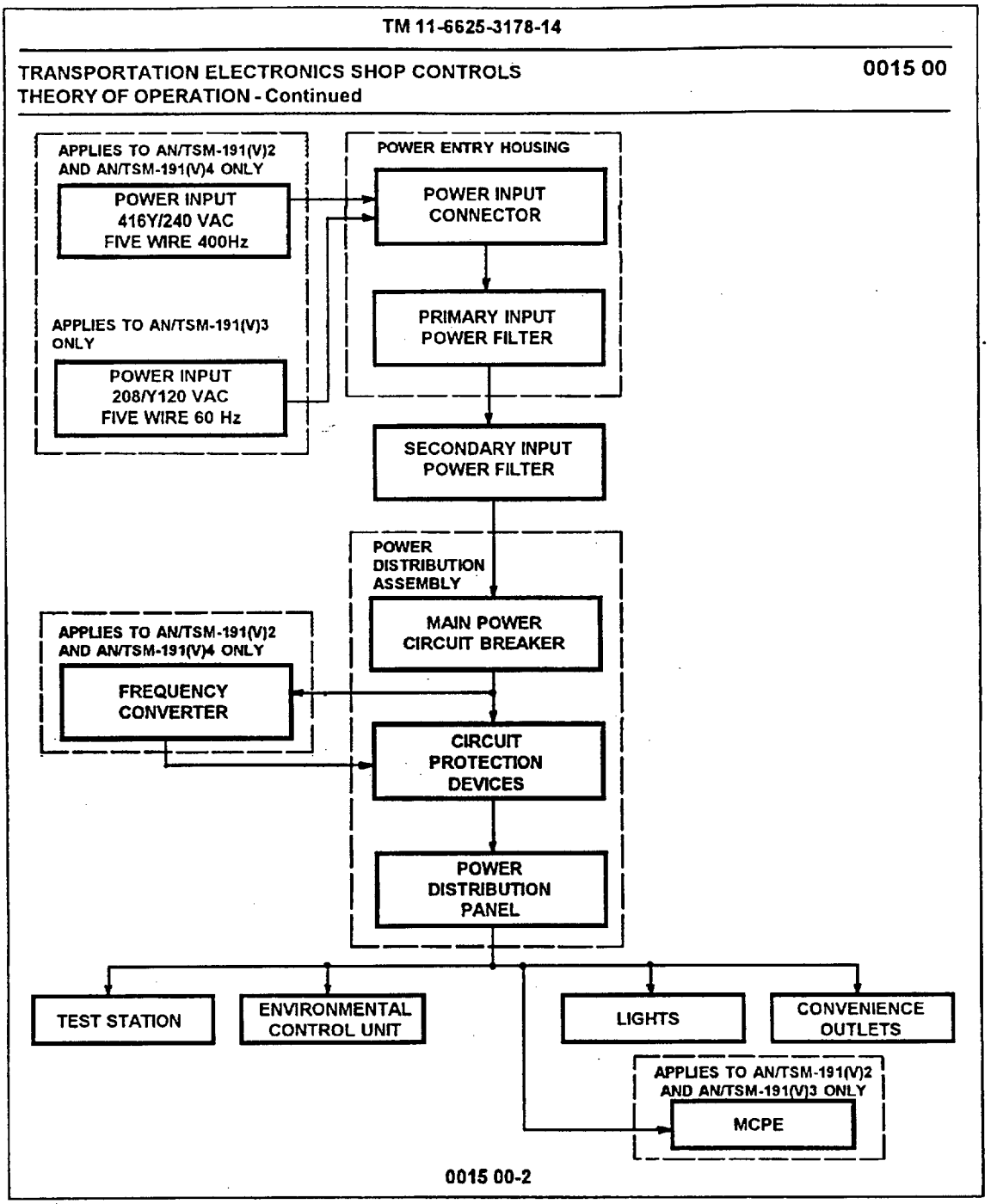


FIGURE 4. Example of system theory of operation - Continued

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TM 1-1520-238-T-9

APU SUBSYSTEM THEORY OF OPERATION 0022 00

GENERAL

The pilot center circuit breaker panel, APU fire panel assembly and pilot caution/warning panel provide the controls and indications for the APU.

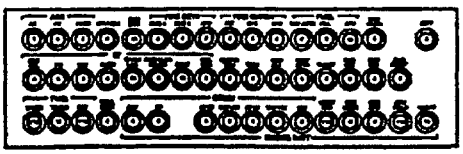
1. Pilot Center Circuit Breaker Panel. The pilot center circuit breaker panel FUEL APU, FUEL BST, and APU HOLD circuit breakers provide circuit protection and 28 VDC for APU operation.

2. Pilot Caution/Warning Panel. The pilot caution/warning panel provides the pilot with APU ON and APU FAIL indications.

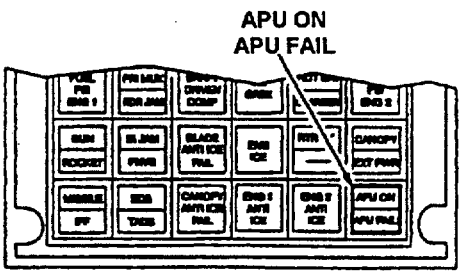
   a. APU ON indicator. The APU ON indicator lights when the APU reaches 95% rpm.

   b. APU FAIL indicator. The APU FAIL indicator lights when oil pressure drops below 75 psig, except when the START/RUN/OFF switch is in the START position.

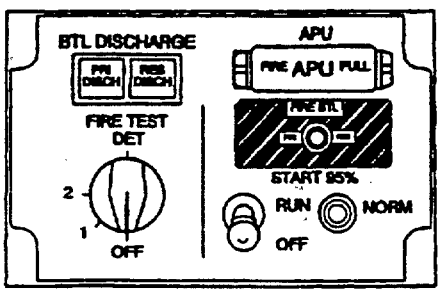
3. APU Fire Panel Assembly. The APU fire panel assembly controls APU start/run operation and APU fire indication.



PILOT CENTER CIRCUIT BREAKER PANEL



PILOT CAUTION/WARNING PANEL



APU FIRE PANEL ASSEMBLY

0022 00-1

FIGURE 5. Example of subsystem theory of operation.

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TM 1-1520-238-T-9

APU SUBSYSTEM THEORY OF OPERATION - Continued 0022 00

GENERAL - Continued

a. APU FIRE PULL handle. The APU FIRE PULL handle wams the pilot that a fire exists in the APU shroud assembly or SDC area. When FIRE TEST DET switch is set to 1, the APU FIRE PULL handle lights to indicate the upper flame detector circuits and components are functioning. When FIRE TEST DET switch is set to 2, the APU FIRE PULL handle lights to indicate the lower flame detector circuits and compo nents are functioning. When the APU FIRE PULL handle is pulled, the APU fuel shutoff valve is closed and the selected fire bottle is discharged.

b. 55%/NORM switch. The 55% position is used during cold weather starts when the outside air temperature is 0°F (-18°C) or below. The NORM position is used during normal starts when the outside air tempera ture is above 0°F (-18°C).

c. START/RUN/OFF switch. The START position initiates the APU start sequence. The START position is spring-loaded to the RUN position. The RUN position allows steady state automatic APU operation. In the OFF position, no voltage is supplied to the APU.

THEORY OF OPERATION

APU circuit breaker (CB148) and APU HOLD circuit beaker (CB10) protect the APU start system circuitry from excessive current and provide 24VDC from the battery to the APU fire panel assembly for START/RUN power.

When the START/RUN/OFF switch is set to the START position, FUEL APU circuit breaker (CB9) supplies 28 VDC to the APU fuel boost pump and APU fuel shutoff valve.

1. The APU shutoff valve provides automatic control of fuel flow from the aft fuel cell to the APU fuel boost pump.

2. The APU fuel boost pump provides a positive fuel flow to the APU fuel solenoid valve.

3. The APU fuel solenoid valve is opened by the APU controller when the APU reaches 5% rpm, providing fuel to the APU fuel control.

FUEL BST circuit breaker (CB57) supplies 28 VDC to the time delay relay and to the APU controller. The SDC time delay relay de-energizes the shaft driven compressor throttle valve for 60 seconds. The APU controller initiates, controls and monitors the start sequence, normal operation and automatic shutdown as follows.

1. The hydraulic start solenoid valve is energized open by the APU controller. The hydraulic start solenoid valve opens to release 3000 psi hydraulic pressure from the utility hydraulic accumulator through the hydraulic starter. APU rotation begins.

0022 00-2

FIGURE 5. Example of subsystem theory of operation - Continued

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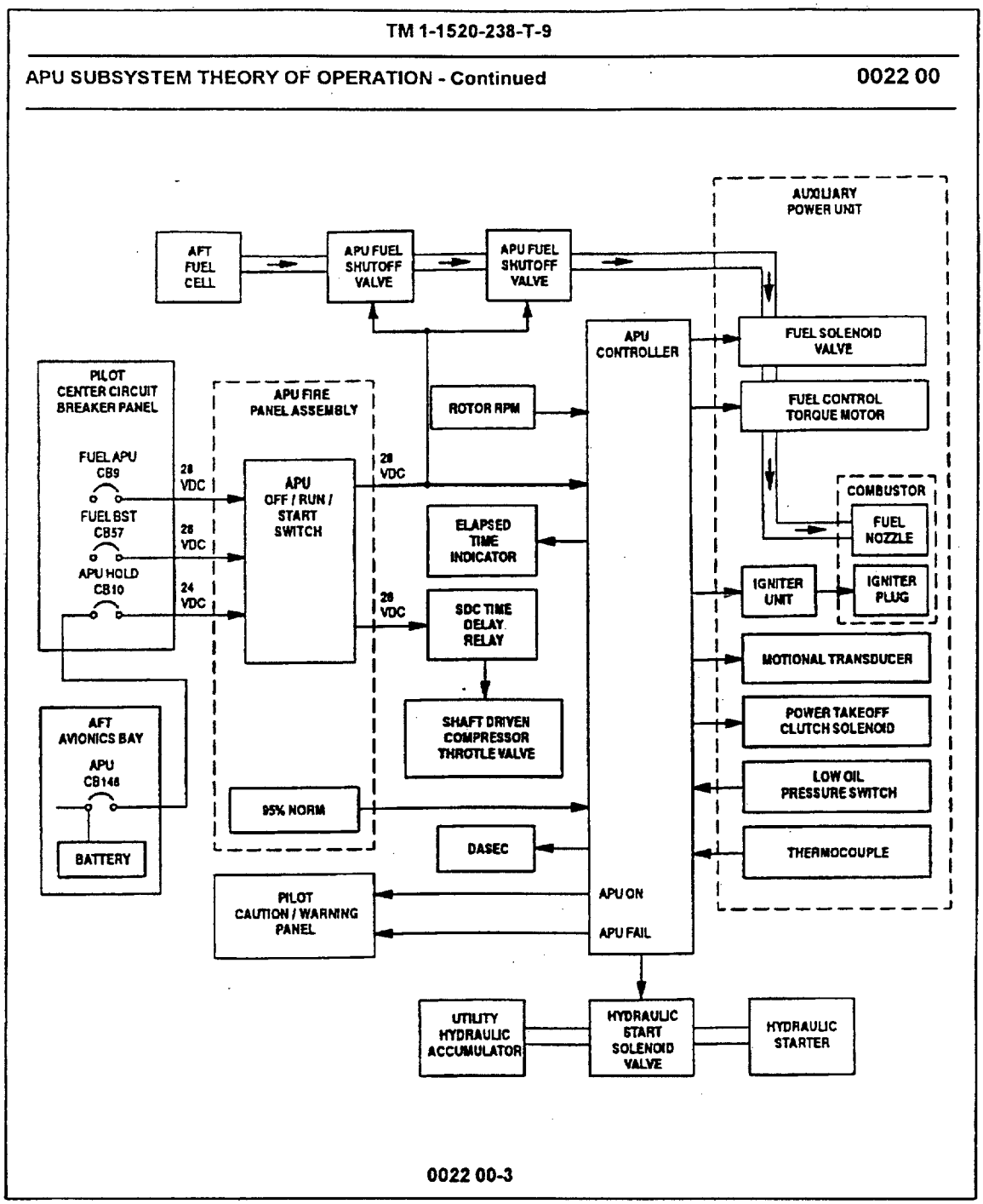


FIGURE 5. Example of subsystem theory of operation - Continued

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TM 11-6625-3278-40

AN/USM-620 DETAILED THEORY OF OPERATION **0411 00**

POWER MODULE SECTION

The Power Module Section provides DC voltages and forced air cooling for TR 503 Tracking Generator. This section describes each module within TM 5003 Power Module. Refer to figures 23 through 26 for schematic diagrams.

A3A10 Main Interface CCA

Sends DC and AC voltages, and the PWR signal, to each of three plug-in sockets. PWR indicates when the main power supply is operating.

DC Voltages are +8, +26, and -26 V. Transformer A3T500 couples AC power from A3A14 Line Filter CCA to secondary voltage taps. AC voltages are 25 and 18 V. A 16 VAC feedback signal connects to A3A10J1110. This signal is rectified by A3A11CR1561 to supply power for A3A11U1550. The current sense input to A3A11U1550 also uses this supply.

The PWR signal from A3A10Q1125 drives pin 6 of each plug-in socket. The signal is normally high when power is on. A3A11U1850 pin 7 controls the status of PWR. Pin 7 is normally low. A3A11U1850 pin 7 toggles high when triggered by A3A11U1720A or by A3A11U1550.

A3A11 Secondary Supply CCA

Contains a secondary rectifier circuit, a control logic and drivers circuit, an over-voltage and over-current detectors circuit, and a control circuit regulator. The secondary rectifier circuit produces DC voltages. The control logic and drivers circuit produces a 20 kHz drive signal. The output regulator circuit produces a 40 kHz clock signal and sets the +8 V output level. The over-voltage and over-current detectors circuit limits duty cycle when output voltage or current is too high. The control circuit regulator produces reference voltages and the PWR signal.

The secondary rectifier circuit produces +26, +8, -26 VDC supplies. Each supply connects to plug-in sockets on A3A10 Main Interface CCA. Transformer A3A11T1210 couples the 20 kHz drive signal to diode rectifiers. Diodes A3A11CR1010 and A3A11CR1020 produce the +26 V supply. Diode A3A11CR500 provides the +8 V supply. Diodes A3A11CR1021 and A3A11CR1120 produce the -26 V supply. Transformer A3A11T1000 regulates power by producing over-voltage feedback that controls the duty cycle of the 20 kHz drive signal. Diodes A3A11CR1502, A3A11CR1510, A3A11CR1511, and A3A11CR1512 rectify the output of A3A11T1000. The resulting DC voltage drives A3A11U1540B, a comparator within the over-voltage and over-current detectors circuit.

The control logic and drivers circuit produces a 20 kHz drive signal for the 20 kHz output amplifier. Transistors A3A11Q1400 and A3A11Q1401 produce the 20 kHz output. Current is supplied by the +10 V source (A3A11U1550 pin 10). Duty cycle of the 20 kHz signal changes to control regulation. Gates A3A11U1610A and A3A11U1610B receive four input signals to control 20 kHz switching. A high on any input causes A3A11Q1400 and A3A11Q1401 to conduct. During power on or power off, A3A11Q1650 also controls A3A11Q1400 and A3A11Q1401 through A3A11R1511 and A3A11R1520. Flip-flop A3A11U1720A controls A3A11U1610 pins 5 and 10. This signal is normally high. Refer to over-voltage and over-current detectors circuit description for details of operation. Multivibrator A3A11U1720B drives A3A11U1610 pins 3 and 12. A3A11U1720B ensures that A3A11Q1400 and A3A11Q1401 never conduct at the same time. The minimum dead time (time when A3A11Q1400 and A3A11Q1401 are both off) is controlled by the time constant of A3A11R1610 and A3A11C1710. The voltage charge on A3A11C1730 and A3A11C1831 also affects dead time. During turn on, current from the supply is limited until these capacitors reach full charge. The 40 kHz clock signal drives A3A11U1720B pin 11 to provide timing. Monostable A3A11U1600A drives A3A11U1610 pins 4 and 11. A3A11U1600A limits duty cycle when output voltages are too high. Refer to output regulator circuit description for details of operation. Divide by two flip-flop A3A11U1600B drives A3A11U1610 pins 2 and 9. A3A11U1600B converts the drive signal from 40 kHz to 20 kHz.

0411 00-1

FIGURE 6. Example of detailed theory of operation

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NOT MEASUREMENT SENSITIVE

MIL-STD-40051-3

DEPARTMENT OF DEFENSE STANDARD PRACTICE

TECHNICAL MANUALS

OPERATOR INSTRUCTIONS

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1. SCOPE.

1.1 Scope. This standard establishes the technical content requirements for the preparation of operator instructions for weapon systems and equipment Technical Manuals (TMs). These requirements are applicable for both paper and digital page-oriented TMs. Electronic delivery of the TMs is accomplished through the use of the Operator Instructions modular Document Type Definition (DTD). The DTD is available in a digital format. Refer to MIL-STD-40051 for information on obtaining this DTD. Operator instructions provide the operator/crew with data for operating the weapon system/equipment including location and use of controls and indicators, how to prepare the weapon system/equipment for use, and how to operate the weapon system/equipment under ordinary, unusual, and emergency conditions.

2. APPLICABLE DOCUMENTS.

The applicable documents in Section 2 of MIL-STD-40051 apply to this Part.

3. DEFINITIONS.

The definitions in Section 3 of MIL-STD-40051 apply to this Part.

4. GENERAL REQUIREMENTS.

4.1 General. Operator instructions shall be prepared for weapon systems, major equipment, components and applicable support and interface equipment. Operating instructions shall describe the operation authorized for the operator/crew. Procedures and supporting illustrations shall be prepared so that personnel can prepare the weapon system/equipment for operation, identify and locate operational controls and indicators, and operate the weapon system/equipment safely and efficiently in both normal and emergency conditions. This information shall be contained in work packages and become a part of an Operator Instructions Information chapter.

4.2 Standard tables. Various standard tables required are noted throughout the text of this standard in bold and in parentheses (i.e., (**standard table**)). The formats and table heading names of these standard tables shall have no deviations.

4.3 Preparation of digital data for electronic delivery. Technical manual data prepared in work package format and delivered digitally in accordance with this standard shall be SGML tagged and assembled using the modular Assembly DTD and Formatting Output Specification Instance (FOSI). The DTD and FOSI has been developed in accordance with MIL-PRF-28001 and ISO 8879. Refer to MIL-STD-40051 for information on obtaining or accessing the modular DTD and FOSI. SGML tags used in the modular DTD are noted throughout the text of this standard in bracketed, bold characters (i.e., **<opim>**) as a convenience for the TM author and to ensure that the tags are used correctly when developing a document instance.

4.3.1 Use of the DTDs / FOSIs. The modular DTDs referenced in this standard interpret the technical content and structure for the functional requirements contained in this standard and are mandatory for use. The modular FOSIs referenced herein interpret the style and format. As specified by the contracting activity, FOSIs or style sheets may be used to produce final reproducible paper copy for all TMs prepared in accordance with this standard.

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4.4 Content structure and format. The examples provided at the rear of this Part are an accurate representation of the content structure and format requirements contained herein and shall be followed to permit the effective use of the modular DTD for Operator Instructions.

4.5 Style and format. Style and format requirements for the preparation of Department of Army TMs are contained in MIL-STD-40051-1; they are considered mandatory and are intended for compliance. Preferred general style and format requirements for Army TMs shall be provided by the procuring activity.

4.6 Work package development. Technical manual data developed in accordance with this standard shall be divided into individual, stand alone units of information work packages. A work package shall consist of descriptive, operational, maintenance, troubleshooting, support, or parts information for the weapon system or equipment.

4.7 Safety devices and interlocks. Information shall be prepared pertaining to the purpose and location of all safety devices and interlocks in conjunction with the pertinent procedures.

4.8 Electrostatic Discharge (ESD) sensitive parts. If the equipment contains ESD sensitive parts, components, or circuits, cautions and ESD labels shall be incorporated into the applicable tasks and procedures to ensure ESD sensitive parts are not damaged or degraded during handling or operation. Refer to MIL-STD-40051-1 for requirements on labeling ESD. Actions which could damage ESD sensitive parts, but which are not directly related to handling or operation of ESD sensitive parts, shall not be annotated with the ESD acronym, but shall be preceded by a caution statement.

4.9 Nuclear hardness. If the weapon system/equipment has nuclear survivability requirements (for example, over pressure and burst, thermal radiation, electromagnetic pulse, or transient radiation effects on electronics), cautions and Hardness-Critical Processes (HCP) labels shall be incorporated into the applicable tasks and procedures to ensure the hardness of the equipment is not degraded during handling or operation. Refer to MIL-STD-40051-1 for requirements on labeling with HCP. Actions which could degrade hardness, but which are not directly involved in establishing nuclear hardness, shall not be annotated with the acronym, but shall be preceded by a caution statement.

4.10 Selective application and tailoring. MIL-STD-40051 contains some requirements that may not be applicable to the preparation of all technical manuals. Selective application and tailoring of requirements contained in MIL-STD-40051 are the responsibility of the contracting activity and shall be accomplished through the use of Appendix A, Technical Manual Content Selection Matrixes, of MIL-STD-40051. The applicability of some requirements is also designated by one of the following statements: unless specified otherwise by the contracting activity; as/when specified by the contracting activity; or when specified by the procuring activity.

5. DETAILED REQUIREMENTS.

5.1 Preparation of operator instructions. Operator instructions shall be prepared as an Operator Instructions Information Chapter **<opim>**. This chapter shall be subdivided into individual work packages that provide the operator of the weapon system/equipment with descriptions and use of controls and indicators and operation of the weapon system/equipment under usual, unusual and emergency conditions. This information shall be divided into the following work packages.

a. Description and use of controls and indicators work package **<ctrlindwp>**.

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b. Operation under usual conditions work package(s) **<opusualwp>**.

c. Operation under unusual conditions work package(s) **<opunuwp>**.

NOTE: In cases where operating instructions are divided by crew station assignment (or auxiliary equipment), work packages shall be developed to support each crew-served station.

The words **END OF OPERATING PROCEDURE**shall be placed below the last data item (i.e., text, illustration, etc.) in any work package containing procedures.

5.1.1 Description and use of controls and indicators work package **<ctrlindwp>**. Information shall be prepared for the description and use of all system or equipment controls and indicators. A description and use of controls **<contdesc>** and indicators **<inddesc>** shall be prepared for each equipment, assembly, or control panel having controls and indicators. Illustrations shall be prepared for all operator controls and indicators. Each control and indicator shall be clearly labelled as it appears on the equipment. Controls and indicators that are not labelled, such as the accelerator or brake pedals, shall be identified. The functional use of each control and indicator shall be explained. A table **<ctlindtab> (standard table)** or list may be used to explain the use of the controls and indicators. (Refer to figure 1.)

5.1.2 Operation under usual conditions work package **<opusualwp>**. Instructions to operate the weapon system/equipment and auxiliary equipment in all modes of operation shall be prepared. Any combination of control settings that will create a hazard to personnel or cause damage to equipment shall be preceded by a warning or caution. Instructions to ensure proper grounding of equipment shall be prepared. The operational tasks **<opertsk>** described in 5.1.2.1 through 5.1.2.8 shall be included, as applicable. (Refer to figure 2.)

5.1.2.1 Initial setup information **<wpinfo>**. Information required by the user before starting a detailed operating procedure or maintenance task shall be listed at the beginning of each work package. This initial setup information is intended to provide the user with essential information to operate the equipment properly or perform the maintenance task correctly. Setup data from the following list shall be included as determined by the individual operational procedure or maintenance task being performed.

a. Maintenance level **<maintlvl>**. The level of maintenance authorized to perform the maintenance contained in the work package (in accordance with the approved MAC) shall be stated. This shall be included for operator or crew, unit, AVUM, direct support, general support, AVIM, and depot levels of maintenance, as applicable. For example,

Maintenance Level
Unit

b. Applicable configurations **<appconfig>**. When the work package does not apply to all configurations of the weapon system/equipment, the applicable configurations **<name>** covered by the work package shall be listed. Omit this requirement if the same tasks/procedures apply to all configurations. (If certain configurations require different tasks/procedures, separate work packages shall be prepared.) For example,

Applicable Configurations
Serial Numbers 12345 through 12399

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c. Test equipment **<testeqp>**. All test equipment required to perform the procedure shall be listed by name **<name>** and part **<partno>** or model number **<modelno>** designation if this information is not contained in an overall list elsewhere in the TM. If such a list exists, refer to it by name **<name>**, item number, and work package number **<simref>** instead of repeating the information throughout the TM. For example,

Test Equipment
Multimeter (Item 4, WP 0108 00)
Oscilloscope (13057)

d. Tools and special tools **<tools>**. The tool kit (box) assigned to the mechanic (on a 1-per-mechanic-by-MOS basis) to be used in maintenance of a particular equipment shall be listed by name **<name>**, tool kit number (**<partno>** or **<nsn>**), supply catalog (SC) **<sc>**, or TM number **<tmno>** if this information is not included in the Tool Identification List work package **(-20/AVUM and above only)** contained in the TM. If such a list exists, refer to it by name **<name>,** item number and work package number **<simref>** instead of repeating the information throughout the TM. No tool in the kit shall be further identified. Other tools required for performance of all tasks for the maintenance levels covered in the work package shall also be identified in the initial setup and shall be referenced to the Tool Identification List work package **(-20/AVUM and above only)**. "Other tools" includes tools which are part of/components of shop sets authorized to sections/teams; tools authorized by RPSTL and CTA-50-970; special and fabricated tools; and items of TMDE. For example,

Tools and Special Tools
Fixed Open End Wrench Set (Item 47, WP 0110 00)
Screw Threading Set (SC number)
Vehicle Tool Kit (407425)

e. Materials/parts **<mtrlpart>**. All expendable items and support materials shall be listed **<name>**. The item number and supporting information work package **<simref>** which lists these items shall be given. Mandatory replacement parts shall be listed by name **<name>** (and part number **<partno>**, if any). The number, quantity **<qty>**, or size necessary to complete the task shall be listed, when applicable. When a mandatory replacement parts work package **(-20/AVUM and above only)** exists, it shall be referred to in lieu of the part number. For example,

Materials/Parts
Grease (Item 5, WP 0112 00)
Wiping Rags (Item 38, WP 0112 00)
Range Lock (P/N 8675309)
Range Lock Flange Kit (P/N 8675310)

f. Personnel required **<persnreq>**. Personnel **<name>** and the number of personnel **<qty>** shall be identified if the task requires more than one. The Military Occupational Specialty (MOS) designation **<nameid>** is not necessary, but it may be included. For example,

Personnel Required
Artillery Mechanic 68M10 (1)
Artillery Mechanic 66J30 (1)

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g. References <**ref**>. Other work packages, TMs, and other sources (<**extref**>/<**xref**>/<**simref**>) that are needed to complete the maintenance tasks shall be listed here. Only references not listed in equipment conditions shall be listed. For example,

References
TM 9-1015-252-20&P
WP 0100 00

h. Equipment conditions <**eqpconds**>. Any special equipment conditions required before the procedure can be started shall be listed here and cross-referenced to the appropriate source (<**extref**> or <**xref**>) for setting up the condition <**condition**>. For example,

Equipment Condition
Firing mechanism removed (WP 0010 00)

i Special environmental conditions <**specenv**>. Any special environmental conditions (such as ventilation, lighting, or temperature) <**condition**> that are required shall be listed here. The reason that such conditions are needed shall be explained. For example,

Special Environmental Condition
Darkened area required for testing lights.

j. Drawings required <**dwgreq**>. All drawings (which are not included in the work package) required to complete the maintenance tasks shall be listed here. Drawings shall be listed by title <**dwgname**> and drawing number <**dwgno**>. For example,
Drawings Required
Power Supply Schematic (132E470092)

5.1.2.2 Siting requirements <**site**>. Siting instructions peculiar to the equipment shall be prepared. Operational features shall be considered, such as the following.

a. Location.

b. Proximity to power sources.

c. Effective ranges.

d. Terrain requirements to avoid screening reflections, ground clutter, and other poor operational conditions due to terrain.

e. Technical requirements.

f. Shelter locations.

g. Compensating for adverse siting conditions.

h. When the equipment contains large components, such as towers and antennas, that require orientation to a baseline during siting.

i. Mobile equipment oriented during installation.

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5.1.2.3 Shelter requirements <**shelter**>. For equipment normally housed in a permanent or semipermanent shelter (other than a military truck, van, or transportable shelter) during use, the following information shall be prepared.

a. Amount of floor, wall, and height space required.

b. A plan for a typical layout.

c. Required weight capacity of the building floor.

d. Dimensions required for installed equipment.

e. Total weight that the floor must support and the area in square feet over which the total weight will be distributed.

f. Environmental conditions (e.g., venting).

g. Power requirements.

h. Unusual requirements specific to equipment, such as air-conditioning.

i. Architectural and engineering data on beam sizes, lengths, bending moments, and required supports shall not be included.

5.1.2.4 Assembly and preparation for use.<**assem**>.

a. Procedures shall be prepared for unpacking, assembly, and installation. When the equipment is shipped or delivered in specially designed containers, unpacking instructions shall be prepared. If the containers are to be used again, kept for future use, turned in to supply, or if any special disposition is required, the necessary procedures shall be prepared. Assembly and installation procedures shall be prepared when needed. These instructions shall be supported by illustrations. As applicable, power requirements, connections, and initial control settings needed for installation purposes shall be included.

b. For security measures for electronic data, instructions shall be prepared for handling, loading, scrubbing, overwriting, or unloading classified electronic data under usual conditions. Instructions shall meet the requirements as they pertain to automation security.

5.1.2.5 Initial adjustments, before use, and self-test <**initial**>. Procedures shall be prepared for any routine checks, self-test, or adjustments that the operator must make before putting the equipment in operation.

5.1.2.6 Operating procedures <**oper**>. The following operating instructions shall be prepared, as applicable.

a. All steps necessary to bring the equipment from OFF through STANDBY condition to full operation, including all necessary warnings and cautions.

b. Procedures for each mode of operation, e.g., manual, automatic, local, remote, etc. The use and relative advantage of each mode shall also be described.

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c. Description of the equipment anti-jamming and interference reduction features, the advantage of each feature, and the operating procedures to be followed. Supporting illustrations (such as indicator displays, waveforms, etc.) shall be included which provide typical observations of jamming and interference for evaluation by the operator.

d. Operator turn-off procedures, including all steps necessary to bring the equipment from full operation through STANDBY to OFF condition.

e. Procedures covering operation of the equipment during emergency conditions (control failure, air failure, lube oil failure, loss of cooling water, etc.). Emergency operating instructions shall be included. Warning or caution to return the equipment to proper operation when the emergency is over shall also be included.

f. Procedures to turn the equipment off during an emergency (fire, water, smoke, hazard to personnel, loss of coolant, normal power, etc.).

g. Operating instructions for misfire, hangfire, and other procedures applicable to ammunition.

h. Operating procedures explaining how the equipment is operated in conjunction with auxiliary equipment or how it operates when integrated with other equipment.

5.1.2.6.1 Operating procedure considerations. The following considerations should be taken into account when preparing operating procedures.

a. Initial safety requirements (actions, inspections, and emergency turn-off procedures).

b. If a particular operating procedure or step is assigned to a specific crew-served position (e.g., gunner), the assignment must be indicated.

c. Connection of any accessory equipment not permanently connected.

d. Instructions for obtaining or confirming the presence of all critical inputs such as power, coolant, air, signal, air-conditioning, etc. Specific values for critical inputs (power, coolant, air, etc.) shall also be included.

e. Procedures for setting controls and making adjustments which must be accomplished by the operator prior to equipment turn-on.

f. Procedures for determining operational readiness and the acceptable indications expected from built-in indicators, such as meters, lamps, gages, displays, and recorder readouts.

g. Milestones in the operational status of the equipment, indicated by brief statements, such as "The generator is now in STANDBY."

h. Visual or audible observations which occur as a result of an operator action, such as boom lowering, sweep rotation, blower motor running, etc.

5.1.2.6.2 Summary operating procedures. Procedural steps may be preceded by a summary. If the summary is used, it shall be written entirely in capital letters.

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5.1.2.6.3 Decals and instruction plates **<instructplt>**. Decals and operating instruction plates located on the equipment, which are essential for operation, shall be clearly illustrated, so that all information is legible. Related warning and caution decals and plates shall be included. An illustration(s) shall be prepared to show the location of all applicable decals and plates.

5.1.2.7 Operating auxiliary equipment **<operaux>**. If applicable, procedures shall be prepared for putting the auxiliary equipment into operation, operating it, and putting it in standby or shutdown status. If these procedures are published in another TM covering the auxiliary equipment, reference shall be made to that TM in accordance with the requirements of MIL-STD-40051-1.

5.1.2.8 Preparation for movement **<prepmove>**. Preparation for movement procedures shall be prepared if the equipment is designed for movement and it can be readied for movement by the operator. Procedures shall be prepared for actions such as disassembly, folding, and telescoping. Illustrations shall be prepared, as required, to support the text. This information shall not duplicate the "assembly and preparation for use" requirements contained in 5.1.2.4.

5.1.3 Operation under unusual conditions work package **<opunuwp>**. Instructions shall be prepared for operation under unusual conditions. Preventive or protective measures to be taken beyond the operator's capabilities shall be identified. Instructions to ensure proper grounding of equipment shall be prepared. For security measures for electronic data, instructions shall be prepared for handling, loading, purging, overwriting, or unloading classified electronic data under unusual conditions. Instructions shall meet current security regulations as they pertain to automation security. The operational tasks **<opunutsk>** described in 5.1.3.1 through 5.1.3.6 shall be included, as applicable. (Refer to figure 3.)

5.1.3.1 Initial setup information **<wpinfo>**. Initial setup information requirements are provided in 5.1.2.1.

5.1.3.2 Unusual environment / weather **<unusualenv>**. Procedures shall be prepared for operation under conditions of extreme moist heat, extreme dry heat, extreme cold, salt air, sea spray, dust storms, sand storms, high altitudes, snow, mud, and other similar conditions. Ranges of environmental/weather operating conditions considered for the system addressed shall be defined.

5.1.3.3 Fording and swimming **<fording>**. If applicable, procedures for fording and swimming the equipment, shall be provided.

5.1.3.4 Interim Nuclear, Biological, and Chemical (NBC) decontamination procedures **<decon>**. As applicable and specified by the contracting activity, interim general NBC decontamination procedures to be performed until NBC decontamination facilities are available shall be prepared. Other decontamination TMs shall be referenced only when necessary.

5.1.3.5 Jamming and Electronic Countermeasures (ECM) procedures **<ecm>**. As applicable, procedures shall be prepared for operation of the equipment in an ECM environment through transmitted and reflected deception signals and through transmitted and reflected jamming.

5.1.3.6 Emergency procedures **<emergency>**. Procedures shall be prepared for temporarily adapting the equipment and the operating procedures to meet the reduction of power, partial failure, failure of a portion of the equipment, or similar conditions when continued equipment use is required.

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6. NOTES.

The notes in section 6 of MIL-STD-40051 apply to this Part.

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TM-6625-3178-14

TRANSPORTABLE ELECTRONICS SHOP CONTROLS AND INDICATORS 0202 00

GENERAL

The following paragraphs contain illustrations that show the location of each control and indicator for operation of the AN/TSM–191(*). Each control and indicator is clearly labeled as it appears on the equipment. Find numbers on the illustration are keyed to the tabular listing which contains the name, based on the panel markings, and at the functional description of each control and indicator.

POWER DISTRIBUTION PANEL CONTROLS AND INDICATORS

Table 1 describes the controls and indicators for the power distribution panel.

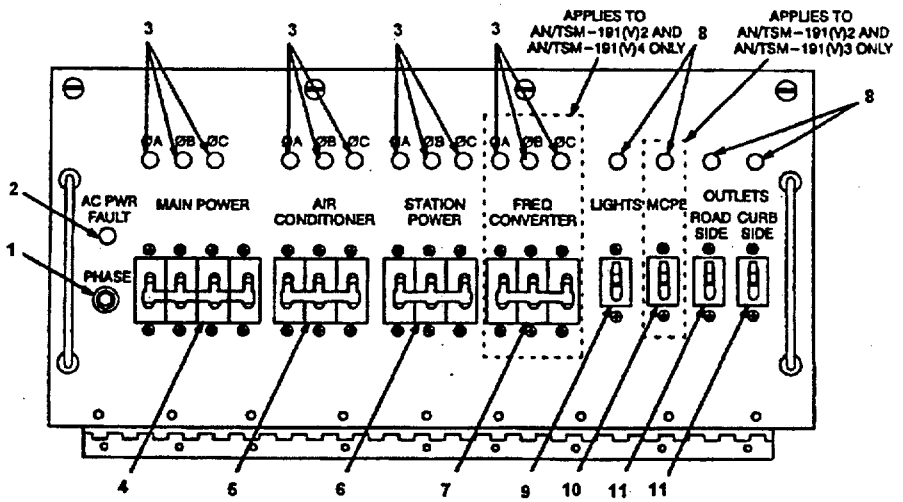


Table 1. Power Distribution Panel Controls and Indicators.

KEY	CONTROL OR INDICATOR	FUNCTION
1	PHASE indicator	When lit, indicates power applied to the input side of the MAIN POWER circuit breaker has the correct phase sequence of A-B-C.
2	AC PWR FAULT indicator	When lit, indicates power is applied to the input side of the MAIN POWER circuit breaker and the MAIN POWER circuit is the OFF (down) position.
3	ØA, ØB, ØC indicators	When lit, indicates circuit breaker; either MAIN POWER , AIR CONDITIONER , STATION POWER, or FREQ CONVERTER (applies to AN/TSM-191(V)2 and AN/TSM-191(V)4 is the ON (up) position and power applied to the input side of the breaker is now applied to the outputs of the breaker.

0202 00-1

FIGURE 1. Example of an operator's controls and indicators work package.

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TM-6625-3178-14

TRANSPORTABLE ELECTRONIC SHOP OPERATION UNDER USUAL CONDITIONS **0304 00**

INITIAL SETUP:

Maintenance Level	Personnel Required
Unit	Two

GENERAL

The equipment contained within this test facility is normally shipped assembled. The assembly paragraph provides the data and procedures for the steps to be taken when AN/TSM-191(*) is emplaced. The items included are ladder installation, ground rod installation, primary power connection and air conditioner drain plug removal.

SITING REQUIREMENTS

WARNING

The power source must be placed at least 75 feet from the test facility.

CAUTION

If the outside ambient temperature is expected to be 90 degrees or above, position the shelter with roadside shaded, if possible, to minimize the effect of direct sunlight.

1. The site requirements for emplacement of the AN/TSM-191 (*) consist of a three -phase 416 wye/ 240 volt ac, 400 Hz (applies to the AN/TSM-191 (V) 2 and AN/TSM-191 (V) 4 only or three-phase 208 wye/120 volts ac, 60 HZ (applies to the AN/TSM-191(V) 3 only) primary power source.
2. The AN/TSM-191(*) should be placed on a minimum of 15 by 25 feet of firm level ground with a six percent slope or less.
3. The power source should be placed at least 75 feet from the facility.

PREPARATION FOR USE AND ASSEMBLY

Ladder Installation

WARNING

The ladder must be installed to provide access to the AN/TSM-191 (*) Ladder weighs 75 lbs. Two people are required to lift the ladder.

1. Remove ladder (1) from storage brackets on exterior entry end of shelter.

0304 00-1

FIGURE 2. Example of an operation under usual conditions work package.

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TRANSPORTABLE ELECTRONIC SHOP OPERATION UNDER USUAL CONDITIONS-Continued 0304 00

PREPARATION FOR USE AND ASSEMBLY-Continued

2. Remove hooks (5) from retainer slots on both sides of tail gate (4).
3. Grasp top of tail gate and pull it back. Do not allow tail gate to drop.
4. Replace hooks in slots of tail gate to secure tail gate in horizontal position.
5. Pull out retaining pins and remove railings (2) from ladder.

NOTE

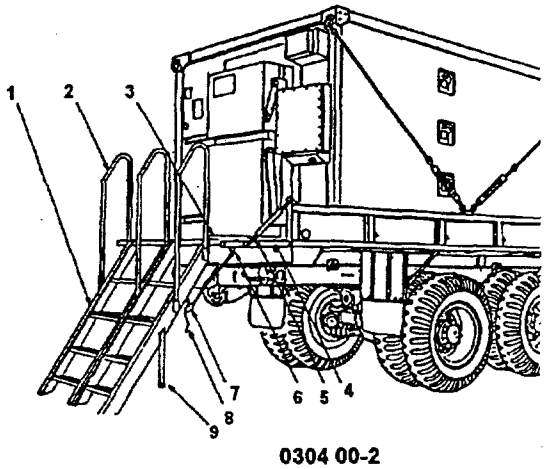
Ladder should be positioned for use while strapped in the folded position.

6. Slide U-brackets (3) at top end of ladder over edge of tail gate.

NOTE

When ladder is unfolded, do not seperate. Ensure hinge pins are engaged.

7. Loosen straps (9) and unfold ladder.
8. To release lanyards (7) from the stored position, pull up on center locking pin and pull cord through.
9. Hook retaining brackets (6) on bottom of tail gate, pull free end of cord (8) and push down on center locking pin to secure ladder in position.
10. Install railing (2) on ladder.



END OF TASK

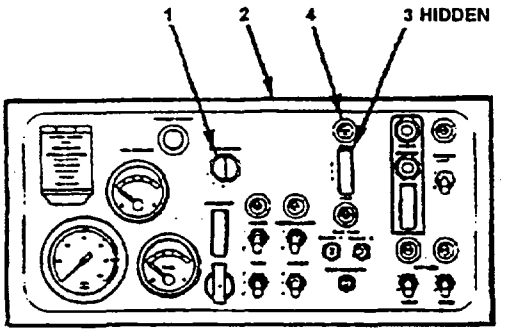
0304 00-2

FIGURE 2. Example of an operation under usual conditions work package - Continued.

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OPERATE SWIM CONTROLS (AMPHIBIOUS OPERATIONS)

1. Set MAIN SWITCH (1) on driver's right instrument panel (2) to position 2.



2. Operate bilge pumps (see WP 0386 00).
3. Operate differential locks (see WP 0394 00).

WARNING

Personnel can be injured by moving propellers. Before turning on amphibious drive, make sure personnel are not closer than 16.4 feet (5m) from propellers.

CAUTION

Before switching on amphibious drive, make sure propellers and drive shafts are not blocked by dirt or frozen mud. Clean to advoid damage to equipment.

Amphibious drive should only be switched on when engine is idling to prevent damage to equipment.

4. Start engine (see WP 0167 00) and run at idle. Open cover and pull out and move SWIM/FREE switch (3) to position 2. Check that SWIM indicator light (4) is lit and have crewmember check that both propellers are operating. If not, do troubleshooting (see WP 0562 00).

NOTE

Swim control handle is stowed behind commander's seat.

FIGURE 3. Example of an operation under unusual conditions work package.

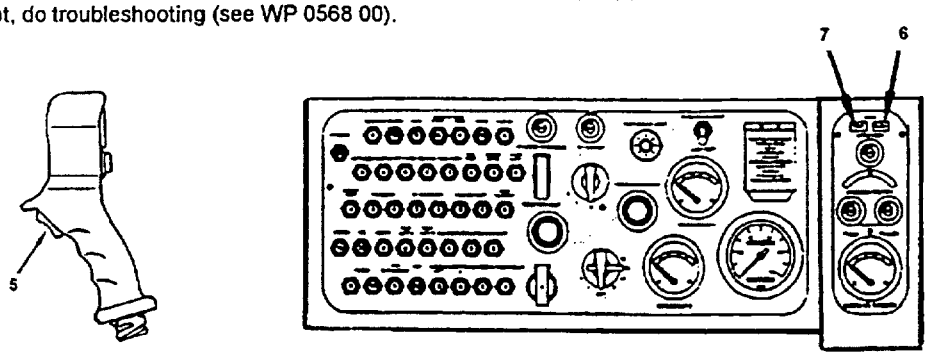
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TM 3-6665-339-10

NBCRS FOX M93A1 OPERATION UNDER UNUSUAL CONDITIONS - Continued 0426 00

OPERATE SWIM CONTROLS (AMPHIBIOUS OPERATIONS) - Continued

5. Move speed rocker switch (5) to position 1. SLOW indicator light (6) should be lit. If not, do troubleshooting (see WP 0568 00).
6. Move speed indicator switch (5) to position 2. FAST indicator light (7) should be lit. If not, do troubleshooting (see WP 0568 00).



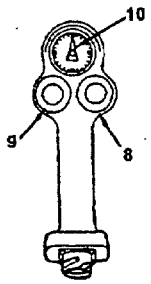
NOTE

Engine output power must be provided by driver while commander operates swim controls. Propeller speed is controlled by the engine speed.

Depressing right or left steering switch to first position will traverse propellers at a slow rate. Depressing right or left steering switch to second position will traverse propellers at a high rate.

Amphibious direction indicator will rotate in direction of steering switch that is depressed. This will indicate direction vechile is heading.

7. Depress right or left hand steering switch (8, 9) as needed to position amphibious direction indicator (10) to 0. This will allow for straight-ahead travel. Have crewmember check that propellers are in the proper position. If not, notify unit maintenance.
8. Operate trim vane (see WP 0331 00).



END OF TASK

0426 00-2

FIGURE 3. Example of an operation under unusual conditions work package - Continued.

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MIL-STD-40051-4

DEPARTMENT OF DEFENSE STANDARD PRACTICE

TECHNICAL MANUALS

TROUBLESHOOTING PROCEDURES

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1. SCOPE.

1.1 Scope. This standard establishes the technical content requirements for the preparation of troubleshooting procedures for weapon systems and equipment Technical Manuals (TMs) and Depot Maintenance Work Requirements (DMWRs). These requirements are applicable for both paper and digital page-oriented TMs. Electronic delivery of the TMs is accomplished through the use of the Troubleshooting Procedures modular Document Type Definition (DTD). The DTD is available in a digital format. Refer to MIL-STD-40051 for information on obtaining this DTD. All troubleshooting requirements necessary to develop Operator (-10), Unit (-20), Aviation Unit Maintenance (AVUM), Direct Support (DS) (-30), Aviation Intermediate Maintenance (AVIM), General Support (GS) (-40), and/or depot level (overhaul) TMs are included.

2. APPLICABLE DOCUMENTS.

The applicable documents in section 2 of MIL-STD-40051 apply to this Part.

3. DEFINITIONS.

The definitions in section 3 of MIL-STD-40051 apply to this Part.

4. GENERAL REQUIREMENTS.

4.1 General. Troubleshooting procedures shall be prepared for weapon systems, major equipment, components and applicable support and interface equipment. Troubleshooting procedures and supporting illustrations shall be prepared so that operator/crew and maintenance personnel can perform all required operator through depot level (overhaul) troubleshooting. This information shall be contained in work packages and become a part of a Troubleshooting Information Chapter.

4.2 Development of troubleshooting instructions. Troubleshooting instructions shall cover all items comprising the weapon system/equipment, such as assemblies, subassemblies, components, wiring, junction boxes, and accessories. Troubleshooting procedures shall isolate faults to the part(s) authorized by the RPSTL for repair or replacement at the maintenance level addressed. Tasks shall be presented in the order in which they are performed. Approved Logistics Support Analysis (LSA), service experience, performance data on similar equipment, and all other reliability, availability, and maintainability (RAM) data available shall be used in the preparation of specific troubleshooting procedures. Troubleshooting procedures shall begin with testing, observed problems, a fault symptom or malfunction and shall diagnose to a single fault/failure. Troubleshooting shall refer to specific maintenance or repair tasks to correct the fault. Instructions, where applicable, shall flow from operator level through unit/AVUM, direct support/AVIM, and general support until the fault is isolated. Procedures shall include schematics and illustrations as needed (or shall reference to required schematics, etc.). Troubleshooting data shall be test and fault-isolation oriented. Troubleshooting instructions shall include detailed inspection and troubleshooting information. Instructions shall include a reference to functional descriptions of subsystems being diagnosed to aid the operator/technician. The method used for identifying system equipment test points, including the requirements and methods of determining defects through visual inspection, shall be explained.

4.3 Maintenance level applicability. Requirements contained in this standard are applicable to all maintenance levels unless specifically noted in bold and in parentheses (i.e.,**-34 only**). The labeled requirements shall be applicable to all TMs containing that maintenance level. For example, a**(-20 only)** requirements would only be applicable to the following TMs: -12, -13, -14, -20, -23, and -24.

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4.4 Depot maintenance work requirements. When the contracting activity specifies that a Depot Maintenance Work Requirement (DMWR) shall be prepared to the best commercial practices, the depot requirements contained in this standard shall be used only as a guide; therefore, the conforming modular DTD for troubleshooting procedures cannot be used.

4.5 Standard tables. Various standard tables required are noted throughout the text of this standard in bold and in parentheses (i.e., **(standard table)**). The formats and table heading names of these standard tables shall have no deviations.

4.6 Preparation of digital data for electronic delivery. Technical manual data prepared in work package format and delivered digitally in accordance with this standard shall be SGML tagged and assembled using the modular Assembly DTD and Formatting Output Specification Instance (FOSI). The DTD and FOSI have been developed in accordance with MIL-PRF-28001 and ISO 8879. Refer to MIL-STD-40051 for information on obtaining or accessing the modular DTD and FOSI. SGML tags used in the modular DTD are noted throughout the text of this standard in bracketed, bold characters (i.e., **<tswp>**) as a convenience for the TM author and to ensure that the tags are used correctly when developing a document instance.

4.6.1 Use of DTDs / FOSIs. The modular DTDs referenced in this standard interpret the technical content and structure for the functional requirements contained in this standard and are mandatory for use. The modular FOSIs referenced herein interpret the style and format. As specified by the contracting activity, FOSIs or style sheets may be used to produce final reproducible paper copy for all TMs prepared in accordance with this standard.

4.7 Content structure and format. The examples provided at the rear of this Part are an accurate representation of the content structure and format requirements contained herein and shall be followed to permit the effective use of the modular DTD for Troubleshooting Procedures.

4.8 Style and format. Style and format requirements for the preparation of Department of Army TMs are contained in MIL-STD-40051-1 and are considered mandatory and are intended for compliance. Preferred general style and format requirements for Army TMs shall be provided by the procuring activity.

4.9 Work package development. Technical manual data developed in accordance with this standard shall be divided into individual, stand alone units of information work packages. A work package shall consist of descriptive, operational, maintenance, troubleshooting, support, or parts information for the weapon system or equipment.

4.10 Safety devices and interlocks. Information shall be prepared pertaining to the purpose and location of all safety devices and interlocks in conjunction with the pertinent procedures.

4.11 Electrostatic discharge (ESD) sensitive parts. If the equipment contains ESD sensitive parts, components, or circuits, cautions and ESD labels shall be incorporated into the applicable tasks and procedures to ensure ESD sensitive parts are not damage or degraded during handling or troubleshooting. Refer to MIL-STD-40051-1 for requirements on labeling ESD. Actions which could damage ESD sensitive parts, but which are not directly related to handling or troubleshooting of ESD sensitive parts, shall not be annotated with the ESD acronym, but shall be preceded by a caution statement.

4.12 Nuclear hardness. If the weapon system/equipment has nuclear survivability requirements (for example, over pressure and burst, thermal radiation, electromagnetic pulse, or transient radiation effects on electronics), cautions and Hardness-Critical Processes (HCP) labels shall be incorporated into the applicable

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tasks and procedures to ensure the hardness of the equipment is not degraded during handling or troubleshooting. Refer to MIL-STD-40051-1 for requirements on labeling with HCP. Actions which could degrade hardness, but which are not directly involved in establishing nuclear hardness, shall not be annotated with the acronym, but shall be preceded by a caution statement.

4.13 Selective application and tailoring. MIL-STD-40051 contains some requirements that may not be applicable to the preparation of all technical manuals. Selective application and tailoring of requirements contained in MIL-STD-40051 are the responsibility of the contracting activity and shall be accomplished through the use of Appendix A, Technical Manual Content Selection Matrixes, of MIL-STD-40051. The applicability of some requirements is also designated by one of the following statements: unless specified otherwise by the contracting activity; as/when specified by the contracting activity; or when specified by the procuring activity.

5. DETAILED REQUIREMENTS.

5.1 Preparation of troubleshooting procedures. Troubleshooting procedures shall be prepared as a Troubleshooting Information Chapter **<tim>**. This chapter shall be subdivided into individual work packages that provide all troubleshooting information to enable a technician to locate a malfunction in the weapon system/equipment to the assembly, subassembly, component or piece part. Troubleshooting tasks and depth of coverage shall be developed in accordance with the Logistics Support Analysis (LSA)/Maintenance Allocation Chart (MAC) or Maintenance Plan and the Source, Maintenance, and Recoverability (SMR) codes developed for the weapon system/equipment. Other factors to be considered in the development of troubleshooting procedures include, but are not limited to, the following.

a. Technical experience (target audience).

b. User environment.

c. System quick-turnaround requirements.

d. Test equipment requirements and availability.

e. Automated versus manual testing.

f. Replaceable component and part reliability.

g. Ease of testing.

h. Test access time.

i. Test time.

5.2 Troubleshooting procedures content. The procedures shall contain all essential and pertinent information that would be included in any other form of maintenance procedure. This includes warnings, cautions, notes, power turn-on procedures, precheckout procedures, reference diagrams, and initial switch settings. In addition to external causes for malfunctions, troubleshooting should also identify symptoms resulting from failure of every spare and repair part authorized for replacement at user level. Troubleshooting procedures shall be prepared assuming one malfunction at a time is being corrected. The operator/technician shall be instructed to perform any applicable self-tests, alignments, and inspections before beginning any other troubleshooting procedures. As applicable, an operational check shall be specified to be performed

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after the fault is corrected to ensure correct operation of the system. Troubleshooting procedural instructions shall be prepared following these general requirements.

a. Assure that procedures are prepared for the operator/technician at an appropriate level of knowledge of the systems operation and that systems relationship to peripheral assemblies is explained.

b. When all probable faults have been determined and described, prepare a malfunction/symptom index work package using the exact description of the fault or symptom as was used in the troubleshooting procedures. Group symptoms to common system areas both in the malfunction/symptom index and in the troubleshooting procedures. For example, if a system has a data link, communications, radar, display, and tracking systems, the symptoms would be grouped into each related area. All fault symptoms of a communications nature would fall into the communications group. The symptoms may be further divided into functions within the communications group that would be common. The same would be done for radar, data link, display, and tracking systems.

c. Develop the troubleshooting procedures based on tests, measurements, and decisions that must be made in order to reach the final outcome of isolating the fault to a replaceable component related to the symptom. All of the most probable faults and causes shall be considered. The troubleshooting procedures shall provide the operator/technician with the means of isolating system failures to the faulty line replaceable item/shop replaceable unit

d. List any self-tests that are associated with the system. Self- test schemes shall be described as the prime troubleshooting tool, with manual troubleshooting prepared to supplement the instructions where the self-test leaves off or fails to locate the malfunction. Build the procedure using system self-tests before using external test equipment. Include any information that will aid the operator/technician, such as waveforms; resistance data; fluid pressures; voltage levels; references to test diagrams, functional diagrams, text, etc.; and alignment procedures, checkout procedures, or other scheduled maintenance procedures. Connector numbers, pin designations, etc., shall be identified.

e. When applicable, the troubleshooting procedures shall take into account the means of fault discovery, such as during operation; during routine maintenance procedures; using nonautomated, semiautomated, or automated test equipment; or using Built-In Test Equipment (BITE).

f. Ensure that the troubleshooting procedures contain all the required references to other troubleshooting aids.

5.3 Testing and troubleshooting. The environment in which testing and troubleshooting procedures will be performed and the volume (bulk) of information required shall be considered. Testing and troubleshooting for a given system shall be presented within a system troubleshooting procedures work package or in a separate assembly troubleshooting procedures work package.

5.3.1 Testing and troubleshooting procedures using test equipment. As applicable, instructions for the use of nonautomated, semiautomated, automated, or BITE in testing and troubleshooting procedures shall be prepared. Procedures for the use of any of this equipment shall be established on the basis of a system test concept. Specific instructions for using any test equipment shall be included in the troubleshooting procedures. Procedures shall cover automated or semiautomated testing performed either to begin a troubleshooting process or to complete a fault isolation step. Malfunction symptoms shall identify specific ports of entry into the testing and troubleshooting cycle. Instructions for repeating time-consuming end-to-end tests shall be avoided. The following procedural content concerning the use of test equipment shall be considered.

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a. An identification of the system under test.

b. A concise explanation of the testing and troubleshooting format.

c. Descriptive text identifying components to be installed and an identification of the access to each. The text shall be accompanied by illustrations that show location for each component, accessory, connector, or junction box in the system under test. The text shall also identify every test connector or other test point to be used in the test.

d. A complete list of test options shall be stipulated by the troubleshooting procedure.

e. Test setup procedures and post-test teardown procedures.

f. Complete step-by-step test and troubleshooting procedures, including instructions required for use and application of installed on-line testing equipment. Procedures shall take into account controls, test point accessibility, indicators, displays, and the feasibility of using BITE or automated test equipment where available.

g. Test procedures (e.g., system turn on, identification of time required to run and complete the system test, and an indication of any possible mid-test interruptions or stoppages and how to respond to them).

h. Backup diagrams showing all test points, input and output signals, logic charts, schematics, signal flow diagrams, tables, and other illustrations as required for comprehensible understanding of the procedures.

5.4 Troubleshooting Information Chapters **<tim>**. Troubleshooting Information Chapters **<tim>** shall be developed for the weapon system/equipment. Chapters shall consist of a title page (refer to MIL-STD-40051-1) and the troubleshooting work packages described in paragraph 5.5. Each Chapter shall be titled "Troubleshooting Procedures" followed by end item/system, subsystem, component, assembly, subassembly name, as applicable (e.g., Troubleshooting Procedures for the M101 Howitzer). When the TM contains more than one level of maintenance, the title of each chapter shall also be preceded by the maintenance level (e.g., Unit Troubleshooting Procedures for the Landing Control Central AN/TSQ-71B).

5.5 Types of troubleshooting work packages. Chapters shall include the following work packages, as applicable: introduction, malfunction/symptom index and troubleshooting procedures.

5.5.1 Introduction work package **<tsintrowp>**. This work package shall consist of descriptive data which explains how to use **<howtouse>** the malfunction/symptom index and troubleshooting procedures work packages. Examples shall be included with the descriptions. This work package may contain any other general information needed to supplement the troubleshooting procedures. (Refer to figure 1.) The following qualifying statements shall be included, as applicable, in the Introduction work package.

a. This table lists all the common malfunctions that you may find with your equipment. Perform the tests, inspections, and corrective actions in the order they appear in the table.

b. This table cannot list all the malfunctions that may occur, all the tests and inspections needed to find the fault, or all the corrective actions needed to correct the fault. If the equipment malfunction is not listed or actions listed do not correct the fault, notify your supervisor.

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5.5.2 Troubleshooting reference index work package **<tsindxwp>**. This work package shall consist of either a system/subsystem index **<tsindx>** or a malfunction/symptom index **<tsindx>**. When applicable, one troubleshooting reference index work package shall be prepared for all troubleshooting chapters. As required, additional malfunction/symptom index work packages may be included. (Refer to figure 2.)

5.5.2.1 Malfunction/symptom index **<tsindx>**. This index shall include the following data.

a. List all fault symptoms or known malfunctions in alphabetical order by malfunction/symptom or by built in test code.

b. For complex systems, list symptoms by subsystem categories, if necessary, and use codes that help identify specific items. (Subsystem categories shall be listed in alphabetical order or by code.)

c. Catalog malfunctions/symptoms by method of detection, if this aids usability.

d. Reference the appropriate troubleshooting procedures work package. Fault symptom descriptions shall be standardized between malfunction/symptom index work packages **<tsindxwp>** and troubleshooting procedures work packages **<tswp>**.

5.5.2.2 System/subsystem index **<tsindx>**. This index shall consist of a list of work packages relating to specific systems, subsystems, assemblies and components.

5.5.3 Troubleshooting testing work package **<testmodulewp>**. As applicable, when troubleshooting procedures reference automated or semiautomated test equipment that is necessary to be used either to begin the troubleshooting process or to complete a fault isolation step, a troubleshooting testing work package may be prepared. The troubleshooting testing work package shall consist of a scope of task, an initial setup, and the test procedures. A troubleshooting testing work package may include test set hookup and disconnect procedures, index of test set message words, a reference index of test set or BIT/BITE fault codes and related actions, and further testing procedures related to the message words and fault codes. When applicable, the information contained in this work package may be included in the troubleshooting procedures work package. (Refer to 5.5.4).

5.5.4 Troubleshooting procedures work packages **<tswp>**. The troubleshooting procedures work packages shall consist of: scope of task (THIS WORK PACKAGE COVERS:), initial setup, and troubleshooting procedures. (Refer to figure 3.) Work packages will relate either to a specific symptom or to a system, assembly, or component. Work packages related to a system of some complexity may contain more than one set of troubleshooting procedures directed to specific subsystems. Work packages may contain supporting technical data as detailed in paragraph 5.3.1h.

5.5.4.1 Scope of task **<wpsum>**. The scope of task (THIS WORK PACKAGE COVERS:) shall be required for all troubleshooting procedures work packages. The scope of task shall list all the tasks that must be performed to complete the procedures. The scope of task allows a technician to select everything needed at the start of the job without checking through the task list. Presentation shall be consistent throughout all work packages. (Refer to figure 3.)

5.5.4.2 Initial setup information **<wpinfo>**. Initial setup information shall be included in each work package and shall immediately follow the scope of task. (Refer to figure 3.) It provides the operator/maintenance technician with general information, equipment, parts, material, and authorized personnel required to perform and complete all the troubleshooting tasks included in the work package. Setup information requirements are described below:

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a. Maintenance level **<maintlvl>**. The level of maintenance authorized to perform the maintenance contained in the work package (in accordance with the approved MAC) shall be stated. This shall be included for operator or crew, unit, AVUM, direct support, general support, AVIM, and depot levels of maintenance, as applicable. For example,

Maintenance Level
Unit

b. Applicable configurations **<appconfig>**. When the work package does not apply to all configurations of the weapon system/equipment, the applicable configurations **<name>** covered by the work package shall be listed. Omit this requirement if the same tasks/procedures apply to all configurations. (If certain configurations require different tasks/procedures, separate work packages shall be prepared.) For example,

Applicable Configurations
Serial Numbers 12345 through 12399

c. Test equipment **<testeqp>**. All test equipment required to perform the procedure shall be listed by name **<name>** and part **<partno>** or model number **<modelno>** designation if this information is not contained in an overall list elsewhere in the TM. If such a list exists, refer to it by name **<name>**, item number, and work package number **<simref>** instead of repeating the information throughout the TM. For example,

Test Equipment
Multimeter (Item 4, WP 0108 00)
Oscilloscope (13057)

d. Tools and special tools **<tools>**. The tool kit (box) assigned to the mechanic (on a 1-per-mechanic-by-MOS basis) to be used in maintenance of a particular equipment shall be listed by name **<name>**, tool kit number (**<partno>** or **<nsn>**), supply catalog (SC) **<sc>**, or TM number **<tmno>** if this information is not included in the Tool Identification List work package contained in the TM. If such a list exists, refer to it by name **<name>**, item number and work package number **<simref>** instead of repeating the information throughout the TM. No tool in the kit shall be further identified. Other tools required for performance of all tasks for the maintenance levels covered in the work package shall also be identified in the initial setup and shall be referenced to the Tool Identification List work package. "Other tools" includes tools which are part of/components of shop sets authorized to sections/teams; tools authorized by RPSTL and CTA-50-970; special and fabricated tools; and items of TMDE. For example,

Tools and Special Tools
Fixed Open End Wrench Set (Item 47, WP 0110 00)
Screw Threading Set (SC number)
Vehicle Tool Kit (407425)

e. Materials/parts **<mtrlpart>**. All expendable items and support materials shall be listed **<name>**. The item number and supporting information work package **<simref>** which lists these items shall be given. Mandatory replacement parts shall be listed by name **<name>** (and part number **<partno>**, if any). The number, quantity **<qty>**, or size necessary to complete the task shall be

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listed, when applicable. When a mandatory replacement parts work package exists, it shall be referred to in lieu of the part number. For example,

Materials/Parts
Grease (Item 5, WP 0112 00)
Wiping Rags (Item 38, WP 0112 00)
Range Lock (P/N 8675309)
Range Lock Flange Kit (P/N 8675310)

f. Personnel required **<persnreq>**. Personnel **<name>** and the number of personnel **<qty>** shall be identified if the task requires more than one. The Military Occupational Specialty (MOS) designation **<nameid>** is not necessary, but it may be included. For example,

Personnel Required
Artillery Mechanic 68M10 (1)
Artillery Mechanic 66J30 (1)

g. References **<ref>**. TMs, other work packages, and other sources (**<extref>**/**<xref>**) that are needed to complete the maintenance tasks shall be listed here. Only references not listed in equipment conditions shall be listed. For example,

References
TM 9-1015-252-20&P
WP 0100 00

h. Equipment conditions **<eqpconds>**. Any special equipment conditions required before the procedure can be started shall be listed here and cross-referenced to the appropriate source (**<extref>** or **<xref>**) for setting up the condition **<condition>**. For example,

Equipment Condition
Firing mechanism removed (WP 0100 00)

i Special environmental conditions **<specenv>**. Any special environmental conditions (such as ventilation, lighting, or temperature) **<condition>** that are required shall be listed here. The reason **<reason>** that such conditions are needed shall be explained. For example,

Special Environmental Condition
Darkened area required for testing lights.

j. Drawings required **<dwgreq>**. All drawings (which are not included in the work package) required to complete the maintenance tasks shall be listed here. Drawings shall be listed by title **<dwgname>** and drawing number **<dwgno>**. For example,

Drawings Required
Power Supply Schematic (132E470092)

5.5.4.3 End of task identifier. The words **END OF TASK** shall be placed below the last item (i.e., text, illustration, etc.) in any work package containing procedures.

5.5.4.4 Troubleshooting procedures. The selection of a troubleshooting type shall be based on the type of equipment or assembly/subassembly being addressed, the target audience description, and the maintenance

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level of the operator/technician. Once selected, the troubleshooting type shall be prepared in accordance with the requirements specified by this document. Troubleshooting procedures work packages shall be prepared in one of the following types:

5.5.4.4.1 Tabular. Tabular troubleshooting procedures shall be prepared in one of two standard tables.

a. When the troubleshooting procedure is based on a test procedure**<opertest>**, a standard troubleshooting table in the format of figure 4 **(standard table)** shall be prepared. The first column of this table shall have an item number and a procedure**<test>** that leads to an indication **<indication>**. The second column shall list the normal indication. The third column of the table shall list the corrective actions **<action>** to follow or reference appropriate maintenance work packages or the applicable TM.

b. When the troubleshooting procedure is based on a known malfunction**<known-malfunc>** that will be diagnosed and corrected through a testing procedure, a standard troubleshooting table**<faulttest>** in the format of figure 4 **(standard table)** shall be prepared. This table shall list item numbers and the malfunctions **<malfunc>** in the first column. The second column shall list the test or inspections **<testing>** (what to check for concerning the malfunction listed). The last column shall give the corrective action **<action>** in detail or shall reference the corrective action.

c. When a known malfunction**<known-malfunc>** exists, a standard troubleshooting table **<faultpath>** may be prepared listing the probable causes **<suspect-fault>** of each malfunction **<malfunc>** with a related corrective action for each cause. The methodology is to diagnose a malfunction's cause by determining what corrective action clears up the malfunction. The first column shall list the item number and malfunction **<malfunc>**. The second column shall list each probable cause**<suspect-fault>** with its corrective action **<action>** or a reference to the action aligned in the third column. (Refer to figure 4.)

5.5.4.4.2 Functional flow tree **<functnl-flow>**. This fault isolation technique shall be composed of a series of operation and decision routines. These troubleshooting procedures shall be presented graphically in functional order and may be supported by functional diagrams. (Refer to figure 5.) Each troubleshooting routine shall be prepared using a standardized set of graphic symbols. The purpose, meaning, and use of the symbols are as follows. Preparation of a functional-flow work package shall be based upon a systematic tracking of faults being tested; each test unit will either isolate or clear the fault(s).

NOTE

If logic tree was prepared as a graphic, treat logic tree as graphic in the SGML document instance.

a. An oval shall be used to provide directive information.

b. A square shall be used to provide instructions for a step to be performed. The instruction shall be prepared in an abbreviated format. For example, CODING SWITCH OFF, shall be used in lieu of "Set the coding switch to the CODING OFF position."

c. A diamond shall be used to propose a question to be answered with "yes" or "no." Arrows shall direct the operator/technician to further instructions depending upon whether a "yes" or "no" was the answer to the previous question.

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d. If the logic tree continues to another page or figure, sufficient information shall be placed on the page or figure of origin to indicate the location of the continuation of the diagram. For example, on the left side of the page, a symbol and explanation shall be placed explaining the origin. On the right side of the page, a symbol and explanation shall be placed explaining the destination. (Refer to figure 5.)

5.5.4.4.3 Narrative. Narrative troubleshooting procedures may be prepared in one of the following formats.

a. The first method **<passfailck>** shall include an item number followed by item or signal names, locations, codes, appropriate data, expected conditions, displays, responses, remarks, and pass/fail instructions, as applicable. Each topic shall be followed by its description, further information, or instructions, etc. (Refer to figure 6.)

b. The second method shall list the malfunction or fault symptom as the work package or task title. The text shall include an item number followed by instructions, checks, tests, and inspections**<test>**, as applicable. Each procedure shall be followed by a question**<query>** concerning the results of the procedure just performed. The operator/technician shall then be given "yes" or "no"**<answer>** answers with corrective actions to follow for the applicable answer. (Refer to figure 7.)

5.5.5 Illustrations. Work packages in the troubleshooting information chapter shall be supported by illustrations to ensure all information needed to perform troubleshooting is complete and comprehensible. In addition to the figures previously referenced, refer to figures 8 and 9 for examples of troubleshooting test setup and test cable interconnection diagrams.

6. NOTES.

The notes in section 6 of MIL-STD-40051 apply to this Part.

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TM 3-6665-339-10

INTRODUCTION **0401 00**

MALFUNCTION/SYMPTOM INDEX

The malfunction/symptom index (WP 0402 00) is a quick reference index for finding troubleshooting procedures. Associated with each symptom name is a work package sequence number representing the starting point in a troubleshooting sequence. Should any one symptom require more than one troubleshooting sequence to arrive at the most likely area of investigation, the additional starting point numbers are presented.

As the troubleshooting activity progresses through to the conclusion of a particular sequence, a reference is made to the next logical troubleshooting sequence by work package sequence number or by referring to the malfunction/ symptom index to locate the next failure symptom work package. This type of activity continues until successful fault isolation is achieved.

TROUBLESHOOTING PROCEDURES

The troubleshooting work packages contain tables listing the malfunctions, tests or inspections, and corrective action required to return the vehicle to normal operation. Perform the steps in the order they appear in the tables.

Each work package is headed by an initial setup. This setup outlines what is needed as well as certain conditions which must be met before starting the task. DON'T START A TASK UNTIL:

You understand the task.

You understand what you are to do.

You understand what is needed to do the work.

You have the things you need.

This manual cannot list all malfunctions that may occur, or all tests or inspections and corrective actions. If a malfunction is not listed or is not corrected by listed corrective actions, notify unit maintenance.

GENERAL INFORMATION

If any circuit breaker pops out after it has been pushed in, notify unit maintenance.

When instructions to lubricate, grease, or oil appear in the troubleshooting work package, refer to LO 9-6665-376-12.

0401 00-1

FIGURE 1. Example of an introduction work package

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TM 3-6665-339-10

MALFUNCTION/SYMPTOM INDEX **1102 00**

Malfunction/Symptom	Troubleshooting Procedure
LEFT DATA WORD DISPLAY WARNINGS	
1. BRAKE comes on	WP 1203 00
2. AIR PRESSURE does not come on - brake pressure is low	WP 1203 00
3. COOLANT comes on and ENG. COOLING gage indicates excessive coolant temperature	WP 1203 00
4. ENG. OIL PRESS comes on and engine oil pressure is low	WP 1203 00
5. HYDRAULIC OIL comes on	WP 1203 00
6. HYDRAULIC OIL does not come on - reservoir level is low	WP 1203 00
DRIVER'S LEFT INSTRUMENT PANEL INDICATOR LIGHTS	
1. STANDBY HYDRAULIC indicator light does not come on with emergency hydraulic system on	WP 1304 00
2. TRUCK TURN SIGN light does not come on when push lever on steering column switch is pulled upward or pushed downward - vehicle flasher lights operate normally	WP 1304 00
3. UPPER BEAM indicator light does not come on - high beam headlights operate normally	WP 1304 00
RIGHT DATA WORD DISPLAY WARNINGS	
1. BATT. CHARGER does not come on when MAIN SWITCH is set to position 1	WP 1405 00
2. Flasher warning lights do not work	WP 1405 00
3. FUEL does not come on and fuel gage does not indicate low fuel level - fuel level is low	WP 1405 00
4. HYDR PRESS comes on	WP 1405 00
5. BRAKE LINING does not come on - brake lines are leaking	WP 1405 00
COMMANDER'S CONTROLS	
1. NORMAL MODE or SIL WATCH does not appear on commander's data word display when NBC CP SYS. switch is set to either NORMAL MODE or SIL WATCH	WP 1506 00
2. PRESSURE GAGE does not read properly when NBC collective protection system is in NORMAL MODE or SIL WATCH	WP 1506 00
3. DUST FILTER appears on commander's data word display when NBC CP SYS. switch is set to either NORMAL MODE or SIL WATCH	WP 1506 00

1102 00-1

FIGURE 2. Example of a troubleshooting reference index work package

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TM 3-6665-339-10

TROUBLESHOOTING PROCEDURES FOR NBCRS FOX M93A1 **1837 00**

THIS WORK PACKAGE COVERS:
Left Data Word Display Warnings

INITIAL SETUP:

Maintenance Level
Operator

Materials/Parts
Oil (Item 20, WP 0270 00)

LEFT DATA WORD DISPLAY WARNINGS

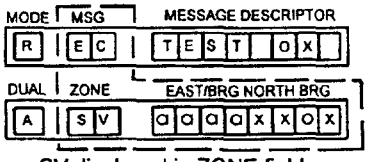
Table 1. Troubleshooting Procedures.

MALFUNCTION	TEST OR INSPECTION	CORRECTIVE ACTION
1. BRAKE IN LEFT DATA WORD DISPLAY COMES ON	1. Shut down engine (see WP 0087 00).	
	2. Check oil level in brake system master cylinder reservoir (see LO 9-6665-376-12).	1. If brake system master cylinder reservoir level is low, add oil (see LO 9-6665-376-12) and continue normal operation.
		2. If brake system master cylinder level is above minimum level mark, go to step 3.
	3. Check oil level in hydraulic brake system expansion tank (see LO 9-6665-376-12).	1. If oil level in hydraulic brake expansion tank is low, add oil (see LO 9-6665-376-12) and continue normal operation.
		2. If oil level in hydraulic brake expansion tank is above minimum level mark, notify unit maintenance.
2. AIR PRESSURE IN LEFT DATA WORD DISPLAY DOES NOT COME ON - BRAKE PRESSURE IS LOW	1. Shut down engine (see WP 0087 00)	

1837 00-1

FIGURE 3. Example of a troubleshooting work package, scope of task, initial setup and troubleshooting procedure.

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ITEM/PROCEDURE	NORMAL INDICATION	CORRECTIVE ACTION
3. On URO, press RCVD key.	URO displays the following:  SV displayed in ZONE field	1. If URO MESSAGE DESCRIPTOR displays any of the following, replace BUU: FA FAULT BAR FAIL CSC LIM. 2. If URO MESSAGE DESCRIPTOR displays SDU ALRM, perform the

TABULAR TROUBLESHOOTING PROCEDURE BASED ON TEST PROCEDURES

MALFUNCTION	TEST OR INSPECTION	CORRECTIVE ACTION
16. WATER PUMP FAILS TO ROTATE	1. Check for broken or cracked motor, damaged shaft threads, and bent shaft.	1. Replace motor if defective (WP 1520 00 or WP 1521 00). Remove the motor, using illustrated instructions below. 2. Tag and disconnect electrical leads as necessary.

TABULAR TROUBLESHOOTING PROCEDURE BASED ON KNOWN MALFUNCTIONS

MALFUNCTION	PROBABLE CAUSE	CORRECTIVE ACTION
1. With generator power applied to the Shelter, MAIN POWER circuit breaker is off, Phase Sequence light is out.	1. Generator Power 2. Phase Sequence indicator 3. Secondary Filter (FL2) 4. Surge Protector 5. Power Input Filter (FL1) 6. Power Input Cable	1. Check generator output power. Adjust as required. 2. Check connections at generator and shelter. **WARNING** Use extreme caution when making power on checks 3. Check secondary filter (FL2) voltage: a. Loosen screws and open cover of FL2 **Note** Use extreme caution when making power on checks Measure LOAD voltage for FL2 terminals

TABULAR TROUBLESHOOTING PROCEDURE BASED ON KNOWN MALFUNCTION THAT EXISTS

FIGURE 4. Example of a tabular troubleshooting procedure

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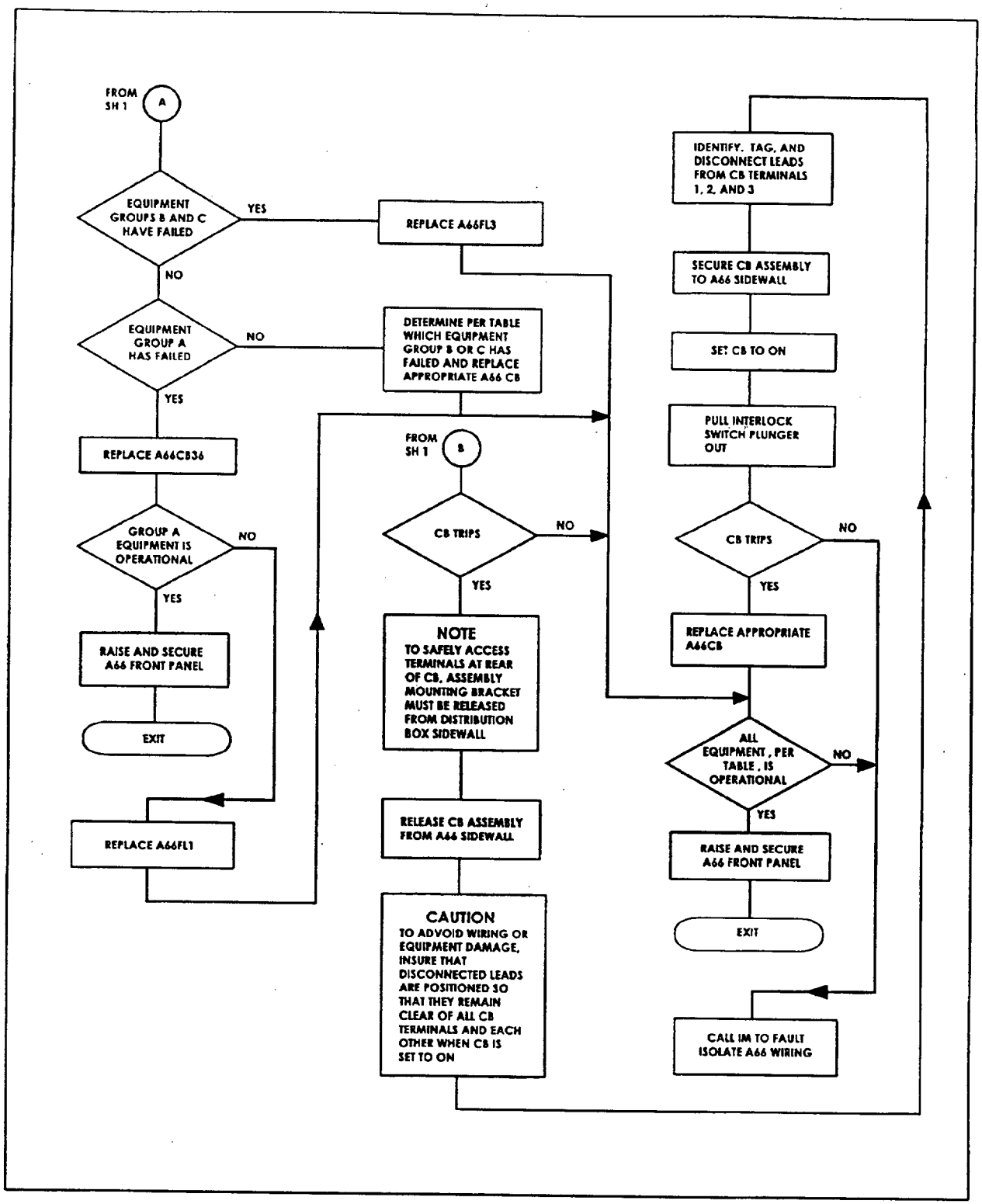


FIGURE 5. Example of a functional flow logic tree troubleshooting procedure.

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TM 55-1520-238-T-3

AIR DATA SYSTEM TROUBLESHOOTING PROCEDURES - Continued 0323 00

RANGE DATA CANNOT BE DISPLAYED - *Continued*

23. **SIGNAL NAME:** DASEC STATUS WORD DC ANALOG OUTPUT BIT
MEMORY LOCATION: 002150
MEMORY DATA BIT(S): 15 (BINARY)
CONDITION: (None)
SIGNAL FUNCTION: Indicates status of DC analog circuits.
REMARKS: From DASEC to FCC.
PASS: If second digit displayed on HOD is 3 or 7, go to step 24.
FAIL: Location of fault: replace DASEC (TM 1-1520-238-23 Series).

24. **SIGNAL NAME:** DASEC STATUS WORD AD/DA BIT
MEMORY LOCATION: 002150
MEMORY DATA BIT(S): 13 (BINARY)
CONDITION: (None)
SIGNAL FUNCTION: Indicates status of analog-to-digital and digital-to-analog circuits.
REMARKS: From DASEC to FCC.
PASS: If third digit displayed on HOD is 1, 3, 5, or 7, go to step 25.
FAIL: Location of fault: replace DASEC (TM 1-1520-238-23 Series).

25. **SIGNAL NAME:** DASEC STATUS WORD FD/LS TEST
MEMORY LOCATION: 002150
MEMORY DATA BIT(S): 12 (BINARY)
CONDITION: (None)
SIGNAL FUNCTION: Indicates FD/LS ground test is being run.
REMARKS: From DASEC to FCC.
PASS: If third digit displayed on HOD is 1 or 5, go to step 26.
FAIL: Location of fault: replace DASEC (TM 1-1520-238-23 Series).

26. **SIGNAL NAME:** DASEC STATUS WORD ASE BIT
MEMORY LOCATION: 002150
MEMORY DATA BIT(S): 11 (BINARY)
CONDITION: (None)
SIGNAL FUNCTION: Indicates last FD/LS test ASE bit status.
REMARKS: From DASEC to FCC.
PASS: If third digit displayed on HOD is 1, go to step 27.
FAIL: Location of fault: replace DASEC (TM 1-1520-238-23 Series).

END OF TASK

0323 00-3

FIGURE 6. Example 1 of a narrative troubleshooting procedure

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TM 55-1520-238-T-6

ELECTRICAL SYSTEM TROUBLESHOOTING - Continued **2732 00**

28 VDC - IS NOT PRESENT AT P769-A AND P769-J - Continued

8. Detach wire at CB-1. Check for short between CB2-1 and ground.

   Does short exist?
   YES Go to step 9.
   NO Go to step 10.

9. Remove bus bar between CB1 and CB2. Check for short between CB1 and ground.

   Does short exist?
   YES Replace shorted MUX L PYL INBD circuit breaker (CB1) (TM 55-1520-238-23 series).
   NO Replace shorted MUX L PYL OUTBD circuit breaker (CB2) (TM 55-1520-238-23 series).

10. Attach CB2 and CB1 wire. Detach wire at CB5-1. Check for short between (A77) J4-A and ground.

    Does short exist?
    YES Repair shorted wire between CB1-1 and CB5-1. Go to WP 3641 00.
    NO Go to step 11.

11. Detach wire at CB6-1. Check for short between CB5-1 and ground.

    Does short exist?
    YES Go to step 12.
    NO Go to step 13.

12. Remove bus bar between CB5 and CB6. Check for short between CB5 and ground.

    Does short exist?
    YES Replace MUX R PYL INBD circuit breaker (CB5) (TM 55-1520-238-23 series).
    NO Replace MUX R PYL OUTBD circuit breaker (CB6) (TM 55-1520-238-23 series).

13. Attach CB5 and CB6 wire. Detach wire at CB10-1. Check for short between (A77) J4-A and ground.

    Does short exist?
    YES Go to step 14.
    NO Repair shorted wire between CB10-1 and CB6-1. Go to WP 3641 00.

14. Remove bus bar between CB5 and CB6. Check for short between CB5 and ground.

    Does short exist?
    YES Replace MUX FAB R circuit breaker (CB9) (TM 55-1520-238-23 series).
    NO Replace MUX FAB L circuit breaker (CB10) (TM 55-1520-238-23 series).

2732 00-3

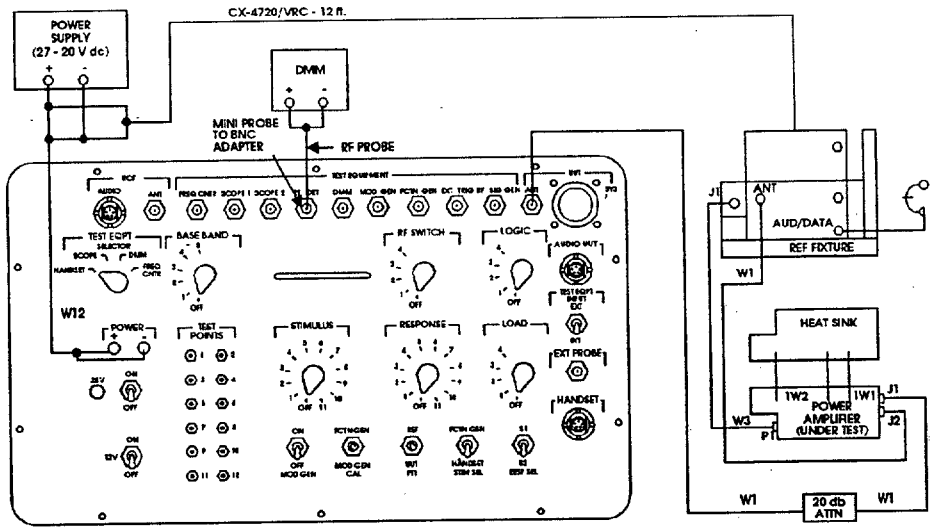
FIGURE 7. Example 2 of a narrative troubleshooting procedure

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EQUIPMENT PRESETS (RF OUTPUT (dBm))

WARNING

High voltage and high RF energy is present in the power amplifier and the test setup. Use caution to avoid personal injury.



REF FIXTURE:

CB1:	ON
FCTN:	SQ ON
MODE:	SC
DATA:	OFF
CHAN:	Do not change from operational check.

DMM:

Set for dBm, 50 Ω REF.

POWER AMPLIFIER:

Disassemble to gain access to test points. Do not disconnect cables.

FIGURE 8. Example of a troubleshooting test setup diagram

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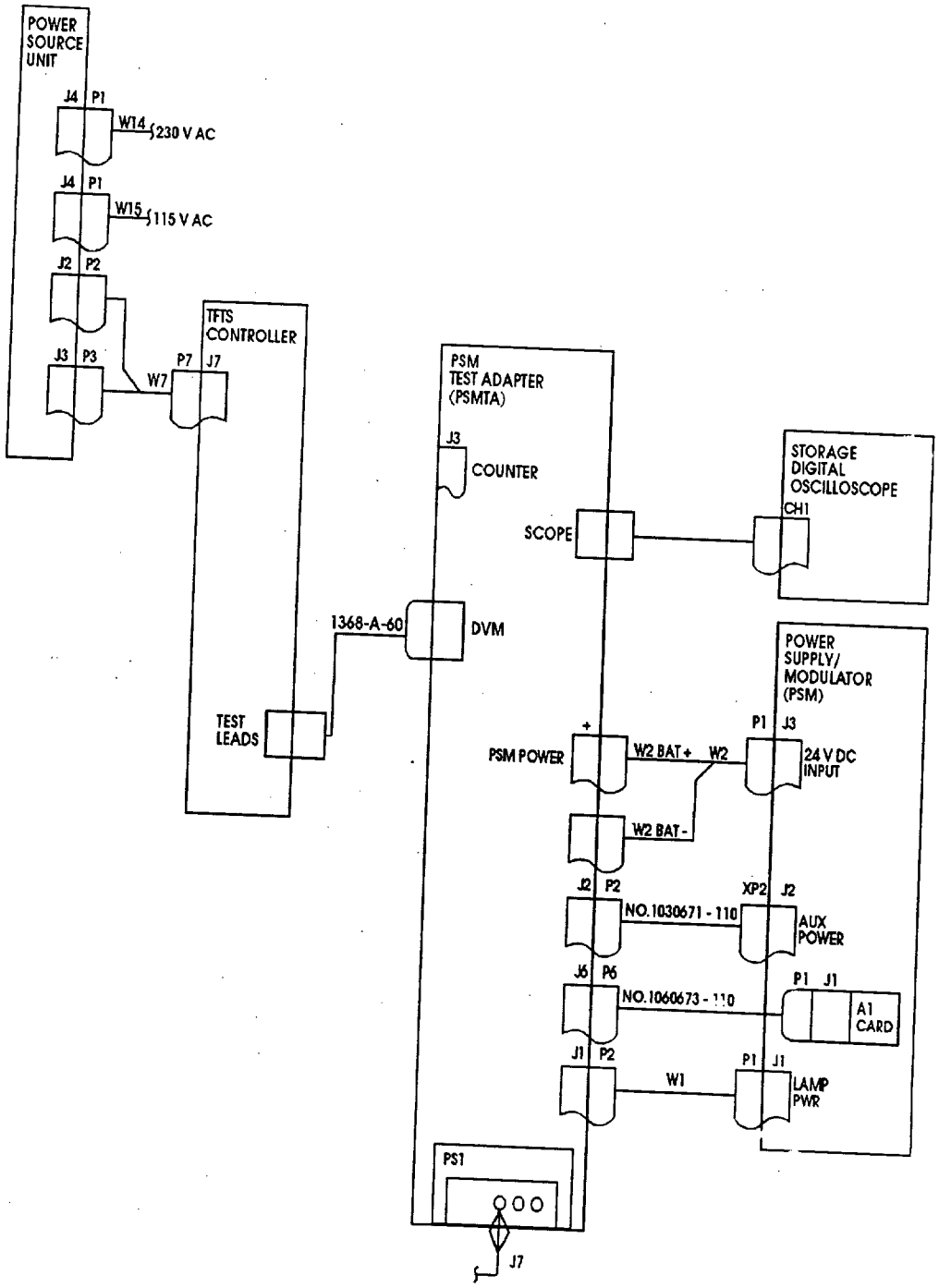


Figure 21. Power Supply/Modulator Unit Test Cable Interconnection Diagram.

FIGURE 9. Example of a test cable interconnection diagram

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NOT MEASUREMENT SENSITIVE

MIL-STD-40051-5

DEPARTMENT OF DEFENSE STANDARD PRACTICE

TECHNICAL MANUALS

MAINTENANCE INSTRUCTIONS

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1. SCOPE.

1.1 Scope. This standard establishes the technical content requirements for the preparation of maintenance instructions for weapon systems and equipment Technical Manuals (TMs) and Depot Maintenance Work Requirements (DMWRs). These requirements are applicable for both paper and digital page-oriented TMs. Electronic delivery of maintenance instructions for TMs is accomplished through the use of the Maintenance Instruction modular Document Type Definition (DTD). The DTD is available in digital format. Refer to MIL-STD-40051 for information on obtaining this DTD. All maintenance requirements necessary to develop Operator (-10), Unit (-20), Aviation Unit Maintenance (AVUM), Direct Support (DS) (-30), Aviation Intermediate Maintenance (AVIM), General Support (GS) (-40), and/or depot level (overhaul) TMs are included.

2. APPLICABLE DOCUMENTS.

The applicable documents in section 2 of MIL-STD-40051 apply to this Part.

3. DEFINITIONS.

The definitions in section 3 of MIL-STD-40051 apply to this Part.

4. GENERAL REQUIREMENTS.

4.1 General. Maintenance instructions shall be prepared for weapon systems, major equipment, components and applicable support and interface equipment. Maintenance procedures and supporting illustrations shall be prepared so that maintenance personnel can perform all required operator through depot level (overhaul) maintenance. This information shall be contained in work packages and become a part of a Maintenance Instructions Information Chapter.

4.2 Development of maintenance instructions Maintenance instructions shall be prepared for all items comprising the weapon system/equipment, such as assemblies, subassemblies, components, wiring, junction boxes, and accessories. Tasks shall be presented in the order in which they are performed. Sound engineering principles and techniques, approved Logistics Support Analysis (LSA), service experience, performance data on similar equipment, and all other reliability, availability, maintainability (RAM) data available shall be used in the preparation of specific maintenance instructions.

4.3 Maintenance level applicability Requirements contained in this standard are applicable to all maintenance levels unless specifically noted in bold and in parentheses (i.e., **-34 only**). The labeled requirements shall be applicable to all TMs containing that maintenance level. For example, **(-20 only)** requirements would only be applicable to the following TMs: -12, -13, -14, -20, -23, and -24. Refer to the TM content selection matrixes in Appendix A of MIL-STD-40051.

4.4 Depot maintenance work requirements When the contracting activity specifies that a Depot Maintenance Work Requirement (DMWR) shall be prepared to the best commercial practices, the depot requirements contained in this standard shall be used only as a guide, therefore, the conforming modular DTD for maintenance instructions cannot be used.

4.5 Standard tables. Various standard tables required are noted throughout the text of this standard in bold and in parentheses (i.e., **(standard table)**). The formats and table heading names of these standard tables shall have no deviations.

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4.6 Preparation of digital data for electronic delivery Technical manual data prepared and delivered digitally in accordance with this standard shall be Standard Generalized Markup Language (SGML) tagged using the modular Document Type Definition (DTD) for Maintenance Instructions and the Formatting Output Specification Instance (FOSI). The DTD and FOSI have been developed in accordance with MIL-PRF-28001 and ISO 8879. Refer to MIL-STD-40051 for information on obtaining or accessing this modular DTD and FOSI. SGML tags used in the modular DTD are noted throughout the text of this standard in bracketed, bold characters (i.e., **<maintwp>**) as a convenience for the TM author and to ensure that the tags are used correctly when developing a document instance.

4.6.1 Use of the DTDs / FOSIs. The modular DTDs referenced in this standard interpret the technical content and structure for the functional requirements contained in this standard and are mandatory for use. The modular FOSIs referenced herein interprets the style and format. As specified by the contracting activity, FOSIs or style sheets may be used to produce final reproducible paper copy for all TMs prepared in accordance with this standard.

4.7 Content structure and format. The examples provided at the rear of this Part are an accurate representation of the content structure and format requirements contained herein and shall be followed to permit the effective use of the modular DTD for Maintenance Instructions.

4.8 Style and format. Style and format requirements for the preparation of Department of Army TMs are contained in MIL-STD-40051-1; they are considered mandatory and are intended for compliance. Preferred general style and format requirements for Army TMs shall be provided by the procuring activity.

4.9 Work package development Technical manual data developed in accordance with this standard shall be divided into individual, stand alone units of information work packages. A work package shall consist of descriptive, operational, maintenance, troubleshooting, support, or parts information for the weapon system or equipment.

4.10 Safety devices and interlocks. Information shall be prepared pertaining to the purpose and location of all safety devices and interlocks in conjunction with the pertinent procedures.

4.11 Electrostatic discharge (ESD) sensitive parts If the equipment contains ESD sensitive parts, components, or circuits, cautions and ESD labels shall be incorporated into the applicable tasks and procedures to ensure ESD sensitive parts are not damaged or degraded during handling or operation. Refer to MIL-STD-40051-1 for requirements on labeling ESD. Actions which could damage ESD sensitive parts, but which are not directly related to handling or operation of ESD sensitive parts, shall not be annotated with the ESD acronym, but shall be preceded by a caution statement.

4.12 Nuclear hardness. If the weapon system/equipment has nuclear survivability requirements (for example, over pressure and burst, thermal radiation, electromagnetic pulse, or transient radiation effects on electronics), cautions and Hardness-Critical Processes (HCP) labels shall be incorporated into the applicable tasks and procedures to ensure the hardness of the equipment is not degraded during handling or operation. Refer to MIL-STD-40051-1 for requirements on labeling with HCP. Actions which could degrade hardness, but which are not directly involved in establishing nuclear hardness, shall not be annotated with the acronym, but shall be preceded by a caution statement.

4.13 Selective application and tailoring MIL-STD-40051 contains some requirements that may not be applicable to the preparation of all technical manuals. Selective application and tailoring of requirements contained in MIL-STD-40051 are the responsibility of the contracting activity and shall be accomplished through the use of Appendix A, Technical Manual Content Selection Matrixes, of MIL-STD-40051. The

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applicability of some requirements is also designated by one of the following statements: unless specified otherwise by the contracting activity; as/when specified by the contracting activity; or when specified by the procuring activity.

5. DETAILED REQUIREMENTS.

5.1 Preparation of maintenance instructions. Maintenance instructions shall be prepared as a Maintenance Information Chapter <**mim**>. This chapter shall be subdivided into individual work packages that provide maintenance information to enable a technician to receive, process, inspect, clean, service, test and repair the weapon system/equipment and associated components to an acceptable performance standard. Maintenance tasks shall be developed in accordance with the Logistics Support Analysis (LSA)/Maintenance Allocation Chart (MAC) or Maintenance Plan, and the Source, Maintenance, and Recoverability (SMR) codes developed for the weapon system/equipment and components. Maintenance work packages shall be arranged to coincide with the Functional Group Code (FGC) sequence followed in the MAC or Repair Parts and Special Tools List (RPSTL).

5.2 Types of maintenance chapters. Depending on the type and complexity of the weapon system/equipment, the TM may contain any of the following maintenance chapters.

a. Weapon system/equipment.

b. Component maintenance.

c. Assembly maintenance.

d. Subassembly maintenance.

e. Auxiliary equipment maintenance.

f. Software maintenance.

g. Ammunition maintenance.

h. AVUM/AVIM (-23) maintenance.

i. Preventive maintenance services (**aircraft only**).

j. Phased maintenance inspections.

5.3 Content requirements for maintenance information chapters<**mim**>. Maintenance information chapters shall consist of a title page and the maintenance work packages described in 5.4 through 5.4.5.10.

5.3.1 Maintenance instruction chapter title page<**titlepg**>. Every maintenance chapter in a TM shall have a title page. The title page shall include a chapter number. It shall be titled "Maintenance Instructions" preceded by the maintenance level and followed by the weapon system/equipment, subsystem, assembly, subassembly, or component name, as applicable. (i.e., Unit Maintenance Instructions for the 155MM, M10946 Howitzer, Aviation Intermediate Maintenance (AVIM) Instructions for the AC Power System, etc). (Refer to figure 1.)

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5.4 Maintenance work packages Maintenance work packages shall be developed for the weapon system/equipment and for all maintainable components/assemblies for each applicable maintenance level as indicated in the approved MAC or maintenance plan.

5.4.1 Work package content Individual maintenance work packages shall be developed for the overall weapon system/equipment and each maintainable subsystem, assembly and component. These work packages shall include a scope of work, initial setup information, and all maintenance tasks, such as remove, inspect, service, test, install, replace, disassemble, assemble, repair, clean, adjust, align, etc. When initial setup information differs for specific maintenance tasks, additional work packages shall be developed. Work packages shall stand-alone and contain complete start-to-finish maintenance procedures. The words "END OF TASK" shall be placed below the last data item (i.e., text, illustration, etc.) of the maintenance procedures. The maintenance work packages described in 5.4.5.1 through 5.4.5.10 shall be prepared, as applicable.

5.4.2 Work package size. To facilitate useability or the revision process, work packages should not exceed 30 pages. A series of maintenance tasks can be divided into two or more work packages unless it is determined that separating the task information would degrade useability (i.e., removal and installation of the gun turret in one work package, disassembly and reassembly of the gun turret in a second work package).

5.4.3 Scope of task **<wpsum>**. All maintenance tasks that are included in an individual work package shall be summarized at the top of the work package. All maintenance tasks shall be identified immediately following the statement "THIS WORK PACKAGE COVERS:". (Refer to figure 2).

5.4.4 Initial setup information **<wpinfo>**. Initial setup information shall be included in each work package and shall immediately follow the scope of task. (Refer to figure 2). It provides the maintenance technician with general information, equipment, parts, material, and authorized personnel required to perform and complete all the maintenance tasks included in the work package. Setup information requirements are described below.

a. Maintenance level **<maintlvl>**. The level of maintenance authorized to perform the maintenance contained in the work package (in accordance with the approved MAC) shall be stated. This shall be included for operator or crew, unit, AVUM, direct support, general support, AVIM, and depot levels of maintenance, as applicable. For example,

Maintenance Level
Unit

b. Applicable configurations **<appconfig>**. When the work package does not apply to all configurations of the weapon system/equipment, the applicable configurations **<name>** covered by the work package shall be listed. Omit this requirement if the same tasks/procedures apply to all configurations. (If certain configurations require different tasks/procedures, separate work packages shall be prepared.) For example,

Applicable Configurations
Serial Numbers 12345 through 12399

c. Test equipment **<testeqp>**. All test equipment required to perform the procedure shall be listed by name **<name>** and part **<partno>** or model number **<modelno>** designation if this information is not contained in an overall list elsewhere in the TM. If such a list exists, refer to

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it by name <**name**>, item number, and work package number<**simref**> instead of repeating the information throughout the TM. For example,

Test Equipment
Multimeter (Item 4, WP 0108 00)
Oscilloscope (13057)

d. Tools and special tools <**tools**>. The tool kit (box) assigned to the mechanic (on a 1-per-mechanic-by-MOS basis) to be used in maintenance of a particular equipment shall be listed by name <**name**>, tool kit number (<**partno**> or <**nsn**>), supply catalog (SC) <**sc**>, or TM number <**tmno**> if this information is not included in the Tool Identification List work package contained in the TM. If such a list exists, refer to it by name<**name**>, item number and work package number<**simref**> instead of repeating the information throughout the TM. No tool in the kit shall be further identified. Other tools required for performance of all tasks for the maintenance levels covered in the work package shall also be identified in the initial setup and shall be referenced to the Tool Identification List work package. "Other tools" includes tools which are part of/components of shop sets authorized to sections/teams; tools authorized by RPSTL and CTA-50-970; special and fabricated tools; and items of Test, Measurement and Diagnostic Equipment (TMDE). For example,

Tools and Special Tools
Fixed Open End Wrench Set (Item 47, WP 0110 00)
Screw Threading Set (SC number)
Vehicle Tool Kit (407425)

e. Materials/parts <**mtrlpart**>. All expendable items and support materials shall be listed <**name**>. The item number and supporting information work package<**simref**> which lists these items shall be given. Mandatory replacement parts shall be listed by name<**name**> (and part number <**partno**>, if any). The number, quantity<**qty**>, or size necessary to complete the task shall be listed, when applicable. When a mandatory replacement parts work package exists, it shall be referred to in lieu of the part number. For example,

Materials/Parts
Grease (Item 5, WP 0112 00)
Wiping Rags (Item 38, WP 0112 00)
Range Lock (P/N 8675309)
Range Lock Flange Kit (P/N 8675310)

f. Personnel required<**persnreq**>. Personnel<**name**> and the number of personnel<**qty**> shall be identified if the task requires more than one. The Military Occupational Specialty (MOS) designation <**nameid**> is not necessary, but it may be included. For example,

Personnel Required
Artillery Mechanic 68M10 (1)
Artillery Mechanic 66J30 (1)

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g. References <**ref**>. Other work packages, TMs, and other sources (<**extref**>/<**xref**>) that are needed to complete the maintenance tasks shall be listed here. Only references not listed in equipment conditions shall be listed. For example,

References
TM 9-1015-252-20&P
WP 0100 00

h. Equipment conditions <**eqpconds**>. Any special equipment conditions required before the procedure can be started shall be listed here and cross-referenced to the appropriate source (<**extref**> or <**xref**>) for setting up the condition <**condition**>. For example,

Equipment Condition
Firing mechanism removed (WP 0010 00)

i. Special environmental conditions <**specenv**>. Any special environmental conditions (such as ventilation, lighting, or temperature) <**condition**> that are required shall be listed here. The reason <**reason**> that such conditions are needed shall be explained. For example,

Special Environmental Condition
Darkened area required for testing lights.

j. Drawings required <**dwgreq**>. All drawings (which are not included in the work package) required to complete the maintenance tasks shall be listed here. Drawings shall be listed by title <**dwgname**> and drawing number <**dwgno**>. For example,

Drawings Required
Power Supply Schematic (132E470092)

5.4.5 Types of maintenance work packages.

5.4.5.1 Service upon receipt work package <**surwp**>.

a. This work package (refer to figure 3) shall contain information required for the user to ensure that the equipment will be adequately inspected, serviced, and operationally tested before it is subjected to use <**geninfo**>.

b. Procedures shall be prepared for performing visual inspection of ammunition received from the ammunition supply facility. This inspection shall include verification that ammunition received was that requisitioned. Instructions shall be prepared for a condition check of the shipment (pallets, containers, boxes, and legibility of markings). Instructions shall be prepared to note the quantity of each lot for recording purposes.

c. For equipment that requires extensive service upon receipt, this work package shall be further subdivided into the following tasks <**surtsk**> described in 5.4.5.1.1 through 5.4.5.1.12.

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5.4.5.1.1 Siting <**siting**>. Siting instructions peculiar to the equipment shall be prepared, as applicable. In preparing the instructions, operational and maintenance features shall be considered, such as the following:

a. Location.

b. Proximity to power sources.

c. Effective ranges.

d. Terrain requirements to avoid screening, reflections, ground clutter, and other poor operational conditions due to terrain.

e. Technical requirements.

f. Shelter locations.

g. Compensating for adverse siting conditions.

h. When the equipment contains large components such as towers and antennas that require orientation to a baseline during siting.

i. Mobile equipment oriented during installation.

5.4.5.1.2 Shelter requirements <**shltr**>. For equipment normally housed in a permanent or semipermanent shelter (other than a military truck, van, or transportable shelter) during use, the following information shall be prepared.

a. Amount of floor, wall, and height space required.

b. A plan for a typical layout.

c. Required weight capacity of the building floor.

d. Dimensions required for installed equipment.

e. Total weights that the floor must support and the area in square feet over which the total weight will be distributed.

f. Environmental conditions (e.g., venting).

g. Power requirements.

h. Unusual requirements specific to equipment, such as air-conditioning.

i. Architectural and engineering data on beam sizes, lengths, bending moments, and required supports shall not be included.

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5.4.5.1.3 Service upon receipt of materiel<**surmat**>. The following instructions shall be prepared.

a. Unpacking <**unpack**>. As a minimum, the following information shall be prepared.

(1) Any special sequence of action necessary to protect the equipment.

(2) If a special design reusable container is involved for either the end item or components which are authorized for replacement, instructions shall be prepared to report or reenter the empty container through supply channels. Instructions shall be prepared on how to package the unserviceable component in the empty container in the same manner that the new component was packaged if a component is being replaced.

(3) Man-hour requirements and total man-hours required for unpacking the equipment.

b. Checking unpacked equipment<**chkeqp**>. Instructions shall be prepared for a condition check of the shipment (including that of pallets, containers, boxes, and legibility of markings). (Refer to figure 4, sheet 1 **(standard table)**.) Instructions shall be prepared regarding any inspection procedures peculiar to the equipment. (Refer to figure 4, sheet 2**(standard table)**). In addition, the following shall be inserted exactly as stated here.

"Inspect the equipment for damage incurred during shipment. If the equipment has been damaged, report the damage on SF 361, Transportation Discrepancy Report.

Check the equipment against the packing slip to see if the shipment is complete. Report all discrepancies in accordance with applicable service instructions (e.g., for Army instructions, see DA PAM 738-750).

Check to see whether the equipment has been modified."

c. Processing unpacked equipment<**processeqp**>. Instructions shall be prepared for processing the unpacked equipment (e.g., removing excess lubricant from a new rifle), as long as they do not conflict with any warranty provisions. The following information shall be prepared, as applicable.

(1) Any special skills required by processing personnel.

(2) All caustic, corrosive, and/or toxic material used during processing shall be identified and applicable warnings and cautions given.

(3) Instructions on safe disposal of waste products generated during processing actions.

(4) Man-hour requirements and total man-hours required for processing the equipment.

5.4.5.1.4 Installation instructions <**install**>. Instructions shall be prepared to install the equipment properly, including use of tools; to make the necessary interconnections; and to lubricate, calibrate, and adjust the equipment.

a. Cable diagrams shall be referenced or prepared as necessary. When cable assemblies are not supplied but are required for bench test setup, instructions shall be prepared for fabricating interconnecting cable assemblies from spares and bulk supplies. The part number, drawing number,

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and manufacturer or designer for each part of the cable assembly shall be shown, and wires, connectors, pin connections, and letters or other designators shall be identified.

(1) Instructions and cautions shall be prepared for any mating connectors that call for a special procedure either to make the proper connection or to prevent damage to the connector.

(2) A wiring diagram shall be prepared which fully identifies each wire to be connected, by color code or wire number if applicable. This diagram shall show the location of each pertinent terminal, which shall be identified by number or other marking, if available, or by position if neither is available. Where appropriate, voltage readings shall be annotated.

(3) All alternate connection patterns required for various modes of operation shall be shown and explained.

(4) Only one diagram shall be used to illustrate interconnection patterns which appear more than once within the same equipment.

b. For installation of plug-in items, diagrams shall be prepared or referenced showing the location of items that are not installed in the equipment when received. Instructions shall be prepared whenever special techniques or connections are required.

5.4.5.1.4.1 Assembly of equipment.

a. Instructions shall be prepared for assembling equipment that has been shipped unassembled. When the equipment is to be shelf or rack mounted, instructions shall also be prepared for assembly of the rack, if necessary, and installation of the equipment in the rack.

b. When the equipment is shipped or delivered in specially designed containers, unpacking instructions shall be prepared. If the containers are to be used again, kept for future use, turned in to supply, or require a special disposition method, the necessary procedures shall be prepared. Assembly and installation procedures shall be prepared when needed. These instructions shall be supported by illustrations. As applicable, power requirements, connections, and initial control settings needed for installation purposes shall be prepared.

c. For security measures for electronic data, instructions shall be prepared for handling, loading, purging, overwriting, or unloading classified electronic data under usual conditions. Instructions shall meet current security regulations as they pertain to automation security.

5.4.5.1.4.2 Installation.

a. Installation instructions shall be prepared for all actions (including placing, mounting, and attaching) for the following.

(1) Cable and wiring interconnections.

(2) Proper use of special tools.

b. Installation instructions shall identify all dimensions that must be maintained in placing, mounting, or attaching items.

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c. When initial adjustments can be made efficiently during installation, such adjustments shall be included.

d. For equipment designed and intended for use in more than one type of installation (e.g., field, fixed station, and mobile), instructions shall be prepared for each type of installation involved.

e. If performance of any step in the installation instructions requires the assistance of personnel from a higher level of maintenance, this shall be stated in a note similar to that below.

"**NOTE**
The following installation procedure must be made with the assistance of (insert level) maintenance personnel (include Military Occupational Specialty (MOS), if applicable)."

f. Installation instructions shall be considered complete only when they include instructions for:

(1) All required installation options (e.g., Electrostatic Discharge (ESD) control requirements).

(2) Accessory items.

(3) Auxiliary items (those that extend or increase equipment capability).

(4) Grounding of the equipment for both safety and proper operation.

(5) Torque requirements.

5.4.5.1.4.3 Special applications. Installation instructions which are common to all special applications of a system shall be prepared. Details resulting from the installation but peculiar only to the equipment into which the system is being installed shall be omitted (e.g., special treatment required when installing the system in a vehicle or aircraft).

5.4.5.1.4.4 Van and shelter installations. The following information shall be prepared only to the extent required for the applicable level of maintenance.

a. Instructions shall be prepared for the removal and replacement of each nonpermanent unit.

b. Installation instructions shall not be prepared when the equipment is permanently installed in vans or shelters.

c. Diagrams and instructions shall be prepared which pertain to electrical and interconnection wiring, exclusive of wiring peculiar to the equipment on which the installation is being made (e.g., headlight, ignition wiring).

d. Instructions shall be prepared for cable run locations, equipment locations, circuit breaker panels, and other similar details.

5.4.5.1.5 Preliminary servicing of equipment<**preserv**>. Instructions for all lubrication required on newly installed equipment shall be prepared.

5.4.5.1.6 Preliminary checks and adjustment of equipment<**prechkadj**>. Instructions for all checks and adjustments to be made on newly installed equipment shall be prepared. Information on the location of items

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such as controls and check points shall be prepared or referenced. Instructions shall be prepared for checks and adjustments that must be made before equipment is put into operation and for all other checks required to ensure proper operation of the equipment. These instructions shall include the following, as applicable:

a. Checks for interconnections.

b. Checks for grounding, including earth ground connections, earth conditioning for conduction, as well as a check of the grounding circuit for negligible resistance.

c. Checks for adequate clearance for rotating or moving devices.

d. Checks of initial settings of all controls that must be preset before power is to be applied.

e. All other checks needed to determine that power can be applied without injuring personnel or damaging the equipment.

f. Firm seating and connection of all plug-in parts, mating connectors, jacks, and plugs.

g. Cable and wire harness routing, dressing, and fastening.

h. Cautions against damaging transistors, diodes, and other electrically sensitive items.

i. Replacement of all covers, inspection and access doors, and plates.

j. Operation of safety interlocks and switches.

k. Operation of ventilating louvers and intake and exhaust ports.

l. Operation and content of liquid cooling systems.

m. Lubricants and Corrosion Prevention Control (CPC) procedures.

n. Switch and control settings that are preset at installation (installer's adjustments).

o. Presetting and adjustment of automatic controls.

p. Terminal connections.

q. Required terminal or capacitor strapping.

r. Preliminary test measurements.

s. Presetting operator's controls. (Reference the operating instructions, as applicable.)

t. Normal operating checks. (Reference the operating instructions, as applicable.)

u. After-installation orientation.

v. Burn-in of parts.

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w. ESD control standards.

x. After operations, shutdown, checks, and inspections.

5.4.5.1.7 Preliminary calibration of equipment **<precal>**. Instructions for all calibration to be made on newly installed equipment shall be prepared.

5.4.5.1.8 Circuit alignment **<calign>**. Instructions shall be prepared for circuit alignment procedures. Applicable instructions shall be prepared in the following order.

a. External connections **<extconn>**. Connections to external lines required for each installation option shall be included. Connection instructions shall conform to the requirements for installing wiring and cabling interconnections.

b. Switch settings, patch panel connections, and internal control settings **<setconn>**. Instructions shall be prepared for all switch settings, patch panel connections, and internal control settings required for each installation option and mode of operation.

c. Alignment procedures **<alignproc>**. Instructions shall be prepared for all alignment procedures, including any variations required for different installation options and modes of operation.

5.4.5.1.9 Ammunition markings **<markings>**. Instructions shall be prepared for marking ammunition and ammunition containers.

5.4.5.1.10 Classification of defects **<defect>**. Procedures shall be prepared for performing visual inspection of ammunition/containers (pallets, boxes, etc.) and shall include classification and disposition of defective ammunition/containers.

5.4.5.1.11 Handling **<handling>**. Procedures shall be prepared for handling ammunition.

a. Unpacking **<unpacking>**. As a minimum, the following information shall be prepared.

(1) Any special sequence of action necessary to protect the ammunition.

(2) If a special design reusable container is involved for either the end item or components which are authorized for replacement, instructions shall be prepared to report or reenter the empty container through supply channels.

(3) Man-hour requirements and total man-hours required for unpacking the ammunition.

b. Packing **<packing>**. As a minimum, the following information shall be prepared.

(1) Any special sequence of action necessary to protect the ammunition.

(2) Instructions shall be prepared on how to package defective ammunition.

(3) Man-hour requirements and total man-hours required for packing the ammunition.

5.4.5.1.12 Procedures needed to activate ammunition, mine, etc **<arm>**. Procedures shall be prepared for activation of ammunition, mines, etc., preparatory to detonation.

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5.4.5.2 Equipment / user fitting instructions work package<**perseqpwp**> **(personal use equipment)**. As applicable, equipment/user fitting instructions for personal use equipment shall be prepared. (Refer to figure 5.)

5.4.5.3 Preventive maintenance checks and services (PMCS), including lubrication instructions, work package <**pmcswp**>. This work package shall consist of PMCS information and periodic lubrication <**lubricant**> instructions. (This work package is not required for aircraft TMs.) The content of PMCS shall be based upon the principles of Reliability Centered Maintenance (RCM) logic and shall include applicable scheduled corrosion inspections. The PMCS procedures shall be arranged in a logical sequence requiring minimum time and motion on the part of the person(s) performing them and shall be so arranged that there will be minimum interference between persons performing the checks simultaneously on the same end item. Instructions shall be prepared in the following order.

5.4.5.3.1 Introduction <**pmcsintro**>. The introduction paragraph shall explain the purpose and use of the PMCS table. (Refer to figure 6.)

a. An explanation shall be prepared for each column of the table and any general checks/services that are common to the entire piece of equipment. The explanation for the item number column shall detail how the item numbers are used when recording results of PMCS on DA Form 2404, Equipment Inspection and Maintenance Worksheet.

b. General statement(s) shall be prepared which apply to the overall understanding of lubrication requirements.

c. Lubricants shall be identified by standard military symbols in accordance with MIL-HDBK-113 and MIL-HDBK-275. The following lubrication interval symbols shall be used, as applicable.

D—daily
W—weekly
M—monthly
Q—quarterly
S—semiannually
A—annually
MRA—maintenance repair action
B—biennially
H—hours (operated)
MI—miles (operated)
KM—kilometers (operated)
RDS—rounds fired
OC—on-condition

d. A statement concerning Corrosion Prevention and Control (CPC) shall be prepared. This statement shall contain maintenance instructions or reference CPC requirements contained in the applicable maintenance instructions. In addition, if the inclusion of such instructions are applicable, a statement shall be prepared which states that the instructions are mandatory.

(1) Oil filter statement. As applicable, the following statement shall be included verbatim:

"Oil filters shall be serviced/cleaned/changed, as applicable, when:

They are known to be contaminated or clogged,

Service is recommended by AOAP laboratory analysis, or

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At prescribed hardtime intervals."

(2) AOAP sampling interval statement. The following statement shall be inserted:

"Engine oil/transmission oil/hydraulic fluids must be sampled at (insert applicable hour/mileage timeframe) as prescribed by (insert TB 43-0106, Aeronautical Equipment Army Oil Analysis Program (AOAP), or DA PAM 738-750, Functional Users Guide for the Army Maintenance Management System (TAMMS))."

(3) AOAP not available/non-enrolled statement When a component/equipment is not enrolled in the AOAP or oil analysis support is not available, the following statement shall be inserted:

"This (enter name of component/equipment) is not enrolled in the Army Oil Analysis Program. HARDTIME INTERVALS APPLY."

(4) Warranty hardtime statement. The following statement shall be used, as applicable:

"For equipment under manufacturer's warranty, hardtime oil service intervals shall be followed. Intervals shall be shortened if lubricants are known to be contaminated or if operation is under adverse conditions (such as longer-than-usual operating hours, extended idling periods, extreme dust)."

e. When the equipment contains fluids, such as lubrication oil or hydraulic fluid, leakage criteria shall be prepared for the PMCS introduction as follows and referred to in the NOT READY/AVAILABLE IF: column.

"It is necessary for you to know how fluid leakage affects the status of the (enter component/equipment name). Following are types/classes of leakage you need to know to be able to determine the status of the (enter component/equipment name). Learn these leakage definitions and remember - when in doubt, notify your supervisor.

CAUTION

- Equipment operation is allowed with minor leakages (Class I or II). Consideration must be given to fluid capacity in the item/system being checked/inspected. When in doubt, notify your supervisor.

- When operating with Class I or II leaks, continue to check fluid levels as required in the PMCS.

- Class III leaks should be reported immediately to your supervisor.

(1) Class I — Seepage of fluid (as indicated by wetness or discoloration) not great enough to form drops.

(2) Class II — Leakage of fluid great enough to form drops but not enough to cause drops to drip from item being checked/inspected.

(3) Class III — Leakage of fluid great enough to form drops that fall from item being checked/inspected."

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f. When specified by the procuring activity, the first page following the introduction shall consist of an illustration of the equipment. This illustration shall include a routing diagram by which the PMCS will be performed.

5.4.5.3.2 PMCS procedures. The checks and services shall be in tabular form as shown in figures 7 and 8 **(standard tables)** and shall be prepared in accordance with the instructions in 5.4.5.3.2.1.

5.4.5.3.2.1 PMCS table preparation **<pmcstable>**

a. Item number column**<itemno>**. Item numbers shall be assigned to procedures.

b. Interval column**<interval>**. Information in this column shall show the designated interval when each check is to be performed. Procedures done first or most frequently (i.e., "before" checks and services) shall appear prior to "during" and "after" checks and services. (Refer to figures 7 and 8 for examples of completed interval column.) When more advantageous to the user, intervals shall be subgrouped by crewmember(s) as shown on figure 8.

c. Man-hour column**<manhours>**. Man-hours required to complete all prescribed lubrication services shall be stated to the nearest 10th of an hour. (Refer to figures 7 and 8.)

d. Item to be checked or serviced column**<checked>**. The items listed in this column shall be identified in as few words as possible to clearly identify the item. Usually the common name (e.g., bumper, gas can and mounting bracket, front axle, etc.) will be enough. (Refer to figures 7 and 8.)

e. Procedure column**<pmcsproc>**. This column shall contain a brief description of the procedure by which each check is to be performed, as well as any information required to accomplish each check or service, including lubrication, appropriate tolerances, adjustment limits, and instrument gage readings. Illustrations shall be prepared to identify the location or the process of the task being performed and shall be integrated with the procedures. (Refer to figures 7 and 8.) Whenever replacement or repair is recommended, this column shall reference the applicable maintenance instructions.

(1) Lubrication procedures shall be prepared including information on authorized lubricants, lubrication intervals, man-hour requirements, and the AOAP. Lubrication instructions shall be prepared so as to enable the user to receive, lubricate, and return to an acceptable performance standard all components of the equipment in a minimum of time with the skills, tools, test equipment, and spare parts authorized by the LSA/MAC. Information shall be included for any special lubrication required under extreme temperature, altitude, and humidity conditions within the limits established by the design specification for the equipment.

(2) Lubricant types and abbreviations for flight vehicles and components shall be identified by standard military symbols as specified in MIL-HDBK-275; lubricant types and abbreviations for ground equipment systems, lubricants, functional fluids, preservatives, and specialty products shall be identified by standard military symbols in accordance with MIL-HDBK-113. (Required abbreviations not covered in the appropriate handbook will be provided by the procuring activity.)

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(3) Lubrication instructions shall be prepared to present all applications, procedures, lubricants, and lubrication points. When grouped lubrication points require the same lubricant at the same interval, the type and number of points shall be identified and described by one of the following methods.

(a) Multi-headed arrows. Multi-headed, solid-shafted arrows shall point to each of the lubrication points.

(b) Lubrication point notes. Lubrication point notes shall contain instructions for applying lubricants, taking into account the following factors: type, grade, availability, and properties of the prescribed lubricant; expected temperature; lubrication guns and tools available to authorized maintenance level; types of lubrication fittings; and possible ill effects of excessive or insufficient lubrication. Caution shall be stressed where over- or under-lubrication of a part will damage that part or closely associated parts.

(4) Disassembly and hand-packing instructions shall be prepared for medium- and high-speed antifriction bearings that are sensitive to the amount of lubrication applied and do not have bleed holes or relief valves.

(5) Cleaning, disassembling, and assembling instructions required before or after lubrication shall be prepared or referenced.

(6) Instructions shall be prepared for washing and natural drying of finely machined and dirt-sensitive parts before relubricating. Use of compressed air jets or temperatures above 212° F shall not be prescribed.

(7) Instructions shall not specify a coating of preservative material, either before or after packing parts that are lubricated with grease; nor shall they specify an application of oil, solvent, or additional grease to a "sealed-for-life" or prepackaged antifriction bearing.

(8) Where applicable, the statement "For Arctic operation, refer to FM 9-207, Operation and Maintenance of Ordanance Materiel in Cold Weather (0 degrees to minus 65 degrees F)." shall be inserted as a note. When specific restrictions, preferred grades of lubricant, and other conditions exist, notes shall be made in the table. For example,

NOTE

When MIL-L-2104 lubricant is authorized, use 15W-40 (OE/HDO-15/40) when available and applicable temperature range exists.

or

NOTE

15W-40 oil is not authorized in this particular (enter component name).

f. Equipment not ready/available if: column<**eqpnotavail**>. This column shall contain a brief statement of the condition (e.g., malfunction, shortage) that would cause the equipment to be less than fully ready to perform its assigned mission. (Refer to figures 7 and 8.)

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5.4.5.3.2.2 Mandatory replacement parts **<mrplpart>**. This paragraph shall contain a table listing all items that must be replaced during PMCS whether they have failed or not. The table shall reflect the interval at which these items must be replaced whether hardtime maintenance or on condition maintenance is the determining factor. (Refer to figure 9.) This table shall be the last page(s) of the PMCS.

5.4.5.3.3 Preventive Maintenance Checklist (PMC)**(operator only)**. When specified by the contracting activity, a PMC shall be prepared as a separate document. Information for a PMC shall come from the applicable operator's PMCS.

5.4.5.4 Preventive maintenance inspections work package**(aircraft only) <pmiwp>**. This work package shall contain the requirements outlined in 5.4.5.4.1 through 5.4.5.4.3.

5.4.5.4.1 General information and introduction**(aircraft only)**. The following paragraph shall be inserted.

"**GENERAL INFORMATION**

This section contains complete requirements for special inspections, overhaul and retirement schedule, and standards of serviceability applicable to the aircraft. The inspections prescribed in this section shall be accomplished at specified periods by AVUM activities, with the assistance of AVIM activities when required. Complete Daily, Intermediate, Periodic, or Phased inspections are contained in the (insert applicable aircraft inspection checklist TM)."

5.4.5.4.2 Standards of serviceability **(aircraft only)**. The following paragraph shall be inserted.

"Standards of serviceability to be utilized in the day-to-day inspection and maintenance of the aircraft can be found as fits, tolerances, wear limits, and specifications in the aircraft maintenance manuals. Standards of serviceability for transfer to aircraft are contained in TM 1-1500-328-23."

5.4.5.4.3 Special inspections **(aircraft only)**.

a. Definition and general information The following paragraph shall be inserted.

"This information supplements scheduled inspections as outlined in the applicable aircraft inspection checklists. Inspection of items which are required to be inspected at intervals not compatible with airframe operating time or airframe inspection intervals is also included. Refer to DA PAM 738-751 (Functional Users Manual for the Army Maintenance Management System-Aviation (TAMMS-A)) for applicable forms, records, and worksheets required for these inspection intervals. Typical examples of this type of inspection are as follows.

(1) Inspections which are solely contingent upon specific conditions or incidents that occur (e.g., hard landings, overspeed, or sudden stoppage), wherein immediate inspection is required to ensure safe flight.

(2) Inspection of components or airframe on a calendar basis: e.g., first aid kits, weight and balance check, aircraft inventory."

b. Requirements. Requirements shall include items which qualify under the criteria for special inspections, e.g., hard landings, sudden stoppage, overspeed. These requirements shall be grouped under area headings only and shall be inserted in a columnar list on the inspection check sheet format in such a manner as to permit local reproduction of the entire section. A line drawing of

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aircraft or accessory showing sequence for inspection by area shall be included. The area identified shall include all surfaces, materials, components, and equipment pertaining to that specific location. The legend shall identify the title or heading of each numbered area of the diagram.

5.4.5.5 Aircraft lubrication instructions work package**(aircraft only) <lubewp>**. This work package shall contain the requirements outlined in 5.4.5.5.1 and 5.4.5.5.2.

5.4.5.5.1 Lubrication instructions **(aircraft only)**. Lubrication schedules shall be prepared to present all applications and procedures, lubricants, and lubrication points to completely lubricate aircraft.

5.4.5.5.2 Lubrication charts **(aircraft only)**.

a. Lubrication charts shall consist of a main drawing prepared as a three-dimensional diagram, and such enlarged or detailed views as are considered necessary to identify items which otherwise would be obscured. They shall show all lubrication requirements for all parts of the aircraft requiring periodic lubrication, other than those lubricated by the main engine oil system. The charts shall also indicate type of lubricant, method of application, and frequency. This information shall be presented through the use of a standard symbol. (Refer to figure 10.)

b. Use of black silhouette figures representing a likeness of the tool used in the application (oil can, grease gun, brush, or hand) shall be the accepted means of presenting application methods on the lubrication chart.

c. Abbreviations, as specified in MIL-HDBK-275, shall be used to present lubricant types. In the event a lubricant does not have an abbreviation listed in MIL-HDBK-275, the abbreviation shall be provided by the procuring activity. Assigned application symbols, type abbreviations, and frequency shall be placed within the standard lubrication symbols.

d. A key shall be prepared for the chart to define each application symbol and lubricant abbreviation used. Consecutively numbered notes may be used to specify requirements other than normal. A statement "SEE NOTE 1." shall be placed adjacent to the symbol, as appropriate.

5.4.5.6 Ammunition maintenance work package**<ammowp>**. This work package shall reference or contain all procedures required for care and handling of ammunition**<handling>**, including hazard distances, storage, special requirements, prevention of deterioration due to rough handling, exposure to adverse weather conditions or other hazards. Visual inspection criteria shall be prepared to determine item serviceability. Procedures shall be prepared for disposition of defective ammunition**<defect>**. These procedures shall include use of cleaning materials and paint authorized for use in the specified maintenance operations **<service>**. Reference shall be made to the RPSTL for replacement parts and any special tools required and to the Expendable and Durable Items List Appendix for the list of expendable and durable materials. (Refer to figure 11).

5.4.5.7 Auxiliary equipment maintenance work package**<auxeqwp>**. When auxiliary equipment (e.g., Modified Tables of Organization and Equipment (MTOE) items, etc.) maintenance TMs or maintenance requirements cards are not procured for peculiar equipment furnished by the contractor, maintenance instructions shall be prepared. Instructions shall be prepared for required fabrication of special tools, when such fabrication is approved by the contracting activity, and included in the proper supporting information work package. Concise step-by-step procedures shall be prepared for proper care of auxiliary equipment while in and out of service. These procedures shall include instructions for store, preventive maintenance, lubrication, operating checks, and adjustments, as applicable. (Refer to figure 12).

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5.4.5.8 Maintenance work packages<**maintwp**>. Maintenance information shall be functionally divided into individual maintenance work packages<**maintwp**>. (Refer to figure 13.) The technical content structure for these work packages shall be consistent from work package to work package. Illustrations shall be prepared to identify the location or the process of the task being performed and shall be integrated with the procedures.

a. Each maintenance work package shall include all authorized maintenance tasks<**maintsk**>. Tasks shall consist of complete start-to-finish maintenance procedures<**proc**> in a logical sequence of occurrence. Task titles<**title**> shall be identical to FGC titles as used in the applicable MAC and RPSTL. Maintenance tasks are described in 5.4.5.8.1.

b. Maintenance instructions shall reference or contain all procedures required for any unusual or critical steps such as specifying Quality Assurance (QA) checks**(depot and aviation only)**, care and handling of ESD sensitive items and all hazardous material, e.g., ammunition, radioactive components or materials, including prevention of deterioration due to rough handling, exposure to adverse weather conditions, or other hazards. Visual inspection and safety criteria shall be prepared to determine item serviceability. Instructions shall also contain procedures for disposition of defective ammunition. Procedures shall be prepared for use of cleaning materials and paint authorized for use in the specified maintenance operations. Tools shall be described only when the tool or procedure is peculiar or abnormal.

c. When peculiar to the equipment, applicable Corrosion Prevention and Control (CPC) procedures shall be included, or the work package shall reference applicable CPC publications.

d. National Stock Numbers (NSNs) shall not be used in procedural steps, illustrations, or legends of maintenance work packages. Part numbers shall not be used in procedural steps, illustrations, or legends, except when essential for identification.

e. Aviation maintenance TMs shall reference procedures in TM 1-1500-204-23, as applicable.

f. The maintenance instructions shall be prepared to include required environmental control data and information. Instructions shall be prepared for information on any special maintenance required under extreme temperature, altitude, and humidity conditions within the limits established by the design specification for the equipment.

g. For **depot only**, a Reliability, Availability, and Maintainability (RAM) table shall be prepared listing the pertinent measurable RAM ranges for the major overhauled components. The RAM requirements shall be prescribed by maintenance engineering of the contracting activity and when established by maintenance engineering shall include critical measurement factors, such as Meantime Between Failures (MTBF), Meantime Between Corrective Maintenance (MTBCM), Maximum Time to Repair (MTR), availability, and maintenance ratio. The reliability and availability portion of the table shall give the minimum acceptable values while the maintainability portion shall provide the maximum allowable rates. Availability may be expressed as a probability versus a qualified number. When specified by maintenance engineering of the contracting activity, the RAM information may also be prepared in a narrative format. (Refer to figure 14.)

5.4.5.8.1 Maintenance tasks <**maintsk**>. Maintenance tasks shall be prepared for each authorized maintenance level in the general order listed below. For each maintenance task illustrations shall be used to support or clarify the text, including schematics, wiring diagrams, parts location drawings and other visual aids.

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Preshop analysis **(depot only) <pshopanal>**.
Assembly and preparation for use **(aviation only) <assem>**.
Servicing **<service>**.
Ground handling.
Operational check **<opchk>**.
Inspection of installed items **<inspinstitm>**.
Removal **<remove>**.
Disassembly **<dissam>**.
Cleaning **<clean>**.
Inspection-acceptance and rejection criteria **<acptrejinsp>**.
Nondestructive Testing Inspection (NDTI) **<ndti>**.
Repair or replacement **<repair-rplc>**.
Alignment **<align>**.
Painting **<paint>**.
Lubrication **<lube>**.
Assembly **<assem>**.
Test and inspection **<test-inspect>**.
Installation **<install>**.
Adjustment **<adjust>**.
Calibration **<calibration>**.
Radio interference suppression **<ris>**.
Placing in service **<pis>**.
Testing **<test-pass>**.
Quality assurance requirements **(depot only)** (reference only).
Preservation, packaging, and marking **(depot only) <ppm>**.
Overhaul and retirement schedule **(aircraft only) <orsch>**.
Preparation for storage or shipment **<pss>**.
Ammunition markings **<markings>**.
Classification of ammunition defects **<defect>**.
Handling ammunition **<handling>**.
Procedures for ammunition activation **<arm>**.

5.4.5.8.1.1 Maintenance task requirements Additional mandatory or unique technical information or additional explanations may be required to be included in the maintenance tasks listed in 5.4.5.8.1, above. This information is described in 5.4.5.8.1.2 through 5.4.5.8.1.19. The following general requirements apply to most of the maintenance tasks in 5.4.5.8.1, above.

a. Peculiar instructions shall be prepared for lockwiring, installing cotter pins, use of sealing compounds, lubricants, or corrosion prevention compounds and similar operations with applicable references to the expendable and durable items list.

b. Procedures shall not be prepared for separation of bonded, press-fitted, soldered, welded, or riveted parts, or the removal of electronic circuitry parts, unless such removal is necessary to clean, inspect, or test separately.

c. If servicing (i.e., pressurizing and charging with gas, lubrication, etc.) is required upon completion of a maintenance task, include this information as part of the task.

d. Warnings and cautions shall be prepared whenever chemicals or cleaning compounds are used or combined which may result in a dangerous or hazardous mixture. Whether the danger is to

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personnel or equipment, it shall be identified and the effect (e.g., gases, fumes, caustic, and fire) shall be stated.

e. For aircraft, instructions shall be prepared for cleaning and washing the entire aircraft. Instructions shall be prepared for the removal of the battery, relief tube, power plant, and armament exhaust deposits, or other items or material as necessary. Instructions shall also be prepared regarding components which require relubrication after the aircraft has been washed or steam cleaned.

f. Torque requirements, values, and sequences shall be indicated. Only critical torques**<torque>** shall be indicated in task steps. All noncritical torques will be covered by the Torque Limits work package and a reference thereto. Torque values shall be given for all structural attaching hardware, fluid couplings (fuel, oil, hydraulic, pneumatic, etc.), and connections. Torque values shall include torque correction factors when crowfoot extensions, thread lubricants, and cadmium-plated screws or nuts are used. Torque values identified in the tasks must reflect torque wrenches authorized to personnel targeted to perform tasks. Upon completion of torque action, instructions shall be prepared on use of an orientation mark (striping).

g. Such terms as "reverse the disassembly procedures" or "installation is the reverse of removal" shall not be used in any maintenance task.

h. Maintenance procedures or steps that have a major quality assurance effect shall be prefixed with the boldface letters **"QA"** to identify them.

5.4.5.8.1.2 Preshop analysis **(depot only) <pshopanal>**. Preshop analysis shall apply when data indicates that an inspection or test is more effective in determining useful life of a system, subsystem, or component than a mandatory disassembly. The following requirements shall apply.

a. Scope. The purpose and coverage of the preshop analysis shall be stated.

b. Unpacking and special handling Procedures shall be prepared for removing the item, assemblies, subassemblies, or components from the shipping containers and packaging material. Instructions shall be prepared on any needed handling requirements for hazardous material, electrostatic sensitive devices, precious metal content, classified material, or critical material. Instructions shall also be prepared for any special condemnation procedures for the item and its assemblies and subassemblies.

c. Checking attached documents Instructions shall be prepared for checking all tags, forms, and documents attached to the item to determine the reason for its return and to identify any other obvious faults or damage.

d. External inspection. Procedures shall be prepared for external inspection of the item to determine if it is complete and if there is any obvious external damage.

e. Cleaning and preservation Instructions shall be prepared for cleaning the item to prepare it for preshop analysis testing. They shall include the procedures for any temporary preservation or corrosion protection measures needed to protect the item until the work required is started.

f. Preshop analysis procedures. Detailed procedures shall be prepared for performing a preshop analysis.

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(1) Preshop analysis text shall be presented in procedural format. Test and analysis procedures shall be presented in a logical sequence—not to cause any unnecessary disassembly—and in the order in which they should be done. Each procedure shall be identified by a step number. Procedures shall be arranged in groups by major components, assemblies, and subassemblies. Each group shall be headed by an applicable title.

(2) When specified by the contracting activity, procedures shall be presented as a checklist **<chklist>**, or as a supporting information work package. (Refer to MIL-STD-40051-7). This work package shall be referenced in the preshop analysis task. If the checklist is included as part of the task, the following shall be included.

(a) Cover sheet **<coverpage>**. The cover sheet shall contain spaces to record the following item information: part number **<partno>**; serial number **<serialno>**; NSN **<nsn>**, modifications required **<modreq>**; reason for overhaul or repair **<reason>**; unpacking of secondary items required **<secitem>**; review of tags **<revtag>** or forms **<revform>** with the item, name **<name>** and signature **<sig>** of person doing the analysis; and date **<date>**. (Refer to figure 15.)

(b) Table of tests and inspections. This table shall have an entry for each test and inspection procedure. Each entry shall have, as a minimum, the following information: inspection point, condition, action, remarks, and identification of the personnel performing the inspection. (Refer to figure 16**(standard table)**.)

5.4.5.8.1.3 Assembly and preparation for use **(aviation only) <assem>**.

a. Procedures shall be prepared for unpacking, assembly, and installation. When the equipment is shipped or delivered in specially designed containers, unpacking instructions shall be prepared. If the containers are to be used again, kept for future use, turned in to supply, or require a special disposition method, the necessary procedures shall be prepared. Assembly and installation procedures shall be prepared when needed. These instructions shall be supported by illustrations. As applicable, power requirements, connections, and initial control settings needed for installation purposes shall be prepared.

b. For security measures for electronic data, instructions shall be prepared for handling, loading, purging, overwriting, or unloading classified electronic data under usual conditions. Instructions shall meet current security regulations as they pertain to automation security.

5.4.5.8.1.4 Servicing **<service>**.

a. Instructions shall be prepared for replenishment of fuel; oil; hydraulic or other fluids; oxygen, nitrogen, other gases; and tire pressure, plus any other such items and materials (except for lubricants) required for complete servicing of the equipment.

b. Servicing instructions shall be supplemented with a diagram showing locations of regular and emergency servicing points. Items located on each side of the equipment which require servicing will be illustrated and identified as right and left side. NO STEP areas on walkways leading to any tank (in an aircraft) shall be indicated and necessary cautions included.

c. Reference shall be made to the servicing diagram and to the list of expendable and durable items for identification of any fuel, oil, or other materials used. All expendable items used in the

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servicing instructions shall be identified by military and federal standard nomenclature and designation number.

d. The warnings and cautions to observe in servicing a particular tank or reservoir (e.g., grounding and prevention of fire hazards) shall be stated clearly.

e. Instructions shall be prepared regarding access to any out-of-the-way or unusual places requiring service.

f. Reference shall be made to graphs or data in other parts of the text pertinent to servicing, such as tire pressure as opposed to gross takeoff weight for aircraft and capacities of tanks.

5.4.5.8.1.5 Ground handling. Descriptions, instructions, and necessary cautions and warnings for ground handling of the aircraft/equipment, including any information needed in extreme cold, heat, humidity, dust, or other unusual or extreme conditions shall be prepared. Instructions for folding and unfolding appropriate parts such as rotor blades or wings, rudders, and fans shall also be included. For aircraft, instructions shall be prepared that are required for blocking and supporting the aircraft during performance of the operation or procedure involved. The following ground handling procedures shall be provided in the order listed below.

a. Towing **<tow>**.

b. Jacking **<jack>**.

c. Parking **<park>**.

d. Mooring **<moor>**.

e. Covering **<cover>**.

f. Hoisting **<hoist>**.

g. Sling loading **<sling>**.

h. External power **<extpwr>**.

5.4.5.8.1.6 Operational check **<opchk>**. Operational checks required to ensure the satisfactory performance of the weapon system/equipment, subsystem or components shall be prepared. Starting, runup, and shutdown procedures contained in operator instruction work packages may be referenced. Corrective actions shall be included as a means of referencing the troubleshooting procedures.

5.4.5.8.1.7 Inspection acceptance and rejection criteria **<acptrejinsp>**. For **depot level and aviation only**, inspection procedures that have a quality impact shall be identified with the boldface letters **"QA"** preceding them. Instructions shall be prepared for tagging all rejected parts, other than mandatory replacement parts, and listing the reasons for rejection. Visual inspection procedures shall be prepared to detect defects such as burrs, cracks, bends, or dents. Accurate and measurable accept or reject requirements and standards shall be prepared which allow the user to determine if the item under inspection conforms to the tolerances, wear limits, fit, or other standards and requirements presented. Overhaul Inspection Procedure (OIP) tables **(standard table)** shall be prepared for items with critical characteristics or tolerances. Illustrations shall be integrated within the text of the OIP table. (Refer to figure 17.)

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5.4.5.8.1.8 Nondestructive Testing Inspection (NDTI)<**ndti**>.

a. **(Aircraft only)** When specified by the contracting activity, TM 55-1500-335-23 shall be the only NDTI document referenced in the NDTI procedures, and technical provisions of this TM shall be followed.

b. **(Aircraft only)** Individual NDTI procedures shall be specified for each part requiring NDTI. In order to satisfy this requirement, the following shall be prepared for aircraft TMs.

(1) If penetrant is required, identification of the particular TM 55-1500-335-23 process that is applicable.

(2) If magnetic particle inspection is required, the specific TM 55-1500-335-23 method, the type of magnetization, and amount of current or ampere turns.

c. The reject criteria shall be specified in all cases. This shall be done by means of a blanket statement, individual criteria for a part, or a combination of both.

d. Unless specified otherwise by the proponent activity, instructions for use of visible dye penetrants shall not be included as part of NDTI instructions. When required, refer to TM 55-1500-335-23 for preparation of those instructions.

e. When several NDTI methods are permitted, the relative order of preference shall be specified.

f. Instructions shall be prepared for removing primer and/or paint for TMs that require the removal process as part of NDTI procedures. If a part requires a special process, this procedure must be contained within the NDTI procedure for that part.

g. Cleaning requirements prior to, during, and after NDTI shall be specified. If a part has a built-in bearing, then a procedure shall be prepared to ensure protection of the bearing for the NDTI procedure.

5.4.5.8.1.9 Repair or replacement<**repair-rplc**>. Instructions shall be prepared for repair or replacement to restore an item to a completely serviceable or fully mission capable status. Depot level maintenance shall include data plate replacement data. For data plates which require replacement, the type of material shall be indicated. Detailed preparation and attachment instructions shall be prepared. The instructions for stamping data plates shall include the initials of the facility performing the overhaul or modification, the contact number (if applicable), the date of overhaul or modification, the part number, and the total operating time since new (if applicable). The instructions shall specify the letter and figure sizes and indicate their placement (adjustment to manufacturer's data). The following statement shall be inserted.

"When sufficient space is not available on the existing data plate to add information, the plate shall be replaced and all pertinent data transferred to the new plate. Data shall not be stamped directly on any part, assembly, or item of equipment except when approved by the Government."

5.4.5.8.1.10 Test and inspection<**test-inspect**>. Procedures shall be prepared for testing and inspection during or after assembly to ensure proper assembly of the item. Correct methods of testing; procedures for making tolerance checks; and procedures for inspection of distance measurements (e.g., clearance, end play, backlash) shall be prepared. Measurement criteria and tolerances shall reflect the Test Measurement and Diagnostic Equipment (TMDE) available to the user. **(Depot and aviation only)** Test and inspection

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procedures which have a major quality assurance effect shall be prefixed by the boldface letters **QA**" to identify them.

5.4.5.8.1.10.1 Inspection and test of conventional and chemical ammunition or components containing radioactive materials **(-30, -40, and AVIM only)**. The following information shall be prepared.

a. A statement to the effect that inspection criteria are provided to assure that maintenance performed will restore the items to an acceptable quality level shall be included. Regulations and technical publications relating to policy responsibility and procedures applicable to ammunition stockpile reliability, ammunition surveillance, radioactive materials procedures, and quality evaluation programs shall be referenced. The types of inspection procedures shall, at a minimum, include a premaintenance inspection to be conducted during unpacking, in-process inspections, and final acceptance inspection.

b. Instructions shall be prepared for inspection methods or techniques used to detect defective components or end items being processed. Classification of Material Defects tables **(standard table)** shall be prepared for ammunition components and packaging and packing material. The tabulated data shall include a list of components, categories of defects (minor, major, critical) attributable to each component, a list of defects by category, a reference to the work package or the TM containing corrective action, the inspection methods used to determine if corrective action was accomplished, and the acceptable quality level established for each defect. (Refer to figure 18.)

c. Visual inspection criteria shall be prepared for the packing of the items in conformance with the inspection criteria noted in subparagraph a. above.

d. Detailed procedures and criteria shall be prepared for function testing. When test fixtures must be fabricated, diagrams and instructions for the fabrication shall be prepared. Where ammunition is required for function testing weapons, it shall be identified in a table by Department of Defense Ammunition Code (DODAC), NSN, and nomenclature, to include dummy rounds.

e. Instructions shall be prepared to establish a uniform system of examination for deterioration or damage. Definitions shall be prepared to explain minor, major, and critical defects. Lower maintenance levels shall be included, when appropriate.

f. A classification of defects (i.e., minor, major, or critical) for both functioning and nonfunctioning categories shall be included. The criteria shall conform to the publications noted in subparagraph b. above.

g. Instructions for disposition of lots shall be prepared and shall be as specified by the contracting activity.

5.4.5.8.1.10.2 Pre-embarkation inspection of material in units alerted for overseas movement Pre-embarkation inspection procedures shall be prepared, if applicable, and shall be as specified by the procuring activity.

5.4.5.8.1.11 Placing in service<**pis**>. Instructions shall be prepared for actions not previously noted that may be required for an assembly, component, or end item. Instructions shall be prepared such as removal of

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an item from storage and preparation for installation on an end item. Final servicing checks, calibration, leak checks, charging, pressurizing, and operational checks shall be prepared.

5.4.5.8.1.12 Testing **<test-pass>**. Instructions shall be prepared, as applicable, to test the performance of components, assemblies, and subassemblies prior to installation in the end item. The following instructions are required for depot and aviation maintenance.

a. **(Depot and aviation only)** Instructions shall be prepared for recording the results of the testing. All testing procedures that have a major quality assurance effect shall be prefixed by the boldface letters "**QA**" to identify them.

b. (**Depot only**) Information shall be prepared for final testing of the highest assembly or equipment/end item involved to assure that the parameters of reliability, availability, maintainability, and durability are met. The following procedures shall be prepared.

(1) Inspection. Inspection procedures shall be prepared that are required prior to final testing to assure that the item is complete and ready for final testing. Instructions shall be prepared for any minor preparation tasks needed prior to final testing.

(2) Lubrication. Any final lubrication procedures that need to be done prior to final testing shall be prepared.

(3) Final test procedures. Test procedures, performance standards, and tolerances shall be prepared to establish that the equipment is adequately overhauled and ready for issue without qualifications. The procedures shall list all tools, TMDE, jigs, fixtures, and other support items required for the test. Operating instructions shall be prepared for special test equipment where necessary. Procedures shall be prepared for minor adjustments that can be done without disassembling equipment. Complete procedures shall be prepared for burn-in or run-in tests.

(4) Final painting, refinishing, and marking Procedures shall be prepared for any final painting, refinishing, and marking that could not be done during the overhaul procedures. The materials and tools required to do the job shall be identified.

5.4.5.8.1.13 Quality assurance requirements **(depot only)**. Quality assurance requirements shall be prepared as a supporting information work package. (Refer to MIL-STD-40051-7.) A reference to the work package shall be included here.

5.4.5.8.1.14 Preservation, packaging, and marking**(depot only) <ppm>**. The following instructions shall be prepared.

a. Packaging information. The following statement shall be included verbatim.

"Refer to the packaging segment of the Army Master Data File Retrieval Microform System (AMDF) file for detailed packaging information. If AMDF does not contain the information that you need, you must contact the AMC Packaging, Storage, and Containerization Center, Tobyhanna Army Depot, for packaging information."

b. Special instructions. Instructions shall be prepared for any special or unique preservation, packaging, or marking instructions that apply to the equipment. These instructions shall

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include warnings, cautions, or references concerning ESD, nuclear material, hazardous substances, special marking instructions, or any other instructions required that are not covered in the standard packaging and preservation information.

5.4.5.8.1.15 Overhaul and retirement schedule**(aircraft only) <orsch>**. This maintenance task shall include the following statement and associated list.

"OVERHAUL AND RETIREMENT SCHEDULE

Units of operating equipment that are to be overhauled or retired at the period specified are listed here. Unless otherwise specified in TM 1-1500-328-23, Aeronautical Equipment Maintenance Management Policies and Procedures, removal of equipment for overhaul may be accomplished at the inspection nearest the time when overhaul is due."

5.4.5.8.1.16 Preparation for storage or shipment **<pss>**. As applicable, the following instructions shall be prepared.

a. Security procedures and special transportation requirements for sensitive items (security, terrorists, etc.).

b. Special preservation, packaging, packing, marking, ESD protective and control measures, and shipping instructions, including use of special design reusable containers.

c. Instructions on special use of corrosion-preventive compounds, moisture barriers, and desiccant materials.

d. Directions or reference to instructions for applying special identifying, shipping, and cautionary markings to shipping containers; including security classification, special temperature requirements, and shelf life.

e. Instructions will be provided by the procuring activity for placing equipment in, and for removing it from, administrative storage.

f. Procedures for proper handling, blocking, and bracing of basic load ammunition when being transported in trucks and other tactical vehicles.

g. Basic load storage, quantity-distance, class, and storage compatibility groupings, storage temperatures, stacking limits, and other pertinent storage requirements (for conventional and chemical ammunition only).

h. For aviation ground support equipment, a reference to TM 1-1500-204-23 for general technical information for preparation for storage or shipment.

5.4.5.8.1.17 Classification of defects **<defect>**. Procedures shall be prepared for performing visual inspection of ammunition/containers (pallets, boxes, etc.) and shall include classification and disposition of defective ammunition/containers.

5.4.5.8.1.18 Handling ammunition **<handling>**. Procedures shall be prepared for handling ammunition.

a. Unpacking **<unpacking>**. As a minimum, the following information shall be prepared.

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(1) Any special sequence of action necessary to protect the ammunition.

(2) If a special design reusable container is involved for either the end item or components which are authorized for replacement, instructions shall be prepared to report or reenter the empty container through supply channels.

(3) Man-hour requirements and total man-hours required for unpacking the ammunition.

b. Packing **<packing>**. As a minimum, the following information shall be prepared.

(1) Any special sequence of action necessary to protect the ammunition.

(2) Instructions shall be prepared on how to package defective ammunition.

(3) Man-hour requirements and total man-hours required for packing the ammunition.

5.4.5.8.1.19 Procedures needed to activate ammunition, mine, etc.**<arm>**. Procedures shall be prepared for activation of ammunition, mines, etc.

5.4.5.9 Phased maintenance inspection work package**(aircraft phased maintenance checklist only) <pmi-cklistwp>**. This work package shall include the following information, in a phased maintenance checklist. (Refer to figure 19.)

a. The work package shall begin with the following note:

"NOTE

Prior to start of the Phased Maintenance Inspection, it is recommended that a pre-inspection maintenance test flight (MTF) be conducted. Accomplishment of the MTF shall be determined by the unit maintenance officer. The pre-inspection MTF should be conducted by a maintenance test pilot following a review of the aircraft forms and records and a briefing from the crew of the aircraft. The MTF is recommended to assess the aircraft performance and identify deficiencies that should be corrected while the aircraft is undergoing phased maintenance inspections."

b. General inspection **<geninspec>** items specified by the procuring activity.

c. Aircraft area inspection **<areainspec>** items approved by the procuring activity. List shall be a logical sequence requiring a minimum of time and motion on the part of the individual performing the inspection.

d. Aircraft Power On Checks **<pwron-inspec>** as approved by the procuring activity.

e. Aircraft final inspection **<finalinspec>** requirements as specified by the procuring activity.

5.4.5.10 Preventive maintenance services inspection work package**(aircraft only) <pms-inspecwp>**. All aircraft items requiring inspection and the inspection procedures shall be included in this work package. The checklist data **<pms-form>** shall be included in a table as provided in figure 20 and numbered in sequence (e.g., area 1 and item 3 will appear at 1.3).

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a. The elements to be inspected daily shall be listed under area and sequence number. (See figure 20). Followed by any additional element required for intermediate inspection (when applicable) and then by the additional elements required for periodic inspection. Since the periodic inspection requirements include all intermediate inspection requirements and the intermediate inspection requirements include all daily inspections requirements, an X in the D column will result in an X in the I and P columns. The additional intermediate requirements will likewise be identified by an X in the I column and also result in an X in the P column. When the inspection items are not to be performed during each inspection (I, P), the interval/frequency shall be stated in the applicable column (i.e., 2nd, 3rd, etc.).

b. Power On and Power Off inspection requirements shall be separated so that power will only have to be applied to the aircraft one time during the entire inspection. The Power On items shall be grouped together under the heading "Power On" and shall follow after completion of the last Power Off requirement for the entire aircraft. Power On items shall fall in the proper numerical area and the first item number shall be the next higher number following the last item number of the Power Off requirement.

c. The title or heading of each inspection area, relating to the area diagram (paragraph 5.4.5.10.3) shall begin 1-1/2 inches below last item or procedure of the preceding inspection area.

d. All communications and navigation electrical equipment inspections (black boxes, etc., which are the inspection and repair responsibility of avionics personnel) shall be included in a separate grouping following Power On inspection requirements and shall not appear in the normal inspection sequence.

NOTE

For avionics equipment inspection the following statement shall be included.

"Perform avionics inspections, check and test electrical equipment as required in TM 11-*Insert TM number*). DA Form 2404, Equipment Inspection and Worksheet, shall also be used to record results of inspections by avionics personnel."

5.4.5.10.1 Mandatory safety-of-flight inspection items. Mandatory safety-of-flight inspection items shall be emphasized by using bold type print.

5.4.5.10.2 Work time. Work times for each item in the checklist shall be included in the work time W/T column. The total time for each I and P inspection shall be indicated at top of checklist (figure 20).

5.4.5.10.3 Area diagram. Area diagram of the aircraft, showing sequences for inspection by area shall be included. The area identified shall include all surfaces, material, components and equipment pertaining to that specific location (figure 21). The legend shall identify the title or heading of each numbered area of the diagram.

5.4.5.10.4 Standard checklist statements. The following statements shall be included in the applicable inspection section and in proper sequence.

a. The first item of each inspection for each aircraft shall read: "Inspect aircraft forms and records for recorded discrepancies (DA PAM 738-750, Functional User's Manual for the Army Maintenance Management System (TAMMS))."

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b. One of the last items in each preventive maintenance services manual shall be the lubrication requirements and shall be entered as follows.

LUBRICATION.

All area. Lubricate in accordance with lubrication chart contained in TM 55-(insert TM number)-20 or -23.

c. Include the following statements, as applicable:

(1) Windshields and windows for cleanliness.

(2) Tire for proper pressure (visual), no cuts, blisters or slippage.

(3) Pitot tubes and static ports for freedom of obstructions and cleanliness.

(4) Portable fire extinguisher for broken or missing seal, pressure indicator in green, and extinguisher and brackets secure. Fixed fire extinguisher for accessibility, broken, or missing seal, and extinguisher and bracket for security.

(5) First aid kits for designated location, presence of inspection date tag, condition of seal, and security.

(6) All visible and all accessible electrical wiring for chafing, or damaged insulation and for security of connections.

(7) Battery for leakage and security (nicad and lead acid). Battery compartment interior for cleanliness and corrosion. Battery and connections for security, cleanliness and corrosion. Battery vent for obstructions, proper positioning.

(8) Instrument gage markings for accuracy and legibility; all gage lens for cracks, cleanliness, looseness, and slippage.

(9) Static discharge wicks for overall length of 6 inches maximum and 1 inch minimum exposed wick beyond plastic sheath.

(10) Static discharge ground for condition and security.

d. Include the following statements as applicable in the intermediate inspection requirements. If the intermediate inspection is not required, the inspections listed below shall be included as periodic inspection requirements as determined by the procuring activity.

(1) 2nd - Compass correction card for availability and legibility.

(2) 2nd - Safety belts and shoulder harness for damage, corrosion, cuts, fraying and security.

(3) 2nd - Inertial reels for damage, security, and positive locking and unlocking.

(4) 2nd - Seats for damage, security, and positive movement and locking in all positions.

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[redacted]

(6) 2nd Battery - (nicad and lead acid) for leakage. Battery compartment interior for cleanliness and corrosion. Battery and connections for security, cleanliness and corrosion. Battery vent for obstructions, proper positioning.

e. Include the following statements as applicable in the periodic inspection requirements.

(1) Spark plugs (fine wire), remove and inspect.

(2) Instrument range markings for accuracy and legibility; all gage lens for cracks, cleanliness, looseness, and slippage.

(3) Static discharge wicks for overall length of 6 inches maximum and 1 inch minimum exposed wick beyond plastic sheath.

(4) Static discharge ground for condition and security.

(5) Spark plugs (massive electrode), remove and inspect.

(6) 2nd - Starter brushes for wear (specify) and freedom of movement in brush holder. Brush leads for deterioration or evidence of chafing. Commutator for evidence of arcing and presence of oil or metal particles. Electrical connections for security.

(7) Magneto breaker compartment for cleanliness. Breaker points for pitting, ventilator plugs or screens for cleanliness, and cam wiper for lubrication.

(8) Pitot and static system for absence of foreign material and moisture.

(9) Fixed fire extinguishers for accessibility, broken or missing seal, and extinguisher and bracket for security. Weight check cylinder, less valve.

f. Include the following statement as the last item of the checklist.

"Forms and Records Completion: Ensure that all entries on forms, records and work sheets have been completed or updated and new forms initiated as required (DA PAM 738-750, Functional User's Manual for the Army Maintenance Management System (TAMMS))."

6. **NOTES.**

The notes in section 6 of MIL-STD-40051 apply to this Part.

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TM 9-2350-314-24

CHAPTER 6

UNIT MAINTENANCE INSTRUCTIONS FOR 155MM, M109A6 HOWITZER

FIGURE 1. Example of a maintenance instruction chapter title page

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TM 11-6625-3178-14

FREQUENCY CONVERTER ASSEMBLY REPLACEMENT **1744 00**

THIS WORK PACKAGE COVERS:
Removal, Installation

INITIAL SETUP:

Maintenance Level
Unit

Personnel Required
Maintainers, Two (one to assist)

Tools and Special Tools
Tool Kit, General Mechanics
(Item 37, WP 3201 00)

General Safety Instructions

WARNING

Materials/Parts
Frequency Converter, A31U29270-1

Remove all power from the Frequency Converter circuit.

Frequency converter requires a two person lift.

1744 00-1

FIGURE 2. Example of scope of task and initial setup information

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TM 9-1090-208-23-1-1

SERVICE 30MM AUTOMATIC GUN UPON RECEIPT 0067 00

THIS WORK PACKAGE COVERS:
Unpacking, Cleaning, Inspection

INITIAL SETUP

Tools and Special Tools
Tool Set, NSN 5180-00-987-9816

Personnel Required
Aircraft Armament/Missle Systems Repairer 68J (3) (Two to assist)
Aircraft Armament Technical Inspector 68J3T

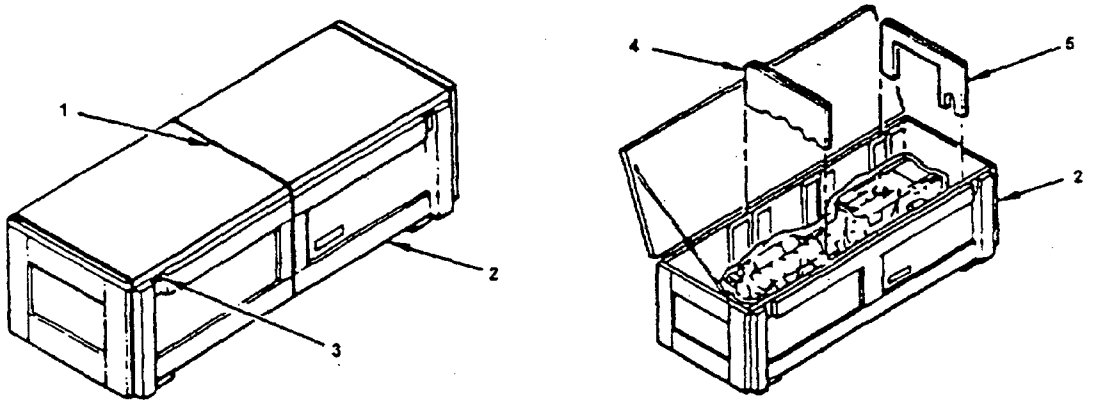
Materials/Parts
Grease (Item 6, WP 0596 00)

WARNING

The gun shipping/storage container weighs approximately 163 pounds, when loaded. Three people are required to lift or move the container.

UNPACKING

1. Cut and remove three straps (1) around grith of storage container (2).
2. Open two link-lock fasteners (3) and lift lid.
3. Remove two cradles (4) and (5). Grasp top of cradle (4) or (5) and lift straight up.



0067 00-1

FIGURE 3. Example of a service upon receipt work package

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TM 9-1090-208-23-1-1

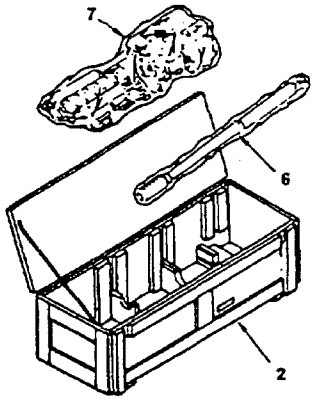
SERVICE 30MM AUTOMATIC GUN UPON RECEIPT - Continued 0067 00

UNPACKING - Continued

4. Remove automatic gun barrel assembly (6) from shipment and st age container (2).

   a. Remove gun barrel (6) from envelope package.

   b. Remove pressure tape and barrier material from gun barrel (6).



WARNING

The gun firing mechanism weighs approximately 76 pounds. Two people are required to remove it from the shipping/storage container.

5. Remove gun firing mechanism (7) from shipping and storage container (2).

   a. Remove gun mechanism (7) from envelope package.

   b. Remove pressure tape and cushioning material from gun mechanism (7).

CLEANING

1. Clean automatic gun firing mechanism (WP 0089 00).

2. Clean automatic gun barrel assembly (WP 0093 00).

INSPECTION

1. Visually check gun firing mechanism for missing or damaged parts.

END OF TASK

0067 00-2

FIGURE 3. Example of a service upon receipt work package - Continued

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TM 9-1410-644-10

SERVICE UPON RECEIPT - Continued 0114 00

Table 2. Inspection Criteria for Packaging.

COMPONENT	ACCEPTABLE	REPARABLE	NONREPARABLE
Wooden Boxes and Crates			
Hardware	Operative and tight. Nails, screws, and fasteners present and in good condition.	Inoperative or loose. Nails, screws, and fasteners that can be replaced or properly sealed.	None None
Ends	Free from damage.	Broken or missing cleats and handles.	Damage that requires disassembly of box.
Wood	Splits less than 3 inches long, no closer than 1 inch to edge of board or adjoining split. The board must be secured by at least one nail on each side of the split when it extends to the end of the board.	Splits more than 3 inches but no closer than 1 inch to edge of board or adjoining split, or 1/2-inch wide, that can be repaired by use of corrugated fasteners.	Splits closer than 1 inch to edge of board or adjoining split or over 1/2-inch wide.
Fiber Containers			
Metal Ends	Minor rust, cracks, indentations, or splits that would not impair water proofing or serviceability of container.	None	Perforations, excessive rust, or ends which are crushed or not securely crimped to body.
Body and Cap	No tears, cuts or gouges.	Cuts, tears, or gouges not closer than 1 inch to closure, less than 1/2 square inch in area, and unpenetrated layers that can be spot painted.	Cuts, tears, or gouges closer than 1 inch to closure, more than 1/2 square inch in area, or through all impregnated layers.

0114 00-2

FIGURE 4. Example of packaging inspection/service upon receipt

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Table 3. M29 and M30 Control Surfaces and Containers.

LOCATION	ITEM	ACTION	REMARKS
1. Container	Components	1. Inspect for rust, fungus, paint damage, and deformation.	WP 0125 00
		2. Reject container if damage prevents it from functioning properly.	--
2. M29	Control Surfaces	1. Inspect for dents and scratches on post, trailing edge phenolic, skin, and closure plate.	WP 0128 00
		2. Reject control surface:	--
		a. If post dents or scratches exceed 0.002 in. (0.051 mm).	
		b. If trailing edge phenolic dents exceed 0.040 in. (10.160 mm).	
		c. If skin dents exceed 0.030 in. (7.620 mm) within 2 in. (50.800 mm) of post.	
		d. If closure plate dents exceed 0.030 in. (7.620 mm) within 2 in. (50.800 mm) of post.	
3. M30	Control Surfaces	1. Inspect for dents and scratches on post and skin.	--
		2. Skin dents or scratches up to 0.050 in. (12.700 mm) are allowable, but should be blended.	--
		3. Reject control surface if post dents or scratches exceed 0.002 in. (0.051 mm).	--

FIGURE 4. Example of packaging inspection/service upon receipt - Continued

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TM 1-4220-250-12&P

DONNING AND FITTING THE SRU-37/P **1324 00**

THIS WORK PACKAGE COVERS:
Donning, Fitting

INITIAL SETUP:

Maintenance Level
Aviation Unit

GENERAL

The container assembly is designed to be worn on the user's chest, upper back, or lower back. If the SARVIP is to be worn, it shall be donned first. The SRU-37/P shall then be donned over the SARVIP.

CHEST MOUNTING OPTION.

WARNING

FLIGHT CONTROL RESTRICTION

When the SRU-37/P is worn on the chest, certain size individuals in certain helicopters may experience aft cyclic control restriction.

1. Position the SRU-37/P across the chest and place both shoulder straps over the shoulders and back. The right shoulder strap with male buckle shall be passed under the left armpit and securely inserted into the female buckle of the left cross strap. The left shoulder strap with male buckle shall be passed under the right armpit and securely inserted into the female buckle of the right cross strap. Ensure that both buckle assemblies are securely fastened. This is essential for the prevention of inadvertent release while performing aircrew duties or during emergencies.
2. Adjust the straps, taking up the slack until snug against the body. Be sure to allow for unobstructed access to the beaded inflation handle.

HIGH BACK MOUNTING OPTION.

WARNING

When the SRU-37/P is worn in the high back position, the straps must be loosened to allow the aviator/crewmembers to reach the beaded activation handle.

1. Place the SRU-37/P across the top of the back (between shoulder blades). Place the shoulder straps over the chest and attach buckles in the same manner as step 1, above - right to left and left to right. Ensure that the buckle assemblies are securelyfastened.
2. Adjust the straps, taking up the slack until snug against the body.

END OF TASK

1324 00-1

FIGURE 5. Example of an equipment/user fitting instruction work package

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TM 3-6665-339-10

NBCRS FOX M93A1 PREVENTIVE MAINTENANCE CHECKS AND SERVICES 0062 00

THIS WORK PACKAGE COVERS:

Introduction, PMCS Procedures

INITIAL SETUP:

Maintenance Level
Operator

INTRODUCTION

General

Preventive Maintenance Checks and Services (PMCS) are performed to keep the vehicle in operating condition. The checks are used to find, correct, or report problems. Crewmembers are to do the PMCS jobs as shown in the PMCS table. PMCS are done every day the vehicle is operated, using the PMCS table. Pat attention to WARNING and CAUTION statements. A WARNING means someone could be hurt. A CAUTION means equipment could be damaged.

Before you begin operating vehicle equipment, do Before PMCS.

During operation, do During PMCS.

After operation, do After PMCS.

Once a week do Weekly PMCS. If vehicle has not been operated in a week, also do Before PMCS at the same time.

Do Monthly PMCS once a month. If vehicle has not been operated in a month, also do After PMCS at the same time.

If you are operating the vehicle for the first time, do your Weekly and Monthly PMCS the first time you do your Before PMCS.

If you find something wrong when performing PMCS, fix it if you can, using troubleshooting procedures and/or maintenance procedures.

The right-hand column of the PMCS table lists conditions that make the vehicle not fully mission capable. Write up items not fixed on DA Form 2404 for unit maintenance. For further information on how to use this form, see DA PAM 738-750.

If tools required to perform PMCS are not listed in WP 1479 00, notify unit maintenance.

Leakage Definition

CAUTION

Equipment operation is allowable with minor leakages (Class I or II) except for fuel leaks. Of course, consideration must be given to the fluid capacity of the item or system being checked. When in doubt, ask your supervisor.

When operating with Class I or II leaks, continue to check fluid levels as required in your PMCS.

Class III leaks should be reported immediately to your supervisor.

FIGURE 6. Example of a PMCS introduction

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TM 3-6665-339-10

NBCRS FOX M93A1 PREVENTIVE MAINTENANCE CHECKS AND SERVICES - Continued **0062 00**

INTRODUCTION - Continued

It is necessary to know how fluid leakage affects the status of the vehicle. The following are definitions of the classes of leakage an operator or crewmember needs to know to be able to determine the condition of the leak. Learn and then be familiar with them, and REMEMBER - WHEN IN DOUBT, ASK YOUR SUPERVISOR.

Leakage Definitions for Crew/Operator PMCS.

CLASS I - Seepage of fluid (as indicated by wetness or discoloration) not great enough to form drops.

CLASS II - Leakage of fluid great enough to form drops but not enough to cause drops to drip from item being checked.

CLASS III - Leakage of fluid great enough to form drops that fall from the item being checked.

INSPECTION

Look for signs of a problem or trouble. Senses help here. You can feel, smell, hear, or see many problems. Be alert when on the vehicle.

Inspect to see if items are in good condition. Are they correctly assembled, stowed, secured, excessively worn, leaking, corroded, or properly lubricated? Correct any problems found or notify unit maintenance.

There are some common items to check all over the vehicle. These include the following:

1. Bolts, clamps, nuts, and screws: Continuously check for looseness. Look for chipped paint, bear metal, rust, or corrosion around bolt and screw heads and nuts. Tighten them when you find them loose. If tools are not available, notify unit maintenance.
2. Welds: Many items on the vehicle are welded. To check these welds, look for chipped paint, rust, corrosion, or gaps. When these conditions exist, notify unit maintenance on DA Form 2404
3. Electrical wires, connectors, and harnesses: Tighten loose connectors. Look for cracked or broken insulation, bare wires, and broken connectors. If any are found, notify unit maintenance.
4. Hoses and fluid lines: Look for wear, damage and leaks, and make sure clamps and fittings are tight. Wet spots mean a leak. A stain by a fitting or connector can also mean a leak. When you find a leak, notify unit maintenance.

LUBRICATION SERVICE INTERVALS - NORMAL CONDITIONS

For safer, more trouble free operations, make sure that your vehicle is serviced when it needs it. For the proper lubrication and service intervals, see LO 9-6665-376-12.

LUBRICATION SERVICE INTERVALS - UNUSUAL CONDITIONS

Your vehicle will require extra service and care when you operate under unusual conditions. High or low temperatures, long periods of hard use, or continued use in sand, water, mud, or snow will break down the lubricant, requiring you to add or change lubricant more often.

CLEANING AND LUBRICATION

Proper cleaning and lubrication can aid in avoiding possible problems or trouble, so make it a habit to do the following:

CAUTION

Follow all cleaning and lubricating instructions carefully. Failure to do so can result in damage to equipment.

1. Expose canvas covers and tarpaulins to fresh air during semiannual service. Reduce interval as required in a rainy climate. Do not stow canvas items when wet.

0062 00-2

FIGURE 6. Example of a PMCS introduction - Continued.

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TM 3-6665-330-10

NBCRS FOX M93A1 PREVENTIVE MAINTENANCE CHECKS AND SERVICES 0132 00

THIS WORK PACKAGE COVERS:
PMCS Checks and Services

INITIAL SETUP:

Maintenance Level
Operator

Table 1. Preventive Maintenance Checks and Services for NBCRS FOX M93A1.

ITEM NO.	INTERVAL	MAN-HOUR	ITEM TO BE CHECKED OR SERVICED	PROCEDURE	EQUIPMENT NOT READY/ AVAILABLE IF:
				NOTE Driver, commander and crew will inspect for damaged or missing items while performing checks in walkaround sequence.	
1	Before		Vehicle Exterior	COMMANDER Check for damaged and missing items. Check lighting fixtures. Make sure all stowed items (pioneer equipment, etc.) are secured to vehicle for travel. Secure stowed items. Replace missing items. Report major damage of lighting fixtures. Report minor damage after operations.	Any damage or missing items that will prevent operation
2	Before		Left Side Exterior	DRIVER **NOTE** If leakage is detected, further investigation is needed to determine the location and cause of the leak. 1. Check underneath vehicle for evidence of fluid leakage.	Any class III leak of oil, fuel, or coolant.
3	Before		Left Side Tires	DRIVER **WARNING** Operating a vehicle with an improperly inflated tire or with a questionable defect may lead to premature tire failure and cause equipment damage, injury, or death to personnel. Check for missing or damaged tires and wheels. Visually check for proper tire inflation.	Missing or damaged tires or wheels. Flat or deflated tires.

0132 00-1

FIGURE 7. Example of a PMCS table format.

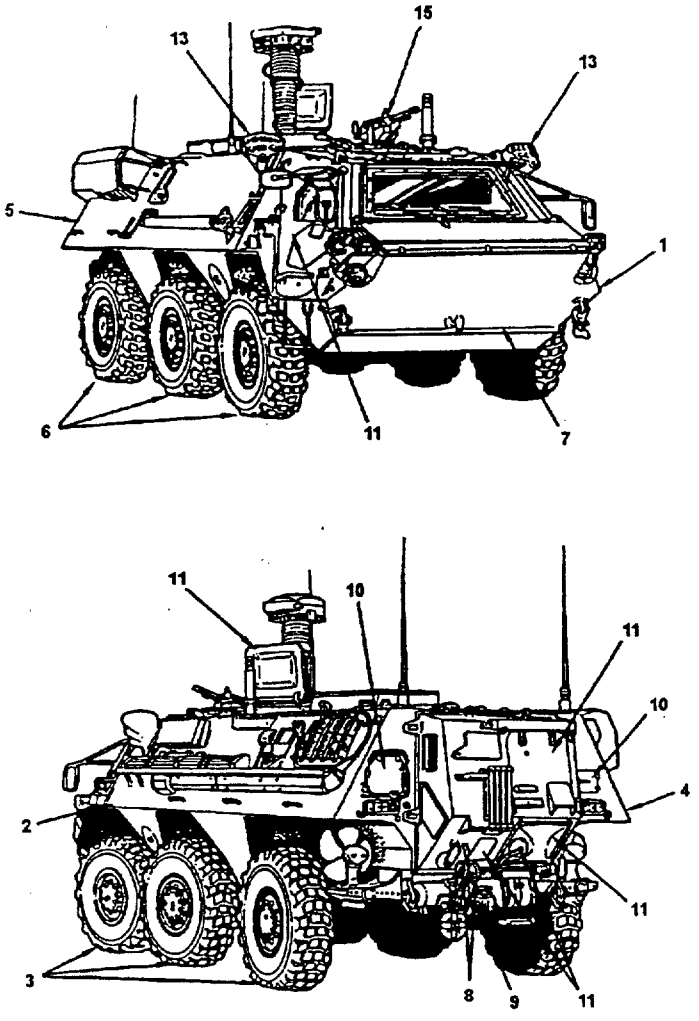
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TM 3-6665-330-10

NBCRS FOX M93A1 PREVENTIVE MAINTENANCE CHECKS AND SERVICES - Continued **0132 00**

Before "Vehicle Exterior Walkaround" PMCS Procedures

These illustrations will help you perform BEFORE vehicle exterior walkaround PMCS. The callouts match PMCS item number/procedures.



0132 00-2

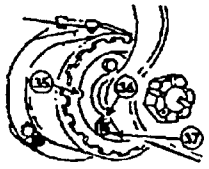
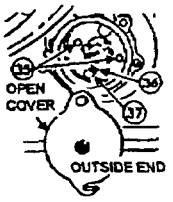
FIGURE 7. Example of a PMCS table format - Continued

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Check the source to verify that this is the current version before use.

TM-9-1015-256-23

M198 HOWITZER PREVENTIVE MAINTENANCE CHECKS AND SERVICES, INCLUDING LUBRICATION INSTRUCTIONS - Continued 0632 00

Table 1. Preventive Maintenance Checks and Services for (XXX) - Continued

ITEM NO.	INTERVAL	MAN-HOUR	ITEM TO BE CHECKED OR SERVICED	CREWMEMBER PROCEDURE	EQUIPMENT NOT READY/ AVAILABLE IF:
7	Before		Recuperator	CREWMEMBER 1 **WARNING** Do not remove cap (37) when engine is hot 1. Check pins (35) before firing. Pins must not extend over 3/4 inch (19.1mm); 1/8 inch (3.2mm) is correct. If pins (35) do extend beyond 3/4 inch, remove cap (37) from valve (36) and add hydraulic fluid (item 5, WP 0746 00) until pins extend 1/8 inch. **NOTE** The recuperator is charged with a pressure of 550 psi (38.5 kg/cm²) in the M109 and M109A1. With temperature of 100° F or over, 600 psi (42.0 kg/cm²) in the M109A1 is recommended. RECUPERATOR: INSIDE END  	Pins (35) extend over 3/4 inch

0632 00-2

FIGURE 8. Example of a PMCS table format with crew member grouping

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SEMI-ANNUAL (1500 MILE) PMCS MANDATORY REPLACEMENT PARTS LIST (AO/A1)

ITEM NO.	PART NUMBER	NSN	NOMENCLATURE	QTY
1	D5-19-2353	4240-01-026-3112	PRECLEANER AND PART (A1 ONLY)	01
2	MS24665-285	5315-01-061-2060	PIN, COTTER	01
3	MS35333-42	5310-00-595-7237	WASHER, LOCK	03
4	MS35338-43	5310-00-045-3296	WASHER, LOCK	01
5	MS35338-44	5310-00-582-5965	WASHER, LOCK	16
6	MS35338-46	5310-00-004-5033	WASHER, LOCK	09
7	MS51922-1	5310-00-088-1251	NUT, SELF-LOCKING	04
8	MS51922-17	5310-00-087-4652	NUT, SELF LOCKING	17
9	11628247	5330-01-109-1925	GASKET	02
10	12294872	5310-01-107-3356	WASHER, FLAT	02
11	2585163-57	5306-00-163-2850	BOLT, SELF-LOCKING	01

ANNUAL (3000 MILE) PMCS MANDATORY REPLACEMENT PARTS LIST (A0/A1)

ITEM NO.	PART NUMBER	NSN	NOMENCLATURE	QTY
1	D5-19-2353	4240-01-026-3112	PRECLEANER AND PART (A1 ONLY)	01
2	MS24665-283	5315-00-842-3044	PIN, COTTER	02
3	MS24665-285	5315-01-061-2060	PIN, COTTER	01
4	MS35333-42	5310-00-595-7237	WASHER, LOCK	03
5	MS35338-43	5310-00-045-3296	WASHER, LOCK	01
6	MS35338-44	5310-00-582-5965	WASHER, LOCK	16
7	MS35338-46	5310-00-004-5033	WASHER, LOCK	09
8	MS51922-1	5310-00-088-1251	NUT, SELF-LOCKING	08
9	MS51922-13	5310-00-984-3807	NUT, SELF-LOCKING	02
10	MS51922-17	5310-00-087-4652	NUT, SELF LOCKING	17
11	MS51922-21	5310-00-959-1488	NUT, SELF-LOCKING	01
12	MS51922-5	5310-09-959-7600	NUT, SELF, LOCKING	01
13	11628247	5310-01-109-1925	GASKET	02
14	11664669	2910-00-152-2033	FILTER ELEMENT, FLUID	01
15	12294872	5310-01-107-3356	WASHER, FLAT	02
16	254608S	5330-01-050-2624	GASKET	02
17	255815S	5330-01-328-9920	GASKET	02
18	2585163-57	5306-00-163-2850	BOLT, SELF-LOCKING	01
19	3302180S	5330-01-328-9921	GASKET	02

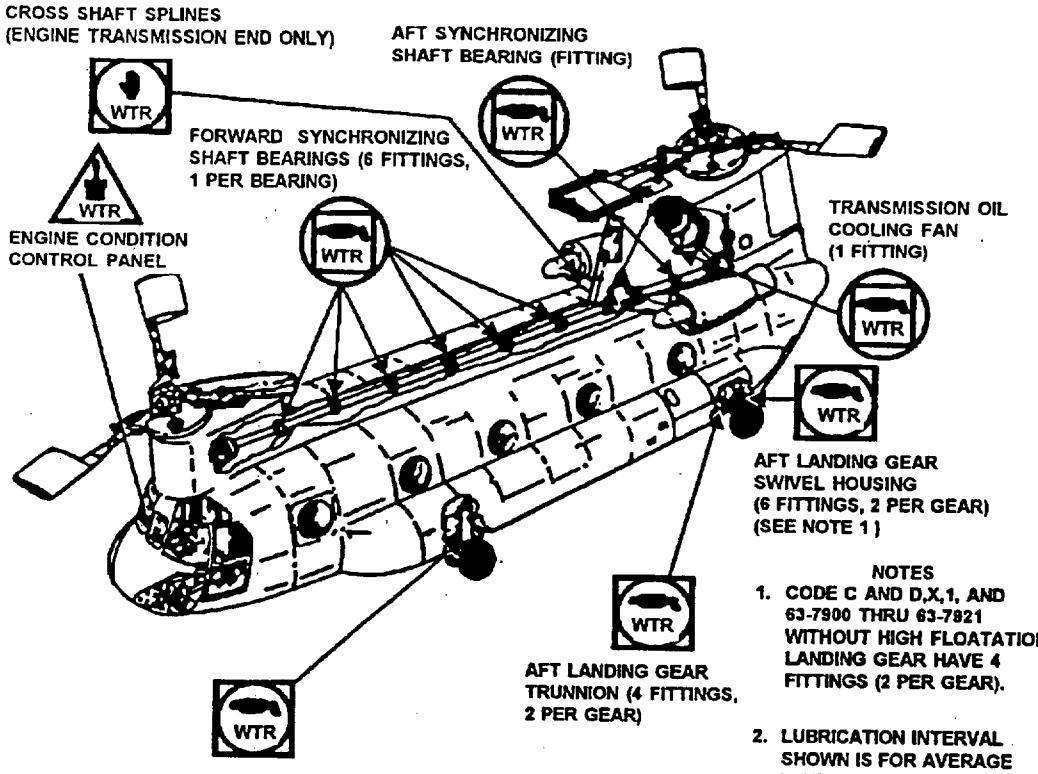
FIGURE 9. Example of a PMCS mandatory replacement parts list.

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TM 55-1520-238-23

AIRCRAFT LUBRICATION INSTRUCTIONS -Continued 0796 00

LUBRICATION INSTRUCTIONS - Continued



NOTES

1. CODE C AND D,X,1, AND 63-7900 THRU 63-7921 WITHOUT HIGH FLOATATION LANDING GEAR HAVE 4 FITTINGS (2 PER GEAR).
2. LUBRICATION INTERVAL SHOWN IS FOR AVERAGE CONDITIONS. MORE FREQUENT LUBRICATION MAY BE NECESSARY UNDER EXTREMELY DUSTY CONDITIONS.

TABLE OF LUBRICANTS		
IDENTIFICATION LETTER	SPECIFICATION	TYPE OF LUBRICANT
WTR	MIL-G-81322B (C221)	GREASE, AIRCRAFT, GENERAL PURPOSE, WIDE TEMPERATURE RANGE

SYMBOLS	FREQUENCY	METHOD OF APPLICATION
	25 HOURS	GREASE GUN
	100 HOURS	BRUSH
	500 HOURS	HAND

0796 00-3

FIGURE 10. Example of a lubrication chart

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TM 9-2350-314-10

155 MM, M284 HOWITZER CANNON AMMUNITION MAINTENANCE **1567 00**

THIS WORK PACKAGE COVERS:
Care, Handling, Storage, Inspection, and Cleaning

INITIAL SETUP:

Maintenance Level	**Materials/Parts**
Operator	Cloth, wiping (item 31, WP 0479 00)

CARE

1. Ammunition is packaged to withstand conditions ordinarily encountered in the field. Keep packing boxes from becoming broken or damaged.

2. Since ammunition is impaired by moisture, frost, extreme temperature, and foreign matter (mud, oil, etc.), observe the following:

   a. Do not break the moisture-resistant seal on the container until ammunition is to be used.

   b. Shield all ammunition from high temperatures (e.g., the direct rays of the sun). When covering projectiles to provide this shield, cover with tarpaulin keeping 18 in. (45.7 cm) air space over and 6 in. (15.2 cm) of air space on sides. This will ensure free air flow necessary to keep the projectiles cooler in hot weather.

   c. Refer tp WP 1992 00 for information regarding maintenance of the M712 Copperhead (HEAT) projectile.

   d. Store M110 projectile in an upright position at all times.

HANDLING

WARNING

Keep fire and flammable materials out of the ammunition areas. There will be no smoking in the vicinity of ammunition.

Shield all ammunition from high temperatures (e.g., direct rays of the sun). When outside temperature is expected to reach +100°F (+38°C), failure to shade could result in damage to material and loss of life.

1. Do not expose ammunition and components containing explosives to extreme temperatures. Do not expose to direct sunlight, flame, or other sources of heat.

2. Do not expose unpacked propelling charges and fuzes to rain, excessive humidity, or ground moisture.

1567 00-1

FIGURE 11. Example of an ammunition maintenance work package

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:04Z
Check the source to verify that this is the current version before use.

TM 9-2350-314-10

155 MM, M284 HOWITZER CANNON AMMUNITION MAINTENANCE - Continued 1567 00

HANDLING - Continued

3. Prevent rough handling of projectiles and fuzes. Do not strike projectiles together and do not off-load ammunition by dropping projectiles on top of projectiles.

4. Protect fuzes, primers, and flash reducers at all times from foreign matter and impact. A drop of 4 feet (1.20 m) may cause the electrolyte vital in a VT fuze battery to break, thus causing a dud.

5. Do not disassemble fuzes.

STORAGE

WARNING

Ammunition exposed directly to sunlight, or in unventilated containers, enclosures, shelters, freight cars, closed vehicles, and similar structures exposed to direct sunlight may reach temperatures exceeding upper storage limits. Avoid exposure of ammunition components to direct sunlight. Do not store ammunition assembled with tetrytol-loaded bursters (i.e., projectiles, 155mm; smoke, WP, M110; gas H, and HD, persistent, M110) at temperatures exceeding +125°F (+52°C)

Temperature Limits

1. Except as otherwise specified, observe the following limits:

   a. Lower limit is -80°F (-62°C) for a period of not more than 3 days.

   b. Upper limit is +160°F (+71°C) for a period of not more than 4 hours per day.

2. Store or transport M110 projectiles containing WP at a temperature below the melting point (+111.4 °F (+44°C) of the WP filler. If this is not practicable, store or transport such projectiles on their bases so that, should the WP filler melt, it will resolidify with the void in the nose of the projectile. This restriction does not apply to the M825/M825A1 WP projectile.

3. Protect proximity fuzes and proximity rounds from long exposure to high humidity. Store M728 and M732 fuzes in temperatures between -65 °F to +145°F (-54°C to +63°C) and 0°F to +120°F (-18°C to +49°C) for the M514 series.

1567 00-2

FIGURE 11. Example of an ammunition maintenance work package - Continued

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Check the source to verify that this is the current version before use.

TM 9-1005-319-23&P

BLANK FIRING ATTACHMENT (M15A2) MAINTENANCE **0062 00**

THIS WORK PACKAGE COVERS:

Removal, cleaning, inspection, painting, replacement, and installation

INITIAL SETUP:

Maintenance Level
Unit
Materials/Parts
Cleaner, lubricant, and preservative (CLP) (Item 9, WP 1132 00)
Coating compound, enamel (Red-M16A2) (Item 23, WP 1132 00)

General Safety Instructions
Do not keep live ammunition near the work area.
Only blank cartridge M200 is to be used when the blank firing attachment is attached to the weapon.

GENERAL

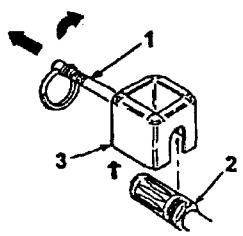
This work package contains information and instructions to keep auxiliary equipment used with your weapon in good repair.

REMOVAL

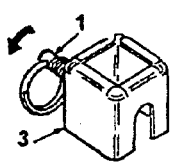
CAUTION

Do not use tools to tighten the blank firing attachment.
USE HANDS ONLY.

1. Unscrew slide (1) to remove from compensator (2).
2. Unhook blank firing attachment (3) from behind the first groove of compensator (2).



3. Screw slide (1) all the way in on blank firing attachment (3).



0062 00-1

FIGURE 12. Example of an auxiliary equipment maintenance work package

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TM 9-1005-319-23&P

BLANK FIRING ATTACHMENT (M15A2) MAINTENANCE - Continued 0062 00

CLEANING

1. Clean blank firing attachment with CLP, wipe dry, and coat with CLP.

INSPECTION

1. Inspect blank firing attachment for cracks or distortion. Be sure the parts in the slide are clear and clean. If blank firing attachment is cracked or distorted, it is unserviceable.

PAINTING

1. Repaint blank firing attachment using enamel coating compound (Red for M16A2 rifle or Yellow for M4/M4A1 carbine). Painting is the only repair authorized.

REPLACEMENT

1. Replace blank firing attachment if unserviceable.

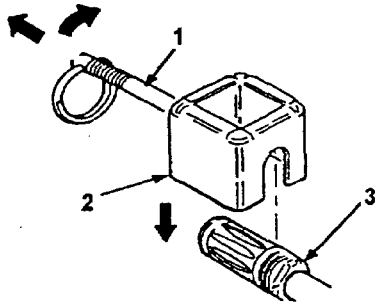
INSTALLATION

NOTE

M23 blank firing attachment is stamped "M4 Carbine Only" painted yellow and may be used on the M4 and M4A1 carbines. M15A2 BFA is painted red and is used on the M16A2 rifle.

1. Unscrew and pull slide (1) all the way out on blank firing attachment (2).

2. Hook blank firing attachment (2) behind the first groove of the compensator (3).



END OF TASK

0062 00-2

FIGURE 12. Example of an auxiliary equipment maintenance work package - Continued

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TM 11-6625-3178-14

CHEMICAL ALARM MOUNT ASSEMBLY REPLACEMENT 0267 00

THIS WORK PACKAGE COVERS:
Removal, Installation

INITIAL SETUP

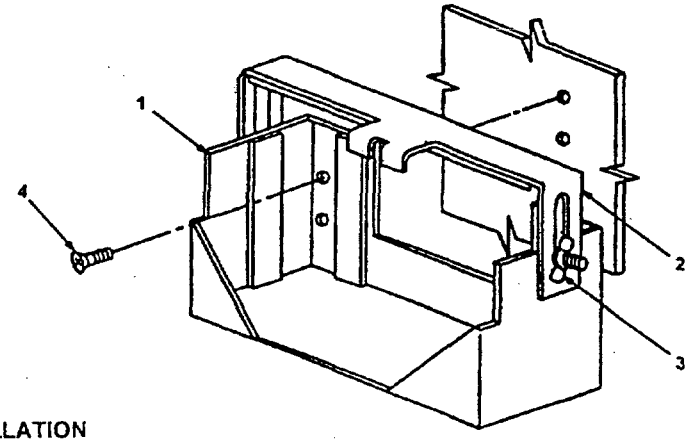
Maintenance Level
Unit

Materials/Parts
Bracket Assembly,Chemical Alarm, A3070119

Tools and Special Tools
Tools Kit, Electrical NSN 5180-01-195-0855

REMOVAL

1. On chemical alarm mount assembly (1) loosen two wing nuts (3) holding clamp assembly (2) and rotate clamp assembly forward.
2. Remove chemical alarm and set aside.
3. Remove four screws (4) and remove chemical alarm mount assembly from shelter wall.



INSTALLATION

1. Position new chemical alarm mount assembly (1) on shelter wall and secure with four screws (4).
2. Replace chemical alarm set aside in removal step 2.
3. Rotate clamp assembly (2) back to upright position and secure by tightening two wing nuts (3).

END OF TASK

0267 00-1

FIGURE 13. Example of a maintenance work package

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Requirements for XXX System

System	MTBF	MTR	A_o
Track	500 mi	30 min	0.89
Engine	170 hr	43 min	0.92
Hull	1,000 mi	80 min	0.86
Radio	400 hr	10 min	0.95
Night Sight	145 hr	10 min	0.88
Gun Tube	10,000 rds	45 min	0.95

Maintenance Ratio for XXX System 0.35

Unit	DS	GS	Depot
0.10	0.05	0.08	0.07

EXAMPLE OF TABULAR RAM DATA

Requirements for XXX System

Maintainability

When maintenance procedures shown in the technical manuals are followed, the mature maintainability data are as follows:

1. Mean Operator Preventive Maintenance Time shall not exceed 0.25 man-hours per mission. This time shall not be included in organizational preventive maintenance time.
2. Maximum operator Corrective Maintenance Time shall not exceed 1.00 man-hours per mission without being classified as a mission failure.
3. The ratio of total corrective and organizational preventive maintenance man-hours to operating hours shall not exceed 0.10.
4. The ratio of total organizational preventive maintenance man-hours to total operating hours shall not exceed 0.04.
5. The ratio of total corrective maintenance man-hours to operating hours shall not exceed 0.06.
6. Mean man-hours to perform a corrective maintenance action shall not exceed 2.5.
7. The Mean Time Between Corrective Maintenance Actions shall not be less than 150 operating hours.
8. The engine shall have an 80 percent probability of not requiring replacement in 20,000 miles of operation.
9. The gun tube shall have an 80 percent probability of not requiring replacement in 50,000 rounds of operation.
10. The track shall have a 92 percent probability of not requiring replacement in 5,450 miles of operation.

EXAMPLE OF NARRATIVE RAM DATA

FIGURE 14. Example of tabular and narrative reliability, availability, and maintainability data

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Check the source to verify that this is the current version before use.

PRESHOP ANALYSIS FOR P/N _ _ _ _ _

Serial No. _ _ _ _ _ _ _

NSN _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _

MWOs Required _

Reason(s) for Overhaul/Repair _____

_____

_____

_____

_____

Unpacking Secondary Items Required ? _ _ _ _

Reviewed Tags ? _ _ _ _

Reviewed Forms ? _ _ _

Name (please print) _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _

Signature _ _ _ _ _ _ _ _ _ _ _ _ Date _ _ _ _ _ _

FIGURE 15. Example of a cover sheet for preshop analysis checklist

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Table 6. Preshop Analysis Checklist.

INSPECTION POINT	CONDITION	ACTION	REMARKS	DONE BY
GENERAL				
Gear Box	Reason(s) for removal from service.	Record:		
	Modification only.	WP 0725 00		
	Burned.	WP 0725 00		
	Crash damage, accident damage, hard landing, sudden stoppage.	WP 0725 00		
	Metal on magnetic plug, contamination, internal failure.	WP 0725 00		
	Dropped.	WP 0725 00		
	No records or historical data.	WP 0725 00		
EXTERIOR INSPECTION				
Case Assembly	Cracks, corrosion, nicks, dents, scratches, pits, broken studs.	Visual, OIP-16		
Output Sleeve	Cracks, corrosion, nicks, dents, burrs, scratches.	Visual, OIP-22		
Output Shaft	Cracks, nicks, dents, or pits on spline; wear over pin dimension of spline not within allowable limits; corrosion.	Visual, OIP-29		
Input Sleeve	Cracks.	Visual, OIP-1		

FIGURE 16. Example of a preshop analysis checklist, table of tests and inspections

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Table 2. Overhaul Inspection Procedures for Gear (Item 15, fig. 21).

QA REQ	NO.	REF LTR	CHARACTERISTIC	INSP METHOD	REQUISITE
	1		Serviceability	Visual	No cracks, chipped grooves, or missing gear teeth
Yes	2		Metal fatigue	Magnetic Particle Inspection	No cracks
Yes	3	A	Tooth wear	Measure with micrometer (599-30-11) over two 0.1920 in. (4.877mm) gage pins	2.256 in. (57.30 mm) min.

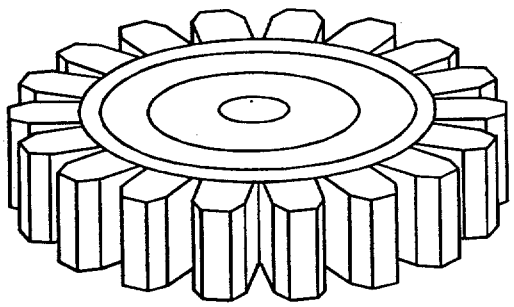


FIGURE 17. Example of an OIP table.

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Table 2. Classification of Materiel Defects.

CATEGORIES OF DEFECTS/COMPONENTS AND DEFECTS	WP NO. OR TM WITH CORRECTIVE ACTION	INSPECTION METHODS AFTER CORRECTIVE ACTION	ACCEPTABLE QUALITY LEVEL
CRITICAL:			
1. Fuze not set on SAFE.	WP 0120 00	Visual	Fuze set on SAFE.
2. Fuze well liner missing.	WP 0120 00	Visual	Fuze well liner in place.
MINOR:			
1. Fuze stake missing.	WP 0120 00	Visual	Fuze stake replaced.
2. Supplementary charge spacer missing.	WP 0120 00	Visual	Supplementary charge spacer replaced.
3. Supplementary charge damaged.	WP 0120 00	Visual	Supplementary charge replaced.
4. Explosive on fuze well threads.	WP 0120 00	Visual	Fuze well threads without caked explosive.
5. Shear or twist pin above flush.	WP 0120 00	Visual	Shear or twist pin flush.

FIGURE 18. Example of a classification of material defects table

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:04Z
Check the source to verify that this is the current version before use.

PHASE NO. _____ PHASED MAINTENANCE CHECKLIST

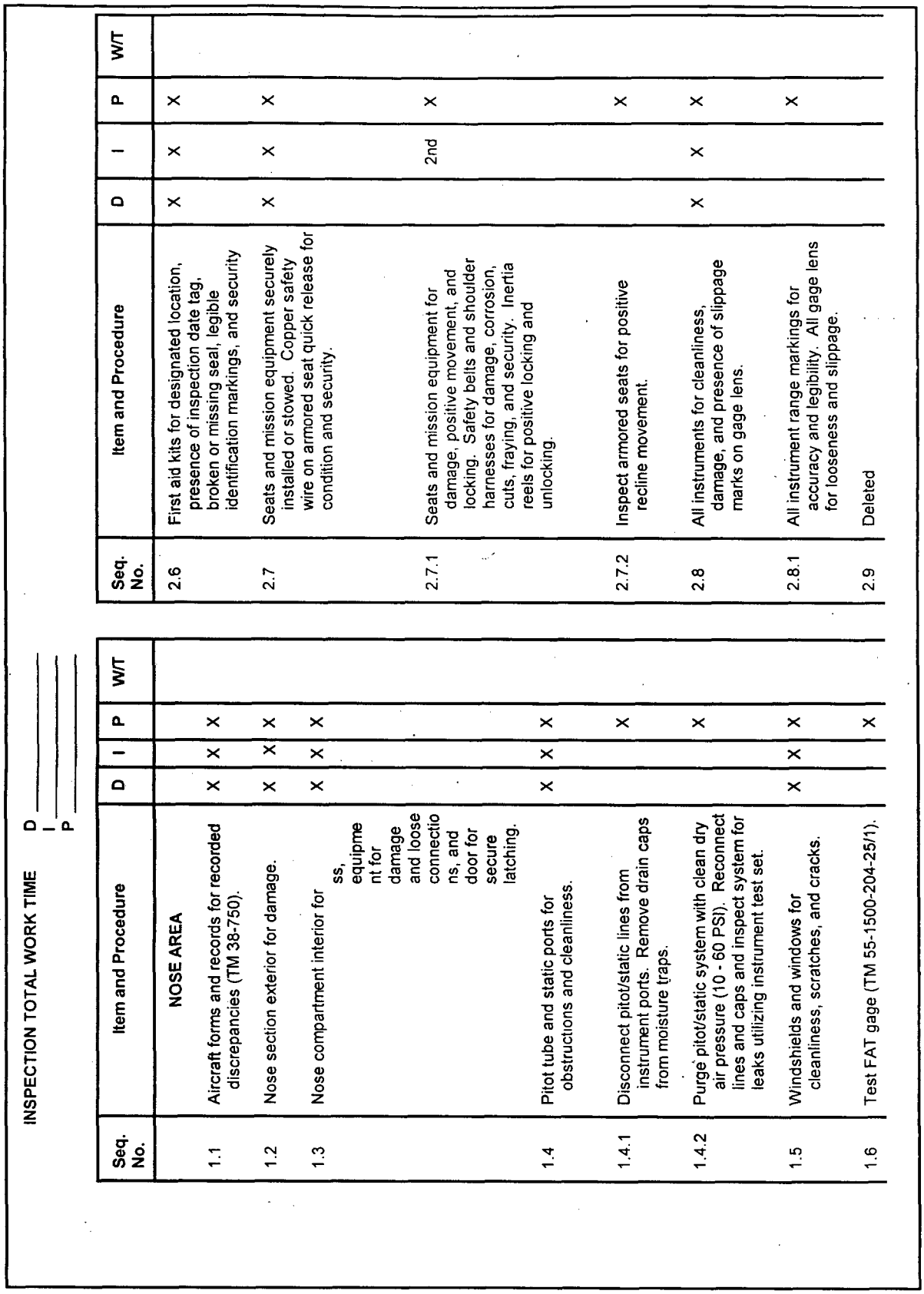
Area Name and No. GENERAL		Aircraft Serial No.		Date:	Total Hrs. This Area	
Inspect Phase Nos	Inspection Requirements	Status	Faults and/or Remarks	Action Taken		Initial
ALL	1. Prior to inspection, check forms and records for recorded deficiencies (Table 1-2).					
ALL	2. Fuel tanks shall be fully serviced prior to start of phased inspection. If maintenance is to be accomplished which requires refueling, this item may be deferred until after such maintenance is completed.					
ALL	3. Perform avionics inspections systems inspection checks and tests as applicable in avionics maintenance manual.					
ALL	4. Perform armament systems inspection checks and tests as applicable in armament maintenance manuals.					

"FOD REMINDER"

Check work area for tools and parts after completion of maintenance and inspections.

FIGURE 19. Example of a phased maintenance checklist

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Check the source to verify that this is the current version before use.

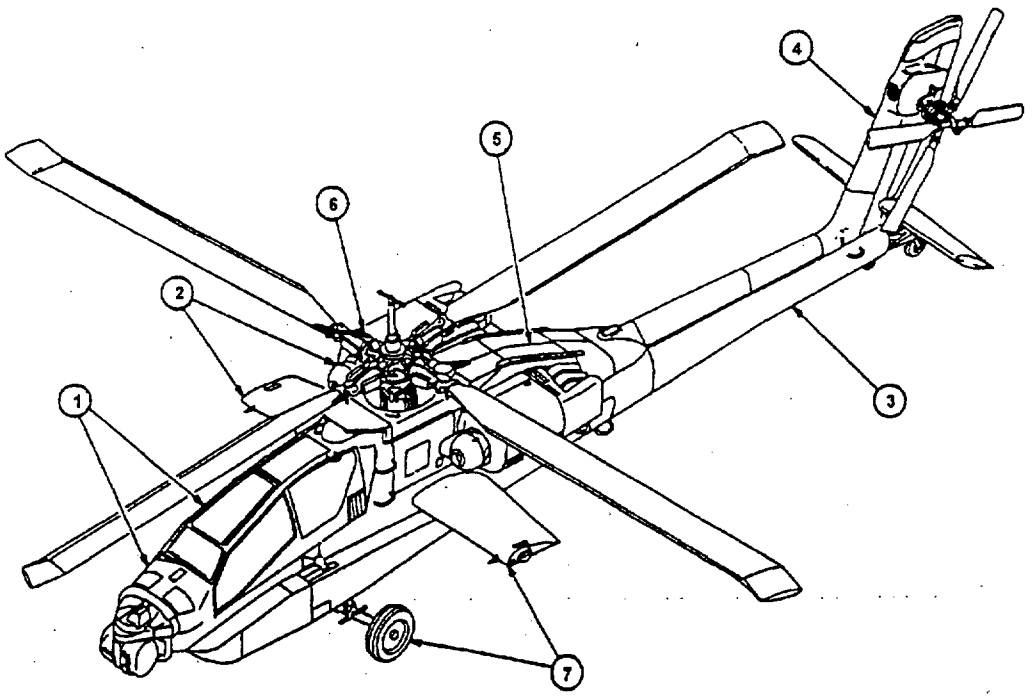


INSPECTION TOTAL WORK TIME D _____ I _____ P _____

Seq. No.	Item and Procedure	D	I	P	W/T
	NOSE AREA				
1.1	Aircraft forms and records for recorded discrepancies (TM 38-750).	X	X	X	
1.2	Nose section exterior for damage.	X	X	X	
1.3	Nose compartment interior for ss, equipment for damage and loose connections, and door for secure latching.	X	X	X	
1.4	Pitot tube and static ports for obstructions and cleanliness.	X	X	X	
1.4.1	Disconnect pitot/static lines from instrument ports. Remove drain caps from moisture traps.			X	
1.4.2	Purge pitot/static system with clean dry air pressure (10 - 60 PSI). Reconnect lines and caps and inspect system for leaks utilizing instrument test set.			X	
1.5	Windshields and windows for cleanliness, scratches, and cracks.	X	X	X	
1.6	Test FAT gage (TM 55-1500-204-25/1).			X	
2.6	First aid kits for designated location, presence of inspection date tag, broken or missing seal, legible identification markings, and security	X	X	X	
2.7	Seats and mission equipment securely installed or stowed. Copper safety wire on armored seat quick release for condition and security.	X	X	X	
2.7.1	Seats and mission equipment for damage, positive movement, and locking. Safety belts and shoulder harnesses for damage, corrosion, cuts, fraying, and security. Inertia reels for positive locking and unlocking.		2nd	X	
2.7.2	Inspect armored seats for positive recline movement.			X	
2.8	All instruments for cleanliness, damage, and presence of slippage marks on gage lens.	X	X	X	
2.8.1	All instrument range markings for accuracy and legibility. All gage lens for looseness and slippage.			X	
2.9	Deleted				

FIGURE 19. Example of a phased maintenance checklist

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AREA NO. 1	CANOPY, CREW STATION, AVIONICS COMPARTMENTS, FOWARD FUSELAGE
AREA NO. 2	RIGHT FUSELAGE, WING, PYLONS, RIGHT MAIN LANDING GEAR, NO. 2 ENGINE AND NACELLE, BATTERY COMPARTMENT
AREA NO. 3	AFT FUSELAGE, TAIL ROTOR DRIVE AND CONTROLS
AREA NO. 4	STABILIZER, STABILATOR,TAIL WHEEL, TAIL ROTOR DRIVES AND CONTROLS
AREA NO. 5	AFT EQUIPMENT BAY (ALL EQUIPMENT)
AREA NO. 6	MAIN ROTOR AND CONTROLS
AREA NO. 7	LEFT FUSELAGE WING, PYLONS, LEFT MAIN LANDING GEAR, NO. 1 ENGINE AND NACELLE AMMO COMPARTMENT

FIGURE 21. Example of an area diagram.

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MIL-STD-40051-6

DEPARTMENT OF DEFENSE STANDARD PRACTICE

TECHNICAL MANUALS

REPAIR PARTS AND SPECIAL TOOLS LIST (RPSTL)

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1. SCOPE.

1.1 Scope. This standard establishes the technical content requirements for the preparation of Repair Parts and Special Tools Lists (RPSTLs) for weapon systems and equipment Technical Manuals (TMs) and Depot Maintenance Work Requirements (DMWRs). These requirements are applicable for both paper and digital page-oriented TMs. Electronic delivery of RPSTL TMs is accomplished through the use of the Repair Parts and Special Tools List modular Document Type Definition (DTD). The DTD is available in a digital format. Refer to MIL-STD-40051 for information on obtaining this DTD. RPSTL requirements include 1) introductory information, 2) tabular listings of all authorized spares and repair parts, special tools, special test, measurement, and diagnostic equipment (TMDE), and other support equipment required for performance of maintenance and 3) illustrations to identify and locate the spare and repair parts.

2. APPLICABLE DOCUMENTS.

The applicable documents in section 2 of MIL-STD-40051 apply to this Part.

3. DEFINITIONS.

The definitions in section 3 of MIL-STD-40051 apply to this Part.

4. GENERAL REQUIREMENTS.

4.1 General. The RPSTL lists and authorizes spares and repair parts; special tools; special Test, Measurement, and Diagnostic Equipment (TMDE); and other special support equipment required for performance of all levels of maintenance of the weapon system/equipment, subsystems, assemblies, and components. It authorizes the requisitioning, issue and disposition of spares, repair parts and special tools in accordance the Source, Maintenance and Recoverability (SMR) codes. When a RPSTL with combined levels of maintenance is authorized, the RPSTL shall contain spares and repair parts data for all levels covered, even though lower levels of maintenance is covered in a separate RPSTL. For example, when both a -20 level (unit) and a -34 level (direct support and general support) RPSTL are published, the -34 RPSTL shall include spare and parts data required for the unit, direct support, and general support level. Illustrations published in the -34 level RPSTL that contain unit authorized items shall be duplicated in the -20 level RPSTL.

4.2 Standard tables. Various standard tables required are noted throughout the text of this standard in bold and in parentheses (i.e., **(standard table)**). The formats and table heading names of these standard tables shall have no deviations.

4.3 Preparation of digital data for electronic delivery. Technical manual data prepared in work package format and delivered digitally in accordance with this standard shall be Standard Generalized Markup Language (SGML) tagged and assembled using the modular DTD and Formatting Output Specification Instance (FOSI). The DTD and FOSI has been developed in accordance with MIL-PRF-28001 and ISO 8879. Refer to MIL-STD-40051 for information on obtaining or accessing this modular DTD and FOSI. SGML tags used in the modular DTD are noted throughout the text of this standard in bracketed, bold characters (i.e., **<plwp>**) as a convenience for the TM author and to ensure that the tags are used correctly when developing a document instance.

4.3.1 Use of the DTDs / FOSIs. The modular DTDs referenced in this Part interpret the technical content and structure for the functional requirements contained in this Part and are mandatory for use. The modular FOSIs referenced herein interprets the style and format. As specified by the contracting activity, FOSIs or

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style sheets may be used to produce final reproducible paper copy for all TMs prepared in accordance with this standard.

4.4 Content structure and format. The examples provided at the rear of this Part are an accurate representation of the content structure and format requirements contained herein and shall be followed to permit the effective use of the modular DTD for RPSTLs.

4.5 Style and format. Style and format requirements for the preparation of Department of Army TMs are contained in MIL-STD-40051-1 and are considered mandatory and are intended for compliance. Preferred general style and format requirements for Army TMs shall be provided by the procuring activity.

4.6 Work package development. Technical manual data developed in accordance with this standard shall be divided into individual, stand alone units of information work packages. A work package shall consist of descriptive, operational, maintenance, troubleshooting, support, or parts information for the weapon system or equipment.

4.7 Selective application and tailoring. MIL-STD-40051 contains some requirements that may not be applicable to the preparation of all technical manuals. Selective application and tailoring of requirements contained in MIL-STD-40051 are the responsibility of the contracting activity and shall be accomplished through the use of Appendix A, Technical Manual Content Selection Matrixes, of MIL-STD-40051. The applicability of some requirements is also designated by one of the following statements: unless specified otherwise by the contracting activity; as/when specified by the contracting activity; or when specified by the procuring activity.

5. DETAILED REQUIREMENTS.

5.1 Preparation of RPSTLs. RPSTLs shall be prepared for weapon systems, major components and applicable support and interface equipment. This information shall be contained in one of the following:

a. A separate RPSTL Technical Manual (TM),

b. RPSTL work packages included in maintenance TM, or

c. RPSTL work packages included in a Depot Maintenance Work Requirement (DMWR).

5.1.1 Separate RPSTL TM. Separate RPSTL TMs shall consist of front and rear matter and Parts Information Chapters **<pim>** containing the work packages described in 5.3.

5.1.2 RPSTL work packages included in a maintenance TM. When a separate RPSTL is not required or authorized, RPSTL data shall be included as part of a maintenance TM. Introduction, repair parts list, special tools list, and cross reference indexes work packages as listed in 5.4 shall be included, as applicable. Front and rear matter requirements shall become part of the maintenance TM that includes the RPSTL work packages.

5.1.3 RPSTL work packages included in a DMWR. If an item of equipment is programmed for depot overhaul and no repair parts (including modules, printed circuits, and components) are authorized for replacement at a level below depot maintenance, authorized repair parts data shall appear in the applicable DMWR.

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5.1.3.1 Depot repair parts. If a figure(s) contains repair parts for both depot level maintenance and a maintenance level(s) below depot, the depot coded repair parts shall be presented in a DMWR as a RPSTL work package or in the separate RPSTL TM as specified by the contracting activity. If the RPSTL TM includes depot repair parts, the statement "Including Depot Maintenance Repair Parts" shall be added to the title of the RPSTL TM.

5.2 Front matter. Front matter for RPSTL TMs shall consist of a cover, title block page, and a table of contents prepared in accordance with MIL-STD-40051-1.

5.3 Parts information chapter **<pim>**. Chapters shall consist of the following applicable work packages.

a. Introduction work package **<introwp>**.

b. Repair parts list work package **<plwp>**.

c. Special tools list work package **<stlwp>**.

d. Cross-reference index work packages

(1) National Stock Number (NSN) index work package **<nsnindxwp>**
(2) Part number index work package **<pnindxwp>**
(3) Reference designator index work package **<refdesindxwp>**

5.4 Introduction work package **<introwp>**. The introduction work package (text below within the quotation marks) shall be prepared and included verbatim, except for the information indicated by italicized text. Italicized text shall be deleted and, as applicable, replaced with the appropriate information. (Refer to figure 1.)

"INTRODUCTION

SCOPE

This RPSTL lists and authorizes spares and repair parts; special tools; special test, measurement, and diagnostic equipment (TMDE); and other special support equipment required for performance of *(enter maintenance level)* maintenance of the *(enter item name)*. It authorizes the requisitioning, issue, and disposition of spares, repair parts, and special tools as indicated by the source, maintenance, and recoverability (SMR) codes.

GENERAL

In addition to the Introduction work package, this RPSTL is divided into the following work packages.

1. Repair Parts List Work Packages. Work packages containing lists of spares and repair parts authorized by this RPSTL for use in the performance of maintenance. These work packages also include parts which must be removed for replacement of the authorized parts. Parts lists are composed of functional groups in ascending alphanumeric sequence, with the parts in each group listed in ascending figure and item number sequence. Sending units, brackets, filters, and bolts are listed with the component they mount on. Bulk materials are listed by item name in FIG. BULK at the end of the work packages. Repair parts kits are listed separately in their own functional group and work package. Repair parts for reparable special tools are also listed in a separate work package. Items listed are shown on the associated illustrations.

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2. Special Tools List Work Packages. Work packages containing lists of special tools, special TMDE, and special support equipment authorized by this RPSTL (as indicated by Basis of Issue (BOI) information in the DESCRIPTION AND USABLE ON CODE (UOC) column). Tools that are components of common tool sets and/or Class VII are not listed.

3. Cross-Reference Indexes Work Packages. There are *(enter applicable number)* cross-reference indexes work packages in this RPSTL: *(enter applicable index titles: the National Stock Number (NSN) Index work package, the Part Number (P/N) Index work package, and the Reference Designator Index work package). (Enter applicable explanations: The National Stock Number Index work package refers you to the figure and item number. The Part Number Index work package refers you to the figure and item number. The Reference Designator Index work package refers you to the figure and item number.)*

EXPLANATION OF COLUMNS IN THE REPAIR PARTS LIST AND SPECIAL TOOLS LIST WORK PACKAGES

ITEM NO. (Column (1)). Indicates the number used to identify items called out in the illustration.

SMR CODE (Column (2)). The SMR code containing supply/requisitioning information, maintenance level authorization criteria, and disposition instruction, as shown in the following breakout:

Source Code	**Maintenance Code**		**Recoverability Code**
xx	xx		x
1st two positions: How to get an item.	3rd position: Who can install, replace, or use the item.	4th position: Who can do complete repair* on the item.	5th position: Who determines disposition action on unserviceable items.

*Complete Repair: Maintenance capacity, capability, and authority to perform all corrective maintenance tasks of the "Repair" function in a use/user environment in order to restore serviceability to a failed item.

Source Code. The source code tells you how you get an item needed for maintenance, repair, or overhaul of an end item/equipment. Explanations of source codes follow:

Source Code	**Application/Explanation**
PA PB PC PD PE PF PG	Stock items; use the applicable NSN to requisition/request items with these source codes. They are authorized to the level indicated by the code entered in the 3rd position of the SMR code. **NOTE** Items coded PC are subject to deterioration.

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KD KF KB	Items with these codes are not to be requested/requisitioned individually. They are part of a kit which is authorized to the maintenance level indicated in the 3rd position of the SMR code. The complete kit must be requisitioned and applied.
MO-Made at unit/ AVUM level MF-Made at DS/ AVIM level MH-Made at GS level ML-Made at SRA MD-Made at depot	Items with these codes are not to be requisitioned/requested individually. They must be made from bulk material which is identified by the P/N in the DESCRIPTION AND USABLE ON CODE (UOC) column and listed in the bulk material group work package of the RPSTL. If the item is authorized to you by the 3rd position code of the SMR code, but the source code indicates it is made at higher level, order the item from the higher level of maintenance.
AO-Assembled by unit/AVUM level AF-Assembled by DS/AVIM level AH-Assembled by GS level AL-Assembled by SRA AD-Assembled by depot	Items with these codes are not to be requested/requisitioned individually. The parts that make up the assembled item must be requisitioned or fabricated and assembled at the level of maintenance indicated by the source code. If the 3rd position of the SMR code authorizes you to replace the item, but the source code indicates the item is assembled at a higher level, order the item from the higher level of maintenance.
XA	Do not requisition an "XA" coded item. Order the next higher assembly. (Refer to NOTE below.)
XB	If an item is not available from salvage, order it using the CAGEC and P/N.
XC	Installation drawings, diagrams, instruction sheets, field service drawings; identified by manufacturer's P/N.
XD	Item is not stocked. Order an XD-coded item through normal supply channels using the CAGEC and P/N given, if no NSN is available.

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NOTE

Cannibalization or controlled exchange, when authorized, may be used as a source of supply for items with the above source codes except for those items source coded "XA" or those aircraft support items restricted by requirements of AR 750-1.

Maintenance Code. Maintenance codes tell you the level(s) of maintenance authorized to use and repair support items. The maintenance codes are entered in the third and fourth positions of the SMR code as follows:

Third Position. The maintenance code entered in the third position tells you the lowest maintenance level authorized to remove, replace, and use an item. The maintenance code entered in the third position will indicate authorization to the following levels of maintenance:

Maintenance Code	Application/Explanation
C	Crew or operator maintenance done within unit/AVUM maintenance.
O	Unit level/AVUM maintenance can remove, replace, and use the item.
F	Direct support/AVIM maintenance can remove, replace, and use the item.
H	General support maintenance can remove, replace, and use the item.
L	Specialized repair activity can remove, replace, and use the item.
D	Depot can remove, replace, and use the item.

Fourth Position. The maintenance code entered in the fourth position tells you whether or not the item is to be repaired and identifies the lowest maintenance level with the capability to do complete repair (perform all authorized repair functions).

NOTE

Some limited repair may be done on the item at a lower level of maintenance, if authorized by the Maintenance Allocation Chart (MAC) and SMR codes.

Maintenance Code	Application/Explanation
O	Unit/AVUM is the lowest level that can do complete repair of the item.
F	Direct support/AVIM is the lowest level that can do complete repair of the item.
H	General support is the lowest level that can do complete repair of the item.
L	Specialized repair activity *(enter specialized repair activity designator)* is the lowest level that can do complete repair of the item.
D	Depot is the lowest level that can do complete repair of the item.

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Z — Nonreparable. No repair is authorized.

B — No repair is authorized. No parts or special tools are authorized for maintenance of "B" coded item. However, the item may be reconditioned by adjusting, lubricating, etc., at the user level.

Recoverability Code. Recoverability codes are assigned to items to indicate the disposition action on unserviceable items. The recoverability code is shown in the fifth position of the SMR code as follows:

Recoverability Code	Application/Explanation
Z	Nonreparable item. When unserviceable, condemn and dispose of the item at the level of maintenance shown in the third position of the SMR code.
O	Reparable item. When uneconomically reparable, condemn and dispose of the item at the unit level.
F	Reparable item. When uneconomically reparable, condemn and dispose of the item at the direct support level.
H	Reparable item. When uneconomically reparable, condemn and dispose of the item at the general support level.
D	Reparable item. When beyond lower level repair capability, return to depot. Condemnation and disposal of item are not authorized below depot level.
L	Reparable item. Condemnation and disposal not authorized below Specialized Repair Activity (SRA).
A	Item requires special handling or condemnation procedures because of specific reasons (such as precious metal content, high dollar value, critical material, or hazardous material). Refer to appropriate manuals/directives for specific instructions.

NSN (Column (3)). The NSN for the item is listed in this column.

CAGEC (Column (4)). The Commercial and Government Entity Code (CAGEC) is a five-digit code which is used to identify the manufacturer, distributor, or Government agency/activity that supplies the item.

PART NUMBER (Column (5)). Indicates the primary number used by the manufacturer (individual, company, firm, corporation, or Government activity), which controls the design and characteristics of the item by means of its engineering drawings, specifications, standards, and inspection requirements to identify an item or range of items.

NOTE

When you use an NSN to requisition an item, the item you receive may have a different P/N from the number listed.

DESCRIPTION AND USABLE ON CODE (UOC) (Column (6)). This column includes the following information:

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1. The federal item name, and when required, a minimum description to identify the item.

2. P/Ns of bulk materials are referenced in this column in the line entry to be manufactured or fabricated.

3. Hardness Critical Item (HCI). A support item that provides the equipment with special protection from electromagnetic pulse (EMP) damage during a nuclear attack.

4. The statement END OF FIGURE appears just below the last item description in column (6) for a given figure in both the repair parts list and special tools list work packages.

QTY (Column (7)). The QTY (quantity per figure) column indicates the quantity of the item used in the breakout shown on the illustration/figure, which is prepared for a functional group, subfunctional group, or an assembly. A "V" appearing in this column instead of a quantity indicates that the quantity is variable and quantity may change from application to application.

EXPLANATION OF CROSS-REFERENCE INDEXES WORK PACKAGES FORMAT AND COLUMNS

1. National Stock Number (NSN) Index Work Package.

STOCK NUMBER Column. This column lists the NSN in National item identification number (NIIN) sequence. The NIIN consists of the last nine digits of the NSN.

NSN (e.g., 5385-01-574-1476) NIIN	When using this column to locate an item, ignore the first four digits of the NSN. However, the complete NSN should be used when ordering items by stock number.

FIG. Column. This column lists the number of the figure where the item is identified/located. The figures are in numerical order in the repair parts list and special tools list work packages.

ITEM Column. The item number identifies the item associated with the figure listed in the adjacent FIG. column. This item is also identified by the NSN listed on the same line.

2. Part Number (P/N) Index Work Package. P/Ns in this index are listed in ascending alphanumeric sequence (vertical arrangement of letter and number combinations which places the first letter or digit of each group in order A through Z, followed by the numbers 0 through 9 and each following letter or digit in like order).

PART NUMBER Column. Indicates the P/N assigned to the item.

FIG. Column. This column lists the number of the figure where the item is identified/located in the repair parts list and special tools list work packages.

ITEM Column. The item number is the number assigned to the item as it appears in the figure referenced in the adjacent figure number column."

NOTE: Include 3, as applicable.

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"3. Reference Designator Index Work Package. Reference designators in this index are listed in ascending alphanumeric sequence (vertical arrangement of letter and number combination which places the first letter or digit of each group in order "A" through "Z," followed by the numbers "0" through "9" and each following letter or digit in like order).

REFERENCE DESIGNATOR Column. Indicates the reference designator assigned to the item.

FIG. Column. This column lists the number of the figure where the item is identified/located in the repair parts list or special tools list work package.

ITEM Column. The item number is the number assigned to the item as it appears in the figure referenced in the adjacent figure number column.

SPECIAL INFORMATION

UOC. The UOC appears in the lower left corner of the Description Column heading. Usable on codes are shown as "UOC: ..." in the Description Column (justified left) on the first line under the applicable item/nomenclature. Uncoded items are applicable to all models. Identification of the UOCs used in the RPSTL are:

Code	**Used On"**
PAA	*Model M114*
PAB	*Model M114A*
PAC	*Model M114B*

NOTE: Include the above UOC content, as applicable.

"Fabrication Instructions. Bulk materials required to manufacture items are listed in the bulk material functional group of this RPSTL. Part numbers for bulk material are also referenced in the Description Column of the line item entry for the item to be manufactured/fabricated. Detailed fabrication instructions for items source coded to be manufactured or fabricated are found in (*enter applicable TM number*).

Index Numbers. Items which have the word BULK in the figure column will have an index number shown in the item number column. This index number is a cross-reference between the NSN / P/N index work packages and the bulk material list in the repair parts list work package."

NOTE: For a combined narrative-RPSTL manual associated publications shall not be included.

"Associated Publications. The publication(s) listed below pertains to the (*enter item name*):

Publication	**Short Title"**

NOTE: The following paragraph shall appear only in the unit maintenance RPSTL special instructions.

"Illustrations List. The illustrations in this RPSTL contain unit authorized items. Illustrations published in (*enter applicable TM number for the higher maintenance level RPSTL, e.g., for direct support, general*

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support, etc.) that contain unit authorized items also appear in this RPSTL. The tabular list in the repair parts list work package contains only those parts coded "O" in the third position of the SMR code, therefore, there may be a break in the item number sequence."

"HOW TO LOCATE REPAIR PARTS

1. When NSNs or P/Ns Are Not Known.

First. Using the table of contents, determine the assembly group to which the item belongs. This is necessary since figures are prepared for assembly groups and subassembly groups, and lists are divided into the same groups.

Second. Find the figure covering the functional group or the subfunctional group to which the item belongs.

Third. Identify the item on the figure and note the number(s).

Fourth. Look in the repair parts list work packages for the figure and item numbers. The NSNs and part numbers are on the same line as the associated item numbers.

2. When NSN Is Known.

First. If you have the NSN, look in the STOCK NUMBER column of the NSN index work package. The NSN is arranged in NIIN sequence. Note the figure and item number next to the NSN.

Second. Turn to the figure and locate the item number. Verify that the item is the one you are looking for.

3. When P/N Is Known.

First. If you have the P/N and not the NSN, look in the PART NUMBER column of the P/N index work package. Identify the figure and item number.

Second. Look up the item on the figure in the applicable repair parts list work package."

NOTE: Include 4 only if the RPSTL has a reference designator index work package.

"4. When Reference Designator Is Known.

First. If you know the reference designator, look in the REFERENCE DESIGNATOR column of the reference designator index work package. Note the figure and item number.

Second. Turn to the figure and locate the item number. Verify that the item is the one you are looking for.

ABBREVIATIONS

Abbreviation	**Explanation**"

NOTE: Include uncommon abbreviations used in the RPSTL. List/define those not found in MIL-STD-12.

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5.5 Repair parts list work package **<plwp>**. Each repair parts list work package shall consist of illustrations and their associated repair parts lists **<pl>**. A separate repair parts list work package shall be prepared for each illustration. (Refer to figures 2 and 3.) Figure 3 is in a **(standard table)** format. Repair parts for special tools listed in special tools list work packages shall be illustrated and listed in a separate work package under a functional group titled "Special Tools (Repair Parts)." The "Special Tools (Repair Parts)" group(s) shall follow the figure and list for the last group of the end item(s) listed on the Maintenance Allocation Chart (MAC) and shall precede the figures and lists for kits and bulk items. (Refer to 5.5.9.)

5.5.1 Figure titles. Figure titles shall be consistent with the titles in the MAC with the nomenclature in right reading order. Unless specified otherwise by the procuring activity, a figure title shall include all the Functional Group Codes (FGCs) illustrated beginning with the highest FGC illustrated. (Refer to figure 2.)

5.5.2 Item number column. Items shall be listed on the repair parts list (in the ITEM NO. column) by the same callout number shown on the associated figure. The items shall be listed in ascending alphanumeric sequence.

5.5.3 Nonconsecutive item numbers. When illustrations contain item callouts that are for a maintenance level higher than the level of the RPSTL, the items not authorized for maintenance at the RPSTL level shall not be listed in the repair parts list; therefore, items may not be listed consecutively. They shall be listed in ascending alphanumeric sequence.

5.5.4 SMR code column. The SMR code column shall include SMR codes assigned to the applicable items.

5.5.5 NSN column. The NSN column shall include the NSN assigned to the applicable item.

5.5.6 Commercial and Government Entity Code (CAGEC) and part number columns. The applicable five-digit CAGEC number, as listed in Catalog Handbook H4/H8, shall appear in the CAGEC column preceding the part number listed in the PART NUMBER column.

5.5.7 Description and UOC column. The DESCRIPTION AND USABLE ON CODE (UOC) column shall include the following information.

a. Header. The header shall consist of the functional group number and title appearing on the top line(s). The next line(s) below shall include the figure number and the figure title (may be the same as the functional group title). The headers for lists shall contain the same wording and information as the associated figures.

b. Item name. The item name shall consist of the federal item name (taken from Federal Supply Cataloging Handbook H6) and, if necessary, a minimum description to further identify the item. If the item is a Hardness Critical Item, the symbol **HCI** shall precede the item name.

c. Indentions. The item name listed in the DESCRIPTION AND USABLE ON CODE (UOC) column shall be indented to show components of assemblies and next higher assemblies. Indentions shall not exceed five positions. (Refer to figure 4.)

d. UOC. When an item has multiconfiguration or multimodel use, the three-position alphanumeric UOC representing the applicable configuration in which the item is used shall be placed on the last line under the item description. The letters "UOC:" followed by the applicable UOC shall be indented. When an item is used on all configurations or when only one configuration is covered by the RPSTL, UOCs shall not be shown.

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e. Serial number application. When P/Ns of spare/repair items are not the same for all serial numbered equipment of the same model, a statement identifying the Usable Effective (USBL EFF) serial numbers shall be made in the DESCRIPTION AND USABLE ON CODE (UOC) column (e.g., USBL EFF SER NOS 1719-1941). (Refer to figure 3 for other examples.)

f. Assembled items. Spare and repair parts that are part of a nonstocked assembled item (source coded "AO", "AF", "AH", or "AD") shall be assigned item numbers on illustrations and shall be listed in item number sequence on the repair parts list. These items/parts shall be listed immediately below the item to be assembled on the repair parts list. When a particular illustration does not show the parts breakdown of the nonstocked assembly, reference shall be made to the breakdown illustration in the RPSTL. (Refer to figure 3.) Instructions, drawings, charts, and tables showing how to assemble assemblies source coded "A( )" shall not appear in the RPSTL, but shall appear in the narrative maintenance TM.

g. Manufactured items. All items source coded "MO", "MF", "MH", or "MD" shall have the statement in the DESCRIPTION AND USABLE ON CODE (UOC) column as follows: "MAKE FROM (*enter applicable bulk material or other replaceable item name, CAGEC, and P/N*)." Material that is used to make items shall also be shown in a separate functional group called BULK MATERIAL and figure to be titled FIG. BULK. Items in the bulk figure shall be listed alphabetically by item name in the DESCRIPTION AND USABLE ON CODE (UOC) column. (Refer to figure 5.) Numbers in the ITEM column of bulk material list apply to the FIG. BULK only and shall not be associated with item numbers (callouts appearing on the illustrations/figures). Instructions, drawings, charts, and tables required to show how items are made shall not be contained in the RPSTL but shall appear in the narrative maintenance TM.

h. Kits and kit repair parts. Kits and repair parts (source coded "KD", "KF", or "KB") shall conform to the format of either option 1 (refer to figure 6) or option 2 (refer to figure 7), as specified by the contracting activity.

(1) Option 1 (kits). Option 1 kits shall appear at the end of the associated parts list. As specified by the contracting activity, the ITEM NO. column for kits shall be either left blank or list an alphabetical character(s). The QTY column for kits shall be a V (variable) when the exact quantity may vary. (Refer to figure 6.)

(2) Option 1 (parts). Option 1 kit repair parts shall be listed with their applicable figure and appear in item number sequence. The statement "part of Kit P/N (*enter kit P/N*)" shall follow item name. Kit repair parts shall also be listed under the kit list at the end of the parts list (refer to figure 6). Parts of the kit list shall be indented and listed alphabetically by item name or in item number sequence immediately below the kit item name. The quantity (in parentheses), figure, and item number shall follow the repair part item name.

(3) Option 2 (kits). Option 2 kits shall be listed in a separate functional group titled "Repair Kits." This functional group shall be located in a repair parts list work package following the figure and list for the last group of the end item(s) listed on the MAC, including the figure and list for Special Tools (Repair Parts). This kit group shall be located before the bulk material functional group. The kits in this group shall be listed alphanumerically in part number sequence. Parts in the kit group shall be indented two positions and listed alphabetically by item name or in item number sequence under their kit name. (Refer to figure 7.) Item names of the parts shall be followed by the quantity (in parentheses) and the

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figure and item numbers that appear in the basic parts list. The QTY column for kits shall contain a V (variable) when the exact quantity may vary.

(4) Option 2 (parts). Option 2 kit repair parts shall appear in the parts list by item number as shown on the associated figure. They shall be listed in item number sequence. The statement "PART OF KIT P/N (*enter kit part number*)" shall follow the item name.

i. End of figure statement. The statement END OF FIGURE shall appear below the last item described in column 6 for each figure of the tabular lists in the repair parts list and the special tools list work packages.

5.5.8 Quantity column. The figure in the QTY column shall represent the number of times the item appears in the illustration/figure with the associated item number. When a definite quantity cannot be determined because the number of uses per equipment or the size/length of an item may vary with each equipment, the letter V shall be placed in the left position of the QTY column.

5.5.9 Basic Issue Items (BII) (repair parts)/special tools (repair parts). Repair parts for reparable BII or special tools that do not have separate TMs, but are authorized for the RPSTL, shall be listed in a functional group titled BASIC ISSUE ITEMS (REPAIR PARTS) or SPECIAL TOOLS (REPAIR PARTS), as applicable. Subfunctional groups shall be assigned, as applicable. (These group(s) precede kits and bulk work packages.) Items listed in functional and subfunctional groups shall be listed and identified with the same basic columnar data required for the end item repair parts. BII and special tools reparable parts shall be supported by illustrations.

5.5.10 Expendable and durable items. Expendable and durable items shall not be listed in the RPSTL. (These items shall appear in the applicable narrative TM.)

5.6 Special tools list work package **<stlwp>**. Each special tools list work package shall consist of illustrations (for special tools, special TMDE, and other special support equipment authorized for maintenance of the end item/assembly) and the associated repair parts list **<pl>**. These tools shall be listed in the format shown in figure 8 **(standard table)**.

5.6.1 Functional grouping. Items shall be listed under a functional group(s) titled SPECIAL TOOLS. Items within the group shall be listed in ascending figure and item number sequence.

5.6.2 Basis of Issue (BOI). The last line entry(s) in the DESCRIPTION AND USABLE ON CODE (UOC) column for individual items, sets, or kits shall be the BOI. The BOI shall indicate the quantity of the items, i.e., sets, or kits authorized to support a quantity of end items/assembly(s) or a specific military unit. For example, BOI: 1 auth for 1-12 equip or BOI: 1 per BN HQ when BN has SVC CO. (For other examples of BOI, refer to figure 8.)

5.6.3 Special tool set/kit line entry(ies). These line entries shall contain complete information in all columns except ITEM NO. and QTY columns. ITEM NO. and QTY columns shall be left blank.

5.6.4 Components list. Components of special tool sets and kits shall be listed in figure and item number sequence immediately following the set or kit entry. The line entry for the components shall be indented under the set or kit entry and shall contain complete information in all columns except the QTY column (which shall be left blank). Quantities of components shall be included in a statement in the DESCRIPTION AND USABLE ON CODE (UOC) column (e.g., qty 1 per set/kit).

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5.6.5 D-coded items. When a depot level RPSTL does not exist and items are maintained at depot level, they shall be identified with a "D" in the third position of the SMR code in the highest level RPSTL prepared.

5.7 Cross-reference index work packages.

5.7.1 NSN index work package <**nsnindxwp**>. This index <**prtindex**> shall list the complete NSN for all NSNs assigned to applicable items. However, the line entries shall be arranged in ascending numeric sequence by National Item Identification Number (NIIN) (the last nine digits of the NSN). The NSN line entry shall identify the first figure/item number for which the stock number is applicable. (Refer to figure 9.) The NSN shall not be repeated on the same page of the index for each additional figure/item number identified by that NSN. When NSN references carry over to another page, the carried over NSN entry shall appear at the top of the list.

5.7.2 P/N index work package <**pnindxwp**>. This index <**prtindex**> shall be arranged in ascending alphanumeric sequence by P/N. The line entry for each P/N listed shall identify the applicable figure and item number. (Refer to figure 10.) When the P/N appears on more than one figure, the P/N shall not be repeated unless it continues on the next page. If the P/N continues to the next page, it shall be repeated at the top of the page.

5.7.3 Reference designator index work package <**refdesindxwp**>. When specified by the procuring activity, a reference designator index work package shall be included. The index <**prtindex**> shall list the reference designators in alphanumeric sequence and shall reference the applicable figure and item number. (Refer to figure 11.)

5.7.4 Bulk figure. When entries in either the NSN or P/N index reference bulk material, the word BULK shall appear in the FIG. column. The numbers in the ITEM No. column shall refer to the item number list in the bulk figure located in the bulk functional group list and shall not refer to item numbers on an illustration.

5.7.5 Sets and kits. P/Ns for sets/kits shall be cross-referenced to NSN, figure, and item number for the set/kit. When Option 1 is selected, the ITEM column shall either be left blank or list an alphabetical character (e.g.,"K" for KIT, "S" for SET, etc.). (Refer to 5.5.7h.) When Option 2 is selected, the FIG. column shall list the word KITS or SETS, as applicable. (Refer to 5.5.7h.)

5.8 Illustrations. Illustrations shall be prepared in accordance with MIL-STD-40051-1. Additional RPSTL specific illustration requirements are described in paragraphs 5.8.1 through 5.8.4.

5.8.1 Arrangement of illustrations. All illustrations prepared for spares, repair parts, special tools, special TMDE, and other special support equipment shall be arranged in figure number sequence. They shall precede their companion parts list (on the left-hand page preceding the parts list or at the top of the same page of the parts list). Illustrations shall not be duplicated to provide facing page illustrations for the second and subsequent pages of the RPSTL. Illustrations shall not be duplicated to show different models or configurations of an assembly when UOCs can be assigned to indicate differences in configurations.

5.8.2 Use of illustrations. Foldout and foldout-foldup illustrations shall not be used in RPSTLs. For clarity, multisheet illustrations may be used. References to illustrations in other TMs or to illustrations in the narrative portion of a combined maintenance TM with a RPSTL shall not be made. Turn pages shall not be prepared except for RPSTLs supporting nuclear weapons (regulated by the Department of Energy/Defense Nuclear Agency).

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5.8.3 Identical parts/item numbers. Identical parts (same part number) appearing in a figure (illustration) having only one FGC shall have the same item number. If a figure has two or more FGCs/assemblies, only the identical parts with identical SMR codes within each FGC/assembly shall have the same item number.

5.8.4 Identical assemblies. When two or more identical assemblies (same part number) exist in different places, i.e., in the equipment, a breakdown of the parts shall be illustrated only once, i.e., the first time the assembly appears in the RPSTL. For subsequent times that the identical assembly appears, the assembly item name shall appear in the description and UOC column and be followed by the statement SEE FIG...FOR BREAKDOWN. (Refer to figure 3.)

5.9 Rear matter. Refer to MIL-STD-40051-1.

6. NOTES.

The notes in section 6 of MIL-STD-40051 apply to this Part.

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TM X-XXXX-XXX-XXP

INTRODUCTION **0441 00**

SCOPE

This RPSTL lists and authorizes spares and repair parts; special tools; special test, measurement, and diagnostic equipment (TMDE); and other special support equipment required for performance of unit maintenance of the M198 howitzer. It authorizes the requisitioning, issue, and disposition of spares, repair parts, and special tools as indicated by the source, maintenance, and recoverability (SMR) codes.

GENERAL

In addition to the Introduction work package, this RPSTL is divided into the following work packages.

1. Repair Parts List Work Packages. Work packages containing lists of spares and repair parts authorized by this RPSTL for use in the performance of maintenance. These work packages also include parts which must be removed for replacement of the authorized parts. Parts lists are composed of functional groups in ascending alphanumeric sequence, with the parts in each group listed in ascending figure and item number sequence. Sending units, brackets, filters, and bolts shall be listed with the component they mount on. Bulk materials are listed by item name in FIG. BULK at the end of the work packages. Repair parts kits are listed separately in their own functional group and work package. Repair parts for reparable special tools are also listed in a separate work package. Items listed are shown on the associated illustrations.

2. Special Tools List Work Packages. Work packages containing lists of special tools, special TMDE, and special support equipment authorized by this RPSTL (as indicated by Basis of Issue (BOI) information in the DESCRIPTION AND USABLE ON CODE (UOC) column). Tools that are components of common tool sets and/or Class VII shall not be listed.

3. Cross-Reference Indexes Work Packages. There are two cross-reference indexes work packages in this RPSTL: the National Stock Number (NSN) Index work package and the Part Number (P/N) Index work package. The National Stock Number Index work package refers you to the figure and item number. The Part Number Index work package refers you to the figure and item number.

EXPLANATION OF COLUMNS IN THE REPAIR PARTS LIST AND SPECIAL TOOLS LIST WORK PACKAGES

ITEM NO. (Column (1)). Indicates the number used to identify items called out in the illustration.

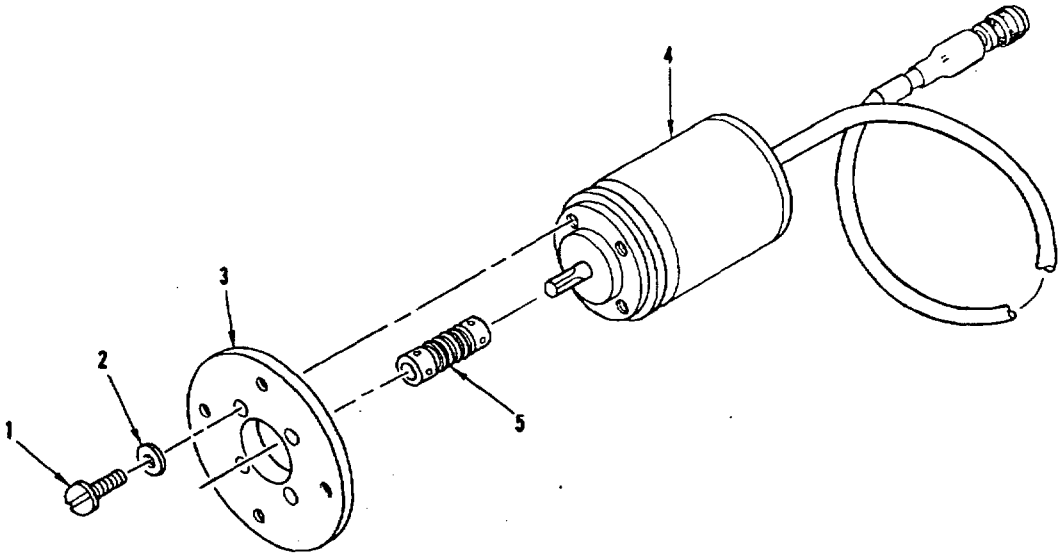
0441 00-1

FIGURE 1. Example of an introduction work package.

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TM 9-1090-208-23P

GROUP 08 ELEVATION RESOLVER ASSEMBLY 7-217221365-3 0031 00



MI 7236

Figure 35. Group 08 Elevation Resolver Assembly 7-217221365-3.

0031 00-1

FIGURE 2. Example of a repair parts list illustration.

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TM 11-1520-238-23P

GROUP 110503 WIRING INSTALLATION 70551-02169-012 AND -014 REPAIR PARTS LIST - Continued **0896 00**

(1) ITEM NO.	(2) SMR CODE	(3) NSN	(4) CAGEC	(5) PART NUMBER	(6) DESCRIPTION AND USABLE ON CODE (UOC)	(7) QTY
					GROUP 110503 WIRING INSTALLATION	
					FIG. 99 WIRING INSTALLATION 70551-02169-012 AND -014	
1	MOOZZ		78286	SS9014J22R	PLATE, INDENT MAKE FROM AL FOIL 0.003 TO 0.005 IN. THK QQ-A-1876 OR GG-P-455 GRADE A	1
2	MOOZZ		78286	SS9014J25R	PLATE, INDENT MAKE FROM AL FOIL 0.003 TO 0.005 IN. THK QQ-A-1876 OR GG-P-455 GRADE A	1
3	AOOOO		78286	70602-02107-041	HARNESS ASSY SEE FIG. 96 FORBREAKDOWN	1
4	AOOOO		78286	70602-02102-041	HARNESS ASSY, ICS USBL EFF 77-22714 THRU 77-22717 SEE FIG. 100 FOR BREAKDOWN	1
4	AOOOO		78286	70602-02102-042	HARNESS ASSY, ICS USBL EFF 77-22718 THRU 83-23866 SEE FIG. 100 FOR BREAKDOWN	1
4	AOOOO		78286	70602-02102-043	HARNESS ASSY, ICS USBL EFF 83-23887 AND SUB SEE FIG. 100 FOR BREAKDOWN	1
5	AOOOO		78286	70602-02103-041	HARNESS ASSY, ICS USBL EFF 77-22714 THRU 83-23886 SEE FIG. 101 FOR BREAKDOWN	1
5	AOOOO		78286	70602-02103-042	HARNESS ASSY USBL EFF 83-23887 AND SUB SEE FIG. 101 FOR BREAKDOWN	1
6	PAOZZ	5340-00-291-5323	96906	MS2191WDG4	CLAMP, LOOP	7
7	MOOZZ		78286	SS9014J204R	PLATE, IDENT MAKE PER SB11-631	1
8	PAOZZ	5310-01-105-7241	88044	AN690JD10L	WASHER, FLAT	16
9	PAOZZ	5310-00-877-5798	96906	MS21044D3	NUT, SELF-LOCKING, HE	23
10	PAOZZ	5340-00-291-5353	96906	MS21919WDG2	CLAMP, LOOP	17
11	PAOZZ	5305-00-947-4282	96906	MS27039DD1-09	SCREW, MACHINE	8
12	PAOZZ	5340-00-598-0529	96906	MS21919WDG28	CLAMP, LOOP	3
13	PAOZZ	5340-00-286-9427	96906	MS21919WDG12	CLAMP, LOOP	1
14	PAOZZ	5305-00-947-4278	96906	MS27039DD1-08	SCREW, MACHINE	9
15	PAOZZ	5365-00-662-3100	82918	BACS18AE3-64	SPACER, SLEEVE	3
16	PAOZZ	5340-00-598-0146	96906	MS21919WDG6	CLAMP, LOOP	6
17	PAOZZ	5340-00-291-5322	96906	MS21919WDG3	CLAMP, LOOP	2
18	PAOZZ	5310-01-134-5794	88044	AN960KD10	WASHER, FLAT	6
19	PAOZZ	5305-00-947-6994	96906	MS27039DD1-24	SCREW, FLAT	2
20	PAOZZ		96906	MS35489-17	GROMMET, NONMETALLIC UOC: 77551-02169-012	1
20	PAOZZ	5325-00-291-9366	96906	MS35489-11	GROMMET, NONMETALLIC UOC: 70511-02169-014	1
					END OF FIGURE	

0896 00-2

FIGURE 3. Example of a repair parts list work package.

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(6) DESCRIPTION AND USABLE ON CODE (UOC)	(7) QTY
GROUP 14 ENGINE ASSEMBLY	
FIG. 24 OIL PUMP ASSEMBLY	
.BOLT, MACHINE	1
.WASHER, LOCK	1
.STRAINER, PUMP	1
.PUMP, ROTARY	1
..REGULATOR PRESS	1
..WASHER, KEY	1
..SPACER, RING	5
..GEAR, OIL PUMP	1
..BOLT, MACHINE CAP SCREW 1/4-20X1-3/8 INCH	2
..WASHER, LOCK, 1/4 IN. MEDIUM SAE	2
..LOCKWASHER, STEEL	2
..SCREW, COVER	6
..COVER, PUMP	1
..PUMP, OIL BSC	1
...GEAR, DR SHAFT	1
...BODY ASSY	1
...SHAFT, IDLER	1
....BODY, PUMP	1
END OF FIGURE	

FIGURE 4. Example of indentions (next higher assembly).

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TM 9-2320-258-34P

GROUP 9501 BULK MATERIAL REPAIR PARTS LIST - Continued **3264 00**

(1) ITEM NO.	(2) SMR CODE	(3) NSN	(4) CAGEC	(5) PART NUMBER	(6) DESCRIPTION AND USABLE ON CODE (UOC)	(7) QTY
					GROUP 95 GENERAL USE STANDARDIZED PARTS	
					GROUP 9501 BULK MATERIAL	
					FIG. 15 BULK	
1	PAOZZ	5330-00-982-5130	81349	MILC7637TYP2	ASBESTOS SHEET, WOVE	1
2	XBOZZ		19207	RRC271	CHAIN, WELDLESS	1
3	PAOZZ	5975-00-285-0907	97030	LOOM 3/8 ID	CONDUIT, NONMETALIC	1
4	PAOZZ	9340-00-142-6860	19207	11633348	GLASS, LAMINATED	1
5	PAOZZ	9340-00-285-6775	19200	8635931	GLASS, LAMINATED	1
6	PFOZZ	4720-00-809-2429	30299	FT3548-5	HOSE, AIR DUCT	1
7	PAFZZ	4720-00-001-0093	81349	MIL-H-13531	NOSE ASSEMBLY, NONME	1
8	XBOZZ		85757	3250-0610	HOSE, NONMETALLIC	1
9	PAFZZ	4720-00-999-8994	01276	303-8	HOSE, NONMETALLIC	1
10	PAOZZ	4720-00-951-2433	96909	MS521301A229R	HOSE, NONMETALLIC	1
11	PAOZZ	4720-01-009-9058	85757	3250-1010	HOSE, NONMETALLIC	1
12	PAOZZ	4720-00-683-8830	81349	MIL-H-8788-4	HOSE, NONMETALLIC	1
13	PAOZZ	4720-00-999-4044	11083	3R7752	HOSE, PREFORMED	1
14	XBOZZ		81349	MIL-I-14511	INSULATION BOARD TH	1
15	PAOZZ	9390-00-488-2106	19207	CPR102201	NONMETALLIC SPECIAL	1
16	PAOZZ	5330-00-333-0313	81348	HHP151	RUBBER SHEET SOLID	1
17	XBOZZ		19207	10287823-7	RUBBER STRIP	1
18	PAFZZ	5330-01-040-8923	19207	CPR104394	SEAL, RUBBER CHANNEL	1
19	PAFZZ	5330-01-082-3792	19207	CPR102235	SEAL, RUBBER, SPECIAL	1
20	PAOZZ	5330-01-082-3793	19207	CPR102232	SEAL, RUBBER, SPECIAL	1
21	PAOZZ	5365-00-944-1871	19204	738942	SPACER, SLEEVE	1
22	PAOZZ	4710-00-234-0701	19207	CPR103203-1	TUBE ASSEMBLY, METAL	1
23	PAOZZ	4710-00-277-5524	19207	7036787	TUBE, METALIC	1
24	PAOZZ	4710-00-277-5526	91340	D11076-4A7	TUBE, METALIC	1
25	PAFZZ	4710-00-006-1647	81348	QQ-T-830	TUBE, METALIC	1
26	PAOZZ	4710-00-203-3174	16236	CS4710-0004GB	TUBE, METALIC	1
27	PAOZZ	4710-00-335-2610	81349	M3520-B70E02G	TUBE, METALIC	1
28	PAOZZ	4710-00-277-4515	81346	ASTM B280	TUBE, METALIC	1
29	PAOZZ	4710-00-203-3172	17590	305087-0116	TUBE, METALIC	1
30	XBOZZ		19207	CPR109328-1	TUBING	1
31	PAOZZ	4720-00-462-7494	19200	8589761-22	TUBING, NONMETALLIC	1
32	XBOZZ		19207	CPR109328-2	TUBING, RUBBER	1
33	PAFZZ		19207	CPR102229	WEATHERSTRIP, DOOR	1
34	PAOZZ	6145-00-705-6674	81349	M13486-1-14	WIRE, ELECTRICAL	1
35	PAOZZ	6145-00-254-6117	81349	M13486-1-15	WIRE, ELECTRICAL	1
36	PAOZZ	6145-00-161-1609	81349	M13486-1-3	WIRE, ELECTRICAL	1
37	PAOZZ	9505-00-555-8648	96906	MS20995C47	WIRE, NONELECTRICAL	1
					END OF FIGURE	

FIGURE 5. Example of a bulk material list.

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TM 55-1520-205-23P

GROUP 15 T62T-2A, T62T-2A1, GAS TURBINE ENGINE REPAIR PARTS LIST - Continued **3034 00**

(1) ITEM NO.	(2) SMR CODE	(3) NSN	(4) CAGEC	(5) PART NUMBER	(6) DESCRIPTION AND USABLE ON CODE (UOC)	(7) QTY
					GROUP 15 AUXILIARY POWER UNIT FIG. 10 T62T-2A, T62T-2A1 GAS TURBINE ENGINE	
1	PAODD	2835-00-906-6766	55820	37688-0	ENGINE, GAS TURBINE T62T-2A UOC:NB4	1
2	PAODD	2835-00-804-8316	55820	37688-1000	ENGINE, GAS TURBINE T62T-2A1 ..... UOC:NB5	1
3	PAOZZ	5310-00-877-5797	96906	MS21044N3	.NUT, SELF-LOCKING UOC:NB4, NB5	2
4	PAOZZ		88044	AN960DD10	.WASHER, FLAT UOC:NB4, NB5, NB6	2
5	KAOZZ	5330-00-263-8030	96906	MS29512-06	.PACKING, PREFORMED PART OF KIT P/N 31766-1	10
6	PAOOO	2910-00-919-2021	58220	28022-4	..NOZZLE ASSEMBLY STATOR UOC:NB4, NB5, NB6	1
7	KDOZZ	5330-00-961-1463	96906	MS35769-5	..GASKET PART OF KIT P/N 31766-1 ..... UOC:NB4, NB5, NB6	1
8	PAOZZ		71895	970HE1UPPH	..NOZZLE, STATOR UOC:NB4, NB5, NB6	1
9	KAOZZ	5330-00-961-1463	55820	26793-1	..GASKET PART OF KIT P/N 31766-1 ..... UOC:NB4, NB5, NB6	1
	PAOZZ		55820	31766-1	SEAL KIT, TURBINE ..... GASKET (1) 10-7 GASKET (1) 10-9 PACKING, PREFORMED (10) 10-5	V
					END OF FIGURE	

3034 00-2

FIGURE 6. Example of kits breakdown option 1.

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TM 9-2330-258-34P

GROUP 9401 REPAIR KITS REPAIR PARTS LIST - Continued **3230 00**

(1) ITEM NO.	(2) SMR CODE	(3) NSN	(4) CAGEC	(5) PART NUMBER	(6) DESCRIPTION AND USABLE ON CODE (UOC)	(7) QTY
					GROUP 94 REPAIR KITS	
					FIG. KITS	
1	PAOZZ	2540-00-255-0775	78385	G704528	PARTS KIT, HEATER, VE PERSONNEL HEATER	1
					BURNER ASSEMBLY (1) 252-6	
					SCREW, MACHINE (1) 252-8	
2	PAOZZ	2540-00-255-0777	78385	G704529	PARTS KIT, HEATER, VE PERSONNEL HEATER	1
					SCREW, MACHINE (1) 252-8	
					VAPORIZER (1) 252-11	
					WASHER, FLAT (1) 252-9	
					WASHER, FIBER (1) 252-10	
					WASHER, FLAT (1) 252-12	
					WICK (1) 252-13	
3	PAFZZ	2990-01-065-7617	19207	12259821	MOUNT, ENGINE TO BE INSTALLED ONLY AS A SET	1
					CAP, ENGINE MOUNT (1) 1-14	
					MOUNT, ENGINE (1) 1-18	
4	PAFZZ	4320-01-133-4069	62983	421242L	PARTS KIT, HYDRAULIC	1
					GASKET (1) 239-5	
					PACKING, PREFORMED (1) 239-4	
					PACKING, PREFORMED (1) 239-6	
					PACKING, SEAL (4) 239-8	
					PARTS KIT, ROTARY PU (2) 239-9	
					PARTS KIT, ROTARY PU (10) 239-10	
					PIN (20) 239-14	
					PLATE, INLET SUPPORT (1) 239-13	
					PLATE, OUTLET SUPPORT (1) 239-7	
					RETAINER, PACKING (1) 239-12	
					RING (1) 239-12	
					ROTOR (1) 239-11	
					SCREW (2) 239-15	
					END OF FIGURE	

3230 00-2

FIGURE 7. Example of kits breakdown option 2.

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TM 11-1520-238-23P

GROUP 30 SPECIAL TOOLS LIST - Continued **3745 00**

(1) ITEM NO.	(2) SMR CODE	(3) NSN	(4) CAGEC	(5) PART NUMBER	(6) DESCRIPTION AND USABLE ON CODE (UOC)	(7) QTY
					GROUP 30 SPECIAL TOOLS FIG. 254	
1	PEODD	6625-01-169-5333	80058	TS-3920A/ASM	TEST SET, STABILIZATION (BOI: 1 AUTH PER 15 AIRCRAFT)	
1	PEODD	6625-01-266-1636	80058	TS-3920B/ASM	TEST SET, STABILIZATION (BOI: 1 AUTH PER 15 AIRCRAFT)	
2	XBOZZ		80063	A3012556	WEDGE, 30/60/90 DEG. (BOI: 1 AUTH PER TEST SET)	
3	XBOZZ		80063	A3012557	WEDGE, 05/85/90 DEG. (BOI: 1 AUTH PER TEST SET)	
4	XBOZZ		80063	A3012558	PIN, ALIGNMENT (BOI: 1 AUTH PER TEST SET)	
5	XBOZZ		80063	A3012559	FIXTURE, PROTRACTOR (BOI: 1 AUTH PER TEST SET)	
					END OF FIGURE	

3745 00-2

FIGURE 8. Example of a special tools list work package.

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TM 9-1090-208-23P

NATIONAL STOCK NUMBER INDEX **1001 00**

STOCK NUMBER	FIG.	ITEM
5365-00-003-6807	4	4
5935-00-005-2826	3	1
5315-00-012-0123	65	1
5310-00-016-7216	28	12
5340-00-021-3495	77	20
	85	15
5310-00-027-7247	8	2
5310-00-030-0580	51	21
3110-00-034-5257	69	5
	72	5
	75	9
	77	8
	79	5
	81	10
	84	5
	85	3
5305-00-038-9048	45	27
5310-00-045-3296	77	25
	85	20
5310-00-045-4007	2	5
5305-00-052-6456	55	2
5310-00-054-0041	24	3
5305-00-054-5637	6	9
5305-00-054-5638	88	11
5305-00-054-5647	33	15
	51	25
5305-00-054-5648	10	1
	33	7
5305-00-054-5649	2	8
	29	13
	33	3
	51	15
	52	8
	89	3
5305-00-054-5650	51	3
5305-00-054-5651	9	5
	28	5
	90	1
5305-00-054-5652	52	1
5305-00-054-5653	28	6
5305-00-054-6653	89	17
5305-00-054-6654	28	23
	29	8
5305-00-054-6655	29	4
	88	21
5305-00-054-6657	29	18
5305-00-054-6566	51	29
5305-00-054-6669	1	10
5305-00-054-9263	60	4
5305-00-056-9961	9	3
	33	16
	88	35
5310-00-057-0573	33	8
	52	2
	90	2
5310-00-058-1823	29	1
	61	4
5315-00-058-6062	45	18
5305-00-059-3657	2	14
5305-00-059-3658	1	3
5305-00-059-3661	51	26
5310-00-061-7326	29	19
5305-00-066-7327	88	37
5305-00-066-7369	34	1
5365-00-067-3836	46	17
5305-00-068-0543	45	9
5365-00-068-8011	70	2
	80	2
5975-00-074-2072	61	12
	88	33
5970-00-074-8780	28	15
5320-00-076-4071	59	19
5360-00-079-1713	11A	2
5305-00-079-5835	51	22
	88	28
5306-00-080-1537	32	28
5305-00-103-2994	45	14
5905-00-104-8368	2	25
5306-00-106-6321	63	5
	77	22

1001 00-1

FIGURE 9. Example of a national stock number index work package.

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TM 11-7021-212-23P

PART NUMBER INDEX 0891 00

PART NUMBER	FIG.	ITEM
AN960C10L	12	20
AN960C4L	8	11
	9	44
	12	24
	17	28
AN960C416L	8	89
AN960C516L	8	66
AN960C6	8	74
	13	5
AN960C6L	27	6
AN960C616	27	22
AN960C8	8	41
	11	10
AN960C816	17	9
AP373-95	8	84
AP373-96	12	15
B3-14	17	12
CA4342	27	23
CA4440-4	16	54
CD2-Z147-1	16	44
CKR05BX102M	10	7
CMR05F201JPDR	14	16
DB-3	16	65
DBM5W5P	15	17
	31	11
	32	2
DBM5W5S	8	29
	31	3
DBM50906-1	8	28
	15	18
	31	5
	32	4
DDM50PE	10	27
DM53744-21	8	32
DM53744-24	31	2
DM53744-25	31	1
DM53745-25	15	16
	31	10
DM53745-27	15	15
	31	9
DM53745-28	32	1
DSC7900-10-C-6	8	19
EP15160X	10	81
EP162996	10	80
	14	91
I/O-100-00000	21	3
	22	5
	23	3
	24	2
	25	2
	26	2
JANTX1N1206A	12	1
	14	3
JANTX1N4102-1	10	89
JANTX1N4106-1	14	98
JANTX1N4109-1	9	31
JANTX1N4150-1	4	2
	8	77
	9	4
	10	1
	14	2
	28	1
JANTX1N4572A-1	10	86
JANTX1N4626-1	10	90
	28	8
JANTX1N4627	10	85
JANTX1N4627-1	14	99
JANTX1N5419	9	1
	12	2
JANTX1N5420	10	3
	14	4
JANTX1N5645A	9	30
JANTX1N5656A	10	87
JANTX1N5806	10	2
JANTX1N5811	9	3
	14	1
JANTX1N6075	9	2
JANTX1N647-1	4	1
JANTX2N2219A	28	5
JANTX2N2222A	10	29
	14	31
JANTX2N2369A	10	30
JANTX2N2907A	10	28
	14	34
JANTX2N3421	10	31
JANTX2N3507	9	11
JANTX2N3737	9	12
JANTX2N3868	9	13
JANTX2N6352	9	10
JANTX2N6764	14	32

0891 00-1

FIGURE 10. Example of a part number index work package.

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TM 9-XXXX-XXX-34P

REFERENCE DESIGNATOR INDEX 0032 00

REFERENCE DESIGNATOR	FIG.	ITEM
S1	1	15
W2	1	3
2AT1	2	309
2AT10	2	552
2AT11	2	699
2AT12	2	699
2AT13	2	479
2AT14	2	479
2AT2	2	309
2AT3	2	558
2AT4	2	564
2AT5	2	705
2AT5	2	479
2AT6	2	494
2AT7	2	675
2AT8	2	624
2AT9	2	552
2A1	2	489
2A1AT2	71	30
2A1AT3	71	33
2A1A1	70	6
2A1A10	70	6
2A1A11	70	12
2A1A13	70	13
2A1A14	70	14
2A1A15	70	14
2A1A16	70	15
2A1A17	70	16
2A1A18	70	13
2A1A19	70	14
2A1A20	70	14
2A1A21	70	15
2A1A22	70	16
2A1A23	70	17
2A1A24	70	13
2A1A25	70	14
2A1A26	70	14
2A1A27	70	15
2A1A28	70	16
2A1A29	70	18

REFERENCE DESIGNATOR	FIG.	ITEM
2A1A4	70	8
2A1A6	70	9
2A1A7	70	9
2A1A8	70	10
2A1A9	70	11
2A1DL1	70	25
2A1DL2	70	25
2A1DL3	70	25
2A1DL4	70	25
2A1DL5	70	25
2A1DL6	70	25
2A1J20	71	27
2A1J25	71	36
2A1J29	71	34
2A1W10	71	46
2A1W12	71	44
2A1W14	71	45
2A1W30	70	31
2A1W31	70	29
2A1W32	70	30
2A1W33	70	28
2A1W34	70	26
2A1W35	70	27
2A1W36	70	35
2A10	2	590
2A10A1	80	2
2A10A10	80	2
2A10A11	80	3
2A10A13	80	4
2A10A14	80	2
2A10A15	80	5
2A10A3	80	2
2A10A5	80	2
2A10A7	80	2
2A10A9	80	2
2A100	3	186
2A100CB1	18	15
2A100CB2	18	37
2A100CB3	18	37
2A100CB4	18	37

0032 00-1

FIGURE 11. Example of a reference designator index work package.

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MIL-STD-40051-7

DEPARTMENT OF DEFENSE STANDARD PRACTICE

TECHNICAL MANUALS

SUPPORTING INFORMATION

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1. SCOPE.

1.1 Scope. This standard establishes the technical content requirements for the preparation of supporting information (previously known as appendixes) for weapon systems and equipment Technical Manuals (TMs) and Depot Maintenance Work Requirements (DMWRs). These requirements are applicable for both paper and digital page-oriented TMs. Electronic delivery of supporting information included in TMs is accomplished through the use of the Supporting Information modular Document Type Definition (DTD). The DTD is available in a digital format. Refer to MIL-STD-40051 for information on obtaining this DTD. Supporting information requirements are included for the preparation of technical data that supplements the maintenance information contained in the body of the TM. This supplemental information includes reference data, general maintenance and parts information and associated illustrations.

2. APPLICABLE DOCUMENTS.

The applicable documents in section 2 of MIL-STD-40051 apply to this Part.

3. DEFINITIONS.

The definitions in section 3 of MIL-STD-40051 apply to this Part.

4. GENERAL REQUIREMENTS.

4.1 General. Supporting information shall be prepared for weapon systems, major equipment, components and applicable support and interface equipment. This information is used to supplement the maintenance information contained in the body of all levels of maintenance TMs. Supporting information shall be contained in work packages and become a part of a Supporting Information chapter.

4.2 Maintenance level applicability. Requirements contained in this standard are applicable to all maintenance levels unless specifically noted in bold and in parentheses (i.e.,**-34 only**). The labeled requirements shall be applicable to all TMs containing that maintenance level. For example, a (**-20 only**) requirement would only be applicable to the following TMs: -12, -13, -14, -20, -23, and -24.

4.3 Standard tables. Various standard tables required are noted throughout the text of this standard in bold and in parentheses (i.e., **(standard table)**). The formats and table heading names of these standard tables shall have no deviations.

4.4 Preparation of digital data for electronic delivery. Technical manual data prepared in work package format and delivered digitally in accordance with this standard shall be Standard Generalized Markup Language (SGML) tagged using the modular Assembly DTD and Formatting Output Specification Instance (FOSI). The DTD and FOSI has been developed in accordance with MIL-PRF-28001 and ISO 8879. Refer to MIL-STD-40051 for information on obtaining or accessing this modular DTD and FOSI. SGML tags used in the modular DTD are noted throughout the text of this standard in bracketed, bold characters (i.e., **<macfwp>**) as a convenience for the TM author and to ensure that the tags are used correctly when developing a document instance.

4.4.1 Use of the DTDs / FOSIs. The modular DTDs referenced in this Part interpret the technical content and structure for the functional requirements contained in this Part and are mandatory for use. The modular FOSIs referenced herein interprets the style and format. As specified by the contracting activity, FOSIs or

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style sheets may be used to produce final reproducible paper copy for all TMs prepared in accordance with this standard.

4.5 Content structure and format. The examples provided at the rear of this Part are an accurate representation of the content structure and format requirements contained herein and shall be followed to permit the effective use of the modular DTD for Supporting Information.

4.6 Style and format. Style and format requirements for the preparation of Department of Army TMs are contained in MIL-STD-40051-1; they are considered mandatory and are intended for compliance. Preferred general style and format requirements for Army TMs shall be provided by the procuring activity.

4.7 Work package development. Technical manual data developed in accordance with this standard shall be divided into individual, stand alone units of information work packages. A work package shall consist of descriptive, operational, maintenance, troubleshooting, support, or parts information for the weapon system or equipment.

4.8 Selective application and tailoring. MIL-STD-40051 contains some requirements that may not be applicable to the preparation of all technical manuals. Selective application and tailoring of requirements contained in MIL-STD-40051 are the responsibility of the contracting activity and shall be accomplished through the use of Appendix A, Technical Manual Content Selection Matrixes, of MIL-STD-40051. The applicability of some requirements is also designated by one of the following statements: unless specified otherwise by the contracting activity; as/when specified by the contracting activity; or when specified by the procuring activity.

5. DETAILED REQUIREMENTS.

5.1 Preparation of supporting information. Supporting information shall be developed as work packages and contained in a Supporting Information chapter**<sim>**. Supporting information work packages are described in 5.2 through 5.23. These work packages shall be placed in the TM in the order in which they are presented herein, as applicable.

5.2 References work package **<refwp>**. This work package shall list all publications referenced in the TM and required by the user to operate and/or maintain the equipment. It shall consist of a scope**<scope>** and publication list **<publist>**.

5.2.1 Scope **<scope>**. Information concerning the use and content of the references work package shall be prepared. (Refer to figure 1.)

5.2.2 Publication list **<publist>**. Individual paragraphs shall be prepared for each publication type. All related/referenced publications, with the exception of those publications that are currently unpublished, shall be listed. This list shall identify the publications by title**<name>** and number **<pubident>**. If the publication is nongovernment, the source shall be given. Titles shall be listed alphabetically under each publication type. (Refer to figure 1.) If a list of Applicable Publications (LOAP) exists, it may be referenced.

5.3 Maintenance allocation chart (MAC) work package**(-20/AVUM level only) <macwp>**.

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a. This work package shall list the applicable maintenance functions assigned to each maintenance level and shall be published only in equipment TMs containing unit or aviation unit maintenance instructions.

b. This work package shall be prepared in Functional Group Code (FGC) sequence to consolidate and identify those groups on the list which involve identified maintenance functions. The MAC shall be prepared according to the approved source data provided by the procuring activity.

c. The data described in 5.3.1 through 5.3.5 shall be prepared for the MAC work package (both standard and three-level for Army aviation formats are included for each).

5.3.1 Introduction for standard format MAC work package**(-20/AVUM only) <intro>**. The following text shall be prepared and included verbatim. (Refer to figure 2 .)

"MAINTENANCE ALLOCATION CHART (MAC)

INTRODUCTION

The Army Maintenance System MAC

This introduction provides a general explanation of all maintenance and repair functions authorized at various maintenance levels under the standard Army Maintenance System concept.

The MAC (immediately following the introduction) designates overall authority and responsibility for the performance of maintenance functions on the identified end item or component. The application of the maintenance functions to the end item or component shall be consistent with the capacities and capabilities of the designated maintenance levels, which are shown on the MAC in column (4) as:

Unit — includes two subcolumns, C (operator/crew) and O (unit) maintenance.

Direct Support — includes an F subcolumn.

General Support — includes an H subcolumn.

Depot — includes a D subcolumn.

The tools and test equipment requirements (immediately following the MAC) list the tools and test equipment (both special tools and common tool sets) required for each maintenance function as referenced from the MAC.

The remarks (immediately following the tools and test equipment requirements) contain supplemental instructions and explanatory notes for a particular maintenance function.

Maintenance Functions

Maintenance functions are limited to and defined as follows:

1. Inspect. To determine the serviceability of an item by comparing its physical, mechanical, and/or electrical characteristics with established standards through examination (e.g., by sight, sound, or feel). This includes scheduled inspection and gagings and evaluation of cannon tubes.

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2. Test. To verify serviceability by measuring the mechanical, pneumatic, hydraulic, or electrical characteristics of an item and comparing those characteristics with prescribed standards on a scheduled basis, i.e., load testing of lift devices and hydrostatic testing of pressure hoses.

3. Service. Operations required periodically to keep an item in proper operating condition; e.g., to clean (includes decontaminate, when required), to preserve, to drain, to paint, or to replenish fuel, lubricants, chemical fluids, or gases. This includes scheduled exercising and purging of recoil mechanisms.

4. Adjust. To maintain or regulate, within prescribed limits, by bringing into proper position, or by setting the operating characteristics to specified parameters.

5. Align. To adjust specified variable elements of an item to bring about optimum or desired performance.

6. Calibrate. To determine and cause corrections to be made or to be adjusted on instruments of test, measuring, and diagnostic equipment used in precision measurement. Consists of comparisons of two instruments, one of which is a certified standard of known accuracy, to detect and adjust any discrepancy in the accuracy of the instrument being compared.

7. Remove/Install. To remove and install the same item when required to perform service or other maintenance functions. Install may be the act of emplacing, seating, or fixing into position a spare, repair part, or module (component or assembly) in a manner to allow the proper functioning of an equipment or system.

8. Replace. To remove an unserviceable item and install a serviceable counterpart in its place. "Replace" is authorized by the MAC and assigned maintenance level is shown as the third position code of the Source, Maintenance and Recoverability (SMR) code.

9. Repair. The application of maintenance services, including fault location/troubleshooting, removal/installation, disassembly/assembly procedures, and maintenance actions to identify troubles and restore serviceability to an item by correcting specific damage, fault, malfunction, or failure in a part, subassembly, module (component or assembly), end item, or system.

NOTE

The following definitions are applicable to the "repair" maintenance function:

Services — Inspect, test, service, adjust, align, calibrate, and/or replace.

Fault location/troubleshooting — The process of investigating and detecting the cause of equipment malfunctioning; the act of isolating a fault within a system or Unit Under Test (ET).

Disassembly/assembly — The step-by-step breakdown (taking apart) of a spare/functional group coded item to the level of its least component, that is assigned an SMR code for the level of maintenance under consideration (i.e., identified as maintenance significant).

Actions — Welding, grinding, riveting, straightening, facing, machining, and/or resurfacing.

10. Overhaul. That maintenance effort (service/action) prescribed to restore an item to a completely serviceable/operational condition as required by maintenance standards in appropriate technical

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publications. Overhaul is normally the highest degree of maintenance performed by the Army. Overhaul does not normally return an item to like new condition.

11. Rebuild. Consists of those services/actions necessary for the restoration of unserviceable equipment to a like new condition in accordance with original manufacturing standards. Rebuild is the highest degree of materiel maintenance applied to Army equipment. The rebuild operation includes the act of returning to zero those age measurements (e.g., hours/miles) considered in classifying Army equipment/components.

Explanation of Columns in the MAC

Column (1) — Group Number. Column (1) lists FGC numbers, the purpose of which is to identify maintenance significant components, assemblies, subassemblies, and modules with the Next Higher Assembly (NHA).

Column (2) — Component/Assembly. Column (2) contains the item names of components, assemblies, subassemblies, and modules for which maintenance is authorized.

Column (3) — Maintenance Function. Column (3) lists the functions to be performed on the item listed in column (2). (For a detailed explanation of these functions refer to "Maintenance Functions" outlined above.)

Column (4) — Maintenance Level. Column (4) specifies each level of maintenance authorized to perform each function listed in column (3), by indicating work time required (expressed as manhours in whole hours or decimals) in the appropriate subcolumn. This work time figure represents the active time required to perform that maintenance function at the indicated level of maintenance. If the number or complexity of the tasks within the listed maintenance function varies at different maintenance levels, appropriate work time figures are to be shown for each level. The work time figure represents the average time required to restore an item (assembly, subassembly, component, module, end item, or system) to a serviceable condition under typical field operating conditions. This time includes preparation time (including any necessary disassembly/assembly time), troubleshooting/fault location time, and quality assurance time in addition to the time required to perform the specific tasks identified for the maintenance functions authorized in the MAC. The symbol designations for the various maintenance levels are as follows:

C — Operator or crew maintenance

O — Unit maintenance

F — Direct support maintenance

L — Specialized repair activity (SRA)

H — General support maintenance

D — Depot maintenance

NOTE

The "L" maintenance level is not included in column (4) of the MAC. Functions to this level of maintenance are identified by a work time figure in the "H" column of column (4), and an associated reference code is used in the REMARKS column (6). This code is keyed to the remarks and the SRA complete repair application is explained there.

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Column (5) — Tools and Equipment Reference Code. Column (5) specifies, by code, those common tool sets (not individual tools), common Test, Measurement and Diagnostic Equipment (TMDE), and special tools, special TMDE and special support equipment required to perform the designated function. Codes are keyed to the entries in the tools and test equipment table.

Column (6) — Remarks Code. When applicable, this column contains a letter code, in alphabetical order, which is keyed to the remarks table entries.

Explanation of Columns in the Tools and Test Equipment Requirements

Column (1) — Tool or Test Equipment Reference Code. The tool or test equipment reference code correlates with a code used in column (5) of the MAC.

Column (2) — Maintenance Level. The lowest level of maintenance authorized to use the tool or test equipment.

Column (3) — Nomenclature. Name or identification of the tool or test equipment.

Column (4) — National Stock Number (NSN). The NSN of the tool or test equipment.

Column (5) — Tool Number. The manufacturer's part number, model number, or type number.

Explanation of Columns in the Remarks

Column (1) — Remarks Code. The code recorded in column (6) of the MAC.

Column (2) — Remarks. This column lists information pertinent to the maintenance function being performed as indicated in the MAC."

5.3.2 Introduction for three-level Army aviation MAC work package(-20/AVUM only) <intro>. The following text shall be prepared and included verbatim. (Refer also to figure 3 .)

"MAINTENANCE ALLOCATION CHART (MAC)

INTRODUCTION

Aviation Maintenance Allocation Chart

This MAC assigns maintenance functions in accordance with the Aviation Maintenance concept for Army aviation. These maintenance levels — Aviation Unit Maintenance (AVUM), Aviation Intermediate Maintenance (AVIM), and depot maintenance — are depicted in the MAC as:

AVUM — corresponds to an "O" code in the Repair Parts and Special Tools List (RPSTL).

AVIM — corresponds to an "F" code in the RPSTL.

DEPOT — corresponds to a "D" code in the RPSTL.

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The maintenance to be performed below depot and in the field is described as follows:

Aviation Unit Maintenance (AVUM). AVUM activities will be staffed and equipped to perform high frequency "On-Aircraft" maintenance tasks required to retain or return aircraft systems to a serviceable condition. The maintenance capability of the AVUM will be governed by the Maintenance Allocation Chart (MAC) and limited by the amount and complexity of Ground Support Equipment (GSE), facilities required, authorized manning strength, and critical skills available. The range and quantity of authorized spare modules/components will be consistent with the mobility requirements dictated by the air mobility concept. (Assignments of maintenance tasks to divisional company size aviation units will consider the overall maintenance capability of the division, the requirement to conserve personnel and equipment resources, and air mobility requirements.)

1. Company Size Aviation Units. Perform those tasks which consist primarily of preventive maintenance and maintenance repair and replacement functions associated with sustaining a high level of aircraft operational readiness. Perform maintenance inspections and servicing to include preflight, daily, intermediate, periodic (or phased), and special inspections, as authorized by the MAC or higher headquarters. Identify the cause of equipment/system malfunctions using applicable technical manual troubleshooting instructions, Built-In Test Equipment (BITE), installed aircraft instruments, or Test, Measurement, and Diagnostic Equipment (TMDE). Replace worn or damaged modules/components that do not require complex adjustments or system alignment and which can be removed/installed with available skills, tools, and ground support equipment. Perform operational and continuity checks and make minor repairs to the electrical system. Inspect, service, and make operational, capacity, and pressure checks to hydraulic systems. Perform servicing, functional adjustments, and minor repair/replacement to the flight control, propulsion, power train, and fuel systems. Accomplish airframe repair that does not require extensive disassembly, jigging, or alignment. The manufacture of airframe parts will be limited to those items which can be fabricated with tools and equipment found in current air mobile tool and shop sets. Evacuate unserviceable modules/components and end items beyond the repair capability of AVUM to the support AVIM.

2. Less than Company Size Aviation Units. Aviation elements organic to brigade, group, battalion headquarters, and detachment size units are normally small and have less than 10 aircraft assigned. Maintenance tasks performed by these units will be those which can be accomplished by the aircraft crew chief or assigned aircraft repairman and will normally be limited to preventive maintenance, inspections, servicing, spot painting, module/component fault diagnosis, and replacement of selected modules/components. Repair functions will normally be accomplished by the support AVIM unit.

Aviation Intermediate Maintenance (AVIM).

1. Provides mobile, responsive "one-stop" maintenance support. (Maintenance functions which are not conducive to sustaining air mobility will be assigned to depot maintenance.)

2. May perform all maintenance functions authorized to be done at AVUM. Repair of equipment for return to user will emphasize support of operational readiness requirements. Authorized maintenance includes replacement and repair of modules/components and end items which can be accomplished efficiently with available skills, tools, and equipment.

3. Establishes the Direct Exchange (DX) program for AVUM units by repairing selected items for return to stock when such repairs cannot be accomplished at the AVUM level.

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4. Inspects, troubleshoots, performs diagnostic tests, repairs, adjusts, calibrates, and aligns aircraft system modules/components. AVIM units will have capability to determine the serviceability of specified modules/components removed prior to the expiration of the Time Between Overhaul (TBO) or finite life. Module/component disassembly and repair will support the DX program and will normally be limited to tasks requiring cleaning and the replacement of seals, fittings, and items of common hardware. Airframe repair and fabrication of parts will be limited to those maintenance tasks which can be performed with available tools and test equipment. Unserviceable reparable modules/components and end items which are beyond the capability of AVIM to repair will be evacuated to depot maintenance.

5. Performs aircraft weight and balance inspections and other special inspections which exceed AVUM capability.

6. Provides quick response maintenance support, including aircraft recovery and air evacuation, on-the-job training, and technical assistance through the use of mobile maintenance contact teams.

7. Maintains authorized operational readiness float aircraft.

8. Provides collection and classification services for serviceable/unserviceable materiel.

9. Operates a cannibalization activity in accordance with AR 710-2 (Supply Policy Below the Wholesale Level) and DA PAM 710-2-2 (Supply Support Activity System Manual Procedures). (The aircraft maintenance company within the maintenance battalion of a division will perform AVIM functions consistent with air mobility requirements and conservation of personnel and equipment resources. Additional intermediate maintenance support will be provided by the supporting nondivisional AVIM unit.)

Use of the MAC

NOTE

Approved item names are used throughout this MAC. Generic terms/nomenclature (if any) are expressed in parentheses and are not to be considered as official terminology.

This MAC assigns maintenance functions to the lowest level of maintenance, based on past experience and the following considerations:

Skills available.

Work time required.

Tools and test equipment required and/or available.

Only the lowest level of maintenance authorized to perform a maintenance function is indicated. If the lowest maintenance level cannot perform all tasks of any single maintenance function (e.g., test, repair), then the higher maintenance level(s) that can accomplish additional tasks will also be indicated.

A maintenance function assigned to a maintenance level will automatically be authorized to be performed at any higher maintenance level.

A maintenance function that cannot be performed at the assigned level of maintenance for any reason may be evacuated to the next higher maintenance level. Higher maintenance levels will perform the maintenance

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functions of lower maintenance levels when required by the commander who has the authority to direct such tasking.

The assignment of a maintenance function will not be construed as authorization to carry the related repair parts or spares in stock. Information to requisition or otherwise secure the necessary repair parts will be as specified in the associated RPSTL.

Normally there will be no deviation from the assigned level of maintenance. In cases of operational necessity, at the request of a lower maintenance level and on a one-time basis, transfer of maintenance functions to the lower level may be accomplished by specific authorization of the maintenance officer of the higher level of maintenance to which the function is assigned. The special tools, equipment, etc., required by the lower level of maintenance to perform this function will be furnished by the maintenance level to which the function is assigned. This transfer of a maintenance function to a lower maintenance level does not relieve the higher maintenance level of the responsibility for the function. The higher level of maintenance will provide technical supervision and inspection of the function being performed at the lower level.

Maintenance Functions

Maintenance functions will be limited to and defined as follows:

1. Inspect. To determine the serviceability of an item by comparing its physical, mechanical, and/or electrical characteristics with established standards through examination (e.g., by sight, sound, or feel).

2. Test. To verify serviceability by measuring the mechanical, pneumatic, hydraulic, or electrical characteristics of an item and comparing those characteristics with prescribed standards.

3. Service. Operations required periodically to keep an item in proper operating condition, i.e., to clean (includes decontaminate, when required), to preserve, to drain, to paint, or to replenish fuel, lubricants, chemical fluids, or gases.

4. Adjust. To maintain or regulate, within prescribed limits, by bringing into proper or exact position, or by setting the operating characteristics to specified parameters.

5. Align. To adjust specified variable elements of an item to bring about optimum or desired performance.

6. Calibrate. To determine and cause corrections to be made or to be adjusted on instruments of test, measuring, and diagnostic equipments used in precision measurement. Consists of comparisons of two instruments, one of which is a certified standard of known accuracy, to detect and adjust any discrepancy in the accuracy of the instrument being compared.

7. Remove/Install. To remove and install the same item when required to perform service or other maintenance functions. Install may be the act of emplacing, seating, or fixing into position a spare, repair part, or module (component or assembly) in a manner to allow the proper functioning of an equipment or system.

8. Replace. To remove an unserviceable item and install a serviceable counterpart in its place. "Replace" is authorized by the MAC and assigned maintenance level is shown as the third position code of the Source, Maintenance, and Recoverability (SMR) code.

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9. Repair. The application of maintenance services, including fault location/troubleshooting, removal/installation, disassembly/assembly procedures, and maintenance actions to identify troubles and restore serviceability to an item by correcting specific damage, fault, malfunction, or failure in a part, subassembly, module (component or assembly), end item, or system.

NOTE

The following definitions are applicable to the "repair" maintenance function:

Services — Inspect, test, service, adjust, align, calibrate, and/or replace.

Fault location/troubleshooting — The process of investigating and detecting the cause of equipment malfunctioning; the act of isolating a fault within a system or Unit Under Test (UUT).

Disassembly/assembly — The step-by-step taking apart (or breakdown) of a spare/functional group coded item to the level of its least component identified as maintenance significant (i.e., assigned an SMR code) for the level of maintenance under consideration.

Actions — Welding, grinding, riveting, straightening, facing, machining, and/or resurfacing.

10. Overhaul. That maintenance effort (service/action) prescribed to restore an item to a completely serviceable/operational condition as required by maintenance standards in appropriate technical publications. Overhaul is normally the highest degree of maintenance performed by the Army. Overhaul does not normally return an item to like new condition.

11. Rebuild. Those services/actions necessary for the restoration of unserviceable equipment to a like new condition in accordance with original manufacturing standards. Rebuild is the highest degree of materiel maintenance applied to Army equipment. The rebuild operation includes the act of returning to zero those age measurements (e.g., hours/miles) considered in classifying Army equipment/components.

Explanation of Columns in the MAC

Columns (1) and (2) - Functional Groups. The functional groupings in the sample below identify maintenance significant components, assemblies, subassemblies, and modules with the next higher assembly.

GROUP NUMBER	DESCRIPTION
04	POWER PLANT
0401	ENGINE, GENERAL
	Servicing, handling inspection requirements, overhaul and retirement schedules. External lines and hoses. (As applicable.)

GROUP NUMBER	DESCRIPTION
0402	COMPRESSOR SECTION (COLD SECTION MODULE)
	Rotor, blades, vanes, impeller, stators, inlet guide vanes, main frame, particle separator, bleed valve, bearings, seals, external lines and hoses.

GROUP NUMBER	DESCRIPTION

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GROUP NUMBER	DESCRIPTION
0403	COMBUSTION SECTION (HOT SECTION MODULE) Liners, nozzles, stators, rotor, seals, couplings, blades.
0404	POWER-TURBINE (POWER TURBINE MODULE) Nozzles, rotors, blades, exit guide vanes, exhaust frame, drive shaft, bearings, seals, external lines and hoses.
0405	ACCESSORY GEAR BOX (ACCESSORY SECTION MODULE) Input and output gears, seals, chip detector, housings, drive shaft, bearings.
0406	FUEL SYSTEM Fuel control, fuel boost pump, governors, fuel filter assembly, sequence valve, fuel manifold, fuel nozzle, external lines and hoses.
0407	ELECTRICAL SYSTEM Electrical control units, exciters, thermocouples, ignition harness, electrical cables, history record, torque overspeed sensor, Np sensor, external lines and hoses.
0408	OIL SYSTEM Tanks, oil filter, oil cooler, lube and scavenger pumps, oil filter bypass sensor, external lines and hoses.

Column (3) — Maintenance Function. Column (3) lists the functions to be performed on the items listed in column (2).

Column (4) — Maintenance Level. The maintenance levels AVUM, AVIM, and DEPOT are listed on the MAC with individual columns that include the work times for maintenance functions at each maintenance level. Work time presentations such as "0.1" indicate the average time (expressed in manhours in whole hours or decimals) it requires a maintenance level to perform a specified maintenance function. If a work time has not been established, the columnar presentation will indicate "--." Maintenance levels higher than the level of maintenance indicated are authorized to perform the indicated function.

Column (5) — Tools and Equipment Reference Code. Column (5) specifies, by code, those common tool sets (not individual tools), common TMDE, and special tools, special TMDE, and special support equipment required to perform the designated function.

Column (6) — Remarks Code. When applicable, this column contains a letter code, in alphabetical order, which is keyed to the remarks.

Explanation of Columns in the Tools and Test Equipment Requirements

Column (1) — Tool or Test Equipment Reference Code. The tool or test equipment reference code correlates with a code used in column (5) of the MAC.

Column (2) — Maintenance Level. The lowest level of maintenance authorized to use the tool or test equipment.

Column (3) — Nomenclature. Name or identification of the tool or test equipment.

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Column (4) — National Stock Number (NSN). The NSN of the tool or test equipment.

Column (5) — Tool Number. The manufacturer's part number.

Explanation of Columns in the Remarks

Column (1) — Remarks Code. The code recorded in column (6) of the MAC.

Column (2) — Remarks. This column lists information pertinent to the maintenance function being performed as indicated in the MAC."

5.3.3 MAC.

5.3.3.1 MAC entries.

a. The basic entries in the MAC shall be a list of functional groups applicable to the end item which require maintenance. The term functional group applies to reparable assemblies and subassemblies, i.e., spares (any reparable component required for the maintenance or repair of an end item), but not to repair parts (any consumable, nonreparable component required for the maintenance or repair of an end item). The end item group shall be numbered "00," or its equivalent "AA."

b. Entries shall be item names (a basic name and a noun word or phrase modifier, e.g., transformer, pulse, low power) and, where applicable, type designators, without stock or part numbers (P/Ns) if possible, in order to minimize need for subsequent change; however, entries shall contain positive identification. Parts that are not subject to maintenance shall not be listed in table 1 of the MAC.

c. All item names of MAC functional groups shall be official nomenclature in accordance with the RPSTL nomenclature or other source as specified by the procuring activity. Reverse word order shall be used in the MAC.

d. The maintenance code entered in the third position of the Source, Maintenance, and Recoverability (SMR) code in the RPSTL shall be used to identify the lowest category of maintenance that is authorized to remove, replace, and use the spare or repair parts. SMR codes are further defined in MIL-STD-40051-6.

e. If the maintenance function is a replace function only for a repair part, the repair part shall not be listed in the MAC, unless not listing the repair part would result in omission of the Next Higher Assembly (NHA) group number; in this case, the part shall be listed in order to list the NHA functional group number.

f. All items in the MAC shall specify the maintenance level(s) to which a function is authorized.

g. Exception is authorized to ammunition MACs to permit use of maintenance function headings that better describe or identify ammunition peculiar maintenance functions. The headings used and their definitions shall be included in the appropriate ammunition TM(s).

5.3.3.2 MAC format. The MAC **<mac> (standard table)** shall be prepared in the format shown in figure 4 (standard) or figure 5 (aviation)**<avmac>**, as applicable.

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a. For an explanation of data to be listed in columns of the MAC, refer to the introduction information presented in 5.3.1 or 5.3.2 as applicable.

b. The group number **<groupno>** shall be entered in column 1, the nomenclature of the spare (component/assembly) **<compassem>** shall be entered in column 2, and the maintenance function **<maintfunc>** shall be listed in column 3 of the MAC.

c. Column 4 of the standard MAC shall be divided into four main headings, one for each level of maintenance **<maintclass>** (i.e., unit **<unit>**, direct support **<direct>**, general support **<gensup>**, and depot **<depot>**). Column 4 of the three-level aviation MAC**<avmaintclass>** shall be divided into three main headings (i.e., AVUM or unit**<avum>**, AVIM or intermediate **<avim>**, and depot **<depot>**).

d. A work time figure must appear in the subcolumn for the maintenance level authorized to perform the maintenance listed in column 3.

e. Reference numbers for all required tools and test equipment **<terefs>** shall be listed in column 5 of the MAC. These reference numbers shall correspond to the appropriate tools/test equipment listed in the tools and test equipment table.

f. Reference letters for applicable remarks **<remarkrefs>** shall be listed in column 6 of the MAC. These reference letters shall correspond to the appropriate remarks listed in the remarks table.

5.3.4 Tools and test equipment requirements **<tereqtab>**. A tabular list **(standard table)** of all tools and test equipment, both special and common, required to maintain the equipment shall be prepared in accordance with the format shown in figure 6 or figure 7 , as applicable. Common tools shall not be included on this list when they are part of an existing set, kit, or outfit authorized to the intended user; however, the authorized set, kit, or outfit which contains the prescribed common tools shall be listed.

5.3.5 Remarks **<remarktab>**. Remarks **(standard table)** pertinent to maintenance functions shall be prepared in accordance with the format shown in figure 6 or 7 , as applicable.

5.4 RPSTL work package **(-20/AVUM level or above only) <rpstlwp>**. This work package shall be prepared in accordance with MIL-STD-40051-6.

5.5 Components of end item (COEI) and basic issue items (BII) lists work package**(operator only) <coeibiiwp>**. This work package shall be prepared as an inventory for the equipment to ensure safe and efficient operation. The data described in 5.5.1 through 5.5.3 shall be prepared.

5.5.1 Introduction for COEI and BII lists work package**(operator only) <intro>**. The following introduction shall be prepared and included verbatim. (Refer also to figure 8.)

"COMPONENTS OF END ITEM (COEI) AND BASIC ISSUE ITEMS (BII) LISTS

INTRODUCTION

Scope

This work package lists COEI and BII for the *(insert the short end item name)* to help you inventory items for safe and efficient operation of the equipment.

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General

The COEI and BII information is divided into the following lists:

Components of End Item (COEI). This list is for information purposes only and is not authority to requisition replacements. These items are part of the *(enter name of end item)*. As part of the end item, these items must be with the end item whenever it is issued or transferred between property accounts. Items of COEI are removed and separately packaged for transportation or shipment only when necessary. Illustrations are furnished to help you find and identify the items.

Basic Issue Items (BII). These essential items are required to place the *(enter name of end item)* in operation, operate it, and to do emergency repairs. Although shipped separately packaged, BII must be with the*(enter name of end item)* during operation and when it is transferred between property accounts. Listing these items is your authority to request/requisition them for replacement based on authorization of the end item by the TOE/MTOE. Illustrations are furnished to help you find and identify the items.

Explanation of Columns in the COEI List and BII List

Column (1) — Illus Number. Gives you the number of the item illustrated.

Column (2) — National Stock Number (NSN). Identifies the stock number of the item to be used for requisitioning purposes.

Column (3) — Description, CAGEC, and Part Number. Identifies the Federal item name (in all capital letters) followed by a minimum description when needed. The stowage location of COEI and BII is also included in this column. The last line below the description is the CAGEC (commercial and Government entity code) (in parentheses) and the part number.

Column (4) — Usable On Code. When applicable, gives you a code if the item you need is not the same for different models of equipment. *(Add the following only as applicable. Replace Xs with appropriate codes and model numbers.) These codes are identified below:*

Code	***Used on***
XXX	*Model XXX*
XXX	*Model XXXX*
XXX	*Model XXXXX*

Column (5) — Unit of Measure (U/M). Indicates the physical measurement or count of the item as issued per the National Stock Number shown in column (2).

Column (6) — Qty Rqr. Indicates the quantity required."

5.5.2 COEI list **<coei>**. This list shall be prepared as an illustrated tabular list of components of the end item (spare/repair parts that are removed from the major end item and separately packaged or stowed for transportation or movement; includes on-board spares). The illustrations shall be placed above the list. The arrangement of the illustrations and list shall be similar to that shown in figure 8.

5.5.2.1 List **<coeitab>**. The COEI list **(standard table)** shall include the headings and basic content shown in figure 8, applicable to the specific equipment. The description of each item shall consist of the approved

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Federal item name **<desc>**, followed by a short description when needed. Items shall be listed alphabetically. The Commercial and Government Entity Code (CAGEC)**<cageno>** shall be located below the item and in parentheses. The part number **<partno>** shall follow the CAGEC. The stowage location of COEI shall also be included in the description column. When more than one model or configuration is applicable and Usable On Codes (UOC) **<uoc>** are assigned, the UOC shall appear in a separate column adjacent to the description column. (Refer to figure 8.) When on-board spares apply, there shall be a break in the text of the list and a new heading ON-BOARD SPARES shall be used. A list of the on-board spares shall appear in the same format as required for the basic COEI list.

5.5.3 BII list **<bii>**. This tabular list **(standard table)** shall be prepared in the same format and include similar content (tailored to the applicable BII) as required for the COEI list. The stowage location of BII shall also be included in the description column. (Refer to 5.5.1 and figure 8 .)

5.6 AAL work package **(operator only) <aalwp>**. This work package shall list all AAL items (i.e., items not issued with the end item; not listed on the end item engineering drawing as part of the end item, National Stock Number (NSN) configuration; not required to be turned in with the end item; separately authorized by MTOE, TDA, CTA, or JTA; and provided for information only). The data described in 5.6.1 and 5.6.2 shall be prepared.

5.6.1 Introduction **<intro>**. The following introduction (text below within the quotation marks) shall be prepared and included verbatim. (Refer also to figure 9.)

"ADDITIONAL AUTHORIZATION LIST (AAL)

INTRODUCTION

Scope

This work package lists additional items you are authorized for the support of the*(enter short item name)*.

General

This list identifies items that do not have to accompany the*(enter short item name)* and that do not have to be turned in with it. These items are all authorized to you by CTA, MTOE, TDA, or JTA.

Explanation of Columns in the AAL

Column (1) — National Stock Number (NSN). Identifies the stock number of the item to be used for requisitioning purposes.

Column (2) — Description, Commercial and Government Entity Code (CAGEC), and Part Number (P/N). Identifies the Federal item name (in all capital letters) followed by a minimum description when needed. The last line below the description is the CAGEC (in parentheses) and the part number.

Column (3) — Usable On Code. When applicable, gives you a code if the item you need is not the same for different models of equipment. *(Add the following only as applicable. Replace Xs with appropriate codes and model numbers.) These codes are identified below:*

Code ***Used on***

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XXX	*Model XXX*
XXX	*Model XXXX*
XXX	*Model XXXXX*

Column (4) — Unit of Measure (U/M). Indicates the physical measurement or count of the item as issued per the National Stock Number shown in column (1).

Column (5) — Qty Recm. Indicates the quantity recommended."

5.6.2 AAL list <**aal**>. A tabular list **(standard table)** of all additional authorized items shall be prepared. The format and general content of the list shall be prepared as shown in figure 9. The headings and subsequent information for this list shall be the same as the COEI and BII lists except the ILLUS NUMBER column required for the COEI and BII lists shall not apply since there are no illustrations used, and the QTY column shall be QTY RECM (quantity recommended). The items shall be listed alphabetically.

5.7 Expendable and durable items list work package<**explistwp**>. This work package shall be prepared to provide the TM user a list of all expendable and durable items called out in the TM text which are necessary to operate and/or maintain the equipment. The following data described in 5.7.1 and 5.7.2 shall be included.

5.7.1 Introduction for expendable and durable items list work package<**intro**>. The following introduction (text below within the quotation marks) shall be prepared and included verbatim. (Refer also to figure 10).

"EXPENDABLE AND DURABLE ITEMS LIST

INTRODUCTION

Scope

This work package lists expendable and durable items that you will need to operate and maintain the*(enter equipment/end item name)*. This list is for information only and is not authority to requisition the listed items. These items are authorized to you by CTA 50-970, Expendable/Durable Items (Except Medical, Class V Repair Parts, and Heraldic Items), or CTA 8-100, Army Medical Department Expendable/Durable Items.

Explanation of Columns in the Expendable/Durable Items List

Column (1) — Item Number. This number is assigned to the entry in the list and is referenced in the narrative instructions to identify the item (e.g., Use brake fluid (item 5, WP 0098 00).).

Column (2) — Level. This column identifies the lowest level of maintenance that requires the listed item *(include as applicable: C = Operator/Crew, O = Unit/AVUM, F = Direct Support/AVIM, H = General Support, D = Depot)*.

Column (3) — National Stock Number (NSN). This is the NSN assigned to the item which you can use to requisition it.

Column (4) — Item Name, Description, Commercial and Government Entity Code (CAGEC), and Part Number (P/N). This column provides the other information you need to identify the item.

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Column (5) — Unit of Measure (U/M). This code shows the physical measurement or count of an item, such as gallon, dozen, gross, etc."

5.7.2 Expendable and durable items list **<explist>**. This list **(standard table)** shall be prepared in tabular format as shown in figure 10. No illustrations shall be prepared for these items. Items appearing in the tabular list shall appear in alphabetical sequence by item name. Items to be listed shall be those approved by the procuring activity.

5.8 Stowage and decal / data plate guide work package **(operator only) <stowagewp>**. This work package shall begin with an introduction **<intro>** which gives the scope of the work package. An illustration detailing the location **<stowinfo>** of applicable COEI, BII, and AAL items shall be prepared for this work package. As applicable, illustrations detailing the locations **<decalinfo>** of all decals and data plates shall be prepared and included. (Refer to figure 11 for an example.)

5.9 On-vehicle equipment loading plan work package **(operator only) <eqploadwp>**. This work package shall be prepared when applicable to the equipment. The loading plan shall include information provided by the procuring activity. The data described in 5.9.1 and 5.9.2 shall be included.

5.9.1 Introduction **<intro>**. A brief scope statement shall be prepared explaining the purpose of the loading plan and identifying the equipment covered by the on-vehicle equipment loading plan work package. (Refer to figure 12.)

5.9.2 Illustrated loading plan list(s) **<loaddesc>**. Lists of the items identified on the illustrations shall be prepared. (Refer to figure 12.) Illustrations shall be prepared to show the location of all applicable equipment. External and internal views shall be used. As applicable, both tactical and nontactical situation loading configurations shall be shown. The list **<loadlist>** shall be located on the same page or the page adjacent to the associated illustration(s). The list shall have two columns: the item number column **<callout>** (corresponding to the illustration item numbers) and the item name **<item>**.

5.10 Tool identification list work package **(-20/AVUM level or above only) <toolidwp>**. This work package shall include a list of the tools authorized to the levels of maintenance covered in the narrative portion of the TM and as referenced by the initial setups. The following shall be prepared. The data described in 5.10.1 and 5.10.2 shall be included.

5.10.1 Introduction for tool identification list work package **(-20/AVUM level or above only) <intro>**. The following introduction (text below within the quotation marks) shall be prepared and included verbatim. (Refer to figure 13.)

"TOOL IDENTIFICATION LIST

INTRODUCTION

Scope

This work package lists all common tools and supplements and special tools/fixtures needed to maintain the *(insert equipment name)*.

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Explanation of Columns in the Tool Identification List

Column (1) — Item Number. This number is assigned to the entry in the list and is referenced in the initial setup to identify the item (e.g., Extractor (item 32, WP 0090 00)).

Column (2) — Item Name. This column lists the item by noun nomenclature and other descriptive features (e.g., Gage, belt tension).

Column (3) — National Stock Number. This is the National Stock Number (NSN) assigned to the item; use it to requisition the item.

Column (4) — Part Number. Indicates the primary number used by the manufacturer (individual, company, firm, corporation, or Government activity) which controls the design and characteristics of the item by means of its engineering drawings, specifications, standards, and inspection requirements to identify an item or range of items.

Column (5) — Reference. This column identifies the authorizing supply catalog or RPSTL for items listed in this work package."

5.10.2 Tool identification list **<avtoolidlist>**. Applicable information for this tabular list **(standard table)** shall be prepared and formatted as shown in figure 13. Item names shall be in alphabetical order.

5.11 Illustrated list of manufactured items work package **(-20/AVUM level or above only) <manuwp>**. This work package shall include technical information for each item authorized to be manufactured or fabricated by unit/AVUM, direct support/AVIM, or general support maintenance personnel (e.g., all "MO," "MF," and "MH" source coded items authorized in the applicable RPSTL). The work package shall include the data described in 5.11.1 through 5.11.3.

5.11.1 Introduction for illustrated list of manufactured items work package **(-20/AVUM level or above only) <intro>**. The following introduction (text below within the quotation marks) shall be prepared and included verbatim. (Refer also to figure 14.)

"ILLUSTRATED LIST OF MANUFACTURED ITEMS

INTRODUCTION

Scope

This work package includes complete instructions for making items authorized to be manufactured or fabricated at the *(enter applicable maintenance level)* .

How to Use the Index of Manufactured Items

A part number index in alphanumeric order is provided for cross-referencing the part number of the item to be manufactured to the page which covers fabrication criteria.

Explanation of the Illustrations of Manufactured Items

All instructions needed by maintenance personnel to manufacture the item are included on the illustrations. *(When applicable, a reference to the associated RPSTL TM or RPSTL work package shall be entered*

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here.) All bulk materials needed for manufacture of an item are listed by part number or specification number in a tabular list on the illustration."

5.11.2 Index of manufactured items **<manuindx>**. A part number description **<partno>** or drawing index **<dwgno>** shall be prepared which lists part numbers, in alphanumeric order, of all items illustrated in this work package. The part number description **<name>**, or drawing of each manufactured item shall be cross-referenced to the applicable figure number **<figno>**. (Refer to figure 14.)

5.11.3 Illustrations of manufactured items. The following information shall be prepared:

a. Illustrations shall be prepared which contain sufficient views to portray all features of the item. More than one illustration may appear on one page to save space. (Refer to figure 14.)

b. All instructions (explanatory text and list of bulk materials) needed by maintenance personnel to manufacture the item (refer to figure 14) shall supplement the illustrations and shall contain the following data.

(1) All dimensional, location, and processing instructions needed to manufacture the item shall be included (e.g., 30-in. long, top surface, primer coating).

(2) A description of the item to be manufactured, including the P/N and name, shall be prepared.

(3) A list of bulk materials needed to manufacture the item shall be prepared. The list of bulk materials shall consist of the PN, NSN, or specification number of the raw bulk material to be used in manufacture of the item and shall cite the technical characteristics (i.e., standards, specifications, conditions, dimensions, and any other pertinent data).

(4) When applicable, reference shall be made to the associated RPSTL TM or RPSTL work package (for combined TMs).

5.12 Torque limits work package **(-20/AVUM level or above only) <torquewp>**. Information shall be prepared to provide applicable torque values **<torqueval>** (expressed in lb-ft or lb-in. terms), data as to bolt grade markings and their proper identification, and specific torque sequencing requirements. The torque limits work package may begin with an introduction **<intro>**. Specific instructions such as torque limits for dry and wet fasteners, fastener sizes and thread patterns, etc., shall be prepared to follow the introduction. Refer to figure 15 for an example of the type of information presented in a torque limits work package.

5.13 Mandatory replacement parts work package **(-20/AVUM level or above only) <mrplwp>**. This work package shall list all mandatory replacement parts referenced in the task initial setups and procedures. This work package shall include an introduction **<intro>** and a tabular list **<mrpl> (standard table)** of mandatory replacement parts. Items in the list shall be identified by an item number **<itemno>**, P/N **<partno>**, NSN **<nsn>**, nomenclature **<name>**, and quantity **<qty>**. Items shall be listed in alphanumeric order by P/N. (Refer to figure 16.)

5.14 Ammunition marking information work package **<ammowp>**. This work package shall be prepared to provide applicable information on ammunition marking **<markings>**, classification, identification **<ammotype>**, care and handling **<handling>**, preservation, transportation, authorized rounds, preparation

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for firing, fuzes, and packing<**packing**>. Reusable original packaging and containers shall be identified for return or temporary storage of ammunition in its original configuration. Information on classifying, identifying, caring for, handling, etc., nonammunition Class V items shall be prepared, when applicable. Individual paragraphs shall be prepared for each ammunition type/classification. (Refer to figure 17 for sample ammunition data presentation.)

5.15 Foreign ammunition (NATO) work package<**natowp**>. A work package to describe foreign ammunition shall be prepared when applicable. The requirements of 5.14 shall apply. (Refer also to figure 18.)

5.16 Aircraft inventory master guide work package **(aircraft only)** <**inventorywp**>. Information shall be prepared on standard inventory procedures to allow determination of inventoriable items of installed and loose equipment authorized and required by the specific aircraft in performance of its mission. The inventory data described in 5.16.1 through 5.16.4 shall be included.

5.16.1 Introduction <**intro**>. A short explanation of the scope and purpose of the work package shall be prepared. Information pertaining to necessary steps to ensure the list is accurate, exact, and complete (e.g., research of authorized changes, Modification Work Orders (MWOs), additions/deletions for special mission requirements) shall be included. The introduction shall include a reference to DA PAM 738-751 for applicable forms and records. (Refer to figure 19.)

5.16.2 Security <**security**>. It shall be stated here that aircraft inventory records should be unclassified but that any classification of the contents, if necessary, should be in accordance with the existing security regulations. (Refer to figure 19.)

5.16.3 Inventoriable items <**inventoriable**>. The selection of inventoriable items to be listed is to be without regard to the agency, governmental or contractual, furnishing the items. (Refer to figure 19.)

a. Items to be listed are as follows.

(1) Items essential to the execution of the designated mission of the aircraft, such as electronic, photographic, armament, special mission instruments, and safety and comfort equipment.

(2) Loose equipment delivered with the aircraft and items subject to pilferage or readily converted to personal use.

(3) Modification kits which are reissued or distributed to using organizations for installation and which are not immediately placed in use. These shall be recorded on the affected aircraft's DA Form 2408-17, Aircraft Inventory Record, and identified as loose equipment until modification is completed.

(4) Equipment required for operation in a specific environment.

b. Items to be excluded are as follows.

(1) Nonaccountable items coded as expendable in the applicable stock lists.

(2) Personal issue or items furnished on unit allowance or other authority.

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(3) Items or components considered as basic or integral parts of the airframe or basic aircraft, such as engines, propellers, wheels, and standard instruments.

(4) Equipment publications, checklists, and aircraft forms.

5.16.4 Periods of inventory **<prdinv>**. The following information shall be included.

"PERIODS OF INVENTORY

Inventoriable items shall be checked against the Aircraft Inventory Record, DA Form 2408-17, at the following periods:

1. Upon receipt.

2. Prior to transfer of the aircraft to another organization.

3. Upon placing aircraft in storage and upon removal from storage. Aircraft need not be inventoried while in storage.

4. Twelve months after last inventory." (Refer to figure 19.)

5.17 Storage of aircraft work package **(aircraft only) <storagewp>**. The data described in 5.17.1 and 5.17.2 shall be included.

5.17.1 General information for storage of aircraft work package **(aircraft only) <geninfo>**. The following text shall be prepared and included verbatim. (Refer to figure 20.)

"STORAGE OF AIRCRAFT

GENERAL INFORMATION

Components Involved in an Accident

Any component removed for reason of accident shall not be preserved, but shall be shipped in the same condition it was in after the accident.

Categories of Storage

1. Flyable storage — no time limit.

2. Short term (administrative storage) — 1 to 45 days.

3. Intermediate storage — 46 to 180 days."

5.17.2 Flyable storage **<flyable>**, short term storage **<short>**, and intermediate storage **<intermediate>**.

a. A general discussion shall be prepared for each category of aircraft storage, to include considerations for selection of the appropriate category (e.g., ground operation, motoring of engines, and other required maintenance for which personnel and materials are needed) and steps to be taken for care of the aircraft during exceptionally wet weather.

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b. All essential information for each of these paragraphs shall be prepared to include all procedures for preparing the complete aircraft for storage and removal from storage, excluding any information on when or why the aircraft are stored. Each paragraph shall make reference to inspection documents and inspection procedures to be conducted before, during, and after storage. (Refer to figure 20.)

5.18 Weighing and loading (AVIM) work package**(aircraft only) <weightbalwp>**. This work package shall provide description, information, and procedures for aircraft weighing, balancing, and loading. The data described in 5.18.1 through 5.18.3 shall be included.

5.18.1 General information **<geninfo>**. The following text shall be prepared and included verbatim.

"WEIGHING AND LOADING AVIM

GENERAL INFORMATION

Scope

This work package contains description, information, and procedures for aircraft weighing and loading. This information replaces the Chart E (Loading Data and Special Weighing Instructions) placed in the individual aircraft weight and balance files by the aircraft manufacturer. Chart E in the aircraft file will no longer be required."

5.18.2 Weighing information. Instructions for preparing the aircraft, weighing the aircraft in the basic weight condition, performing calculations, and using and recording data on DD Form 365-1 (Basic Weight Checklist) and DD Form 365-2 (Aircraft Weighing Record) shall be included. (Refer to figure 21 for an example of DD Form 365-2.) Instructions shall include initial setup requirements, procedures for positioning the aircraft in the weighing area, and assembly of the aircraft weighing equipment. Illustrations shall be prepared to support the text, including a two view chart diagram. (Refer to figure 22.)

5.18.3 Loading information. Descriptions and instructions shall be prepared for aircraft loading, and computing weight and balance information. Sufficient information and data shall be provided so that an aviator, knowing the basic weight and moment of the aircraft, can compute any combination of weight and balance using the prescribed charts and forms. Reference shall be made to AR-93-3 (Aviation: General Provisions, Training, Standardization, and Resource Management), DA PAM 738-751 and TM 55-1500-342-23 for additional information governing weight and balance of aircraft, forms, and records. Data shall include fundamental principles of loading. An illustration of aircraft compartments and stations shall be included. Reference shall be made to DD Form 365-1 for a more complete listing of compartments and equipment that comprise the basic weight of the aircraft. Loading information shall include weight and balance characteristics, center of gravity limits, weight / balance and loading, and weight and moment tables for load items such as crew, fuel, cargo, and armament.

5.19 Depot mobilization requirements work package**(depot only) <mobilwp>**. When specified and provided by the contracting activity, this work package shall include the modifications, deletions, or additions to the preshop analysis or overhaul procedures required during mobilization. The data described in 5.19.1 and 5.19.2 shall be included.

5.19.1 Introduction for depot mobilization requirements work package**(depot only) <intro>**. The following text shall be prepared and included verbatim. (Refer to figure 23.)

"DEPOT MOBILIZATION REQUIREMENTS

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INTRODUCTION

Scope

The purpose of this work package is to streamline and accelerate the overhaul process during the mobilization of the depot.

Explanation of Mobilization Requirements

The mobilization requirements include a list of instructions for modifying preshop analysis and/or overhaul procedures. The pertinent procedures to be modified are referred to by page and work package number, followed by the action to be taken."

5.19.2 Mobilization requirements **<mobilreq>**. A tabular list **(standard table)** of instructions shall be prepared for modifying preshop analysis and/or overhaul procedures. (Refer to figure 23 for an example.)

5.20 Component checklist work package **(depot only) <compchklistwp>**. A component checklist work package shall be prepared when required to support the preshop analysis procedures. (Refer to MIL-STD-40051-5.) This work package shall consist of the data described in 5.20.1 and 5.20.2.

5.20.1 Introduction **<intro>**. The checklist shall be preceded by a brief explanation of its use. (Refer to figure 24.)

5.20.2 Component checklist **<compchklist>**. The checklist shall contain the following data, as applicable (refer to figure 24).

a. Name/nomenclature of the equipment/item **<name>**.

b. Serial number **<serno>**.

c. Date received **<daterec>**.

d. Received from (identify unit) **<recfrom>**.

e. Component name **<compname>**.

f. NSN **<nsn>**.

g. Part number **<partno>**.

h. Quantity required **<qty>**.

i. Quantity received **<qtyrec>**.

j. Visual damage found **<damage>**.

5.21 QA requirements work package **(depot only) <qawp>**. This work package shall include the data described in 5.21.1 through 5.21.7.

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5.21.1 Statement of responsibility <**responsibility**>. The following information shall be included.

"**STATEMENT OF RESPONSIBILITY**

The depot/contractor is responsible for complying with the quality assurance requirements contained in this work package and in accordance with DESCOM-R 702-1 (DESCOM Product Assurance Program). The commodity manager reserves the right to perform inspections or make changes that ensure the depot work being done meets the quality standards of the DMWR and preserves the inherent reliability of the item." (Refer to figure 25.)

5.21.2 Definitions <**def**>. Definitions shall be prepared for all QA terms extensively used in the Depot Maintenance Work Requirement (DMWR). (Refer to figure 25.)

5.21.3 Special requirements for inspection tools and equipment <**specialreq**>. Any special requirements for the maintenance and calibration of tools and test equipment used for QA inspections shall be listed.

5.21.4 Certification requirements <**certreq**>. Any certification or licensing requirements for processes, procedures, materials, equipment, or personnel skills shall be listed. The list shall include appropriate standards, specifications, regulations, or laws that apply. The list shall reference the text in the DMWR where there is a requirement for a soldering, welding, or magnetic particle inspection certification, radioactive substance, or test driver licenses. (Refer to figure 25.)

5.21.5 In-process inspections <**inprocess**>. The following statement shall be included.

"**IN-PROCESS INSPECTIONS**

In-process quality assurance inspections are contained throughout the overhaul procedures of this DMWR. These inspections are prefixed with the boldfaced letters "**QA**" to identify them, and they are the minimum inspections required. Additional quality assurance inspections may be established by the depot or the commodity manager." (Refer to figure 25.)

5.21.6 Acceptance inspections <**acceptance**>. The following statement shall be included.

"**ACCEPTANCE INSPECTIONS**

Items overhauled in accordance with this DMWR will be accepted based on the following criteria:

1. Conformance to quality of material requirements.
2. Conformance to all in-process quality assurance inspections.
3. Conformance to all final assembly testing requirements.
4. Conformance to the preservation, packaging, and marking requirements."

5.21.7 First article inspection <**first**>. When applicable, first article inspection/test shall be prepared for the DMWR in accordance with DESCOM-R 702-1.

5.22 Wiring diagrams work package <**wiringwp**>. This work package shall include wiring and cable provisions contained in the equipment/end item, including all systems or equipment which can be installed or

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removed later (e.g., mission-related systems/equipment). Applicability of diagrams shall be explained in relation to equipment configuration. At a minimum, the wiring data described in 5.22.1 through 5.22.4 shall be included.

5.22.1 Introduction **<intro>**. Information shall be prepared to include the scope of the work package. (Refer to figure 26.) A statement shall be included explaining that wiring diagrams and essential wiring information are provided for all electrical and electronic systems and circuits.

5.22.2 Wire identification **<wireid>**. Identification of wires by number shall be explained. A list of circuit designators and a wire identification diagram shall be prepared. (Refer to figure 26.)

5.22.3 Abbreviations **<abbrev>**. A statement shall be prepared that abbreviations are in accordance with MIL-STD-12, except when the abbreviation stands for a marking actually found in the equipment. (Refer to figure 26.)

5.22.4 Wiring diagrams **<wiringdiag>**. Wiring diagrams shall be prepared for all electrical and electronic systems and circuits. (Refer to figure 26.)

5.23 Additional work packages **<genwp>**. Additional work packages may be prepared when the work packages previously described herein will not support the data/information to be presented.

6. NOTES.

The notes in section 6 of MIL-STD-40051 apply to this Part.

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TM 3-6665-339-10

REFERENCES 0095 00

SCOPE

This work package lists all field manuals, forms, technical manuals and miscellaneous publications referenced in this manual.

FIELD MANUALS

Basic Cold Weather Manual	FM 31-70
Desert Operations	FM 90-3
First Aid for Soldiers	FM 21-11
Mountain Operations	FM 90-6
NBC Decontamination Avoidance	FM 3-3
NBC Reconnaissance	FM 3-19
Northern Operations	FM 31-71
Operation and Maintenance of Ordnance Material in Cold Weather	FM 9-207
Vehicle Recovery Operations	FM 20-22

FORMS

Accident Identification Card	DD 518
Equipment Control Record	DA Form 2408-9
Equipment Inspection and Maintenance Worksheet	DA Form 2404
Equipment Log Assembly (Records)	DA Form 2408
Hand Receipt	DA Form 2062
Motor Vehicle Accident Report	SF 91
Oil Analysis Log	DA Form 2408-20
Product Quality Deficiency Report	SF 368
Recommended Changes to Equipment Technical Publications	DA Form 2028-2
Recommended Changes to Publications and Blank Forms	DA Form 2028
Uncorrected Fault Record	DA Form 2408-14
Weapon Record Data	DA Form 2408-4

TECHNICAL MANUALS

Ammunition, General	TM 9-1300-200
Destruction of TACOM Equipment	TM 750-244-6
Hand Set Microphone	TM 11-5695-286-14

0095 00-1

FIGURE 1. Example of a references work package

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TM X-XXXX-XXX-XX

MAINTENANCE ALLOCATION CHART (MAC) **0133 00**

INTRODUCTION

The Army Maintenance System MAC

This introduction provides a general explanation of all maintenance and repair functions authorized at various maintenance levels under the standard Army Maintenance System concept.

The MAC (immediately following the introduction) designates overall authority and responsibility for the performance of maintenance functions on the identified end item or component. The application of the maintenance functions to the end item or component shall be consistent with the capacities and capabilities of the designated maintenance levels, which are shown on the MAC in column (4) as:

Unit - includes two subcolumns, C (operator/crew) and O (unit) maintenance.

Direct Support - includes an F subcolumn.

General Support - includes an H subcolumn.

Depot - includes a D subcolumn.

The tools and test equipment requirements (immediately following the MAC) list the tools and test equipment (both special tools and common tool sets) required for each maintenance function as referenced from the MAC.

The remarks (immediately following the tools and test equipment requirements) contain supplemental instructions and explanatory notes for a particular maintenance function.

STANDARD FORMAT

0133 00-1

FIGURE 2. Example of a standard MAC work package, introduction

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TM X-XXXX-XXX-XX

MAINTENANCE ALLOCATION CHART (MAC) **0133 00**

INTRODUCTION

Aviation Maintenance Allocation Chart

This Maintenance Allocation Chart (MAC) assigns maintenance functions in accordance with the Aviation Maintenance concept for Army aviation. These maintenance levels - Aviation Unit Maintenance (AVUM), Aviation Intermediate Maintenance (AVIM), and Depot Maintenance - are depicted in the MAC as:

AVUM - corresponds to an O code in the Repair Parts and Special Tools List (RPSTL).

AVIM - corresponds to an F code in the Repair Parts and Special Tools List (RPSTL).

DEPOT - corresponds to a D code in the Repair Parts and Special Tools List (RPSTL).

The maintenance to be performed below depot and in the field is described as follows:

Aviation Unit Maintenance (AVUM). AVUM activities will be staffed and equipped to perform high frequency "On-Aircraft" maintenance tasks required to retain or return aircraft systems to a serviceable condition. The maintenance capability of the AVUM will be governed by the Maintenance Allocation Chart (MAC) and limited by the amount and complexity of ground support equipment (GSE), facilities required, authorized manning strength, and critical skills available. The range and quantity of authorized spare modules/components will be consistent with the mobility requirements dictated by the air mobility concept. (Assignments of maintenance tasks to divisional company size aviation units will consider the overall maintenance capability of the division, the requirement to conserve personnel and equipment resources, and air mobility requirements.)

THREE-LEVEL FOR ARMY AVIATION FORMAT

0133 00-1

FIGURE 3. Example of an aviation MAC work package, introduction

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MAINTENANCE ALLOCATION CHART FOR TSEC/ST-34

Table 1. MAC for TSEC/ST-34.

(1) GROUP NUMBER	(2) COMPONENT/ASSEMBLY	(3) MAINTENANCE FUNCTION	(4) MAINTENANCE LEVEL					(5) TOOLS AND EQUIPMENT REF CODE	(6) REMARKS CODE
			UNIT		DIRECT SUPPORT	GENERAL SUPPORT	DEPOT		
			C	O	F	H	D		
00	TSEC/ST-34	Inspect	0.1						A
		Service	0.2						B
		Replace	0.4					1	C,D
		Test	0.3						E
		Repair			1.5			1-2	F
		Repair				2.0		1-5	G,H
		Repair					2.0	1-9	I
		Overhaul					16.0	1-9	J
01	POWER UNIT, STP-34	Inspect	0.1						A
		Test	0.3						E
		Repair			1.8			1-2	F
		Repair				2.0		1-5	G,J
		Repair					2.0	1-9	H
0101	PRINTED CIRCUIT BOARDS, STP-34								
010101	E-EB0/1	Inspect			0.1				A
		Test			0.5			1,2	E
		Test					1.0	1-3,6-8	I
		Replace			0.5			1	
		Repair					2.0	1-4,6-8	G
010102	SWITCHING ASSEMBLY	Inspect			0.1			1	A
		Replace			0.5			1	
		Test					1.0	1-3,6-8	
		Repair					2.0	1-4,6-8	H
02	LOGIC UNIT, STB-34	Inspect	0.1						A
		Test	0.5						E
		Repair			1.0			1-2	F
		Repair				2.0		1-5	G,J
0201	PRINTED CIRCUIT BOARDS, STB-34	Inspect			1.0				A
		Test			0.2				E
		Test				1.0		1-3,5	J
		Replace			0.3			1	
		Repair				2.0		2-5	G

FIGURE 4. Example of a standard MAC.

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MAINTENANCE ALLOCATION CHART (AVIATION) FOR T-XXX TURBINE ENGINE

Table 1. MAC for T-XXX Turbine Engine.

(1) GROUP NUMBER	(2) DESCRIPTION	(3) MAINTENANCE FUNCTION	(4) MAINTENANCE LEVEL AVUM (O)	AVIM (F)	DEPOT (D)	(5) TOOLS & EQUIP REF CODE	(6) REMARKS CODE
04	POWER PLANT						
0401	ENGINE, TURBINE	Inspect	--			1	A
		Test	--				B
		Test		-- (3)			
		Test			--		
		Service	0.2				C
		Install	--				
		Replace	--				
		Repair	--				
		Repair		-- (4)			A
		Overhaul			--		
040101	EXTERNAL LINES & HOSES	Inspect	--			3	D
		Test		--			
		Install	--				
		Replace	--				
		Repair		--			
0402	COMPRESSOR SECTION (COLD SECTION MODULE)	Inspect	0.1				E
		Inspect		0.2			
		Test		--			
		Service	0.2				
		Repair	0.4				
		Repair		0.6			
		Overhaul			--		

FIGURE 5. Example of an aviation MAC.

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TOOLS AND TEST EQUIPMENT REQUIREMENTS FOR TSEC/ST-34

Table 2. Tools and Test Equipment for TSEC/ST-34.

TOOL OR TEST EQUIPMENT REF CODE	MAINTENANCE LEVEL	NOMENCLATURE	NATIONAL STOCK NUMBER	TOOL NUMBER
1	H	Automatic test system ST-51	5810-00-089-4599	TSEC/ST-51
2	F	Multimeter, digital	6625-01-139-2512	AN/PSM-45
3	D	Multimeter, digital	6625-01-145-2430	AN/USM 486
4	H	Oscilliscope	6625-01-187-7847	AN/USM-488
5	D	Power supply (0-35 VDC 2.4A)	6130-00-006-5224	HP 6434B86
6	D	Power supply tester	N/A	ON502427
7	H	Repair and soldering center (page)	4940-01-031-4541	PRC-350C/equip
8	F	Tool, kit, electronic equipment	5180-00-610-8177	TK-105/6

REMARKS FOR TSEC/ST-34

Table 3. Remarks for TSEC/ST-34.

REMARKS CODE	REMARKS
A	External.
B	Preventive maintenance checks and services (PMCS).
C	Replace rack installed unit, 0.4 hrs.
D	Bench top use only, 0.1 hrs.
E	Self-test.
F	Repair by PMA and authorized component replacement only.
G	Complete unit and subassembly repair (except STP-34 switching assembly and E-EBO/1).
H	Complete unit and subassembly repair.
I	In compliance with TSEC/ST-34 CIDOS.
J	Function performed by specialized repair activity (SRA). (Theater COMSEC Logistics Support Center-Europe or Lexington-Blue Grass Army Depot)

FIGURE 6. Example of standard MAC tools and test equipment and remarks tables

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TOOLS AND TEST EQUIPMENT REQUIREMENTS FOR T-XXX TURBINE ENGINE

Table 2. Tools and Test Equipment for T-XXX Turbine Engine.

TOOL OR TEST EQUIPMENT REF CODE	MAINTENANCE LEVEL	NOMENCLATURE	NATIONAL STOCK NUMBER	TOOL NUMBER
1	AVUM/AVIM	Sling, aircraft maintenance	1730-00-903-5019	LTCT 773
2	AVUM/AVIM	Wrench, crowfoot	5120-00-034-6193	LTCT 4810
3	AVIM	Wrench, socket	5120-00-875-2588	LTCT 393
4	AVIM	Wrench, spanner	5120-00-886-1794	LTCT 9263

REMARKS FOR T-XXX TURBINE ENGINE

Table 3. Remarks for T-XXX Turbine Engine.

REMARKS CODE	REMARKS
A	Diagnostic inspection using borescope.
B	Functional test at AVUM - engine in airframe.
C	Functional test at AVIM - engine in METS.
D	Repair at AVIM includes the engine assembly, individual line replacement units (LRU) (accessories) and modules.
E	Replace seal.
F	Repair limited to replacement of rotor assembly, stator, stage 1 nozzle, face type seal, and combustion liner.
G	Repair limited to replacement of external lines, hoses, and line replacement units (LRU) (accessories).
H	Replacement of carbon seal.
I	Reset button.
J	Water wash.
K	Visible inspection without detailed disassembly.
L	All repair and replacement of parts performed by AVUM is limited to authorized items listed in TM (cite specific TM-20P or -23P).

FIGURE 7. Example of an aviation MAC tools and test equipment and remarks tables

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TM 9-2350-314-10

COMPONENTS OF END ITEM (COEI) AND BASIC ISSUE ITEMS (BII) LISTS **0121 00**

INTRODUCTION

Scope

This work package lists COEI and BII for the M198 howitzer to help you inventory items for safe and efficient operation of the equipment.

General

The COEI and BII information is divided into the following lists:

Components of End Item (COEI). This list is for information purposes only and is not authority to requisition replacements. These items are part of the M198 howitzer. As part of the end item, these items must be with the end item whenever it is issued or transferred between property accounts. Items of COEI are removed and separately packaged for transportation or shipment only when necessary. Illustrations are furnished to help you find and identify the items.

Basic Issue Items (BII). These essential items are required to place the M198 howitzer in operation, operate it, and to do emergency repairs. Although shipped separately packaged, BII must be with the M198 howitzer during operation and when it is transferred between property accounts. Listing these items is your authority to request/requisition them for replacement based on authorization of the end item by the TOE/MTOE. Illustrations are furnished to help you find and identify the items.

Explanation of Columns in the COEI List and BII List

Column (1), Illus Number, gives you the number of the item illustrated.

Column (2), National Stock Number, identifies the stock number of the item to be used for requisitioning purposes.

Column (3), Description, CAGEC, and Part Number, identifies the Federal item name (in all capital letters) followed by a minimum description when needed. The stowage location of COEI and BII is also included in this column. The last line below the description is the CAGEC (commercial and Government entity code) (in parentheses) and the part number.

Column (4), Usable on Code, gives you a code if the item you need is not the same for different models of equipment. These codes are identified below:

Code	Used on
PAA	Model XXX
PAB	Model XXXX
PAC	Model XXXXX

Column (5), U/M (unit of measure), indicates how the item is issued for the National Stock Number shown in column (2).

Column (6), Qty Rqr, indicates the quantity required.

0121 00-1

FIGURE 8. Example of a components of end item and basic issue items lists work package

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TM 9-2350-314-10

COMPONENTS OF END ITEM (COEI) AND BASIC ISSUE ITEMS (BII) LISTS - Continued 0121 00

COMPONENTS OF END ITEM (COEI) LIST

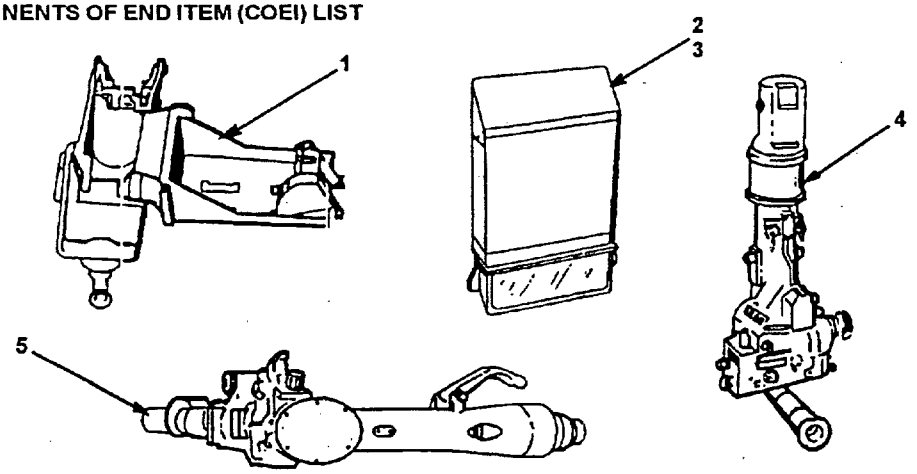


Table 1. Components of End Item List.

(1) ILLUS NUMBER	(2) NATIONAL STOCK NUMBER	(3) DESCRIPTION, CAGEC, AND PART NUMBER	(4) USUABLE ON CODE	(5) U/M	(6) QTY RQR
1	1005-00-706-8880	MOUNT, MACHINE GUN: cal. .50 (in mount on cupola) (19204) 7068880	PAA	EA	1
2	1240-00-344-4643	PERISCOPE: M27 (chief of section) (stowage box cab wall) (19200) 7633132	PAA	EA	1
3	1240-00-509-2743	PERISCOPE: M45 (driver's) (stowage box driver's compartment) (19200) 8213430	PAA	EA	3
4	1240-00-864-2930	TELESCOPE, PANORAMIC M117 (in mount M145 or telescope box) (19200) 7660400	PAA	EA	1
5	1240-00-491-9676	TELESCOPE, ELBOW: M118CA1 (in mount M146) (19200) 10559855	PAB	EA	1

0121 00-2

FIGURE 8. Example of a components of end item and basic issue items lists work package- Continued.

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COMPONENTS OF END ITEM (COEI) AND BASIC ISSUE ITEMS (BII) LISTS - Continued 0121 00

BASIC ISSUE ITEMS (BII) LIST

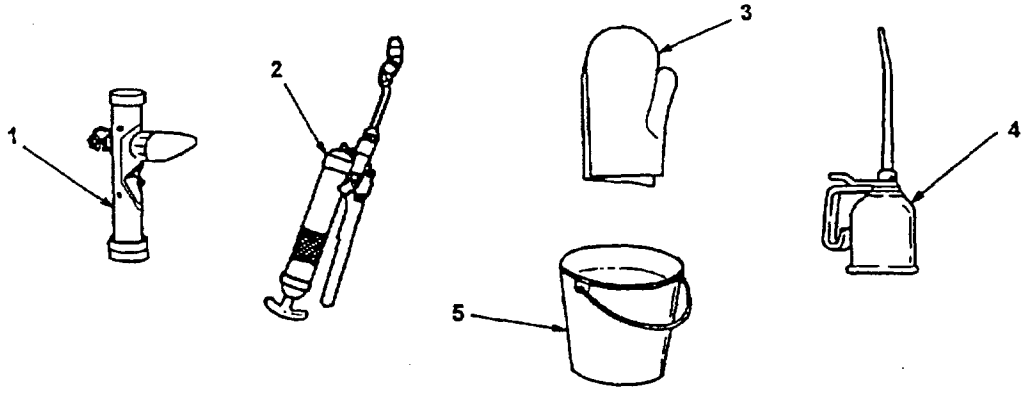


Table 2. Basic Issue Items List.

(1) ILLUS NUMBER	(2) NATIONAL STOCK NUMBER	(3) DESCRIPTION, CAGEC, AND PART NUMBER	(4) USUABLE ON CODE	(5) U/M	(6) QTY RQR
1	1290-00-535-7629	LIGHT, AIMING POST: M14 (19200) 7197188	PAA	EA	2
2	4930-00-766-3545	LUBRICATING GUN, HAND: high pressure (in tool bag) (36251) 102758	PAA	EA	1
3	8415-00-266-8843	MITTENS, CLOTH: (pair) M1942 (in oddment tray) (19207) 11655982	PAA	PR	2
4	4930-00-262-8868	OILER, HAND: steel, pump type, 1 pt, spout 9 lg (in left cab door stowage box) (72798) 328	PAA	EA	1
5	7240-00-160-0455	PAIL, UTILITY: 14-qt capacity (on vehicle floor) (81348) RRP35	PAA	EA	1

0121 00-7

FIGURE 8. Example of a components of end item and basic issue items lists work package- Continued.

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TM 3-6665-339-10

ADDITIONAL AUTHORIZATION LIST (AAL) 5120 00

INTRODUCTION

Scope
This work package lists additional items you are authorized for the support of the NBCRS FOX M93A1.

General
This list identifies items that do not have to accompany the NBCRS FOX M93A1 and that do not have to be turned in with it. These items are all authorized to you by CTA, MTOE, TDA, or JTA.

Explanations of Columns in the AAL
Column (1), National Stock Number, identifies the stock number of the item to be used for requisitioning purposes. Column (2), Description, CAGEC, and Part Number, identifies the Federal item name (in all capital letters) followed by a minimum description when needed. The last line below the description is the CAGEC (Commercial and Government Entity Code) (in parentheses) and the part number.

Column (3), Usable On Code, when applicable, gives you a code if the item you need is not the same for different models of equipment.

Column (4), U/M (unit of measure), indicates how the item is issued for the National Stock Number shown in column (1). Column (5), Qty Recm, indicates the quantity recommended.

ADDITIONAL AUTHORIZED LIST ITEMS

Table 1. Additional Authorization List.

(1) NATIONAL STOCK NUMBER	(2) DESCRIPTION, CAGEC, AND PART NUMBER	(3) USABLE ON CODE	(4) U/M	(5) QTY RECM
	MTOE AUTHORIZED ITEMS			
6665-01-105-5623	ALARM, CHEMICAL AGENT (19200) 8762101		EA	1
1240-01-207-5787	BINOCULARS, MOD, CON M22 (19200) 9370122		EA	1
2590-01-148-7961	CABLE KIT, SPECIAL PURPOSE (19200) 223592-2000		EA	1
1080-00-623-7295	CAMOUFLAGE SCREEN WOODLAND/DST POLES (34623) 11655722		EA	1
1080-00-103-1246	CAMOUFLAGE SCREEN WOODLAND RAD SCT (34623) 11655720		EA	1
6665-01-199-4153	CHEMICAL AGENT MONITOR (CAM) (34623) 11645620		EA	1

5120 00-1

FIGURE 9. Example of an additional authorization list work package

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EXPENDABLE AND DURABLE ITEMS LIST 5230 00

INTRODUCTION

Scope

This work package lists expendable and durable items that you will need to operate and maintain the NBCRS FOX M93A1. This list is for information only and is not authority to requisition the listed items. These items are authorized to you by CTA 50-970, Expendable/Durable Items (Except Medical, Class V Repair Parts, and Heraldic Items), or CTA 8-100, Army Medical Department Expendable/Durable Items.

Explanations of Columns in the Expendable / Durable Items List

Column (1) - Item Number. This number is assigned to the entry in the list and is referenced in the narrative instructions to identify the item (e.g., "Use lubricating oil (Item 25, WP 5230 00).").

Column (2) - Level. This column includes the lowest level of maintenance that requires the listed item (C =Operator/Crew).

Column (3) - National Stock Number. This is the NSN assigned to the item which you can use to requisition it.

Column (4) - Item Name, Description, Commercial and Government Entity Code (CAGE), and Part Number (P/N). This column provides the other information you need to identify the item.

Column (5) - Unit of Measure (U/M). This code shows the physical measurement or count of an item, such as gallon, dozen, gross, etc.

EXPENDABLE AND DURABLE ITEMS LIST

Table 1. Expendable and Durable Items List.

(1) ITEM NUMBER	(2) LEVEL	(3) NATIONAL STOCK NUMBER	(4) ITEM NAME, DESCRIPTION, CAGE, PART NUMBER	(5) U/M
1	C	6810-00-201-0906	Alcohol, denatured, Grade III, 16 ounce bottle (81348) O-E-760	BT
2	C	8030-01-138-1666	Antiseize Compound, 250-gram tube (81349) MIL-T-5544	TU
3	C	6515-00-059-5235	Applicator, disposable, package of 1000 (58536) A-A-30016	PK
4	C	8020-00-224-8024	Brush, artist, MTL ferrule, round, tapered point, Type I, camel hair (81348) H-B-118	EA
5	C	9150-01-054--6453	Cleaner, Lubricant & Preservation (CLP), 1-pint bottle with sprayer (81349) MIL-L-63640	PT
6	C	6850-00-926-2275	Cleaning Compound (81348) O-C-1901	PT

5230 00-1

FIGURE 10. Example of an expendable and durable items list work package

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:04Z
Check the source to verify that this is the current version before use.

TM X-XXXX-XXX-XX

STOWAGE AND DECAL/DATA PLATE GUIDE **0113 00**

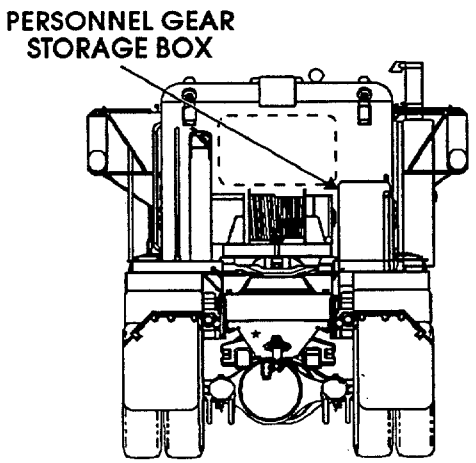
INTRODUCTION

Scope

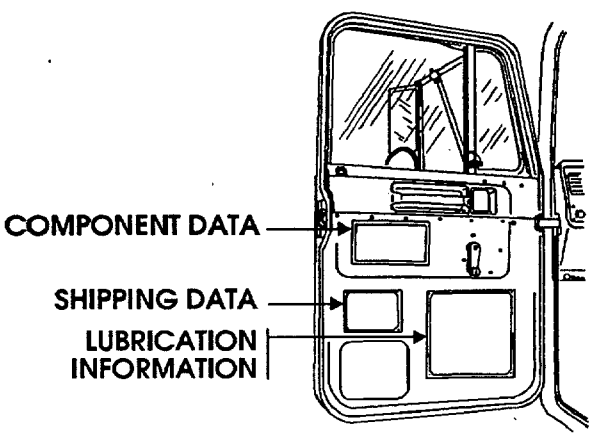
This work package shows stowage location for equipment, metal signs, and stencils that must be in place on the M915A2 and M916A1.

General

The illustrations on the following pages show stowage locations for Basic Issue Items (BII), tire chains, and personnel gear. Also shown are locations for metal signs and stencils that contain cautions or information required to operate the vehicle safely.



M916A1



M915A2 and M916A1

0113 00-1

FIGURE 11. Example of a stowage and decal/data plate guide work package

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TM 3-6665-339-10

ON - VEHICLE EQUIPMENT LOADING PLAN 0532 00

INTRODUCTION

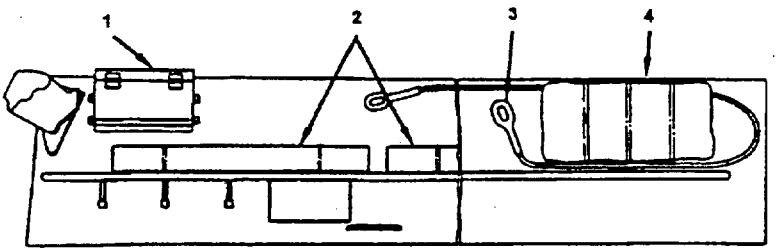
Scope

This work package lists user equipment and its location on the NBCRS FOX M93A1.

General

The following illustrations and tables will assist you in locating user equipment on the NBCRS FOX M93A1.

EXTERNAL LEFT SIDE VIEW



Item No.	Item Name
1	Stowage Box Spare Smoke Grenades (12)
2	Camouflage Screen Equipment
3	Tow Cables (2)
4	Camouflage Screens (2)

0532 00-1

FIGURE 12. Example of tactical on-vehicle equipment loading plan work package

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TM 9-1250-252-20

TOOL IDENTIFICATION LIST **0105 00**

INTRODUCTION

Scope

This work package lists all common tools and supplements and special tools/fixtures needed to maintain the MICLIC.

Explanation of Columns in the Tool Identification List

Column (1) - Item Number. This number is assigned to the entry in the list and is referenced in the initial setup to identify the item (e.g., "Extractor (item 32, WP 0105 00)").

Column (2) - Item Name. This column lists the item by noun nomenclature and other descriptive features (e.g., "Gage, belt tension").

Column (3) - National Stock Number. This is the National Stock Number (NSN) assigned to the item; use it to requisition the item.

Column (4) - Part Number. Indicates the primary number used by the manufacturer (individual, company, firm, corporation, or Government activity) which controls the design and characteristics of the item by means of its engineering drawings, specifications, standards, and inspection requirements to identify an item or range of items.

Column (5) - Reference. This column identifies the authorizing supply catalog or RPSTL for items listed in this work package.

TOOL IDENTIFICATION LIST

Table 1. Tool Identification List.

(1) ITEM NO.	(2) ITEM NAME	(3) NATIONAL STOCK NUMBER	(4) PART NUMBER	(5) REFERENCE
1	Adapter, socket wrench, 1/2 inch-3/4 inch	5120-00-114-5207	11655788-3	TM 9-2350-252-20P-1
2	Adapter, socket wrench, 3/4 inch-1 inch	5120-00-227-8104		SC 4910-95-CL-A72
3	Adapter, socket wrench, 3/8 inch-1/2 inch	5120-00-240-8703	EX503B	TM 9-2350-252-20P-1
4	Adapter, test	4910-01-138-9334	11629693-1	TM 9-1250-252-20P-1
5	Adapter, test	4910-01-138-9335	11629693-2	TM 9-1250-252-20P-1
6	Adapter, torque wrench, 1/2 inch drive, 1/2 inch	5120-00-399-1157	2588756	TM 9-2350-252-20P-1
7	Adapter, torque wrench, 1/2 inch drive, 3/4 inch	5120-00-399-1154	2588757	TM 9-2350-252-20P-1
8	Adapter, torque wrench, 1/2 inch drive, 5/16 inch	5120-01-115-1891	12298105-1	TM 9-2350-252-20P-1
9	Adapter, torque wrench, 1/2 inch drive, 15/16 inch	5120-00-215-8200	11663358-2	TM 9-2350-252-20P-1
10	Adjusting tool, belt	4910-01-128-2670	3375058	TM 9-2350-252-20P-1
11	Automotive electrical tool kit	5180-00-422-8594		SC 4910-95-CL-A74
12	Bit, screwdriver, 1/4 inch drive	5120-00-316-9228	TMC105A	TM 9-2350-252-20P-1

0105 00-1

FIGURE 13. Example of a tool identification list work package

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TM 1-1520-238-23

ILLUSTRATED LIST OF MANUFACTURED ITEMS **0110 00**

INTRODUCTION

Scope

This work package includes complete instructions for making items authorized to be manufactured or fabricated at the direct support maintenance level.

How to Use the Index of Manufactured Items

A part number index in alphanumeric order is provided for cross-referencing the part number of the item to be manufactured to the figure which covers fabrication criteria.

Explanation of the Illustrations of Manufactured Items

All instructions needed by maintenance personnel to manufacture the item are included on the illustrations. Refer to the RPSTL work package 0115 00 for further information on all manufactured items. All bulk materials needed for manufacture of an item are listed by part number or specification number in a tabular list on the illustration.

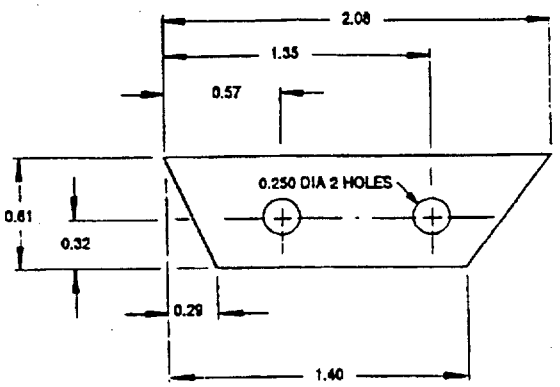
INDEX OF MANUFACTURED ITEMS

Part Number	Figure Number
342-83028-1	1
342-83028-3	2

0110 00-1

FIGURE 14. Example of an illustrated list of manufactured items work package

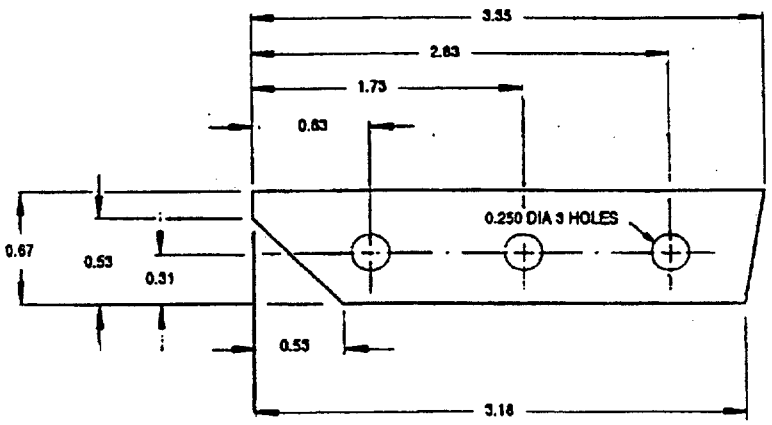
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NOTES:

- SHIM, BLADE - MAKE FROM AL-ALY SHEET 2024-T3, 0.016 THICK, QQ-A-250/5 , NSN 6635-00-232-0543
- FINISH - CHEMICALLY FILM TREAT PER MIL-C-5541 CLASS 1A
- ALL DIMENISIONS ARE IN INCHES

Figure 1. Shim, Blade PN 366-83019-3



NOTES:

- SHIM, BLADE - MAKE FROM AL- ALY SHEET 2024 - T3, 0.016 THICK, QQ-A-250/5 , NSN 6635-00-232-0543
- FINISH - CHEMICALLY FILM TREAT PER MIL-C-5541 CLASS 1A
- ALL DIMENISIONS ARE IN INCHES

Figure 2. Shim, Blade PN 366-83019-5

FIGURE 14. Example of an illustrated list of manufactured items work package- Continued.

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TM 9-2350-XXX-XX

TORQUE LIMITS 0731 00

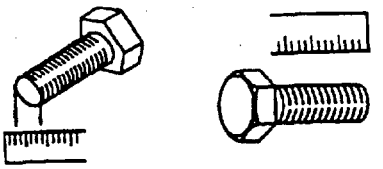
INTRODUCTION

Scope

This work package contains the torque standards for specific types and sizes of hardware. It defines the different types of bolts by grade.

TORQUE TABLES

How To Use Torque Tables



1. Measure the diameter of the screw you are installing.
2. Count the number of threads per inch or use a pitch grade.
3. Under the heading SIZE, look down the left-hand column until you find the diameter of the screw you are installing. (There will usually be two lines beginning with the same size.)
4. In the second column under SIZE, find the numbers of threads per inch that matches the number of threads you counted in step 2. (Not required for metric screws.)



STANDARD METRIC

CAPSCREW HEAD MARKINGS

NOTE

Manufacturer's marks may vary. Standard are all SAE Grade 5 (3- Line). Metric screws are of three grades: 8.8, 10.9, and 12.9. Grades and manufacturer's marks appear on the screw head.

5. To find the grade screw you are installing, match the markings on the head to the correct picture of CAPSCREW HEAD MARKINGS in the illustration preceding the torque table.

0731 00-1

FIGURE 15. Example of a torque limits work package

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Check the source to verify that this is the current version before use.

TM X-XXXX-XXX-XX

MANDATORY REPLACEMENT PARTS **0115 00**

INTRODUCTION

Scope

This work package includes a list of all mandatory replacement parts referenced in the task initial setups and procedures. These are items that must be replaced during maintenance whether they have failed or not. This includes items based on usage intervals such as miles, time, rounds fired, etc.

MANDATORY REPLACEMENT PARTS LIST

Table 1. Mandatory Replacment Parts List Semiannual (1500 Mile).

ITEM NO.	PART NUMBER	NSN	NOMENCLATURE	QTY
1	HC7500Y144	4330-01-217-8184	FILTER ELEMENT, FLUID (M977/M985 ONLY)	1
2	HD223	2940-01-132-4842	FILTER ELEMENT, FLUID	1
3	MS35338-46	5310-00-637-9541	WASHER, LOCK (ALL EXCEPT M984A1)	2
4	MS35802-3	2940-00-580-6283	FILTER ELEMENT, FLUID	1
5	S-268-1	5306-01-084-5390	BOLT, MACHINE (ALL EXCEPT M984A1)	2
6	WA110	5310-01-061-5302	WASHER, LOCK	4
7	10232C	5330-01-168-8707	GASKET (M978 ONLY)	1
8	11007	5330-01-046-1990	GASKET	1
9	1112310	5330-01-225-4803	PACKING, PREFORMED (M977/M985/M984A1 ONLY)	1
10	1124510	5330-01-143-0135	PACKING, PREFORMED (M977/M985/M984A1 ONLY)	1
11	1128242	5330-01-145-1112	PACKING, PREFORMED (M977/M985/M984A1 ONLY)	1
12	11350	5330-01-147-6003	PACKING, PREFORMED	1
13	1199478	5330-01-234-7625	GASKET	1
14	1199738	4330-01-192-7664	SCREEN, BY-PASS FILTER (M977/M985/M984A1 ONLY)	1
15	1300766	4330-01-232-8305	FILTER ELEMENT, FLUID (M984A1 ONLY)	1
16	1300767	4330-01-192-8832	FILTER ELEMENT, FLUID (M977/M985/M984A1 ONLY)	1
17	2020SM	4330-01-046-3399	FILTER ELEMENT, FLUID WASHER,	1
18	2463HX	5310-01-054-5896	FLAT (ALL EXCEPT M984A1)	2
19	25010778	2910-01-022-8183	FILTER ELEMENT, FLUID	1
20	4-1/2IN77/	5330-01-163-5849	GASKET (M978 ONLY)	2

FIGURE 16. Example of a mandatory replacement parts work package

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TM X-XXXX-XXX-XX

AMMUNITION MARKING INFORMATION **0108 00**

GENERAL

1. Ammunition for the M199 cannon is the separate loading type. The loading of each complete round into the cannon requires three separate operations: loading the fuzed projectile, the propelling charge, and the primer.

2. These components are shipped separately; therefore, the cannon crew must know how to store, unpack, inspect, prepare, and load each complete round every time the weapon is fired.

   a. The chief of section supervises the loading and the preparation duties performed by cannoneers.

   b. The chief of section must also see that the cannoneers and driver are cross-trained in the specific duties of the care, handling, unpacking, inspection, preparation, and loading of the ammunition components in order to sustain a 24-hour operation or to operate with a reduced crew.

3. It is planned that future ammo for all new 155mm weapons will be interchangeable. This will enable projectiles and propelling changes of one NATO nation to be fired from the 155mm weapons of all others. Current items of interchangeability are contained in Chapter 5.

4. For maintenance of ammunition, see TM 9-1300-251-20.

WARNING

Until safety and reliability testing is completed, the use of ammo other than prescribed in this manual is prohibited.

5. Refer to Work Package 0011 00 for information about the Loose Projectile Restraint System (LPRS). The LPRS is a divider rack for securing loose unfuzed projectiles for transportation in a field artillery companion vehicle.

0108 00-1

FIGURE 17. Example of an ammunition marking information work package

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TM X-XXXX-XXX-XX

AMMUNITION MARKING INFORMATION - Continued **0108 00**

Table 1. Authorized Projectile Fuze Combinations for 8-Inch Howitzer, SP, M110A2 Cannon M201A1.

TYPE AND MODEL NUMBER OF PROJECTILE	FUZE										
	PD		MT		MTSQ			PROX(VT)		ET	
	M739 SERIES	M557	M572	M565	M564	M557 SERIES	M582 SERIES	M728	M732	M762	M767
Agent GB, VX, M426	X	X	X					X[2]	X[2]		
HE, M106 (Shallow Cavity)	X	X	X		X[3]		X		X		X
HE, M106 (Deep Cavity)	X	X	X		X[3]		X	X[4]	X		X
HE, M404 ICM				X		X				X	
HE, M509A1 ICM						X				X	
HERA, M650 (Rocket-On)	X	X	X								X
HERA M650[1] (Rocket-Off)	X	X	X		X[3]		X		X		X

WARNING

[1] Do not fire the M650 projectile if the obturating band is missing or broken. If the band is displaced and can be repositioned and remain in the groove, the projectile can be fired.

NOTE

[2] M728 and M732 fired only with "VX" projectile and only in combat emergency.

[3] Fuze, MTSQ, M564 is restricted from firing with zone 9 M188A1 propelling charge.

[4] Authorized, requires removal of supplementary charge.

FIGURE 17. Example of an ammunition marking information work package - Continued

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TM X-XXXX-XXX-XX

FOREIGN AMMUNITION (NATO) **0101 00**

GENERAL

1. Agreements between the United States and NATO allies have established the interoperability of weapon systems and ammunition of the nations. The agreements enable the safe and effective firing of major types of ammunition of the same size from the same compatible size and type weapon of the NATO armies.

2. The following pages cover only authorized German (GE), United Kingdom (UK), Canadian (CA), Netherlands (NL), French (FR), Norwegian (NO), Italian (IT), Danish (DA), Greek (GR), or Belgian (BE) 155mm components. If a munitions item has not yet been authorized, it is because it has not yet been determined to be safe to fire or it has been determined that the munitions item cannot be safely fired from the US weapon system.

WARNING

Only under emergency combat conditions will zone 1 of the M3A1 and DM62 propelling charge be fired from the M199 cannon tube of the M198 howitzer weapon system.

Do not mix US, GE, UK, CA, NL, FR, NO, IT, DA, GR, or BE components (i.e., projectile, propelling charge, flash reducer, fuze). Fire only components from one nation, except GE primer DM9141. GE must use DM191A1 when firing US, GE, UK, CA, NL, FR, NO, IT, DA, GR, or BE 155mm munitions. Mixing may cause injury to personnel.

NOTE

At the conclusion of any training excercise, ammunition drawn from a NATO nation and not fired should be returned to the troops of the NATO nation from whom it was obtained.

AUTHORIZED PROJECTILES

The following GE munitions are authorized for use in M198 howitzers:

Projectile	155mm, HE, DM21 (TNT-loaded only)
Charge, propelling	Green bag, DM62, zones 1-5
Charge, propelling	White bag, DM42B1, zones 3-7
Fuze	Point-detonating, DM21 1
Primer	Use only US M82 primer. Do not use GE DM191A1 primer.

0101 00-1

FIGURE 18. Example of a foreign ammunition (NATO) work package

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TM X-XXXX-XXX-XX

FOREIGN AMMUNITION (NATO) - Continued **0101 00**

AUTHORIZED PROJECTILES - Continued

The following GE munitions are authorized for use in M198 howitzers:

Projectile	155mm, HE, M107 (TNT-loaded only)
Charge, propelling	Green bag, M3A1, zones 1-5
Charge, propelling	White bag, M4A2, zones 3-7
Fuze	Point-detonating, M557, L85A2
Primer	M82

NOTE

M3 and M4A1 propelling charges do not have flash reducers.

The following CA munitions are authorized for use in M198 howitzers:

Projectile	155mm, HE, M107
Charge, propelling	M3, M3A1, M4A1, M4A2
Fuze	Point-detonating, M5S7, M564 proximity, M514A1
Primer	M82

NOTE

The M107C1 projectile, M3C1 and M4C3 propelling charges, M557C1 fuze, and M82C1 primer are NL manufacture.

The following NL munitions are authorized for use in M198 howitzers:

Projectile	155mm, HE, M107, M107C1 (TNT-loaded only)
Charge, propelling	M3C1, M4C3, M4A1
Fuze	Point-detonating, M557, M557C1
Primer	M82, M82C1

The following FR munitions are authorized for use in M198 howitzers:

Projectile	155mm, HE, M107
Charge, propelling	Green bag, M3, zones 1-5
Charge, propelling	White bag, M4A1, zones 5-7
Fuze	Point-detonating, M557
Primer	Use US M82 only, not MK2A4.

0101 00-2

18. Example of a foreign ammunition (NATO) work package - Continued

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TM 1-1520-238-23-1

AIRCRAFT INVENTORY MASTER GUIDE 0347 00

INTRODUCTION

Scope

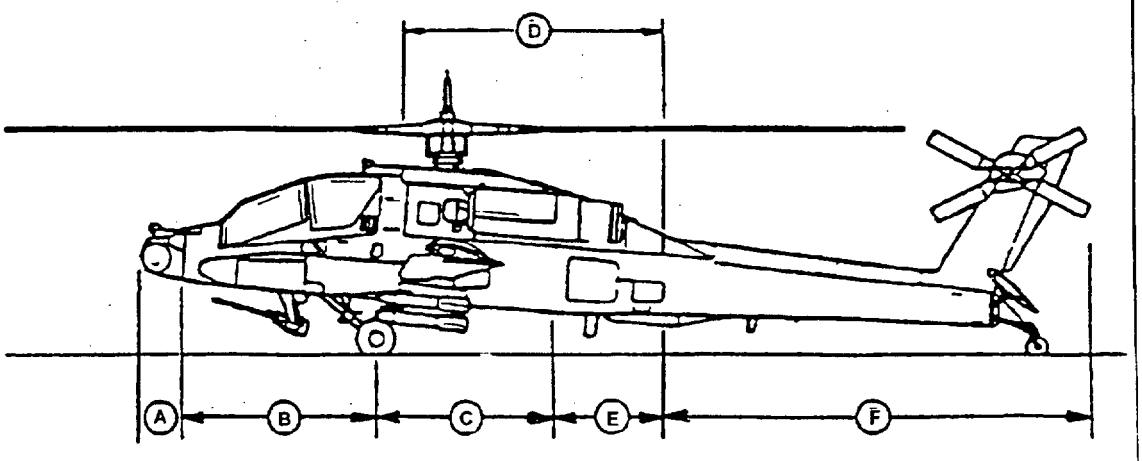
This work package includes a listing of all inventoriable items of installed and loose equipment authorized and required by the aircraft in performance of its mission. Refer to DA PAM 738-751 for applicable forms and records to use in performing the inventory.

SECURITY

Aircraft inventory records should be unclassified, but any classification of the contents, if necessary, should be in accordance with the existing security regulations.

INVENTORIABLE ITEMS

Modification kits which are reissued or distributed to using organizations for installation, and which are not immediately placed in use, shall be recorded on the aircraft's DA Form-2408-17, Aircraft Inventory Record, and identified as loose equipment until modification is complete. Refer to table 1 for a list of inventoriable items for the aircraft.



0347 00-1

FIGURE 19. Example of an aircraft inventory master guide work package

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Check the source to verify that this is the current version before use.

TM 1-1520-238-23-1

AIRCRAFT INVENTORY MASTER GUIDE - Continued 0347 00

INVENTORIABLE ITEMS - Continued

Table 1. Inventoriable Items.

NOMENCLATURE	PART NUMBER	NSN	QTY RQD
SECTION A - NOSE ELECTRONICS			
TADS Turret	130760000	1270-01-188-4138	1
PNVS Turret and Faring	130900000	1090-01-169-9415	1
Multiplex Terminal - Type 3	4032298-956	7025-01-210-7768	1
Signal Data Converter - Eng.	7-311C20019	6620-01-160-3518	1
Radar Jammer Transmit Antenna	SM-A-919685-2		1
Radar Warning Antenna	AS2891/APR-39V	1020-00-024-7608	2
Power Transformer	7-219510089	5950-01-186-8032	1
SECTION B - RIGHT FORWARD AVIONICS BAY			
Gun Control Box	7-317222500-603	1005-01-211-4165	1
Multiplex Terminal - Type 1	4032297-955	7025-01-211-0130	1
Turret Control Box	7-317222004	5930-01-239-2391	1
PNVS Electronic Box	13080274	4931-01-169-9369	1
Display Electronics - IHADSS	7-319430041	1270-01-183-0518	1
Sight Electronic Unit - IHADSS	7-319430031	1270-01-183-0519	1
Fire Control Computer	7-31900001	1430-01-211-0023	1
Remote Electronic Unit	7-317141001	1270-01-187-5778	1
Portable Fire Bottle	7-211180002-501	4210-00-555-8837	1
Electromagnet Relay	7-211B12033	5945-01-160-5639	1

PERIODS OF INVENTORY

Inventoriable items shall be checked against the Aircraft Inventory Record, DA Form 2408-17, at the following periods:

1. Upon receipt.
2. Prior to transfer of the aircraft to another organization.
3. Upon placing aircraft in storage and upon removal from storage. Aircraft need not be inventoried while in storage.
4. Twelve months after last inventory.

0347 00-2

FIGURE 19. Example of an aircraft inventory master guide work package - Continued

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TM 1-1520-238-23

STORAGE OF AIRCRAFT **2100 00**

THIS WORK PACKAGE COVERS:

General Information, Flyable Storage

INITIAL SETUP:

Maintenance Level
Unit

Tools and Special Tools

Aircraft mechanic's tool kit (item 376, WP 2653 00)
Light duty laboratory apron (item 27, WP 2652 00)
Chemical protective gloves (item 154, WP 2652 00)
Adjustable air filtering respirator (item 262, WP 2652 00)

Materials/Parts:

Barrier material (item 32, WP 1846 00)
Cloth (item 52, WP 1846 00)
Dry cleaning solvent (item 74, WP 1846 00)
Lubricating oil (item 117, WP 1846 00)
Tape (item 207, WP 1846 00)

Personnel Required:

Attack Helicopter Repairer, 67R (1)
Attack Helicopter Repairer/Technical Inspector, 67R3F (1)
Maintenance Test Pilot, 152FG (1)

References:

TM 1-1500-204-23
TM 1-1520-238-PMS
TM 9-1090-208-23
TM 11-6140-203-14-1
TM 11-6140-203-14-2
TM 55-1500-344-23
TM 55-1520-238-10
TM 55-2840-248-23

Equipment Conditions:

Helicopter safed (WP 1765 00)
Engine 10-hour/14 day inspection performed (TM 55-2840-248-23)
10-hour/14 day inspection performed (TM 1-1520-238-PMS)

STORAGE OF AIRCRAFT

GENERAL INFORMATION

Components Involved in an Accident

Any component removed for reason of accident shall not be preserved, but shall be shipped in the same condition it was in after the accident.

Categories of Storage

1. Flyable storage - no time limit.
2. Short term (administrative storage) - 1 to 45 days.
3. Intermediate storage - 46 to 180 days.

NOTE

Refer to TM 1-1500-204-23 for general procedures for storage of aircraft.

FLYABLE STORAGE - NO TIME LIMIT

1. Start and operate auxiliary power unit (APU) (WP 0431 00).
   a. Allow APU to run for 10 to 15 minutes.

2100 00-1

FIGURE 20. Example of a storage of aircraft work package

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FORM B - AIRCRAFT WEIGHING RECORD

FOR USE IN T.O 1-1 B-40, NAVAIR 01-1B-40 AND TM-55-405-9

DATE WEIGHED (YYMMDD)	MODEL/DESIGN/SERIES	SERIAL NUMBER

PLACE WEIGHED	WEIGHT AND BALANCE TECHNICIAN (last, first, M.I.)	DUTY PHONE NUMBER

	REACTION (Wheels, jackpoints, etc.)		SCALE READING	TARE	NET WEIGHT	ARM	MOMENT
(1)	LEFT MAIN	1	4102.7		4101.5 (8)		
		2	4100.3				
(2)	RIGHT MAIN	1	3998.6		4002.3 (9)		
		2	4006.0				
	SUB-TOTAL (Both main)				8103.8 (10)	E 120.2 (14)	974076 (17)
(3)	NOSE OR TAIL	1	2984.1		2985.7 (11)	F 450.0 (15)	1343565 (18)
		2	2987.3				
	TOTAL (as weighed) Not to be posted on Chart C				11089.5 (12)	208.9 (16)	317641 (19)

(4) ∝ = 3.5° PLUMB-BOB ANGLE MEASUREMENTS

B= N/A the distance from the jib point, to the center line of the main reactions. Obtain by measurements.

I= N/A the distance from the reference datum to the jib point of the aircraft, from which a plumb bob can be dropped to the ground. Obtain from the aircraft diagram in Chart E.

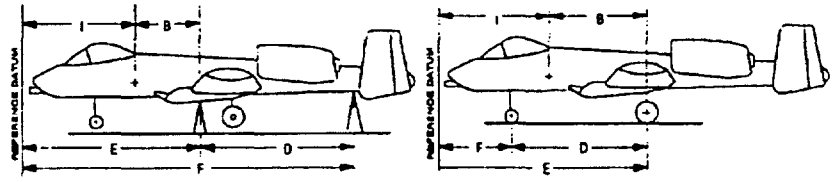
(5) E= 120.2 * the distance from the reference datum to the center line of the main reactions.
E = I + B
E = I − B (If the jib point is aft of the center line of the main reactions.)

D= N/A the distance between the main and nose tail reaction. Obtain by measurement.

(6) F= 450.0 * the distance from the reference datum to the center line of the nose or tail reaction.
F = E − D (for nose reaction)
F = E + D (for tail reaction)

(7) H= 208.9 ARM AS WEIGHED (3.5° NOSE-UP) (13) CORRECTED ARM 206.0 (FOR LEVEL ATTITUDE)

TAIL REACTION



DIAGRAMS FOR MEASURING VARIOUS TYPES OF REACTIONS TO DETERMINE ARM OF SUPPORT POINTS.

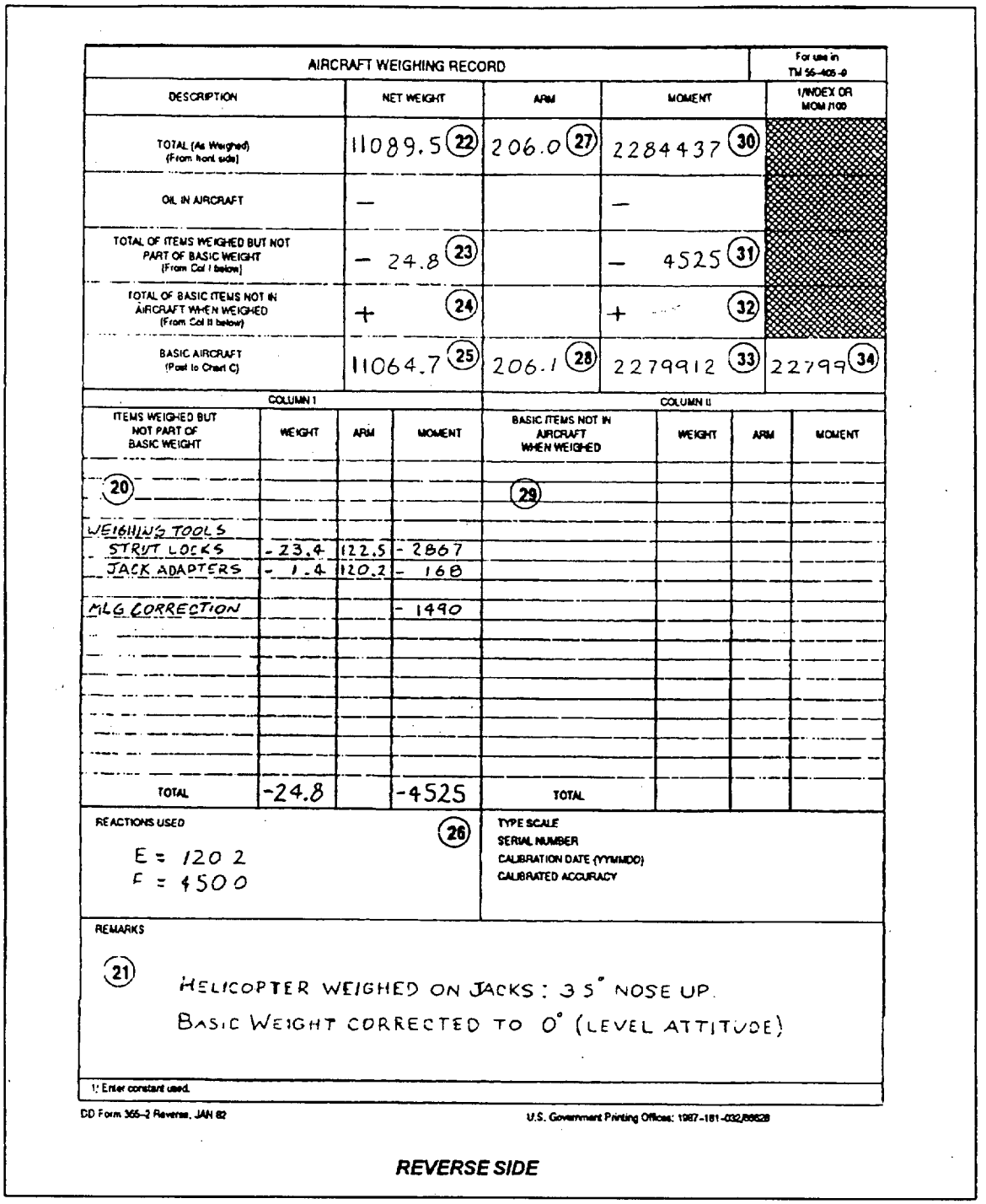
*Check dimensions E and F against approximate dimensions listed on Chart E.

DD FORM 82 JAN 365-2 REPLACES DD FORM 365B, SEP, WHICH WILL BE USED

FRONT SIDE

FIGURE 21. Example of an aircraft weighing record DD Form 365-2

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AIRCRAFT WEIGHING RECORD

For use in TM 55-405-9

DESCRIPTION	NET WEIGHT	ARM	MOMENT	I/INDEX OR MOM /100
TOTAL (As Weighed) (From front side)	11089.5 (22)	206.0 (27)	2284437 (30)	
OIL IN AIRCRAFT	—		—	
TOTAL OF ITEMS WEIGHED BUT NOT PART OF BASIC WEIGHT (From Col I below)	− 24.8 (23)		− 4525 (31)	
TOTAL OF BASIC ITEMS NOT IN AIRCRAFT WHEN WEIGHED (From Col II below)	+ (24)		+ (32)	
BASIC AIRCRAFT (Post to Chart C)	11064.7 (25)	206.1 (28)	2279912 (33)	22799 (34)

COLUMN I				COLUMN II			
ITEMS WEIGHED BUT NOT PART OF BASIC WEIGHT	WEIGHT	ARM	MOMENT	BASIC ITEMS NOT IN AIRCRAFT WHEN WEIGHED	WEIGHT	ARM	MOMENT
(20)				(29)			
WEIGHING TOOLS							
STRUT LOCKS	-23.4	122.5	-2867				
JACK ADAPTERS	- 1.4	120.2	- 168				
MLG CORRECTION			-1490				
TOTAL	-24.8		-4525	TOTAL			

REACTIONS USED (26)

E = 1202
F = 4500

TYPE SCALE
SERIAL NUMBER
CALIBRATION DATE (YYMMDD)
CALIBRATED ACCURACY

REMARKS

(21) HELICOPTER WEIGHED ON JACKS: 3.5° NOSE UP.
BASIC WEIGHT CORRECTED TO 0° (LEVEL ATTITUDE)

1/ Enter constant used.

DD Form 365-2 Reverse, JAN 82

U.S. Government Printing Office: 1987-181-032/86628

REVERSE SIDE

FIGURE 21. Example of an aircraft weighing record DD Form 365-2 - Continued

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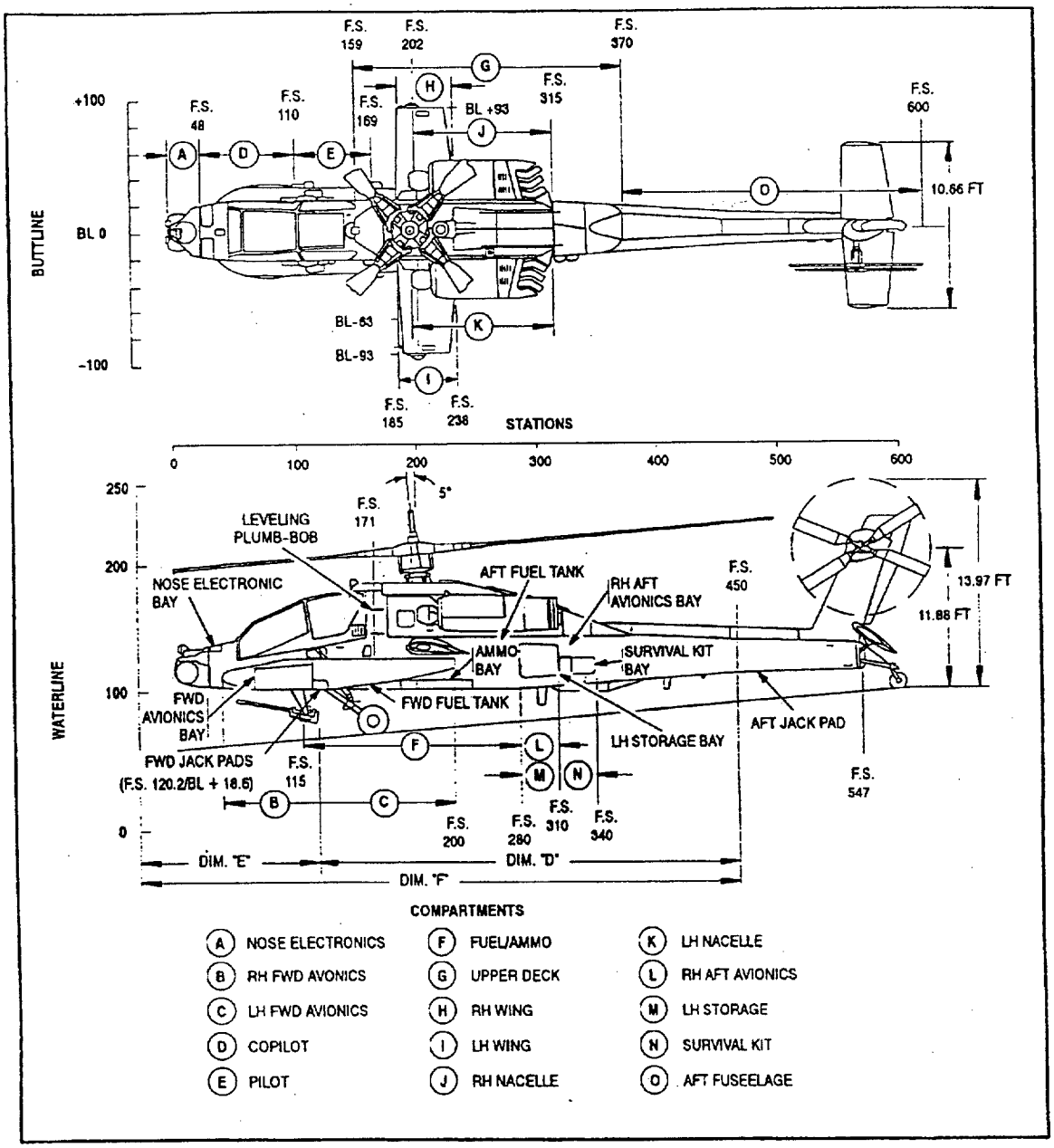


FIGURE 22. Example of a two view chart diagram

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TM X-XXXX-XXX-XX

DEPOT MOBILIZATION REQUIREMENTS **0105 00**

INTRODUCTION

Scope

The purpose of this work package is to streamline and accelerate the overhaul process during the mobilization of the depot.

Explanation of Mobilization Requirements

The mobilization requirements include a list of instructions for modifying preshop analysis and/or overhaul procedures. The pertinent procedures to be modified are referred to by page and work package number, followed by the action to be taken.

MOBILIZATION REQUIREMENTS

The following list of actions (table 1) will be in effect during depot mobilization:

Table 1. Mobilization Requirements.

WORK PACKAGE	ACTION
0088 00	Add "Depending on the urgency of requirements, availability of materials, and fabrication lead time, provisions of this work package may be relaxed. When that occurs, any practical method may be used to inscribe or attach the data to the equipment, i.e., decals."
0090 00	Add "Clean only to the extent necessary to perform preshop analysis."
0092 00	Add "Clean only to the extent necessary to inspect components."
0098 00	Add "Painted surfaces will be treated for corrosion and scratches that expose bare metal. Touch-up painting need not correlate in hue and gloss."
0098 00	Delete.
0098 00	Add "Depending on the urgency of requirements, availability of materials, and fabrication lead time, provisions of this paragraph may be relaxed. When that occurs, any practical method may be used to inscribe or attach the data to the equipment, i.e., decals."

0105 00-1

FIGURE 23. Example of a depot mobilization requirements work package

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TM X-XXXX-XXX-XX

COMPONENT CHECKLIST 0194 00

INTRODUCTION

Scope

This work package includes a list which is to be copied for each item received for a preshop analysis. After copying one list for each item, the information required must be completed on the checklist prior to the preshop analysis.

COMPONENT CHECKLIST

Name/nomenclature of the equipment/item _____.

Serial number _____.

Date received _____.

Received from (identify unit) _____.

Component name _____.

NSN _____.

Part number _____.

Quantity required _____.

Quantity received _____.

Visual damage found _____.

0194 00-1

FIGURE 24. Example of a component checklist work package

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TM X-XXXX-XXX-XX

QUALITY ASSURANCE REQUIREMENTS 0163 00

STATEMENT OF RESPONSIBILITY

The depot/contractor is responsible for complying with the quality assurance requirements contained in this work package and in accordance with DESCOM-R 702-1 (DESCOM Product Assurance Program). The commodity manager reserves the right to perform inspections or make changes that ensure the depot work being done meets the quality standards of the DMWR and preserves the inherent reliability of the item.

DEFINITIONS

For quality assurance terms and definitions, refer to the glossary in this DMWR.

SPECIAL REQUIREMENTS FOR INSPECTION TOOLS AND EQUIPMENT

The overhaul facility is responsible for acquisitions, maintenance, calibration, and disposition of all inspection and test equipment. Test equipment to be used by AMC (Army) elements will be acquired in accordance with AR 750-43. All instrumentation and inspection equipment used in compliance with this DMWR shall be calibrated and controlled in accordance with MIL-I-45607B or DESCOM-R-702-1, Depot Quality System (Army facility), with all standards traceable to the National Bureau of Standards. Descriptions of inspecting and measuring equipment are left to the discretion of the overhauling facility to be considered as good shop practice.

CERTIFICATION REQUIREMENTS

The contractor/depot QA activity shall be responsible for ascertaining and certifying personnel skills, equipment, and materiel meet the requirements of the work to be accomplished. Unless otherwise specified in the contract or by PA/CM representative, the contractor/depot QA activity shall provide the PA/CM with statements or other evidence that specifications for such special processes as welding, nondestructive testing, plating, etc., have been complied with. Personnel performing magnetic particle and penetrant tests shall be certified in accordance with MIL-STD-410.

IN-PROCESS INSPECTIONS

In-process quality assurance inspections are contained throughout the overhaul procedures of this DMWR. These inspections are prefixed with the boldfaced letters "**QA**" to identify them, and they are the minimum inspections required. Additional quality assurance inspections may be established by the depot or the commodity manager.

0163 00-1

FIGURE 25. Example of a quality assurance requirements work package

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TM X-XXXX-XXX-XX

QUALITY ASSURANCE REQUIREMENTS - Continued 0163 00

ACCEPTANCE INSPECTIONS

Items overhauled in accordance with this DMWR will be accepted based on the following criteria:

1. Conformance to quality of material requirements.
2. Conformance to all in-process quality assurance inspections.
3. Conformance to all final assembly testing requirements.
4. Conformance to the preservation, packaging, and marking requirements.

FIRST ARTICLE INSPECTION

The contractor/depot quality assurance activity shall perform a first article inspection on each of the first overhauled assemblies produced in accordance with this DMWR. After contractor/depot acceptance, the procuring agency's quality assurance representative may perform a separate first overhauled article inspection. The first article inspection shall be conducted as follows: (1) Component Inspection (Chapter 3), (2) Final Inspection (Chapter 4), and (3) Processing for Storage and Shipment (Chapter 6).

Submission of Product

The inspection of lot size, lot formation, and presentation of lots shall be as specified by the contracting activity.

Quality Assurance Component Inspection

The component inspection shall be conducted by the procuring activity's quality assurance representative during production of first overhauled item to evaluate conformance of materials and workmanship to drawings and overhaul technical data package.

Final Acceptance

The contractor's/depot's quality assurance activity, along with the procuring activity's quality assurance representative, shall conduct the final acceptance inspection in accordance with the requirements specified herein. During this time, the contractor/depot quality assurance activity shall have available the written inspection system plan, procedures, inspection records, and components.

Processing for Storage and Shipment

Inspection of processing for storage and shipment shall be conducted in accordance with applicable specifications on the first overhauled item and on one of the next ten processed items under the procedure established during the first overhauled item.

0163 00-2

FIGURE 25. Example of a quality assurance requirements work package - Continued

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TM X-XXXX-XXX-XX

WIRING DIAGRAMS **0306 00**

INTRODUCTION

Scope

This work package describes the wiring provisions contained in the aircraft, including all systems or equipment which can be installed or removed later (e.g., mission-related systems/equipment). Wiring diagrams and essential wiring information are provided for all electrical and electronic systems and circuits. All critical wire and cable data has been included.

WIRE IDENTIFICATION

All wires have been identified by number. A tabular list of circuit designators and wire identification diagrams are included. (Refer to table 1.)

ABBREVIATIONS

All abbreviations are in accordance with MIL-STD-12, except when the abbreviation stands for a marking actually found in the aircraft.

WIRING DIAGRAMS

Wiring diagrams have been included for all electrical and electronic systems and circuits.

0306 00-1

FIGURE 26. Example of a wiring diagrams work package

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Concluding Material

Custodian:
Army - TM

Preparing Activity:
Army - TM

Review Activities:
Army - AL, AR, AT, AV, CR, CU, EA, MI, PT, SC3

Project Number:
TMSS A326

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1,2,3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4,5,6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of this form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractural requirements.

I RECOMMEND A CHANGE	1.DOCUMENT NUMBER MIL-STD-40051	2.DOCUMENT DATE *(YYMMDD)* 960731

3.DOCUMENT TITLE
DOD Standard Practice, Technical Manual Preparation

4.NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5.REASON FOR RECOMMENDATION

6.SUBMITTER

a.NAME *(Last, First, Middle Initial)*	b.ORGANIZATION	
c.ADDRESS *(Include Zip Code)*	d.TELEPHONE (Include Area Code) (1)Commercial (2)AUTOVON(If applicable)	7.DATE SUBMITTED *(YYMMDD)*

8.PREPARING ACTIVITY

a.NAME USAMC Logistics Support Activity	b.TELEPHONE *(Include Area Code)* (1)Commercial (2)AUTOVON (205) 955-9860 645-9860
C. ADDRESS *(Include Zip Code)* ATTN: AMXLS-APA Redstone Arsenal, AL 35898-7466	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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